Test Run Report — Run #292

Environment: **Production** • Website: **aepc2.p2agentx.com** • Trigger: Scheduled • Started: Oct 19, 2026 19:00 • Ended: Oct 19, 2026 22:21

Generated: 10/20/2026, 1:20:22 AM

Summary

220	91.8%	202 / 18	0
Total Tests	Pass Rate (excl. skipped)	Passed / Failed	Skipped (connection)

Plant-wise Summary

10 plants

Pass rate is based on completed tests only: passed, failed, and errors. Skipped tests are excluded.

Plant	Environment	Total	Passed	Failed	Errors	Skipped	Pass r
Abdali_Mall_Barza	Production	22	21	1	0	0	95.
Abdali_Mall_Lahoun	Production	22	20	2	0	0	90.
Abdali_Mall_Netmetering	Production	22	20	2	0	0	90.
BF_Wheeling	Production	22	21	1	0	0	95.
Galleria_Mall	Production	22	17	5	0	0	77.
GID	Production	22	21	1	0	0	95.
KUPCO_HQ_DAJEEJ	Production	22	22	0	0	0	100.
MOF	Production	22	20	2	0	0	90.
Noor_Al_aqaba	Production	22	21	1	0	0	95.
SAK_1	Production	22	19	3	0	0	86.

Failures (18)

Failure Cluster	Category	Failures
Tool Execution Failures	Tool Execution	8
LLM Reasoning Issues	Llm Reasoning	5
Data Availability Issues	Data Availability	5

Tool Execution Failures TOOL EXECUTION (8 failures)

1. MOF

Production • 24.5s

Question: Use 'clipping_loss_via_requests' in order to calculate clipping loss if any for July 2026

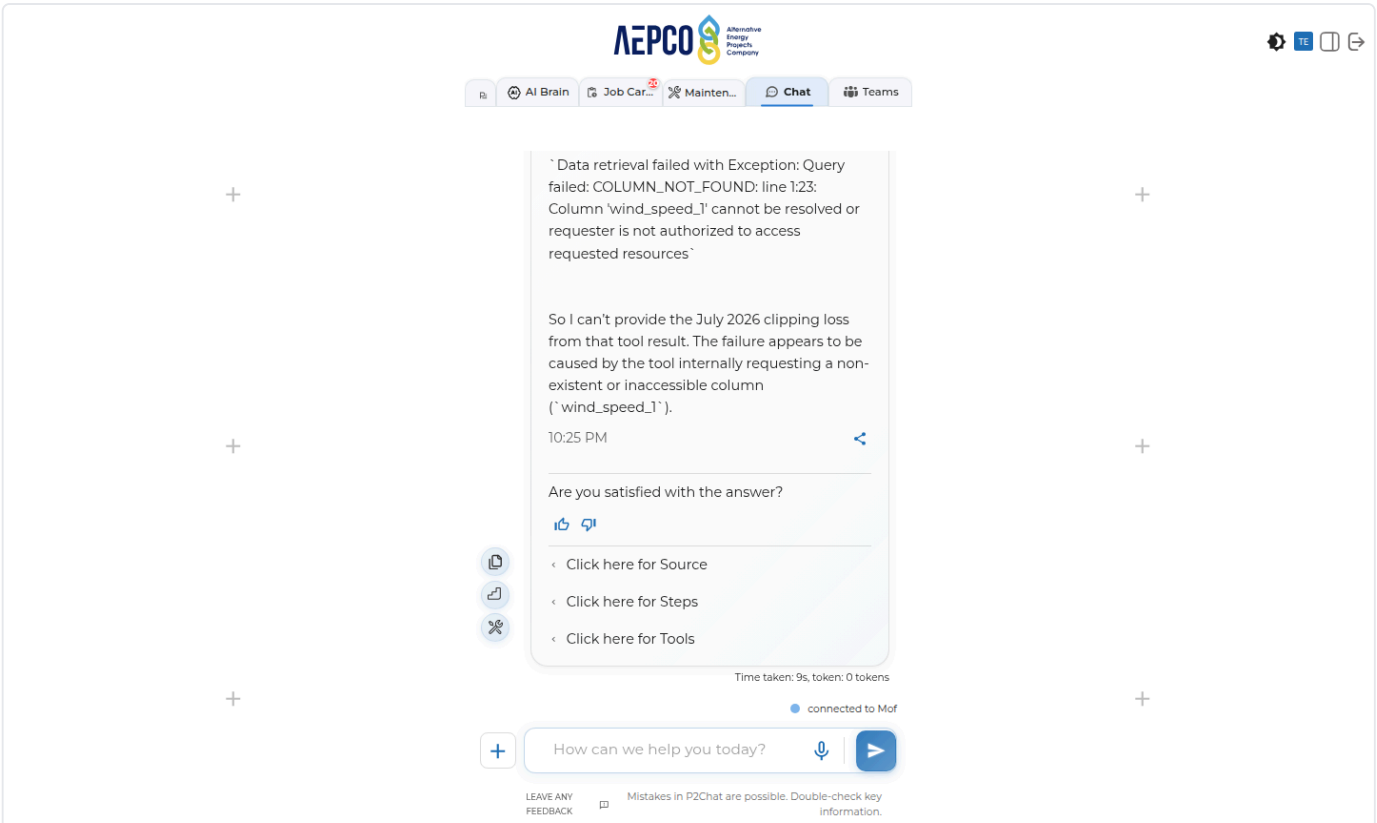
Error reason: AI indicated failure (matched 'i can't'): I used `clipping_loss_via_requests` for July 2026, but it failed and returned no clipping-loss values. Error returned by the tool: `Data retrieval failed with Exception: Query failed: COLUMN_NOT_

TECHNICAL EVIDENCE

API call: `GET /testuser2/ai_agent_chat → 200`

Raw API response:


```
{
  "result": {
    "user_input": "Use 'clipping_loss_via_requests' in order to calculate clipping loss if any for July 202
6",
    "displayed_response": "I used `clipping_loss_via_requests` for July 2026, but it failed and returned no
clipping-loss values.\n\nError returned by the tool:\n`Data retrieval failed with Exception: Query failed: C
OLUMN_NOT_FOUND: line 1:23: Column 'wind_speed_1' cannot be resolved or requester is not authorized to acces
s requested resources`\n\nSo I can’t provide the July 2026 clipping loss from that tool result. The failure
appears to be caused by the tool internally requesting a non-existent or inaccessible column (`wind_speed_1
`).",
    "coding_response": [
      "I used `clipping_loss_via_requests` for July 2026, but it failed and returned no clipping-loss value
s.\n\nError returned by the tool:\n`Data retrieval failed with Exception: Query failed: COLUMN_NOT_FOUND: li
ne 1:23: Column 'wind_speed_1' cannot be resolved or requester is not authorized to access requested resourc
es`\n\nSo I can’t provide the July 2026 clipping loss from that tool result. The failure appears to be cause
d by the tool internally requesting a non-existent or inaccessible column (`wind_speed_1`)."
    ],
    "used_agents": [
      "data_agent"
    ],
    "used_tools": [
      "functions.clipping_loss_via_requests"
    ],
    "urls": [],
    "steps": []
  }
}
```


AEPCO
Alternative Energy Projects Company
AI Brain
Job Car...
Mainten...
Chat
Teams
` Data retrieval failed with Exception: Query failed: COLUMN_NOT_FOUND: line 1:23: Column 'wind_speed_1' cannot be resolved or requester is not authorized to access requested resources `
So I can't provide the July 2026 clipping loss from that tool result. The failure appears to be caused by the tool internally requesting a non-existent or inaccessible column (` wind_speed_1 `).
10:25 PM
Are you satisfied with the answer?
Click here for Source
Click here for Steps
Click here for Tools
Time taken: 9s, token: 0 tokens
connected to Mof
How can we help you today?
LEAVE ANY FEEDBACK
Mistakes in P2Chat are possible. Double-check key information.

2. MOF

Production • 33.7s

Question: How did my plant perform yesterday?

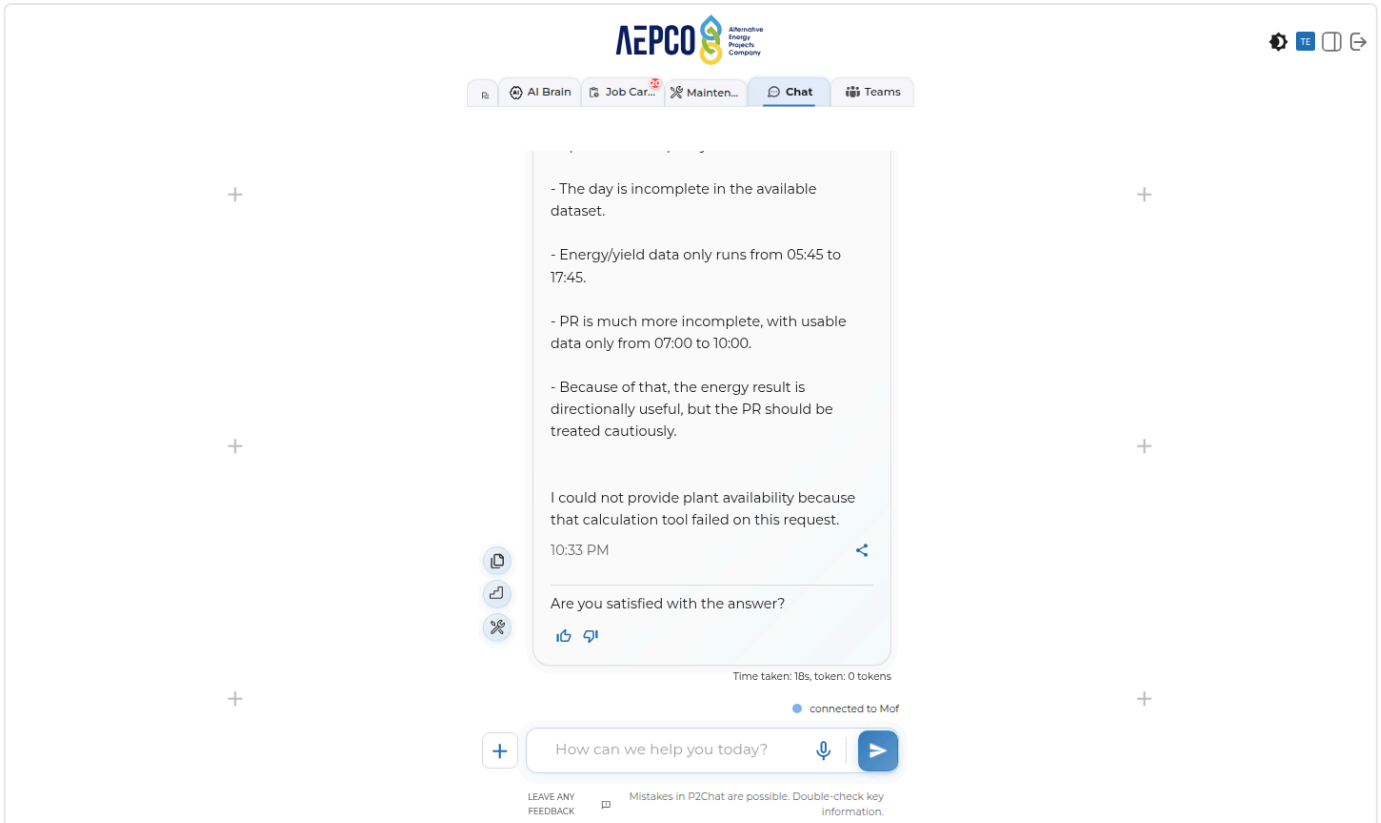
Error reason: AI indicated failure (matched 'tool failed'): Yesterday (2026-10-18), your plant produced 1,461.27 kWh. Key performance: - Specific yield: 2.98 kWh/kWp - Performance ratio: 0.822 (82.24%) - PVsyst expected energy: 1,690.0 kWh - Versus

TECHNICAL EVIDENCE

API call: `GET /testuser2/ai_agent_chat → 200`

Raw API response:


```
{
  "result": {
    "user_input": "How did my plant perform yesterday?",
    "displayed_response": "Yesterday (2026-10-18), your plant produced 1,461.27 kWh.\n\nKey performance:\n- Specific yield: 2.98 kWh/kWp\n- Performance ratio: 0.822 (82.24%)\n- PVsyst expected energy: 1,690.0 kWh\n- Versus PVsyst expectation: -228.73 kWh (-13.5%)\n\nImportant data-quality note:\n- The day is incomplete in the available dataset.\n- Energy/yield data only runs from 05:45 to 17:45.\n- PR is much more incomplete, with usable data only from 07:00 to 10:00.\n- Because of that, the energy result is directionally useful, but the PR should be treated cautiously.\n\nI could not provide plant availability because that calculation tool failed on this request.",
    "coding_response": [
      "Yesterday (2026-10-18), your plant produced 1,461.27 kWh.\n\nKey performance:\n- Specific yield: 2.98 kWh/kWp\n- Performance ratio: 0.822 (82.24%)\n- PVsyst expected energy: 1,690.0 kWh\n- Versus PVsyst expectation: -228.73 kWh (-13.5%)\n\nImportant data-quality note:\n- The day is incomplete in the available dataset.\n- Energy/yield data only runs from 05:45 to 17:45.\n- PR is much more incomplete, with usable data only from 07:00 to 10:00.\n- Because of that, the energy result is directionally useful, but the PR should be treated cautiously.\n\nI could not provide plant availability because that calculation tool failed on this request."
    ],
    "used_agents": [
      "data_agent"
    ],
    "used_tools": [
      "multi_tool_use.parallel"
    ],
    "urls": [],
    "steps": "1. Choose data from October 18, 2026 (including that day).\n2. Use the \"active_power\" column for calculations.\n3. Clean the data by removing or marking any invalid values in the power column.\n4. For each data point, calculate energy as (Power × 0.25 hours), then convert units if needed (e.g., divide by 1000 to get kWh from Watts).\n5. Make sure all days in the selected range are included.\n6. Add up all the energy values for the chosen period to get the total energy.\n\n---\n\n1. Choose data from October 18, 2026 (including that day).\n2. Find all inverter power columns and their DC rated power values from the plant information.\n3. Set calculation rules, such as the irradiance threshold (usually 50 W/m²), power threshold (usually 0.1 kW), and data interval (typically 5 minutes).\n4. For each inverter, filter the data to only include times when irradiance is above the threshold, then count the number of useful and downtime intervals.\n5. Convert these counts into hours based on the data interval.\n6. Calculate Technical Availability for each inverter using: (Useful Time - Downtime) / Useful Time × 100.\n7. Return a dictionary showing the availability percentage for each inverter.\n\n---\n\n1. Choose data from October 18, 2026 (including that day).\n2. Group the data by day (or by minute, hour, month, or year as needed).\n3. Make sure all days are included in the calculation.\n4. Calculate the total energy produced during the selected period using the plant's power data and the 0.25-hour interval.\n5. Divide the total energy by the plant's installed capacity (rated power) to get the specific yield in kWh per kWp.\n6. Return the specific yield for each time interval as a detailed breakdown.\n\n---\n\n1. Choose data from October 18, 2026 (including that day).\n2. Identify the columns for power, irradiance, and temperature.\n3. Clean the data by removing or marking invalid values in the power and irradiance columns.\n4. Calculate the actual energy produced by multiplying power by the 0.25-hour interval and adjusting units as needed.\n5. Calculate the expected energy without considering temperature effects.\n6. If the system is bifacial, apply extra corrections using metadata like ground albedo, area normalization, and shading losses.\n7. Group both actual and expected energy by day.\n8. Calculate the performance ratio for each interval, including bifacial corrections if needed.\n9. Retur
```


AEPCO
Alternative Energy Projects Company
AI Brain
Job Car...
Mainten...
Chat
Teams
- The day is incomplete in the available dataset.
- Energy/yield data only runs from 05:45 to 17:45.
- PR is much more incomplete, with usable data only from 07:00 to 10:00.
- Because of that, the energy result is directionally useful, but the PR should be treated cautiously.
I could not provide plant availability because that calculation tool failed on this request.
10:33 PM
Are you satisfied with the answer?
Time taken: 18s, token: 0 tokens
connected to Mof
How can we help you today?
LEAVE ANY FEEDBACK
Mistakes in P2Chat are possible. Double-check key information.

3. Galleria_Mall

Production • 24.6s

Question: Use 'clipping_loss_via_requests' in order to calculate clipping loss if any for July 2026

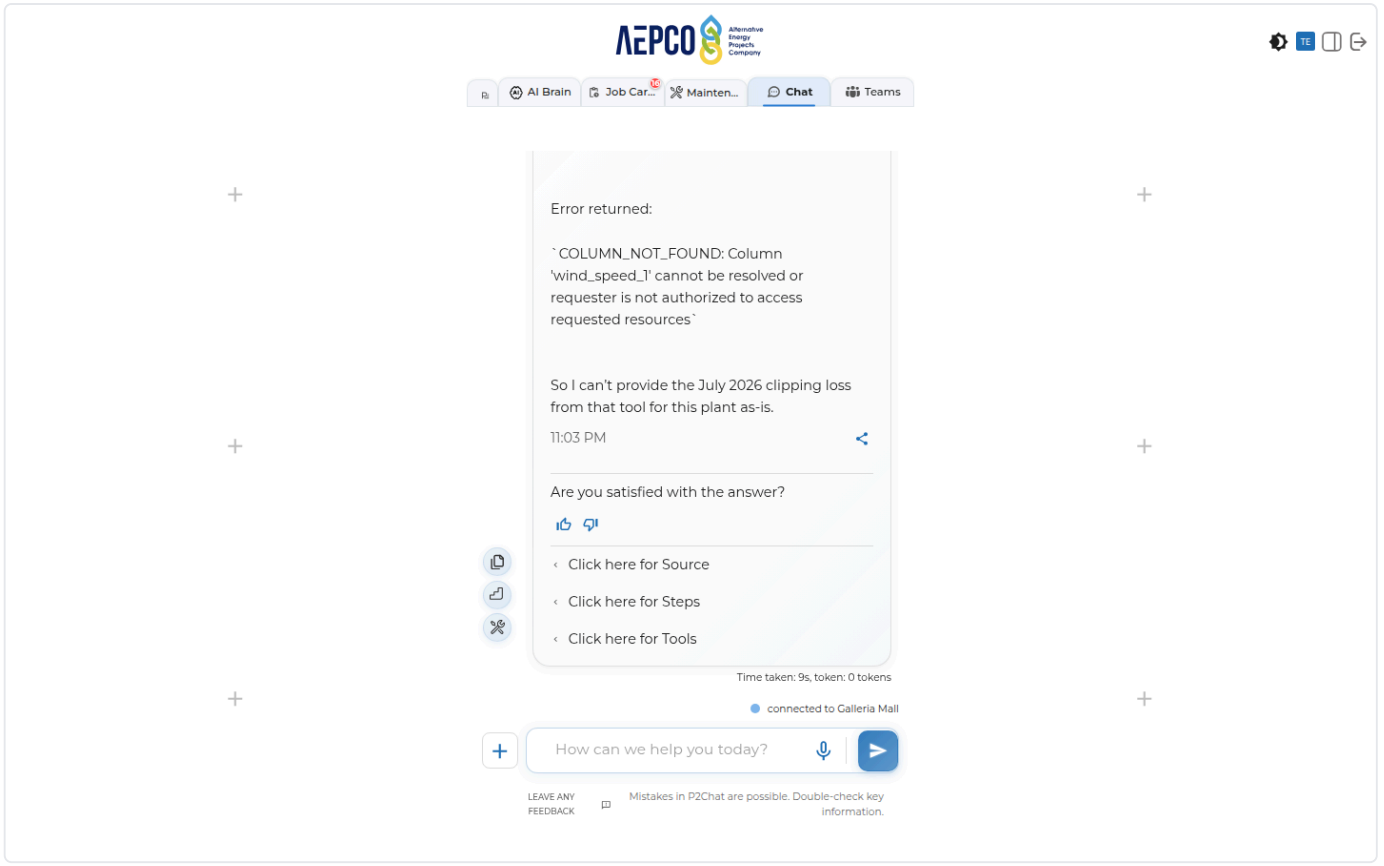
Error reason: AI indicated failure (matched 'i can't'): I used `clipping_loss_via_requests` for 2026-07-01 to 2026-07-31, but it failed internally and did not return clipping loss values. Error returned: `COLUMN_NOT_FOUND: Column 'wind_speed_1' cannot

TECHNICAL EVIDENCE

API call: `GET /testuser2/ai_agent_chat → 200`

Raw API response:


```
{
  "result": {
    "user_input": "Use 'clipping_loss_via_requests' in order to calculate clipping loss if any for July 2026",
    "displayed_response": "I used `clipping_loss_via_requests` for 2026-07-01 to 2026-07-31, but it failed internally and did not return clipping loss values.\n\nError returned:\n`COLUMN_NOT_FOUND: Column 'wind_speed_1' cannot be resolved or requester is not authorized to access requested resources`\n\nSo I can’t provide the July 2026 clipping loss from that tool for this plant as-is.",
    "coding_response": [
      "I used `clipping_loss_via_requests` for 2026-07-01 to 2026-07-31, but it failed internally and did not return clipping loss values.\n\nError returned:\n`COLUMN_NOT_FOUND: Column 'wind_speed_1' cannot be resolved or requester is not authorized to access requested resources`\n\nSo I can’t provide the July 2026 clipping loss from that tool for this plant as-is."
    ],
    "used_agents": [
      "data_agent"
    ],
    "used_tools": [
      "functions.clipping_loss_via_requests"
    ],
    "urls": [],
    "steps": []
  }
}
```


AEPCO
Alternative Energy Projects Company
AI Brain
Job Car...
Mainten...
Chat
Teams
Error returned:
`COLUMN_NOT_FOUND: Column 'wind_speed_1' cannot be resolved or requester is not authorized to access requested resources`
So I can't provide the July 2026 clipping loss from that tool for this plant as-is.
11:03 PM
Are you satisfied with the answer?
Click here for Source
Click here for Steps
Click here for Tools
Time taken: 9s, token: 0 tokens
connected to Galleria Mall
How can we help you today?
LEAVE ANY FEEDBACK
Mistakes in P2Chat are possible. Double-check key information.

4. Abdali_Mall_Netmetering

Production • 30.8s

Question: How did my plant perform yesterday?

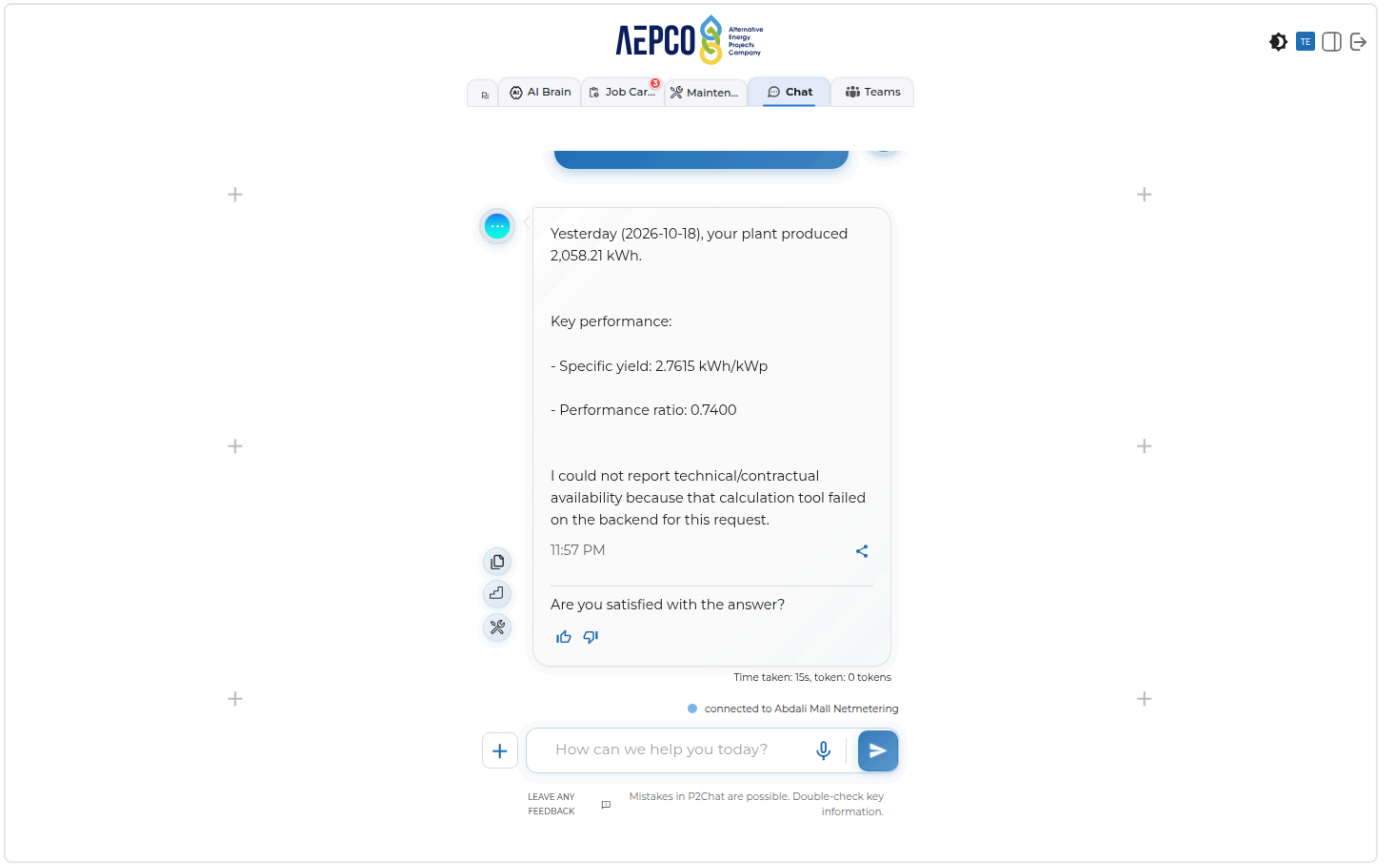
Error reason: AI indicated failure (matched 'tool failed'): Yesterday (2026-10-18), your plant produced 2,058.21 kWh. Key performance: - Specific yield: 2.7615 kWh/kWp - Performance ratio: 0.7400 I could not report technical/contractual availability

TECHNICAL EVIDENCE

API call: `GET /testuser2/ai_agent_chat → 200`

Raw API response:


```
{
  "result": {
    "user_input": "How did my plant perform yesterday?",
    "displayed_response": "Yesterday (2026-10-18), your plant produced 2,058.21 kWh.\n\nKey performance:\n- Specific yield: 2.7615 kWh/kWp\n- Performance ratio: 0.7400\n\nI could not report technical/contractual availability because that calculation tool failed on the backend for this request.",
    "coding_response": [
      "Yesterday (2026-10-18), your plant produced 2,058.21 kWh.\n\nKey performance:\n- Specific yield: 2.7615 kWh/kWp\n- Performance ratio: 0.7400\n\nI could not report technical/contractual availability because that calculation tool failed on the backend for this request."
    ],
    "used_agents": [
      "data_agent"
    ],
    "used_tools": [
      "multi_tool_use.parallel"
    ],
    "urls": [],
    "steps": "1. I choose data from October 18, 2026, including that day.\n2. I find all inverter power columns and their DC rated capacities using the plant's metadata.\n3. I set calculation settings like the irradiance threshold (usually 50 W/m²), power threshold (usually 0.1 kW), and the data interval (typically 5 minutes).\n4. For each inverter, I filter the data to times when irradiance is above the threshold, then count the total useful intervals (T_useful) and the downtime intervals (T_down).\n5. I convert these interval counts into hours based on the data interval length.\n6. I calculate each inverter's Technical Availability using: (T_useful - T_down) / T_useful × 100.\n7. I provide a dictionary showing the availability percentage for each inverter.\n\n---\n\n1. I select data from October 18, 2026, including that day.\n2. I group the data by day (or by minute, hour, month, or year if needed).\n3. I make sure all days are included in the calculation.\n4. I calculate the total energy produced during the selected period using the plant's power data, adjusting for the 5-minute interval (0.0833 hours).\n5. I divide the total energy by the plant's installed capacity (rated power) to get the specific yield in kWh per kWp.\n6. I return the specific yield for each time period as a detailed breakdown.\n\n---\n\n1. I get your question about a specific plant detail, like 'power (kW DC)'.\n2. I open the plant's metadata dictionary, which has all the system information.\n3. I look in the 'System' section of the metadata for key plant details.\n4. I search for the 'power (kW DC)' field within the 'System' section.\n\n---\n\n1. I select data from October 18, 2026, including that day.\n2. I identify the columns for power, irradiance, and temperature.\n3. I clean the data by removing or flagging invalid values in the power and irradiance columns.\n4. I calculate the actual energy produced by multiplying power by the time interval (0.0833 hours) and adjusting for units.\n5. I calculate the expected energy without temperature correction.\n6. If the system is bifacial, I apply extra corrections using metadata like ground albedo, area normalization, and shading losses.\n7. I group the actual and expected energy by day.\n8. I calculate the performance ratio for each day, applying bifacial corrections if needed.\n9. I return the average performance ratio across all days.\n10. I cap the performance ratio at 1.3 to avoid unrealistic results.\n\n---\n\n1. I select data from October 18, 2026, including that day.\n2. I use the active_power column(s) for the calculation.\n3. I clean the data by removing or flagging invalid values in the power column(s).\n4. I calculate energy for each data point as (Power × 0.0833 hours), converting units if needed (e.g., dividing by 1000 to go from Watts to kW).\n5. I include all days in the calculation.\n6. I sum up all the energy values for the selected period to get the total energy produced."
  }
}
```


AEPCO
Alternative Energy Projects Company
AI Brain
Job Car...
Mainten...
Chat
Teams
Yesterday (2026-10-18), your plant produced 2,058.21 kWh.
Key performance:
- Specific yield: 2.7615 kWh/kWp
- Performance ratio: 0.7400
I could not report technical/contractual availability because that calculation tool failed on the backend for this request.
11:57 PM
Are you satisfied with the answer?
Time taken: 15s, token: 0 tokens
connected to Abdali Mall Netmetering
How can we help you today?
LEAVE ANY FEEDBACK
Mistakes in P2Chat are possible. Double-check key information.

5. BF_Wheeling

Production • 34.2s

Question: How did my plant perform yesterday?

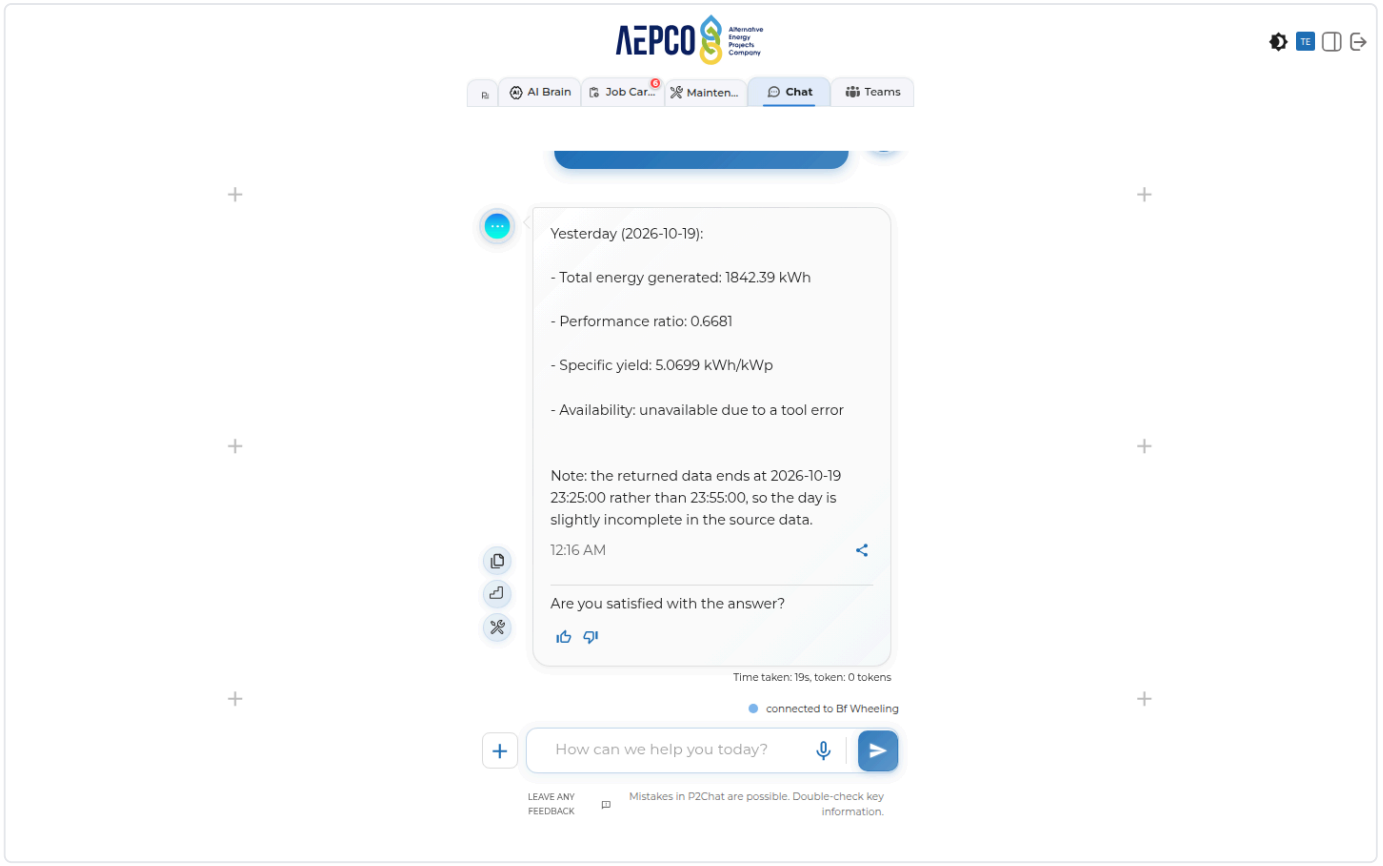
Error reason: AI indicated failure (matched 'tool error'): Yesterday (2026-10-19): - Total energy generated: 1842.39 kWh - Performance ratio: 0.6681 - Specific yield: 5.0699 kWh/kWp - Availability: unavailable due to a tool error Note: the returned

TECHNICAL EVIDENCE

API call: `GET /testuser2/ai_agent_chat → 200`

Raw API response:


```
{
  "result": {
    "user_input": "How did my plant perform yesterday?",
    "displayed_response": "Yesterday (2026-10-19):\n- Total energy generated: 1842.39 kWh\n- Performance rat
io: 0.6681\n- Specific yield: 5.0699 kWh/kWp\n- Availability: unavailable due to a tool error\n\nNote: the r
eturned data ends at 2026-10-19 23:25:00 rather than 23:55:00, so the day is slightly incomplete in the sour
ce data.",
    "coding_response": [
      "Yesterday (2026-10-19):\n- Total energy generated: 1842.39 kWh\n- Performance ratio: 0.6681\n- Specif
ic yield: 5.0699 kWh/kWp\n- Availability: unavailable due to a tool error\n\nNote: the returned data ends at
2026-10-19 23:25:00 rather than 23:55:00, so the day is slightly incomplete in the source data."
    ],
    "used_agents": [
      "data_agent"
    ],
    "used_tools": [
      "functions.energy_based_avaiability_ag",
      "functions.technical_contractual_availability_ag",
      "multi_tool_use.parallel"
    ],
    "urls": [],
    "steps": "1. I filter the data to include only records from October 19, 2026 (including the end date).\n
2. I calculate the total energy produced by the plant during this period.\n3. I identify and sum up all sign
ificant energy losses during this time, excluding baseline and measured energy, using a loss classification
method.\n4. I determine the energy-based availability by dividing the delivered energy by the sum of deliver
ed and lost energy, then express this as a percentage.\n5. I provide the final energy-based availability val
ue, which will be between 0% and 100%."
  }
}
```


AEPCO
Alternative Energy Projects Company
AI Brain
Job Car...
Mainten...
Chat
Teams
Yesterday (2026-10-19):
- Total energy generated: 1842.39 kWh
- Performance ratio: 0.6681
- Specific yield: 5.0699 kWh/kWp
- Availability: unavailable due to a tool error
Note: the returned data ends at 2026-10-19 23:25:00 rather than 23:55:00, so the day is slightly incomplete in the source data.
12:16 AM
Are you satisfied with the answer?
Time taken: 19s, token: 0 tokens
connected to Bf Wheeling
How can we help you today?
LEAVE ANY FEEDBACK
Mistakes in P2Chat are possible. Double-check key information.

6. Abdali_Mall_Lahoun

Production • 39.9s

Question: How did my plant perform yesterday?

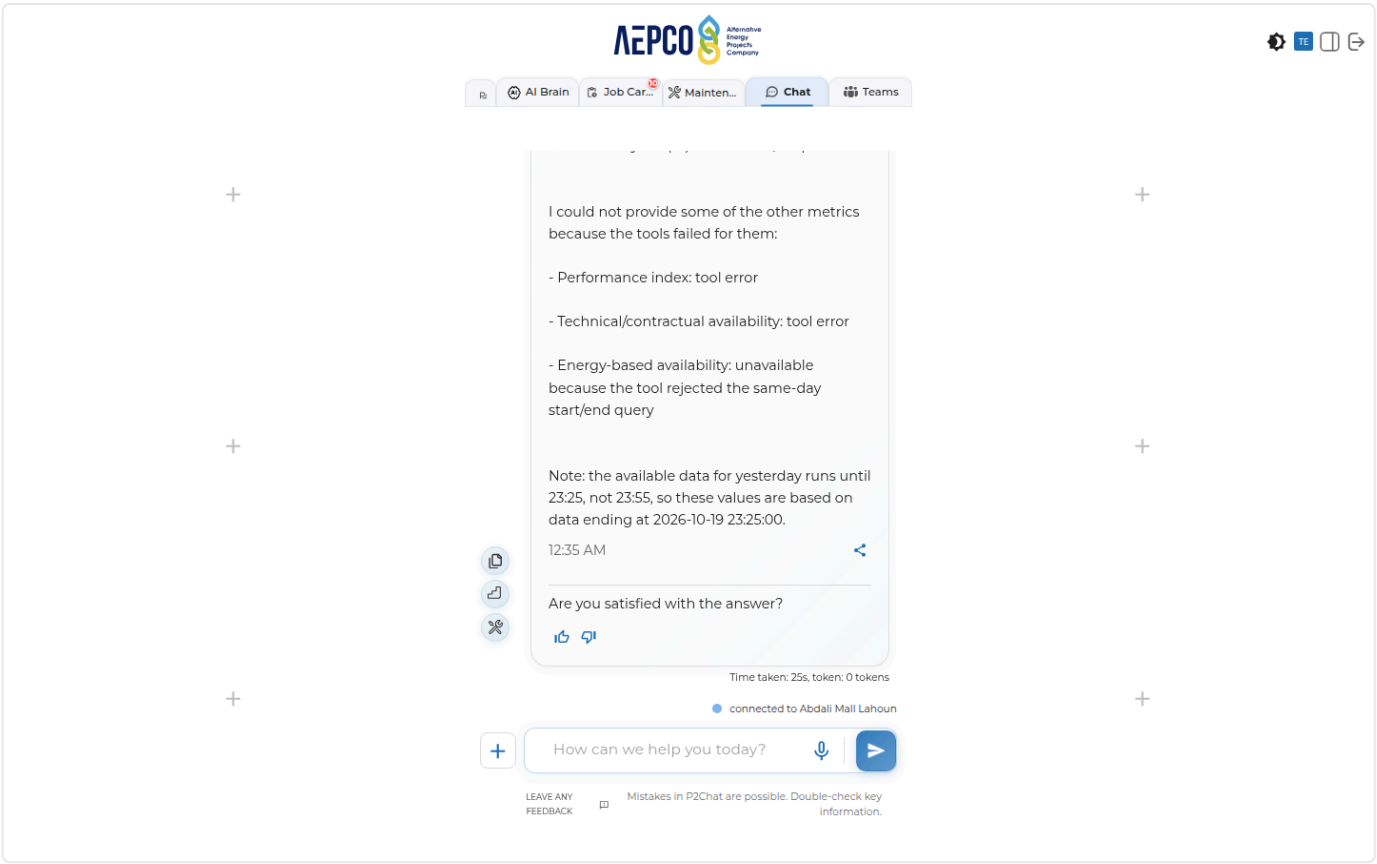
Error reason: AI indicated failure (matched 'tool error'): Yesterday (2026-10-19): - Energy generated: 4,230.37 kWh - Performance ratio (PR): 0.9049 - Specific yield: 5.3821 kWh/kWp - Reference yield (Yr): 4.5831 kWh/kWp I could not provide some of

TECHNICAL EVIDENCE

API call: `GET /testuser2/ai_agent_chat → 200`

Raw API response:


```
{
  "result": {
    "user_input": "How did my plant perform yesterday?",
    "displayed_response": "Yesterday (2026-10-19):\n- Energy generated: 4,230.37 kWh\n- Performance ratio (P
R): 0.9049\n- Specific yield: 5.3821 kWh/kWp\n- Reference yield (Yr): 4.5831 kWh/kWp\n\nI could not provide
some of the other metrics because the tools failed for them:\n- Performance index: tool error\n- Technical/c
ontractual availability: tool error\n- Energy-based availability: unavailable because the tool rejected the
same-day start/end query\n\nNote: the available data for yesterday runs until 23:25, not 23:55, so these val
ues are based on data ending at 2026-10-19 23:25:00.",
    "coding_response": [
      "Yesterday (2026-10-19):\n- Energy generated: 4,230.37 kWh\n- Performance ratio (PR): 0.9049\n- Specif
ic yield: 5.3821 kWh/kWp\n- Reference yield (Yr): 4.5831 kWh/kWp\n\nI could not provide some of the other me
trics because the tools failed for them:\n- Performance index: tool error\n- Technical/contractual availabil
ity: tool error\n- Energy-based availability: unavailable because the tool rejected the same-day start/end q
uery\n\nNote: the available data for yesterday runs until 23:25, not 23:55, so these values are based on dat
a ending at 2026-10-19 23:25:00."
    ],
    "used_agents": [
      "data_agent"
    ],
    "used_tools": [
      "multi_tool_use.parallel"
    ],
    "urls": [],
    "steps": "1. Choose all data from October 19, 2026 (including that day).\n2. Use the \"active_power\" co
lumn for calculations.\n3. Clean the data by removing or marking any invalid power readings.\n4. For each da
ta point, calculate energy by multiplying the power value by 0.0833 hours, then convert units if needed (for
example, divide by 1000 to go from Watts to kilowatts).\n5. Make sure every day in the selected range is inc
luded.\n6. Add up all the calculated energy values for the period to get the total energy produced.\n\n---\n
\n1. I pick data from October 19, 2026 (including the end date).\n2. I calculate the total energy the plant
produced during this time.\n3. I identify and sum up all significant energy losses during the period, except
for baseline and measured energy, using a loss classification method.\n4. I determine energy-based availabil
ity by dividing delivered energy by the sum of delivered and lost energy, then express it as a percentage.\n
5. I provide the energy-based availability value, which will be between 0 and 100%.\n\n---\n\n1. I select da
ta from October 19, 2026 (including the end date).\n2. I group the data by day (or by minute, hour, month, o
r year as needed).\n3. I make sure all days are included in the calculation.\n4. I calculate the total energ
y produced during the period using the plant's power data and the 0.0833-hour time interval.\n5. I divide th
e total energy by the plant's rated capacity to get the specific yield in kWh per kWp.\n6. I return the spec
ific yield for each time interval as a detailed breakdown.\n\n---\n\n1. I select data from October 19, 2026
(including the end date).\n2. I identify the columns for power, irradiance, and temperature.\n3. I clean the
data by removing or flagging invalid power and irradiance readings.\n4. I calculate the actual energy produc
ed by multiplying power by the 0.0833-hour interval and adjusting units as needed.\n5. I calculate the expec
ted energy without temperature correction.\n6. If the system is bifacial, I apply extra corrections for fact
ors like ground reflectivity, area, and shading.\n7. I group actual and expected energy by day.\n8. I calcul
ate the performance ratio for each day, including bifacial corrections if needed.\n9. I return the average p
erformance ratio across all days.\n10. I cap the performance ratio at 1.3 to avoid unrealistic values.\n\n--
-\n\n1. I select data from October 19, 2026 (including the end date).\n2. I find all inverter power columns
and their DC rated powers from the plant's metadata.\n3. I set calculation parameters like the irradiance th
reshold (
```


AEPCO
Alternative Energy Projects Company
AI Brain
Job Car...
Mainten...
Chat
Teams
I could not provide some of the other metrics because the tools failed for them:
- Performance index: tool error
- Technical/contractual availability: tool error
- Energy-based availability: unavailable because the tool rejected the same-day start/end query
Note: the available data for yesterday runs until 23:25, not 23:55, so these values are based on data ending at 2026-10-19 23:25:00.
12:35 AM
Are you satisfied with the answer?
Time taken: 25s, token: 0 tokens
connected to Abdali Mall Lahoun
How can we help you today?
LEAVE ANY FEEDBACK
Mistakes in P2Chat are possible. Double-check key information.

7. Abdali_Mall_Barza

Production • 38.4s

Question: How did my plant perform yesterday?

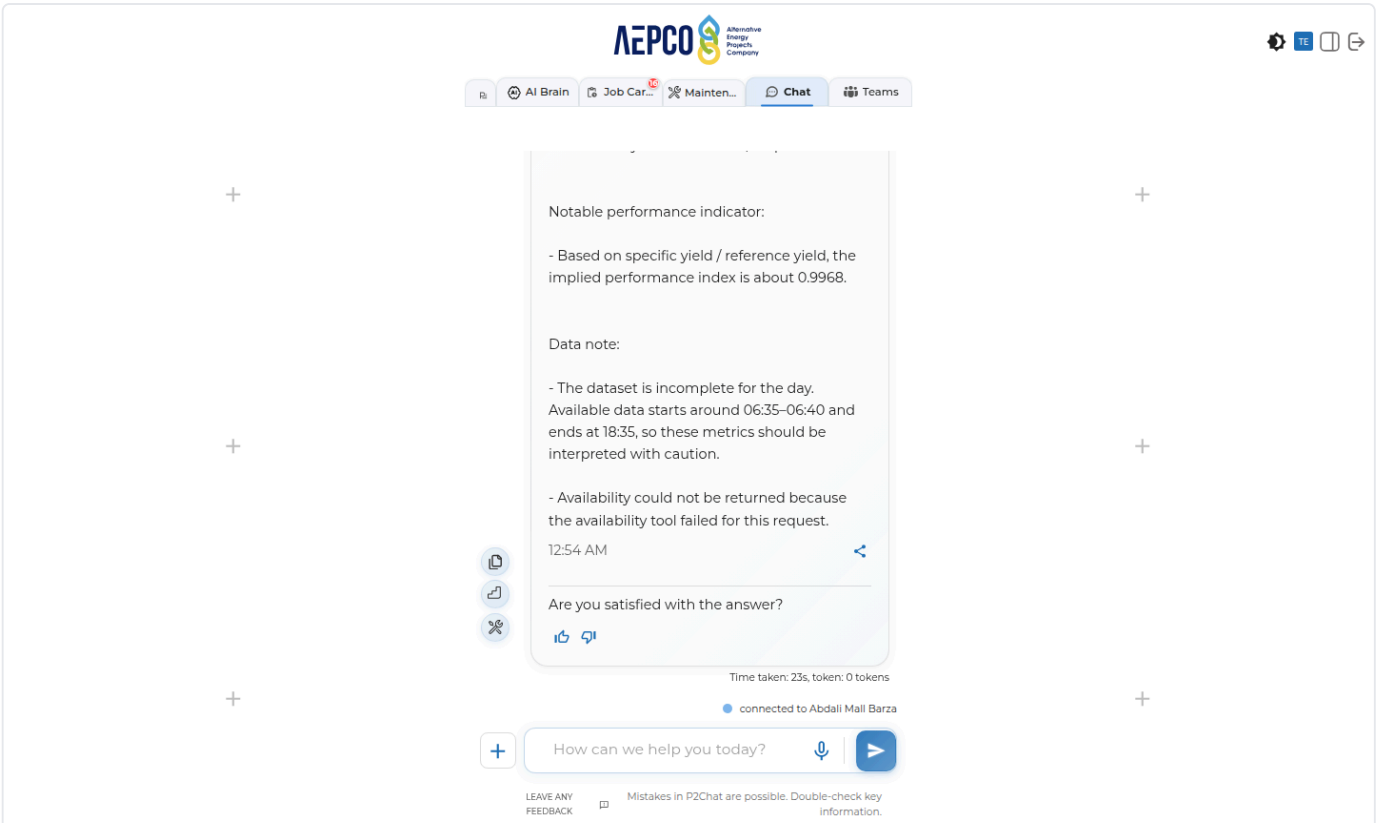
Error reason: AI indicated failure (matched 'tool failed'): Yesterday (2026-10-19): - Total energy generation: 2069.23 kWh - Performance ratio (PR): 0.8133 - Specific yield: 5.2652 kWh/kWp - Reference yield: 5.2819 kWh/kWp Notable performance indica

TECHNICAL EVIDENCE

API call: `GET /testuser2/ai_agent_chat → 200`

Raw API response:


```
{
  "result": {
    "user_input": "How did my plant perform yesterday?",
    "displayed_response": "Yesterday (2026-10-19):\n- Total energy generation: 2069.23 kWh\n- Performance ra
tio (PR): 0.8133\n- Specific yield: 5.2652 kWh/kWp\n- Reference yield: 5.2819 kWh/kWp\n\nNotable performance
indicator:\n- Based on specific yield / reference yield, the implied performance index is about 0.9968.\n\nD
ata note:\n- The dataset is incomplete for the day. Available data starts around 06:35-06:40 and ends at 18:
35, so these metrics should be interpreted with caution.\n- Availability could not be returned because the a
vailability tool failed for this request.",
    "coding_response": [
      "Yesterday (2026-10-19):\n- Total energy generation: 2069.23 kWh\n- Performance ratio (PR): 0.8133\n-
Specific yield: 5.2652 kWh/kWp\n- Reference yield: 5.2819 kWh/kWp\n\nNotable performance indicator:\n- Based
on specific yield / reference yield, the implied performance index is about 0.9968.\n\nData note:\n- The dat
aset is incomplete for the day. Available data starts around 06:35-06:40 and ends at 18:35, so these metrics
should be interpreted with caution.\n- Availability could not be returned because the availability tool fail
ed for this request."
    ],
    "used_agents": [
      "data_agent"
    ],
    "used_tools": [
      "multi_tool_use.parallel"
    ],
    "urls": [],
    "steps": "1. I pick out the data from October 19, 2026, including that day.\n2. I total up all the energ
y the solar plant produced during this time.\n3. I identify and add up all significant energy losses during
this period, except for baseline and actual measured energy, using a specific method to classify losses.\n4.
I figure out the energy-based availability by dividing the delivered energy by the sum of delivered and lost
energy, then convert this to a percentage.\n5. I provide the final energy-based availability value, which wi
ll be between 0% and 100%.\n\n---\n\n1. I select the data from October 19, 2026, including that day.\n2. I g
roup the data by day (or by minute, hour, month, or year, as needed).\n3. I make sure all days in the range
are included.\n4. I calculate the total energy produced during this period using the plant's power data, adj
usting for the time interval of 0.083 hours.\n5. I divide the total energy by the plant's installed capacity
to get the specific yield in kWh per kWp.\n6. I return the specific yield for each time period as a detailed
breakdown.\n\n---\n\n1. I select the data from October 19, 2026, including that day.\n2. I find the columns
for power, irradiance, and temperature.\n3. I clean the data by removing or marking any invalid power or irr
adiance readings.\n4. I calculate the actual energy produced by multiplying power by the time interval (0.08
3 hours) and adjusting the units.\n5. I calculate the expected energy, without adjusting for temperature.\n
6. If the system is bifacial, I apply extra corrections based on things like ground reflectivity, area, and
shading.\n7. I group the actual and expected energy by day.\n8. I calculate the performance ratio for each p
eriod, including bifacial corrections if needed.\n9. I return the average performance ratio for all period
s.\n10. I cap the performance ratio at 1.3 to avoid unrealistic results.\n\n---\n\n1. I select the data from
October 19, 2026, including that day.\n2. I use the active power column for calculations.\n3. I clean the da
ta by removing or flagging any invalid power readings.\n4. I calculate the energy for each data point by mul
tiplying power by 0.083 hours and converting units if needed.\n5. I include all days in the calculation.\n6.
I add up all the energy values for the selected period to get the total energy produced.\n\n---\n\n1. I sele
ct the data from October 19, 2026, including that day.\n2. I filter the data to only include plane of array
(POA) irradiance readings from the pyranometer.\n3. I convert irradiance (W/m²) to irradiation (kWh/m²) by m
ultiplying by the time interv
```


AEPCO
Alternative Energy Projects Company
AI Brain
Job Car...
Mainten...
Chat
Teams
Notable performance indicator:
- Based on specific yield / reference yield, the implied performance index is about 0.9968.
Data note:
- The dataset is incomplete for the day. Available data starts around 06:35–06:40 and ends at 18:35, so these metrics should be interpreted with caution.
- Availability could not be returned because the availability tool failed for this request.
12:54 AM
Are you satisfied with the answer?
Time taken: 23s, token: 0 tokens
connected to Abdali Mall Barza
How can we help you today?
LEAVE ANY FEEDBACK
Mistakes in P2Chat are possible. Double-check key information.

8. SAK_1

Production • 32.3s

Question: How did my plant perform yesterday?

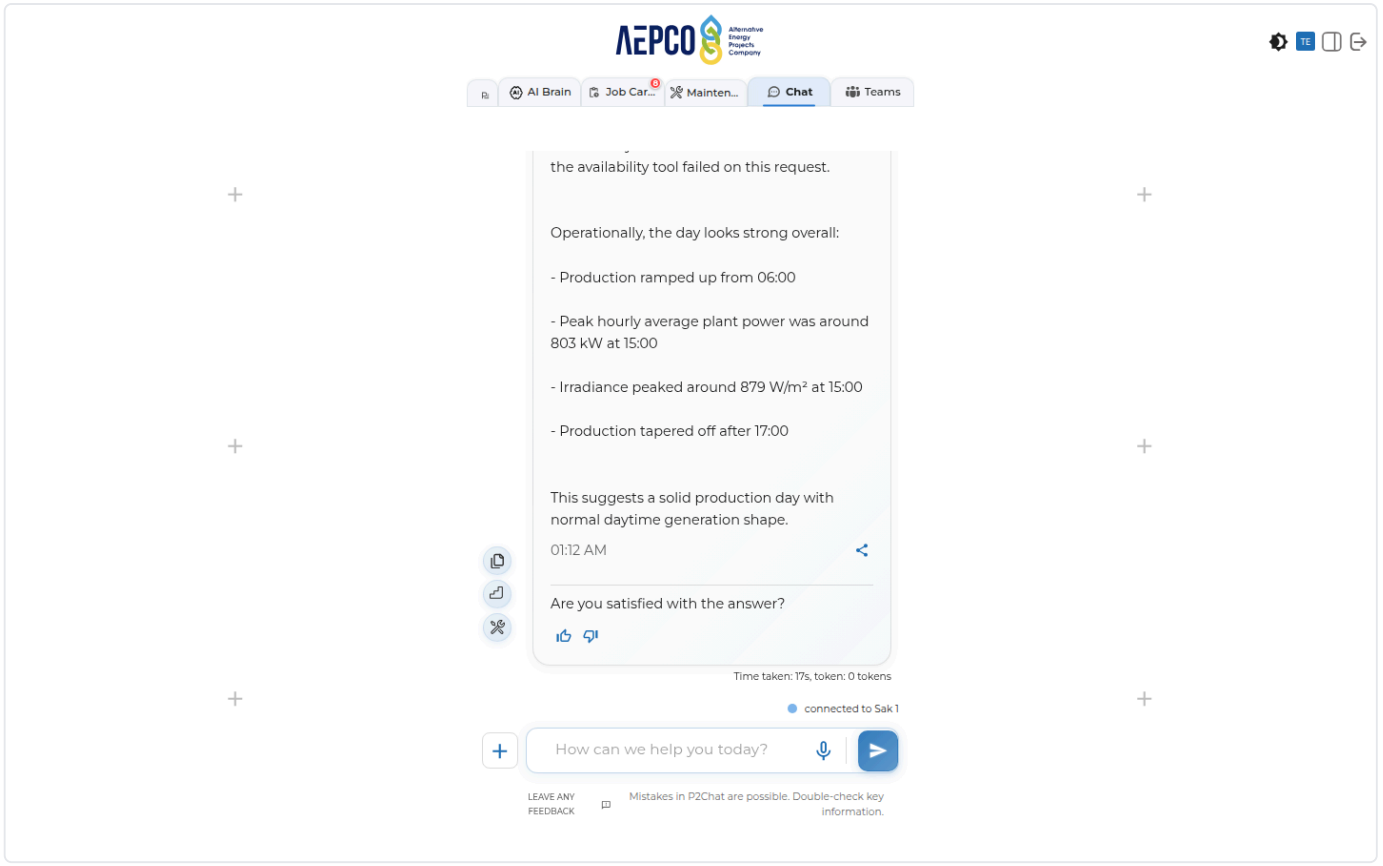
Error reason: AI indicated failure (matched 'tool failed'): Yesterday (2026-10-19), your plant performed as follows: - Total energy: 6,185.71 kWh - Specific yield: 5.91 kWh/kWp - Performance ratio (PR): 0.702 Notes: - The PR calculation used an inc

TECHNICAL EVIDENCE

API call: `GET /testuser2/ai_agent_chat → 200`

Raw API response:


```
{
  "result": {
    "user_input": "How did my plant perform yesterday?",
    "displayed_response": "Yesterday (2026-10-19), your plant performed as follows:\n\n- Total energy: 6,18
5.71 kWh\n- Specific yield: 5.91 kWh/kWp\n- Performance ratio (PR): 0.702\n\nNotes:\n- The PR calculation us
ed an incomplete daytime dataset: data starts at 06:40 and ends at 16:30, so treat the PR with some cautio
n.\n- Availability could not be returned because the availability tool failed on this request.\n\nOperationa
lly, the day looks strong overall:\n- Production ramped up from 06:00\n- Peak hourly average plant power was
around 803 kW at 15:00\n- Irradiance peaked around 879 W/m² at 15:00\n- Production tapered off after 17:00\n
\nThis suggests a solid production day with normal daytime generation shape.",
    "coding_response": [
      "Yesterday (2026-10-19), your plant performed as follows:\n\n- Total energy: 6,185.71 kWh\n- Specific
yield: 5.91 kWh/kWp\n- Performance ratio (PR): 0.702\n\nNotes:\n- The PR calculation used an incomplete dayt
ime dataset: data starts at 06:40 and ends at 16:30, so treat the PR with some caution.\n- Availability coul
d not be returned because the availability tool failed on this request.\n\nOperationally, the day looks stro
ng overall:\n- Production ramped up from 06:00\n- Peak hourly average plant power was around 803 kW at 15:00
\n- Irradiance peaked around 879 W/m² at 15:00\n- Production tapered off after 17:00\n\nThis suggests a soli
d production day with normal daytime generation shape."
    ],
    "used_agents": [
      "data_agent"
    ],
    "used_tools": [
      "multi_tool_use.parallel"
    ],
    "urls": [],
    "steps": "1. Choose data from 2026-10-19, including that date.\n2. Use the \"active_power\" column for c
alculations.\n3. Clean the data by removing or marking any invalid values in the power column.\n4. For each
data point, calculate energy by multiplying the power value by 0.0833 hours, then convert units if needed
(e.g., divide by 1000 to go from Watts to kilowatts).\n5. Make sure all days in the selected range are inclu
ded.\n6. Add up all the calculated energy values for the total energy during this period.\n\n1. Select data
from 2026-10-19, including that date.\n2. Identify the columns for power, irradiance, and temperature.\n3. C
lean the power and irradiance data by removing or flagging invalid values.\n4. Calculate the actual energy p
roduced by multiplying power by the time interval (0.0833 hours) and adjusting units as needed.\n5. Calculat
e the expected energy without considering temperature effects.\n6. If the system is bifacial, apply extra co
rrections using metadata like ground reflectivity, area, and shading losses.\n7. Group the actual and expect
ed energy values by day.\n8. Calculate the performance ratio for each day, including bifacial corrections if
needed.\n9. Return the average performance ratio across all days.\n10. Limit performance ratio values to a m
aximum of 1.3 to avoid unrealistic results.\n\n1. Select data from 2026-10-19, including that date.\n2. Grou
p the data by day (or by minute, hour, month, or year as needed).\n3. Include all days in the calculation.\n
4. Calculate the total energy produced during the selected period using the plant's power data and the 0.083
3-hour interval.\n5. Normalize the total energy by dividing by the plant's rated capacity to get the specifi
c yield in kWh per kWp.\n6. Return the specific yield for each time interval as a detailed breakdown.\n\n1.
Select data from 2026-10-19, including that date.\n2. Identify all inverter power columns and their DC rated
powers from the plant's metadata.\n3. Set calculation parameters: irradiance threshold (usually 50 W/m²), po
wer threshold (usually 0.1 kW), and data interval (typically 5 minutes).\n4. For each inverter, filter the d
ata to times when irradiance is above the threshold, then count the total useful time (T_useful) and downtim
e (T_down).\n5. Convert these counts to hours based on the data interval.\n6
```



LLM Reasoning Issues LLM REASONING (5 failures)

1. Noor_Al_aqaba

Production • 44.5s

Question: Is there shading happening with my plant? Please analyze the pyranometer and check for frequent dips at certain times

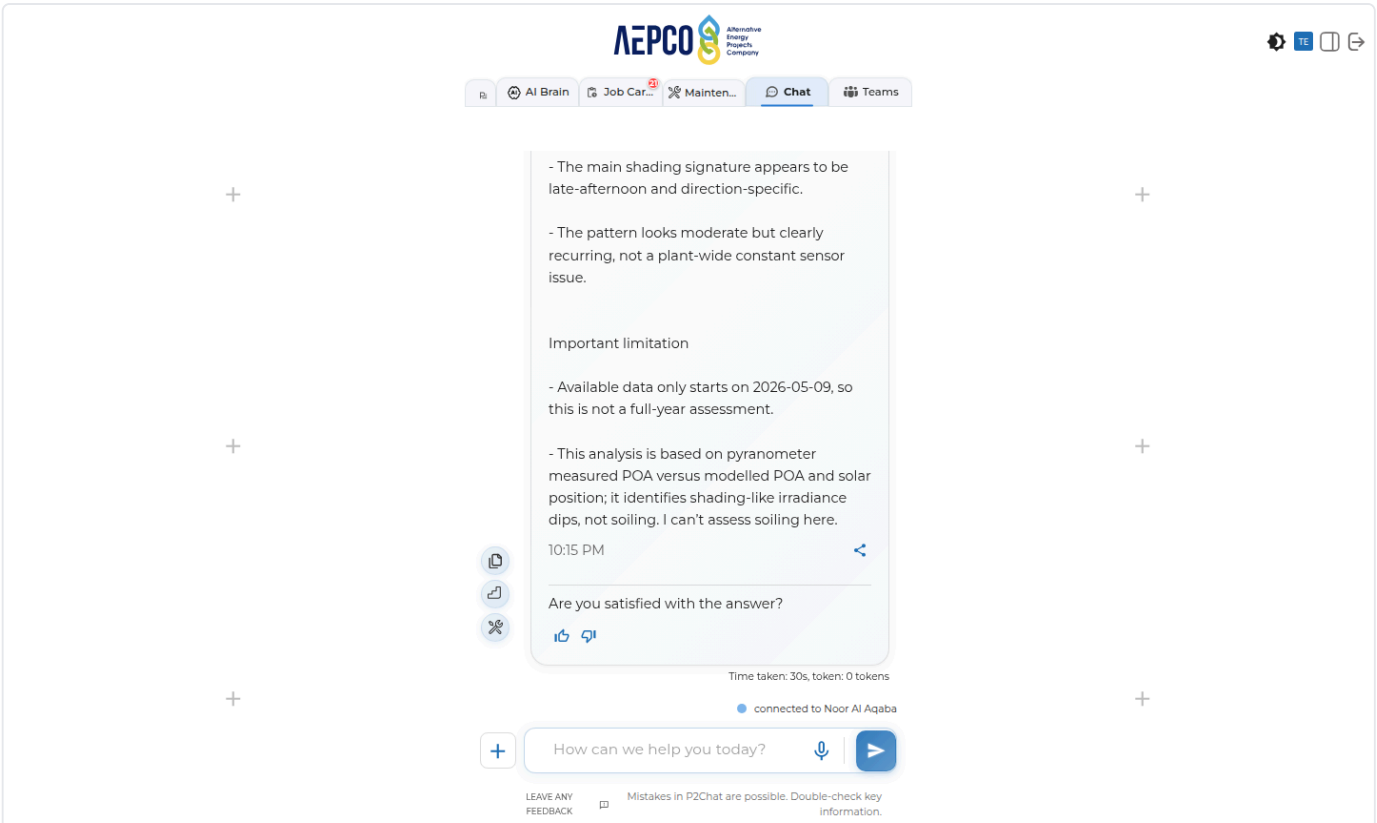
Error reason: AI indicated failure (matched 'i can't'): I found recurring pyranometer irradiance dips that are consistent with shading or repeated obstruction patterns. Key results - Data analyzed: 2026-05-09 10:45 to 2026-10-19 16:55 - Daylight 5-m

TECHNICAL EVIDENCE

API call: `GET /testuser2/ai_agent_chat → 200`

Raw API response:


```
{
  "result": {
    "user_input": "Is there shading happening with my plant? Please analyze the pyranometer and check for fr
equent dips at certain times",
    "displayed_response": "I found recurring pyranometer irradiance dips that are consistent with shading or
repeated obstruction patterns.\n\nKey results\n- Data analyzed: 2026-05-09 10:45 to 2026-10-19 16:55\n- Dayl
ight 5-minute points analyzed: 22,359\n- Overall dip share: 5.7% of analyzed daylight points\n- Median measu
red/modelled POA ratio: 1.033\n  - This means the pyranometer generally tracks the model well overall, so th
e issue is localized in time rather than a persistent whole-day bias.\n\nMost frequent dip times\nThese are
the strongest recurring dip windows, ranked by frequency:\n- 18:35 — 47/101 dips, 46.5%, median ratio 0.770
\n- 18:45 — 35/79 dips, 44.3%, median ratio 0.767\n- 18:40 — 39/89 dips, 43.8%, median ratio 0.772\n- 18:30
— 46/108 dips, 42.6%, median ratio 0.781\n- 18:25 — 45/113 dips, 39.8%, median ratio 0.782\n- 18:50 — 26/69
dips, 37.7%, median ratio 0.775\n- 18:20 — 40/117 dips, 34.2%, median ratio 0.818\n- 18:55 — 16/57 dips, 28.
1%, median ratio 0.785\n- 18:15 — 32/122 dips, 26.2%, median ratio 0.837\n\nInterpretation\n- The dip patter
n is concentrated in the late afternoon, especially around 18:15–18:55.\n- The strongest azimuth bins during
dips are around 270°–290°, which is west to west-northwest solar position.\n- There is also a smaller mornin
g-side signature around 80°–110° azimuth, suggesting a secondary east-side obstruction may also exist.\n- Be
cause the dips recur at specific times/solar positions instead of being randomly distributed, this is more c
onsistent with fixed-object shading or repeated horizon obstruction than normal passing-cloud behavior.\n\nS
easonality of dips\nFlagged dip counts by month:\n- May: 381\n- June: 116\n- July: 100\n- August: 229\n- Sep
tember: 131\n- October: 316\n\nAssessment\n- Yes, shading-like occurrences are present.\n- The main shading
signature appears to be late-afternoon and direction-specific.\n- The pattern looks moderate but clearly rec
urring, not a plant-wide constant sensor issue.\n\nImportant limitation\n- Available data only starts on 202
6-05-09, so this is not a full-year assessment.\n- This analysis is based on pyranometer measured POA versus
modelled POA and solar position; it identifies shading-like irradiance dips, not soiling. I can’t assess soi
ling here.",
    "coding_response": [
      "I found recurring pyranometer irradiance dips that are consistent with shading or repeated obstructio
n patterns.\n\nKey results\n- Data analyzed: 2026-05-09 10:45 to 2026-10-19 16:55\n- Daylight 5-minute point
s analyzed: 22,359\n- Overall dip share: 5.7% of analyzed daylight points\n- Median measured/modelled POA ra
tio: 1.033\n  - This means the pyranometer generally tracks the model well overall, so the issue is localize
d in time rather than a persistent whole-day bias.\n\nMost frequent dip times\nThese are the strongest recur
ring dip windows, ranked by frequency:\n- 18:35 — 47/101 dips, 46.5%, median ratio 0.770\n- 18:45 — 35/79 di
ps, 44.3%, median ratio 0.767\n- 18:40 — 39/89 dips, 43.8%, median ratio 0.772\n- 18:30 — 46/108 dips, 42.
6%, median ratio 0.781\n- 18:25 — 45/113 dips, 39.8%, median ratio 0.782\n- 18:50 — 26/69 dips, 37.7%, media
n ratio 0.775\n- 18:20 — 40/117 dips, 34.2%, median ratio 0.818\n- 18:55 — 16/57 dips, 28.1%, median ratio
0.785\n- 18:15 — 32/122 dips, 26.2%, median ratio 0.837\n\nInterpretation\n- The dip pattern is concentrated
in the late afternoon, especially around 18:15–18:55.\n- The strongest azimuth bins during dips are around 2
70°–290°, which is west to west-northwest solar position.\n- There is also a smaller morning-side signature
around 80°–110° azimuth, suggesting a secondary east-side obstruction may also exist.\n- Because the dips re
cur at specific times/solar positions instead of being randomly distributed, this is more consistent with fi
xed-object shading or repea
```


AEPCO
Alternative Energy Projects Company
AI Brain
Job Car...
Mainten...
Chat
Teams
- The main shading signature appears to be late-afternoon and direction-specific.
- The pattern looks moderate but clearly recurring, not a plant-wide constant sensor issue.
Important limitation
- Available data only starts on 2026-05-09, so this is not a full-year assessment.
- This analysis is based on pyranometer measured POA versus modelled POA and solar position; it identifies shading-like irradiance dips, not soiling. I can't assess soiling here.
10:15 PM
Are you satisfied with the answer?
Time taken: 30s, token: 0 tokens
connected to Noor Al Aqaba
How can we help you today?
LEAVE ANY FEEDBACK
Mistakes in P2Chat are possible. Double-check key information.

2. Galleria_Mall

Production • 30.8s

Question: Is there shading happening with my plant? Please analyze the pyranometer and check for frequent dips at certain times

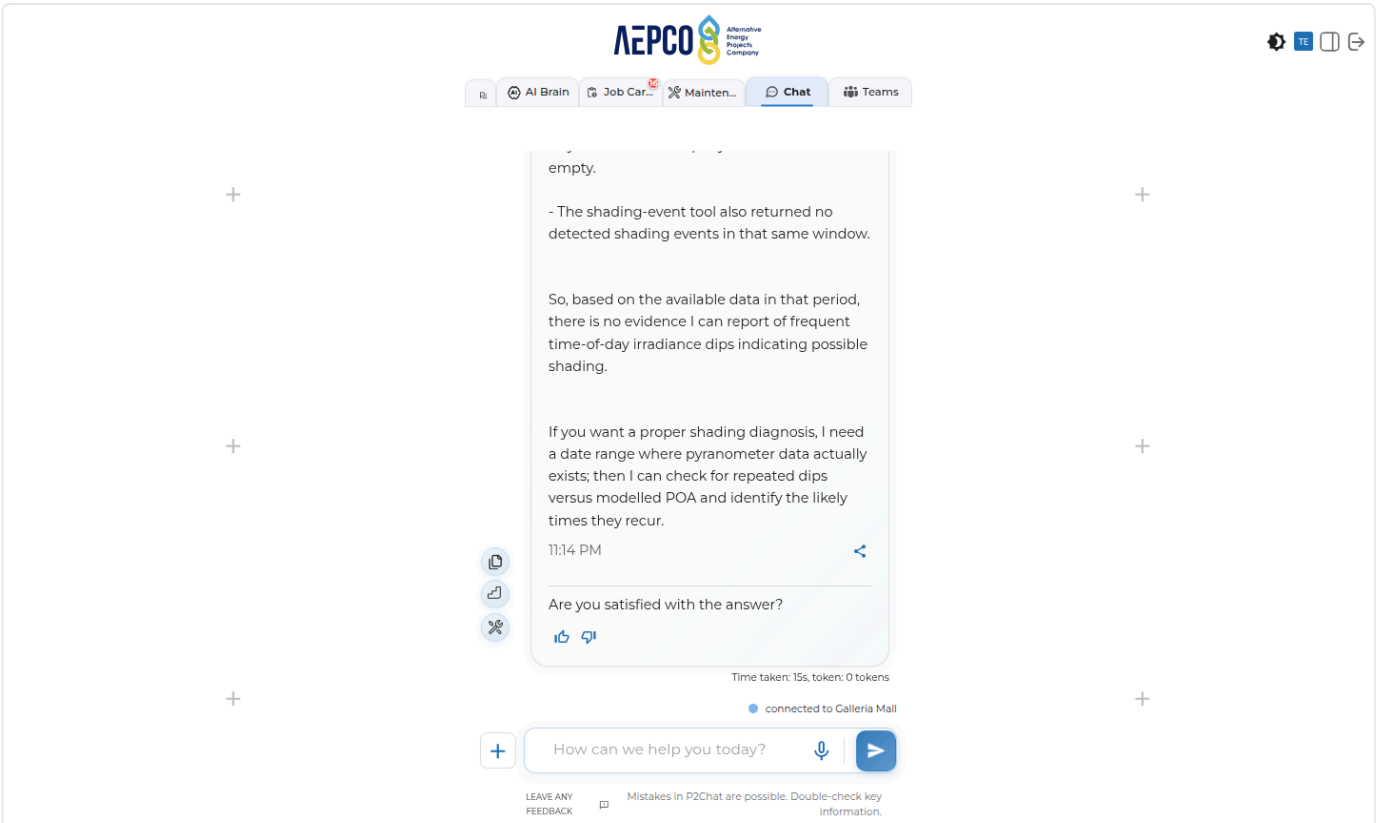
Error reason: AI indicated failure (matched 'i can't'): I checked the most recent month available for this shading-style pyranometer analysis (2026-09-19 to 2026-10-19), but the pyranometer extraction returned empty data for that period, so I can't identif

TECHNICAL EVIDENCE

API call: `GET /testuser2/ai_agent_chat → 200`

Raw API response:


```
{
  "result": {
    "user_input": "Is there shading happening with my plant? Please analyze the pyranometer and check for fr
equent dips at certain times",
    "displayed_response": "I checked the most recent month available for this shading-style pyranometer anal
ysis (2026-09-19 to 2026-10-19), but the pyranometer extraction returned empty data for that period, so I ca
n't identify recurring irradiance dips from the sensor data there.\n\nWhat I could confirm:\n- Pyranometer d
ata query for that window was empty.\n- The shading-event tool also returned no detected shading events in t
hat same window.\n\nSo, based on the available data in that period, there is no evidence I can report of fre
quent time-of-day irradiance dips indicating possible shading.\n\nIf you want a proper shading diagnosis, I
need a date range where pyranometer data actually exists; then I can check for repeated dips versus modelled
POA and identify the likely times they recur.",
    "coding_response": [
      "I checked the most recent month available for this shading-style pyranometer analysis (2026-09-19 to
2026-10-19), but the pyranometer extraction returned empty data for that period, so I can't identify recurri
ng irradiance dips from the sensor data there.\n\nWhat I could confirm:\n- Pyranometer data query for that w
indow was empty.\n- The shading-event tool also returned no detected shading events in that same window.\n\n
So, based on the available data in that period, there is no evidence I can report of frequent time-of-day ir
radiance dips indicating possible shading.\n\nIf you want a proper shading diagnosis, I need a date range wh
ere pyranometer data actually exists; then I can check for repeated dips versus modelled POA and identify th
e likely times they recur."
    ],
    "used_agents": [
      "data_agent"
    ],
    "used_tools": [
      "functions.extract_timeseries_data_ag",
      "functions.calculate_shading_loss_ag"
    ],
    "urls": [],
    "steps": "1. Pull raw string or MPPT-level data from Athena for the plant between 2026-09-19 and 2026-10
-19. Make sure to use string or MPPT data, as inverter-level data isn't detailed enough.\n\n2. Run the propr
ietary shading detection algorithm (rooftop_mppt_shading_from_raw_df) on this raw data to find shading event
s.\n\n3. Filter the results to include only rows where shading was detected (is_shaded == True), then group
by timestamp so each row represents a single shading event. For each event, include the affected strings (as
a list), the number of strings shaded, zenith, azimuth, and date.\n\n4. Skip the loss calculation step and j
ust return the DataFrame of shading events (with timestamp, affected_strings, num_strings_shaded, zenith, az
imuth, and date).\n\n5. Make the complete shading event DataFrame available as a global variable X, so it ca
n be used by other tools or processes.\n\n6. If any required data (like string-level columns, digital twin d
ata, or metadata) is missing, return a clear error message instead of incomplete or misleading results."
  }
}
```


AEPCO
Alternative Energy Projects Company
AI Brain
Job Car...
Mainten...
Chat
Teams
empty.
- The shading-event tool also returned no detected shading events in that same window.
So, based on the available data in that period, there is no evidence I can report of frequent time-of-day irradiance dips indicating possible shading.
If you want a proper shading diagnosis, I need a date range where pyranometer data actually exists; then I can check for repeated dips versus modelled POA and identify the likely times they recur.
11:14 PM
Are you satisfied with the answer?
Time taken: 15s, token: 0 tokens
connected to Galleria Mall
How can we help you today?
LEAVE ANY FEEDBACK
Mistakes in P2Chat are possible. Double-check key information.

3. GID

Production • 33.9s

Question: How did my plant perform yesterday?

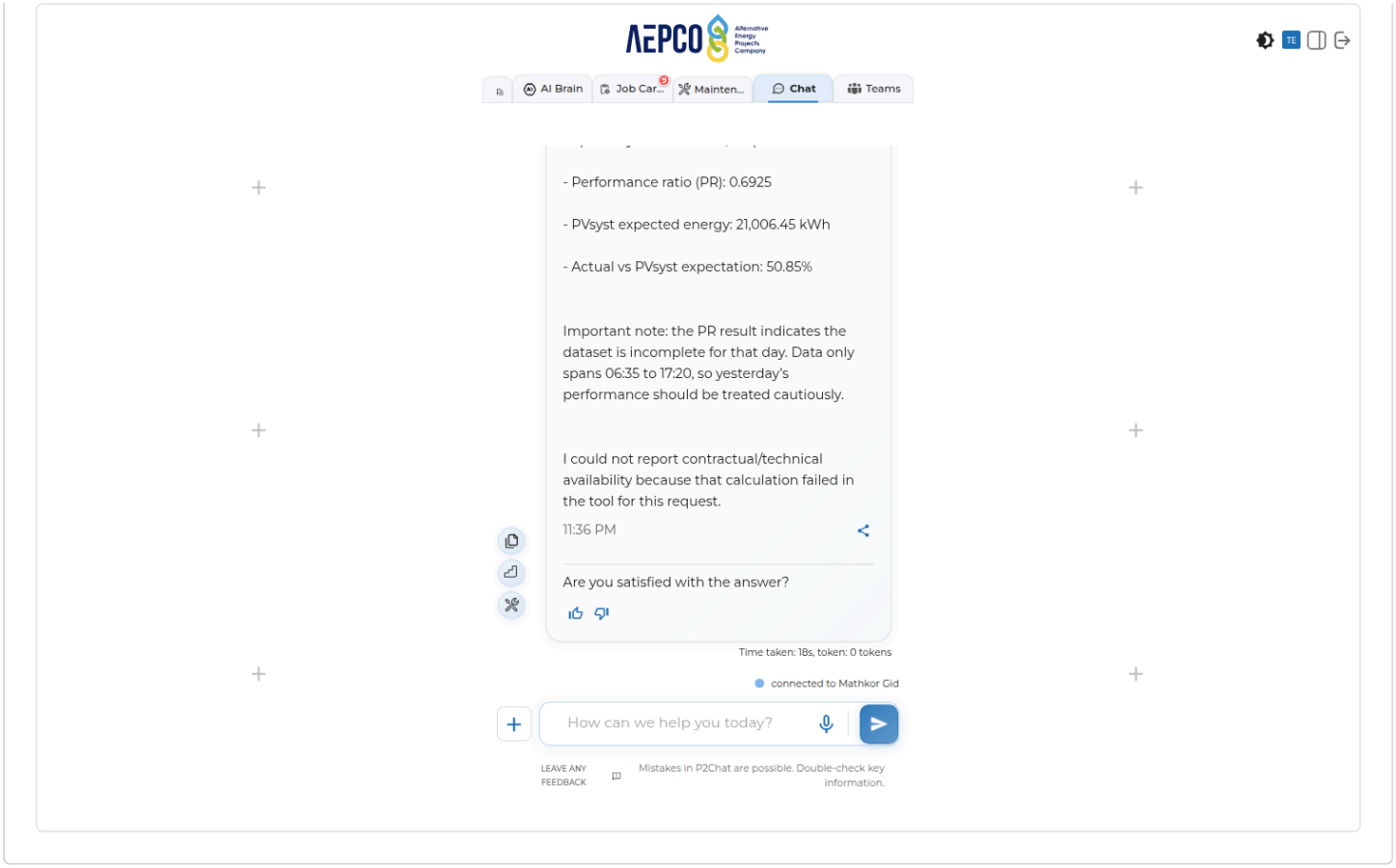
Error reason: AI indicated failure (matched 'calculation failed'): Yesterday (2026-10-18), your plant produced 10,681.68 kWh. Key performance: - Specific yield: 2.90 kWh/kWp - Performance ratio (PR): 0.6925 - PVsyst expected energy: 21,006.45 kWh - Actual

TECHNICAL EVIDENCE

API call: `GET /testuser2/ai_agent_chat → 200`

Raw API response:


```
{
  "result": {
    "user_input": "How did my plant perform yesterday?",
    "displayed_response": "Yesterday (2026-10-18), your plant produced 10,681.68 kWh.\n\nKey performance:\n- Specific yield: 2.90 kWh/kWp\n- Performance ratio (PR): 0.6925\n- PVsyst expected energy: 21,006.45 kWh\n- Actual vs PVsyst expectation: 50.85%\n\nImportant note: the PR result indicates the dataset is incomplete for that day. Data only spans 06:35 to 17:20, so yesterday's performance should be treated cautiously.\n\nI could not report contractual/technical availability because that calculation failed in the tool for this request.",
    "coding_response": [
      "Yesterday (2026-10-18), your plant produced 10,681.68 kWh.\n\nKey performance:\n- Specific yield: 2.90 kWh/kWp\n- Performance ratio (PR): 0.6925\n- PVsyst expected energy: 21,006.45 kWh\n- Actual vs PVsyst expectation: 50.85%\n\nImportant note: the PR result indicates the dataset is incomplete for that day. Data only spans 06:35 to 17:20, so yesterday's performance should be treated cautiously.\n\nI could not report contractual/technical availability because that calculation failed in the tool for this request."
    ],
    "used_agents": [
      "data_agent"
    ],
    "used_tools": [
      "multi_tool_use.parallel"
    ],
    "urls": [],
    "steps": "1. Select data from 2026-10-18 to 2026-10-18, including the end date.  \n2. Find all inverter power columns and their DC rated powers using the plant metadata.  \n3. Set calculation parameters: irradiance threshold (usually 50 W/m²), power threshold (usually 0.1 kW), and data interval (typically 5 minutes).  \n4. For each inverter, filter the data to times when irradiance is above the threshold, then count the total number of valid intervals (T_useful) and the number of downtime intervals (T_down).  \n5. Convert these interval counts into hours based on the data interval length.  \n6. Calculate Technical Availability for each inverter using: (T_useful - T_down) / T_useful × 100.  \n7. Return a dictionary showing the availability percentage for each inverter.\n\n---\n\n1. Select data from 2026-10-18 to 2026-10-18, including the end date.  \n2. Group the data by day (or by minute, hour, month, or year as needed).  \n3. Make sure all days in the range are included in the calculation.  \n4. Calculate the total energy produced during the selected period using the plant's power data, adjusting for the time interval (0.0833 hours).  \n5. Normalize the total energy by dividing by the plant's installed capacity (rated power) to get the specific yield in kWh per kWp.  \n6. Return the specific yield for each time interval as a detailed breakdown.\n\n---\n\n1. Select data from 2026-10-18 to 2026-10-18, including the end date.  \n2. Use the active_power column(s) for the calculation.  \n3. Clean the data by removing or flagging invalid values in the power column(s).  \n4. For each data point, calculate energy as (Power × 0.0833 hours), converting units if needed (e.g., from Watts to kW).  \n5. Include all days in the calculation.  \n6. Add up all the energy values for the selected period to get the total energy produced.\n\n---\n\n1. Select data from 2026-10-18 to 2026-10-18, including the end date.  \n2. Identify the columns for power, irradiance, and temperature.  \n3. Clean the data by removing or flagging invalid values in the power and irradiance columns.  \n4. Calculate the actual energy produced by multiplying power by the time interval (0.0833 hours) and adjusting for units.  \n5. Calculate the expected energy without temperature correction.  \n6. If the system is bifacial, apply extra correction factors using metadata like ground albedo, area normalization, and shading losses.  \n7. Group the actual and expected energy by day.  \n8. Calculate the performance ratio for each interval, applying bifacial corrections if needed.  \n9. Return the average performance ratio across all intervals.  \n10. Cap the performance ratio at a maximum of 1.3 to av
```


AEPCO
Alternative Energy Projects Company
AI Brain
Job Car...
Mainten...
Chat
Teams
- Performance ratio (PR): 0.6925
- PVsyst expected energy: 21,006.45 kWh
- Actual vs PVsyst expectation: 50.85%
Important note: the PR result indicates the dataset is incomplete for that day. Data only spans 06:35 to 17:20, so yesterday's performance should be treated cautiously.
I could not report contractual/technical availability because that calculation failed in the tool for this request.
11:36 PM
Are you satisfied with the answer?
Time taken: 18s, token: 0 tokens
connected to Mathkor Gid
How can we help you today?
LEAVE ANY FEEDBACK
Mistakes in P2Chat are possible. Double-check key information.

4. Abdali_Mall_Netmetering

Production • 43.0s

Question: Is there shading happening with my plant? Please analyze the pyranometer and check for frequent dips at certain times

Error reason: AI indicated failure (matched 'encountered an issue'): Sorry, we encountered an issue processing your request. Our team has been notified and will investigate. 11:58 PM Do you want to report this error? YES NO Time taken: s, token: tokens connected t

TECHNICAL EVIDENCE

Exception	`PydanticSerializationError: Unable to serialize unknown type: <class 'pandas.core.frame.DataFrame'>`
Agent / Tool	`data_agent / unknown`
Endpoint	`/testuser1/ai_agent_chat`
Location	`agent_graph.py:916`

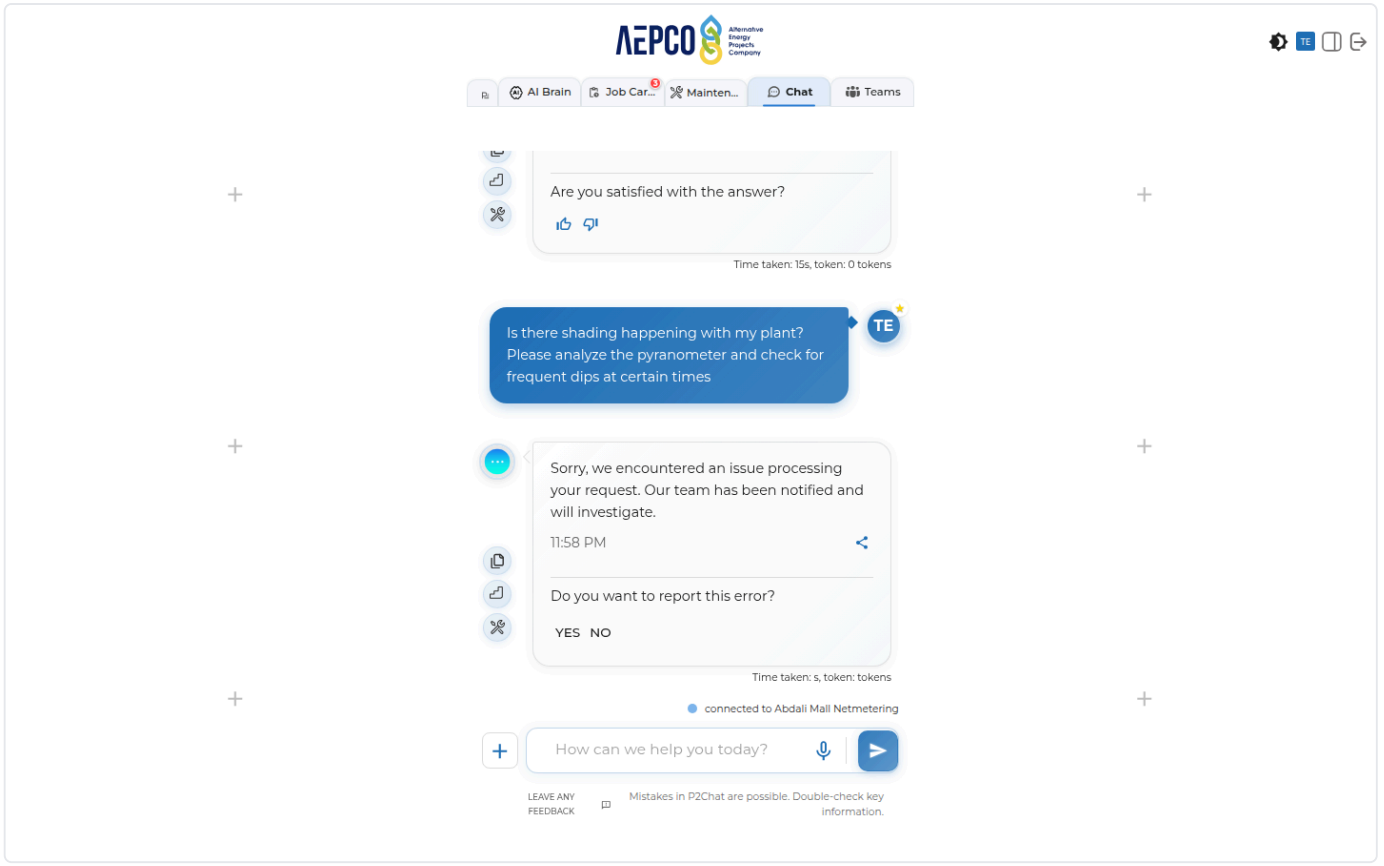
API call: `GET /testuser2/ai_agent_chat → 400`

Raw API response:


```
{
  "displayed_response": "Sorry, we encountered an issue processing your request. Our team has been notified and will investigate.",
  "coding_response": [
    "==================== ERROR SUMMARY ====================\nAPI ENDPOINT: /testuser1/ai_agent_chat\nAGENT: data_agent\nTOOL: unknown\nERROR TYPE: PydanticSerializationError\nERROR MESSAGE: Unable to serialize unknown type: <class 'pandas.core.frame.DataFrame'>\nLOCATION: agent_graph.py:916\n============================================================"
  ]
}
```


```
==================== ERROR SUMMARY ====================
API ENDPOINT: /testuser1/ai_agent_chat
AGENT: data_agent
TOOL: unknown
ERROR TYPE: PydanticSerializationError
ERROR MESSAGE: Unable to serialize unknown type: <class 'pandas.core.frame.DataFrame'>
LOCATION: agent_graph.py:916
============================================================
```

Console errors: Failed to load resource: the server responded with a status of 400 (Bad Request)


AEPCO
Alternative Energy Projects Company
TE
AI Brain
Job Car...
Mainten...
Chat
Teams
Are you satisfied with the answer?
Time taken: 15s, token: 0 tokens
Is there shading happening with my plant? Please analyze the pyranometer and check for frequent dips at certain times
TE
Sorry, we encountered an issue processing your request. Our team has been notified and will investigate.
11:58 PM
Do you want to report this error?
YES NO
Time taken: s, token: tokens
connected to Abdali Mall Netmetering
How can we help you today?
LEAVE ANY FEEDBACK
Mistakes in P2Chat are possible. Double-check key information.

5. Abdali_Mall_Lahoun

Production • 29.2s

Question: I have a COMMUNICATION_FAULT what do I do to my inverters?

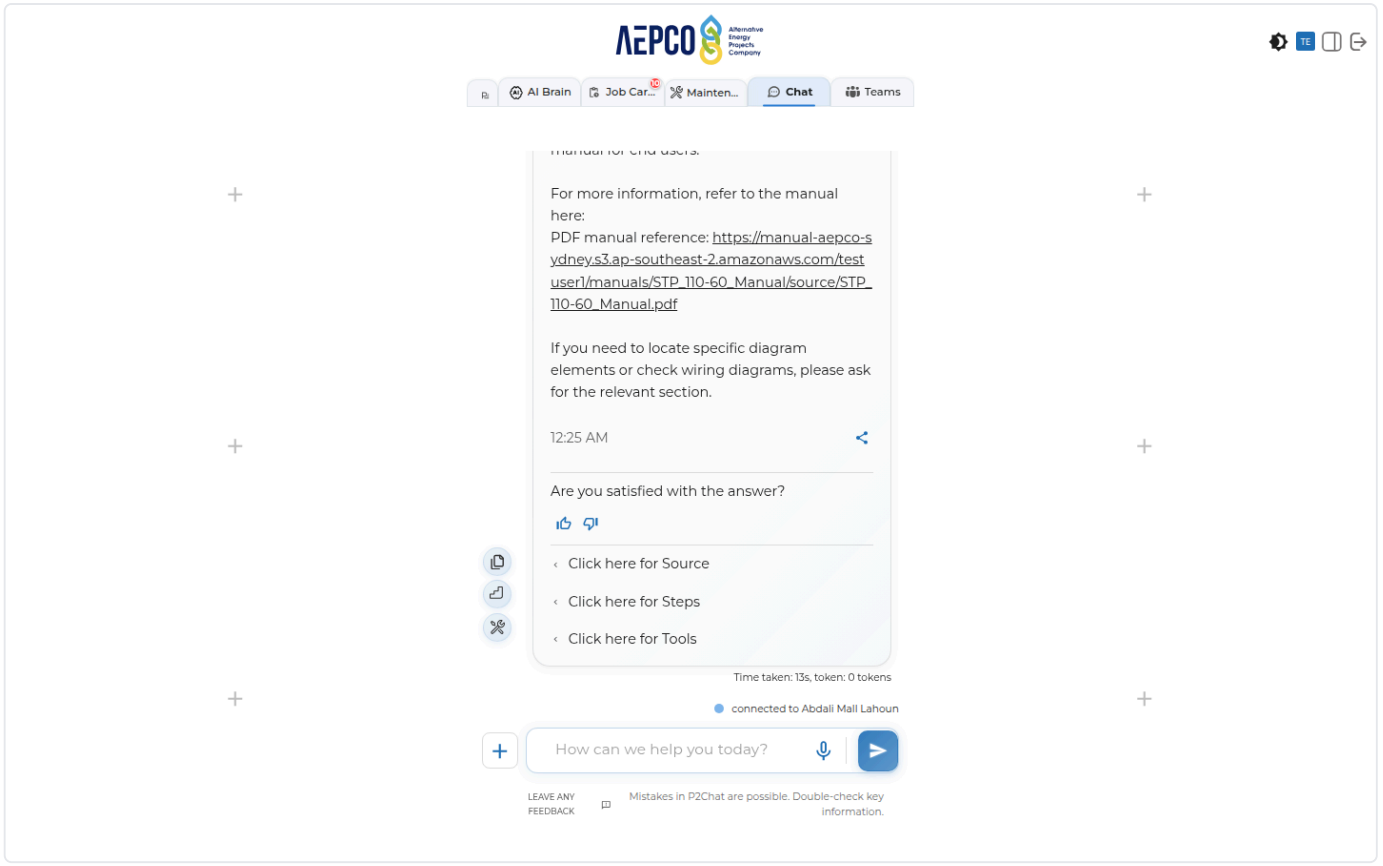
Error reason: AI indicated failure (matched 'an error'): The manual lists COMMUNICATION_FAULT as an error related to the communication processor. The inverter continues feeding in, but the underlying cause must be determined by Service personnel. Troublesh

TECHNICAL EVIDENCE

API call: `GET /testuser2/ai_agent_chat → 200`

Raw API response:


```
{
  "result": {
    "user_input": "I have a COMMUNICATION_FAULT what do I do to my inverters?",
    "displayed_response": "The manual lists COMMUNICATION_FAULT as an error related to the communication pro
cessor. The inverter continues feeding in, but the underlying cause must be determined by Service personne
l.\n\nTroubleshooting Steps from the Manual:\n1. If the COMMUNICATION_FAULT message appears frequently, cont
act the Service department for further diagnosis and support.\n\nNo further corrective actions or troublesho
oting steps are provided in the manual for end users.\n\nFor more information, refer to the manual here:\nPD
F manual reference: https://manual-aepco-sydney.s3.ap-southeast-2.amazonaws.com/testuser1/manuals/STP_110-60
_Manual/source/STP_110-60_Manual.pdf\n\nIf you need to locate specific diagram elements or check wiring diag
rams, please ask for the relevant section.",
    "coding_response": [
      "The manual lists COMMUNICATION_FAULT as an error related to the communication processor. The inverter
continues feeding in, but the underlying cause must be determined by Service personnel.\n\nTroubleshooting S
teps from the Manual:\n1. If the COMMUNICATION_FAULT message appears frequently, contact the Service departm
ent for further diagnosis and support.\n\nNo further corrective actions or troubleshooting steps are provide
d in the manual for end users.\n\nFor more information, refer to the manual here:\nPDF manual reference: htt
ps://manual-aepco-sydney.s3.ap-southeast-2.amazonaws.com/testuser1/manuals/STP_110-60_Manual/source/STP_110-
60_Manual.pdf\n\nIf you need to locate specific diagram elements or check wiring diagrams, please ask for th
e relevant section."
    ],
    "used_agents": [
      "manual_agent"
    ],
    "used_tools": [
      "functions.manual_rag_tool"
    ],
    "urls": [],
    "steps": []
  }
}
```



Data Availability Issues DATA AVAILABILITY (5 failures)

1. Galleria_Mall

Production • 24.6s

Question: Was there any 0 production days in the last 7 days?

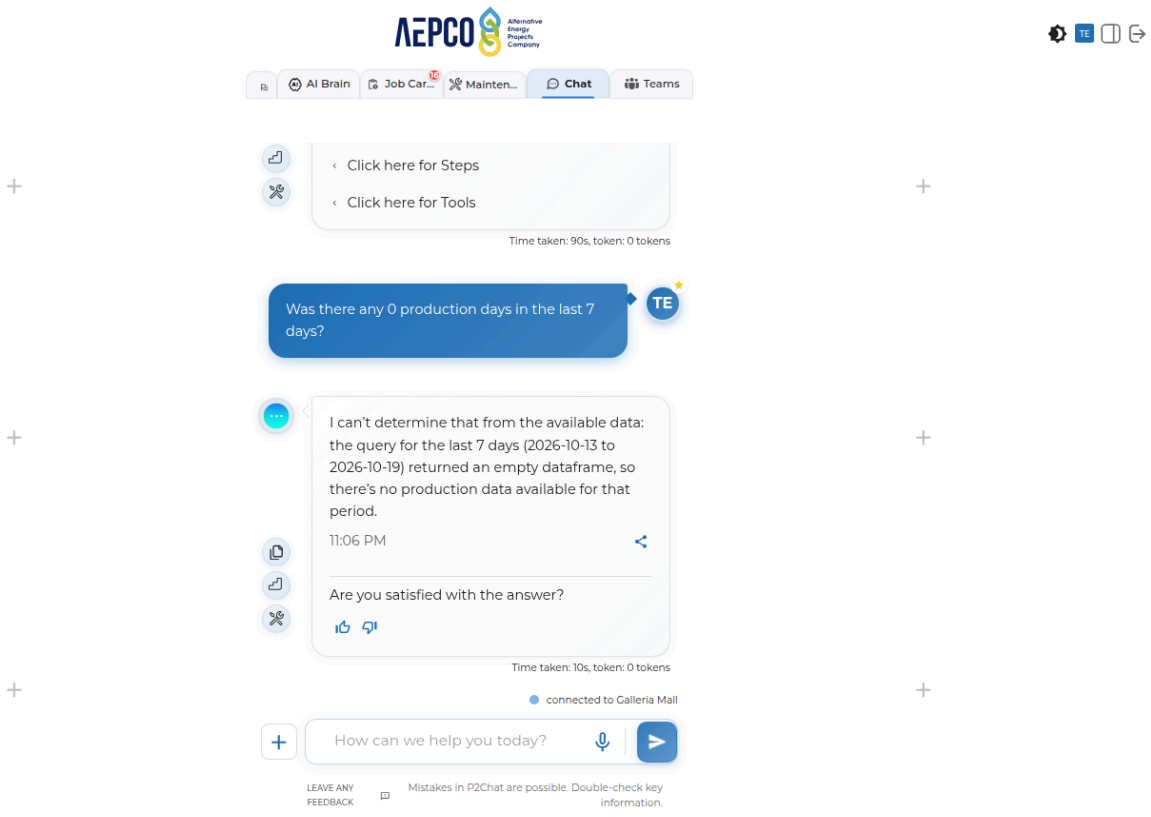
Error reason: AI indicated failure (matched 'i can't'): I can’t determine that from the available data: the query for the last 7 days (2026-10-13 to 2026-10-19) returned an empty dataframe, so there’s no production data available for that period. 11:06 PM

TECHNICAL EVIDENCE

API call: `GET /testuser2/ai_agent_chat → 200`

Raw API response:


```
{
  "result": {
    "user_input": "Was there any 0 production days in the last 7 days?",
    "displayed_response": "I can’t determine that from the available data: the query for the last 7 days (2026-10-13 to 2026-10-19) returned an empty dataframe, so there’s no production data available for that period.",
    "coding_response": [
      "I can’t determine that from the available data: the query for the last 7 days (2026-10-13 to 2026-10-19) returned an empty dataframe, so there’s no production data available for that period."
    ],
    "used_agents": [
      "data_agent"
    ],
    "used_tools": [
      "extract_timeseries_data_ag"
    ],
    "urls": [],
    "steps": []
  }
}
```

2. Galleria_Mall

Production • 26.1s

Question: How is my data quality for POA and power? How many of them show missing data during daytime or mismatches with satellite prediction?

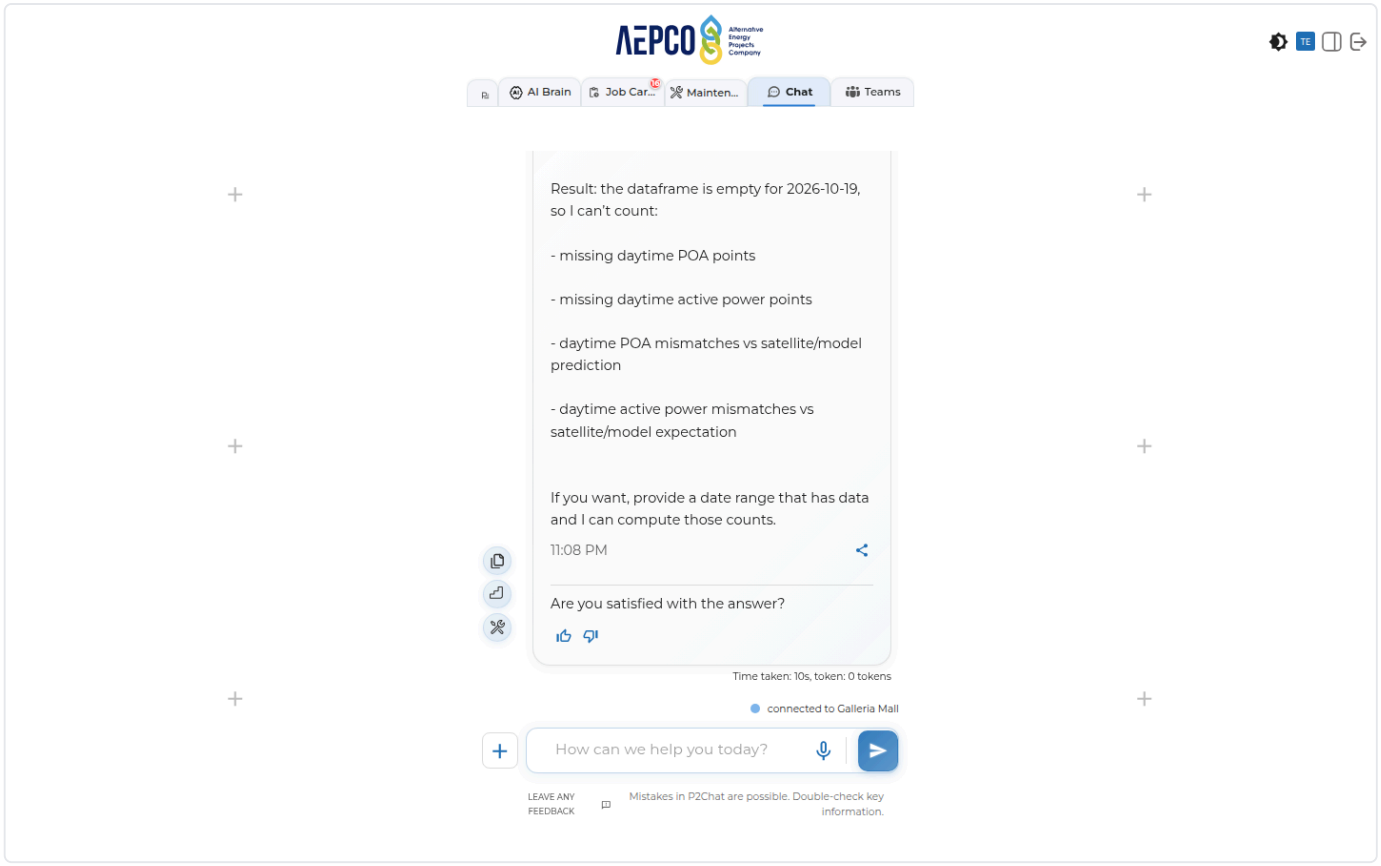
Error reason: AI indicated failure (matched 'i can't'): I can’t assess it for 2026-10-19 because the time-series query returned no data for that date. Result: the dataframe is empty for 2026-10-19, so I can’t count: - missing daytime POA points - mi

TECHNICAL EVIDENCE

API call: `GET /testuser2/ai_agent_chat → 200`

Raw API response:


```
{
  "result": {
    "user_input": "How is my data quality for POA and power? How many of them show missing data during daytime or mismatches with satellite prediction?",
    "displayed_response": "I can’t assess it for 2026-10-19 because the time-series query returned no data for that date.\n\nResult: the dataframe is empty for 2026-10-19, so I can’t count:\n- missing daytime POA points\n- missing daytime active power points\n- daytime POA mismatches vs satellite/model prediction\n- daytime active power mismatches vs satellite/model expectation\n\nIf you want, provide a date range that has data and I can compute those counts.",
    "coding_response": [
      "I can’t assess it for 2026-10-19 because the time-series query returned no data for that date.\n\nResult: the dataframe is empty for 2026-10-19, so I can’t count:\n- missing daytime POA points\n- missing daytime active power points\n- daytime POA mismatches vs satellite/model prediction\n- daytime active power mismatches vs satellite/model expectation\n\nIf you want, provide a date range that has data and I can compute those counts."
    ],
    "used_agents": [
      "data_agent"
    ],
    "used_tools": [
      "extract_timeseries_data_ag"
    ],
    "urls": [],
    "steps": []
  }
}
```


AEPCO
Alternative Energy Projects Company
AI Brain
Job Car...
Mainten...
Chat
Teams
Result: the dataframe is empty for 2026-10-19, so I can't count:
- missing daytime POA points
- missing daytime active power points
- daytime POA mismatches vs satellite/model prediction
- daytime active power mismatches vs satellite/model expectation
If you want, provide a date range that has data and I can compute those counts.
11:08 PM
Are you satisfied with the answer?
Time taken: 10s, token: 0 tokens
connected to Galleria Mall
How can we help you today?
LEAVE ANY FEEDBACK
Mistakes in P2Chat are possible. Double-check key information.

3. Galleria_Mall

Production • 30.8s

Question: How did my plant perform yesterday?

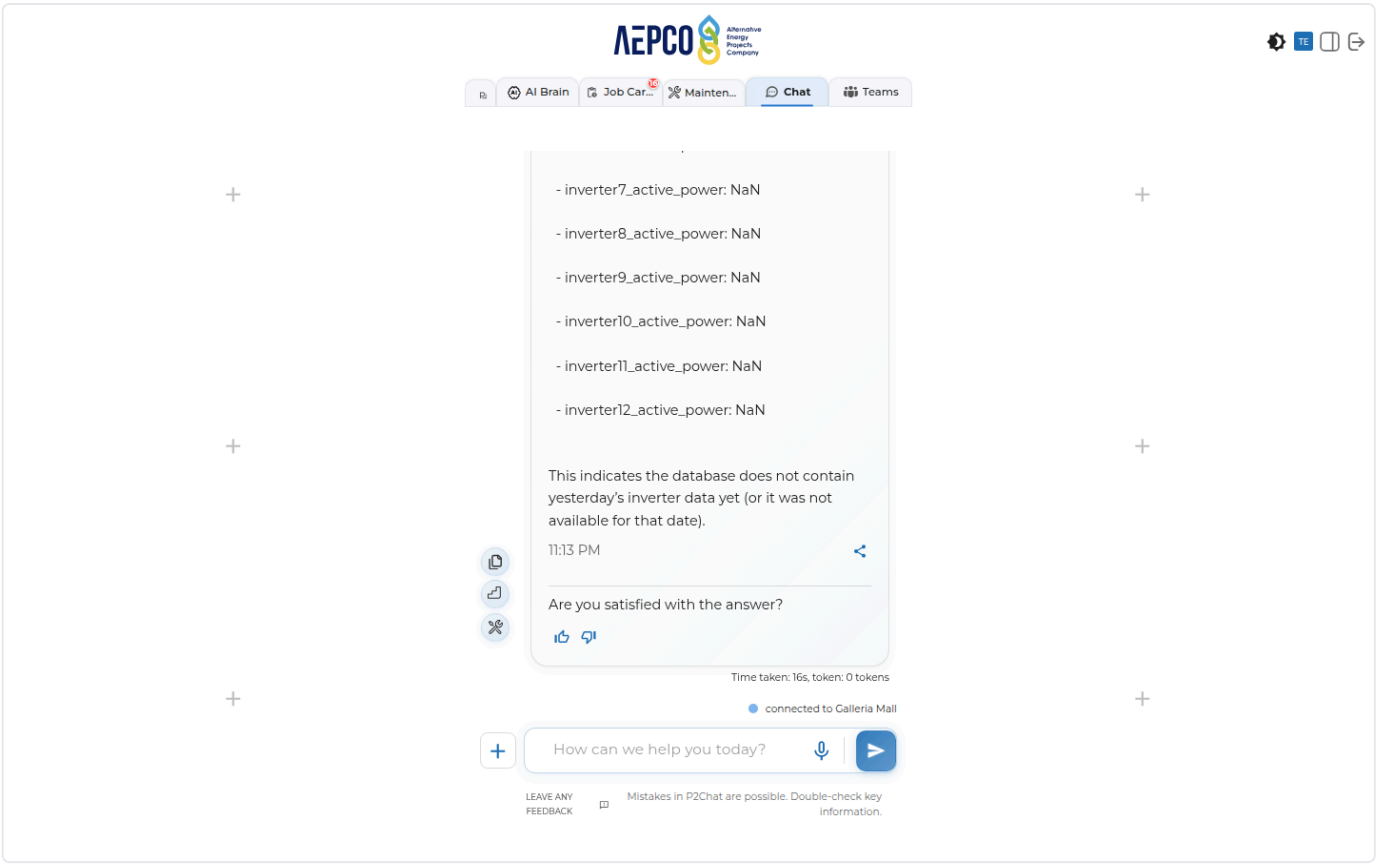
Error reason: AI indicated failure (matched 'i can't'): I can’t assess yesterday’s performance because there is no data available for 2026-10-18. What I found for 2026-10-18: - Energy: 0 kWh for all inverters, with the dataset reported as empty - Sp

TECHNICAL EVIDENCE

API call: `GET /testuser2/ai_agent_chat → 200`

Raw API response:


```
{
  "result": {
    "user_input": "How did my plant perform yesterday?",
    "displayed_response": "I can’t assess yesterday’s performance because there is no data available for 2026-10-18.\n\nWhat I found for 2026-10-18:\n- Energy: 0 kWh for all inverters, with the dataset reported as empty\n- Specific yield: empty for all inverters\n- Performance ratio: empty for all inverters\n- Availability:\n  - inverter1_active_power: NaN\n  - inverter2_active_power: NaN\n  - inverter3_active_power: NaN\n  - inverter4_active_power: NaN\n  - inverter5_active_power: NaN\n  - inverter6_active_power: NaN\n  - inverter7_active_power: NaN\n  - inverter8_active_power: NaN\n  - inverter9_active_power: NaN\n  - inverter10_active_power: NaN\n  - inverter11_active_power: NaN\n  - inverter12_active_power: NaN\n\nThis indicates the database does not contain yesterday’s inverter data yet (or it was not available for that date).",
    "coding_response": [
      "I can’t assess yesterday’s performance because there is no data available for 2026-10-18.\n\nWhat I found for 2026-10-18:\n- Energy: 0 kWh for all inverters, with the dataset reported as empty\n- Specific yield: empty for all inverters\n- Performance ratio: empty for all inverters\n- Availability:\n  - inverter1_active_power: NaN\n  - inverter2_active_power: NaN\n  - inverter3_active_power: NaN\n  - inverter4_active_power: NaN\n  - inverter5_active_power: NaN\n  - inverter6_active_power: NaN\n  - inverter7_active_power: NaN\n  - inverter8_active_power: NaN\n  - inverter9_active_power: NaN\n  - inverter10_active_power: NaN\n  - inverter11_active_power: NaN\n  - inverter12_active_power: NaN\n\nThis indicates the database does not contain yesterday’s inverter data yet (or it was not available for that date)."
    ],
    "used_agents": [
      "data_agent"
    ],
    "used_tools": [
      "functions.perf_ratio_ag",
      "functions.total_energy_func_ag",
      "functions.specific_yield_ag",
      "functions.technical_contractual_availability_ag",
      "multi_tool_use.parallel"
    ],
    "urls": [],
    "steps": "1. I choose data from October 18, 2026, including that day.\n2. I find all inverter power columns and their DC rated power values using the plant’s metadata.\n3. I set calculation parameters like the irradiance threshold (usually 50 W/m²), power threshold (usually 0.1 kW), and the data interval (typically 5 minutes).\n4. For each inverter, I filter the data to only include times when irradiance is above the threshold, then count the total number of valid intervals (T_useful) and the number of downtime intervals (T_down).\n5. I convert these interval counts into hours based on the data interval length.\n6. I calculate the Technical Availability for each inverter using this formula: (T_useful - T_down) / T_useful × 100.\n7. I return a dictionary showing the availability percentage for each inverter."
  }
}
```


AEPCO
Alternative Energy Projects Company
AI Brain
Job Car...
Mainten...
Chat
Teams
- inverter7_active_power: NaN
- inverter8_active_power: NaN
- inverter9_active_power: NaN
- inverter10_active_power: NaN
- inverter11_active_power: NaN
- inverter12_active_power: NaN
This indicates the database does not contain yesterday's inverter data yet (or it was not available for that date).
11:13 PM
Are you satisfied with the answer?
Time taken: 16s, token: 0 tokens
connected to Galleria Mall
How can we help you today?
LEAVE ANY FEEDBACK
Mistakes in P2Chat are possible. Double-check key information.

4. SAK_1

Production • 79.4s

Question: Please generate last months report

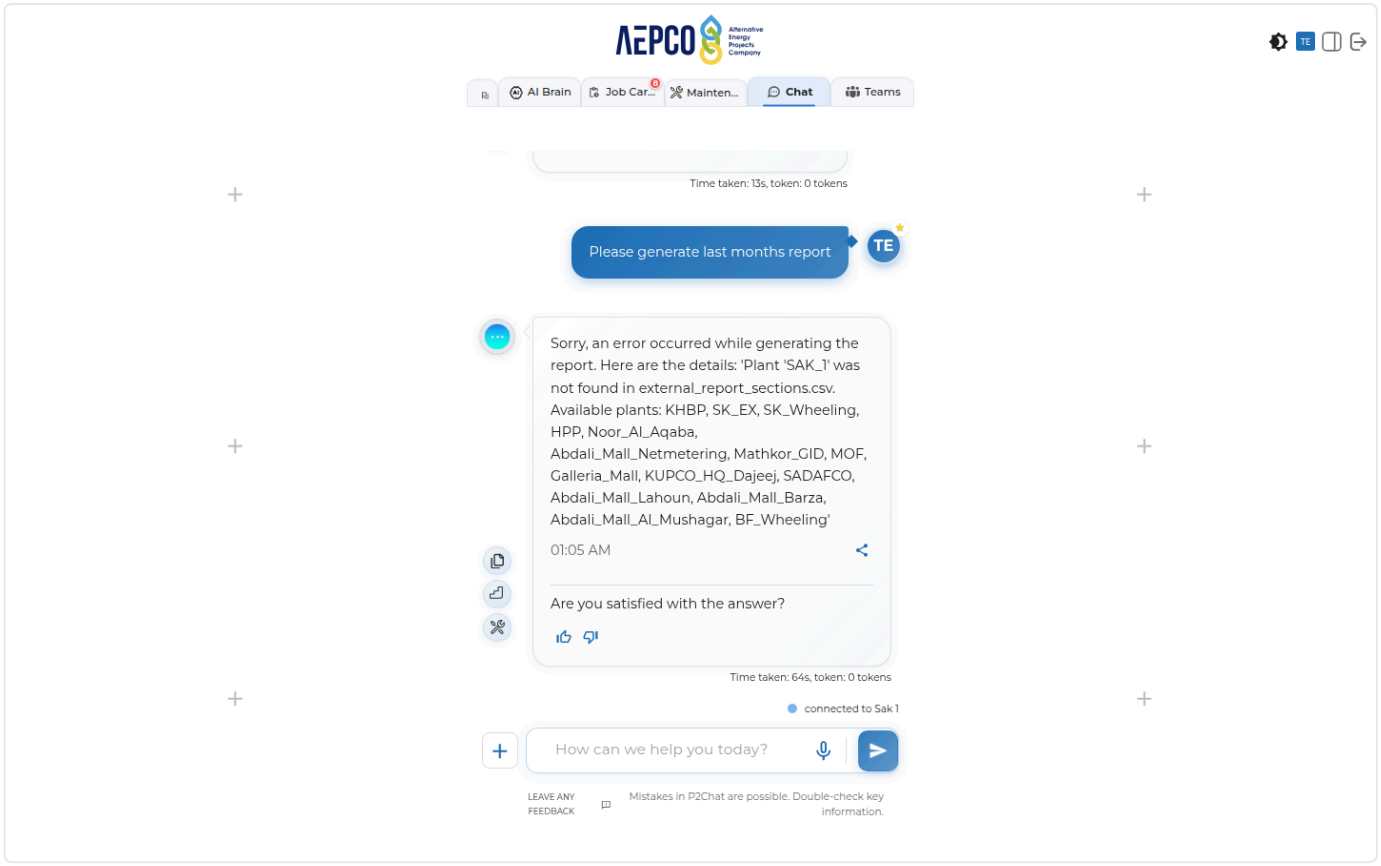
Error reason: AI indicated failure (matched 'error occurred'): Sorry, an error occurred while generating the report. Here are the details: 'Plant 'SAK_1' was not found in external_report_sections.csv. Available plants: KHBP, SK_EX, SK_Wheeling, HPP, Noor_Al_Aqaba

TECHNICAL EVIDENCE

API call: `GET /testuser2/ai_agent_chat → 200`

Raw API response:


```
{
  "result": {
    "user_input": "Please generate last months report",
    "displayed_response": "Sorry, an error occurred while generating the report. Here are the details: 'Plan
t 'SAK_1' was not found in external_report_sections.csv. Available plants: KHBP, SK_EX, SK_Wheeling, HPP, No
or_Al_Aqaba, Abdali_Mall_Netmetering, Mathkor_GID, MOF, Galleria_Mall, KUPCO_HQ_Dajeej, SADAFCO, Abdali_Mall
_Lahoun, Abdali_Mall_Barza, Abdali_Mall_Al_Mushagar, BF_Wheeling'",
    "coding_response": [
      [
        {
          "answer": "Sorry, an error occurred while generating the report. Here are the details: 'Plant 'SAK
_1' was not found in external_report_sections.csv. Available plants: KHBP, SK_EX, SK_Wheeling, HPP, Noor_Al_
Aqaba, Abdali_Mall_Netmetering, Mathkor_GID, MOF, Galleria_Mall, KUPCO_HQ_Dajeej, SADAFCO, Abdali_Mall_Lahou
n, Abdali_Mall_Barza, Abdali_Mall_Al_Mushagar, BF_Wheeling'",
          "action": []
        }
      ]
    ],
    "used_agents": [
      "report_agent"
    ],
    "used_tools": [
      "agent_external_report_tool"
    ],
    "urls": [],
    "steps": []
  }
}
```


AEPCO
Alternative Energy Projects Company
TE
AI Brain
Job Car...
Mainten...
Chat
Teams
Time taken: 13s, token: 0 tokens
Please generate last months report
TE
Sorry, an error occurred while generating the report. Here are the details: 'Plant 'SAK_1' was not found in external_report_sections.csv. Available plants: KHBP, SK_EX, SK_Wheeling, HPP, Noor_Al_Aqaba, Abdali_Mall_Netmetering, Mathkor_GID, MOF, Galleria_Mall, KUPCO_HQ_Dajeej, SADAFCO, Abdali_Mall_Lahoun, Abdali_Mall_Barza, Abdali_Mall_Al_Mushagar, BF_Wheeling'
01:05 AM
Are you satisfied with the answer?
Time taken: 64s, token: 0 tokens
connected to Sak 1
How can we help you today?
LEAVE ANY FEEDBACK
Mistakes in P2Chat are possible. Double-check key information.

5. SAK_1

Production • 53.5s

Question: Please generate December 2025 performance report

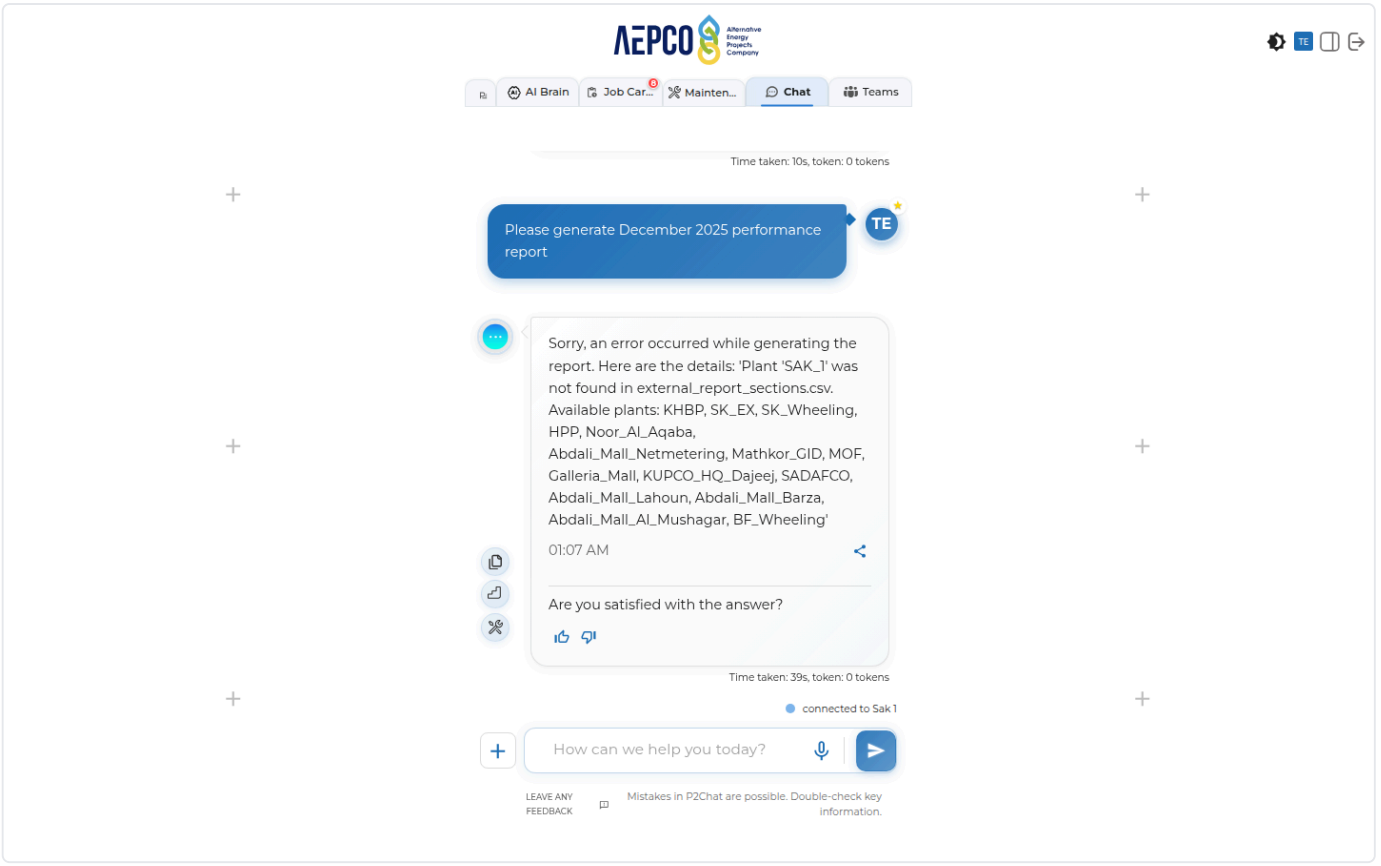
Error reason: AI indicated failure (matched 'error occurred'): Sorry, an error occurred while generating the report. Here are the details: 'Plant 'SAK_1' was not found in external_report_sections.csv. Available plants: KHBP, SK_EX, SK_Wheeling, HPP, Noor_Al_Aqaba

TECHNICAL EVIDENCE

API call: `GET /testuser2/ai_agent_chat → 200`

Raw API response:


```
{
  "result": {
    "user_input": "Please generate December 2025 performance report",
    "displayed_response": "Sorry, an error occurred while generating the report. Here are the details: 'Plan
t 'SAK_1' was not found in external_report_sections.csv. Available plants: KHBP, SK_EX, SK_Wheeling, HPP, No
or_Al_Aqaba, Abdali_Mall_Netmetering, Mathkor_GID, MOF, Galleria_Mall, KUPCO_HQ_Dajeej, SADAFCO, Abdali_Mall
_Lahoun, Abdali_Mall_Barza, Abdali_Mall_Al_Mushagar, BF_Wheeling'",
    "coding_response": [
      [
        {
          "answer": "Sorry, an error occurred while generating the report. Here are the details: 'Plant 'SAK
_1' was not found in external_report_sections.csv. Available plants: KHBP, SK_EX, SK_Wheeling, HPP, Noor_Al_
Aqaba, Abdali_Mall_Netmetering, Mathkor_GID, MOF, Galleria_Mall, KUPCO_HQ_Dajeej, SADAFCO, Abdali_Mall_Lahou
n, Abdali_Mall_Barza, Abdali_Mall_Al_Mushagar, BF_Wheeling'",
          "action": []
        }
      ]
    ],
    "used_agents": [
      "report_agent"
    ],
    "used_tools": [
      "agent_external_report_tool"
    ],
    "urls": [],
    "steps": []
  }
}
```


AEPCO
Alternative Energy Projects Company
AI Brain
Job Car...
Mainten...
Chat
Teams
Time taken: 10s, token: 0 tokens
Please generate December 2025 performance report
TE
Sorry, an error occurred while generating the report. Here are the details: 'Plant 'SAK_1' was not found in external_report_sections.csv. Available plants: KHBP, SK_EX, SK_Wheeling, HPP, Noor_Al_Aqaba, Abdali_Mall_Netmetering, Mathkor_GID, MOF, Galleria_Mall, KUPCO_HQ_Dajeej, SADAFCO, Abdali_Mall_Lahoun, Abdali_Mall_Barza, Abdali_Mall_Al_Mushagar, BF_Wheeling'
01:07 AM
Are you satisfied with the answer?
Time taken: 39s, token: 0 tokens
connected to Sak 1
How can we help you today?
LEAVE ANY FEEDBACK
Mistakes in P2Chat are possible. Double-check key information.

Passed (202)

1. Noor_Al_aqaba

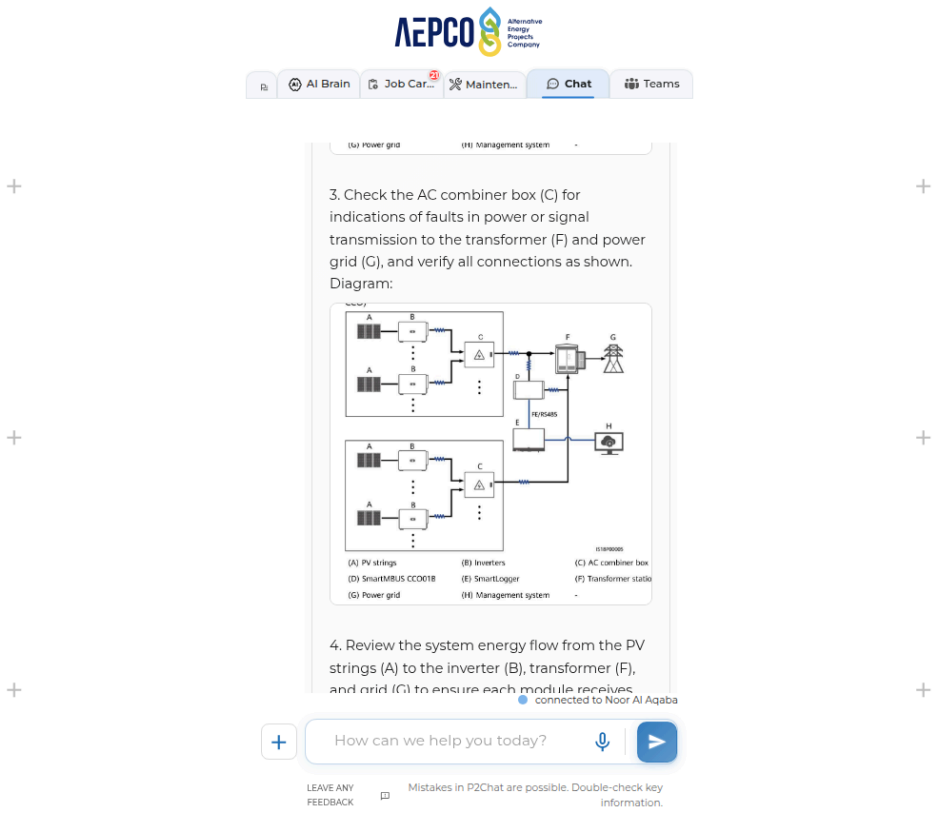
Production • 35.4s

Question: I have a COMMUNICATION_FAULT what do I do to my inverters?

Response: Troubleshooting Steps for COMMUNICATION_FAULT on Inverters: 1. Inspect the PV strings (A), inverter (B), AC combiner (C), and all connections up to the SmartLogger (E), SmartMBUS CCO01B (D), and management system (H) as shown in the system overview. Look for any loose or disconnected communication cables in this pathway. Refer to this diagram: 2. Confirm correct physical wiring and communication lines between the inverter (B) and the SmartLogger (E) and SmartMBUS (D). Check each connector for secure contact and absence of corrosion or debris. Diagram: 3. Check the AC combiner box (C) for indications of faults in power or signal transmission to the transformer (F) and power grid (G), and verify all connections as shown. Diagram: 4. Review the system energy flow from the PV strings (A) to the inverter (B), transformer (F), and grid (G) to ensure each module receives and transmits data and power properly, referencing connector positions in the diagrams. Diagram: 5. For troublesho

TECHNICAL EVIDENCE

Agents used	`manual_agent`
Tools used	`functions.manual_rag_tool`
API calls made	`1`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`1`
Network failures	`0`



2. Noor_Al_aqaba

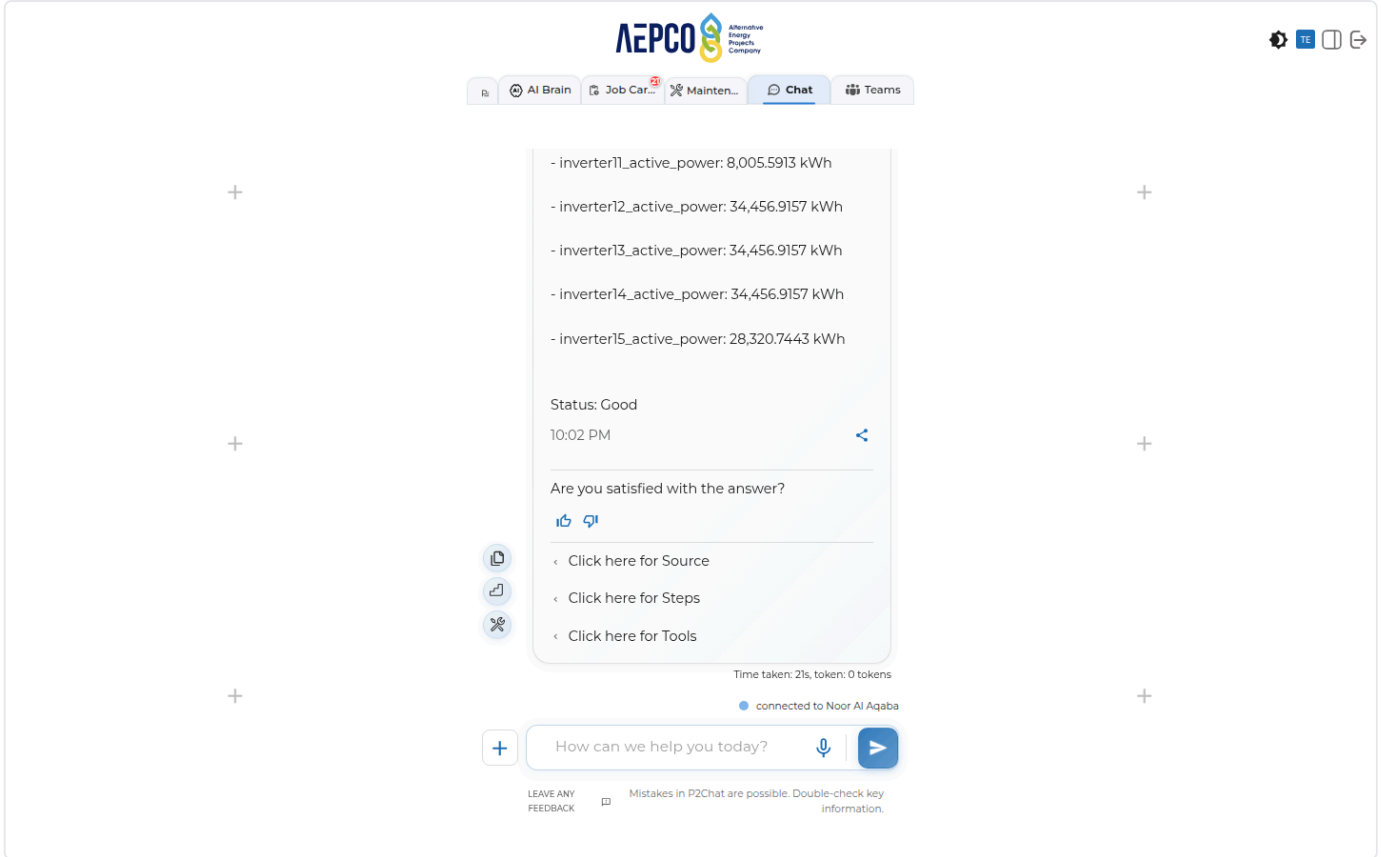
Production • 35.2s

Question: Use 'clipping_loss_via_requests' in order to calculate clipping loss if any for July 2026

Response: Clipping loss for July 2026 via `clipping_loss_via_requests`: - inverter1_active_power: 34,258.2605 kWh - inverter2_active_power: 34,257.4975 kWh - inverter3_active_power: 34,257.4975 kWh - inverter4_active_power: 34,257.4975 kWh - inverter5_active_power: 29,015.9477 kWh - inverter6_active_power: 34,250.2521 kWh - inverter7_active_power: 34,258.2605 kWh - inverter8_active_power: 34,250.2521 kWh - inverter9_active_power: 34,242.4765 kWh - inverter10_active_power: 29,010.0137 kWh - inverter11_active_power: 8,005.5913 kWh - inverter12_active_power: 34,456.9157 kWh - inverter13_active_power: 34,456.9157 kWh - inverter14_active_power: 34,456.9157 kWh - inverter15_active_power: 28,320.7443 kWh Status: Good 10:02 PM Are you satisfied with the answer? Click here for Source Click here for Steps Click here for Tools Time taken: 21s, token: 0 tokens connected to Noor Al Aqaba LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`functions.clipping_loss_via_requests`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



3. Noor_Al_aqaba

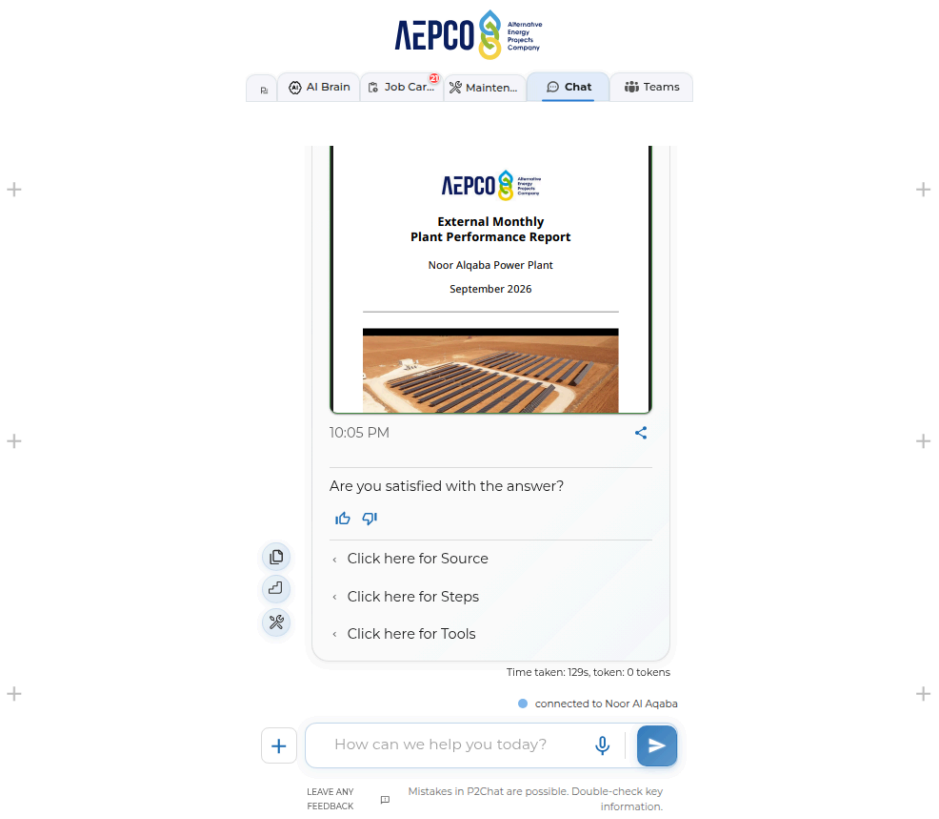
Production • 144.7s

Question: Please generate last months report

Response: I've generated the plant performance report for September 2026. You can view it here: 'https://report-aepco-sydney.s3.ap-southeast-2.amazonaws.com/Noor_Alqaba_Power_Plant_September_2026_external_report.pdf' Noor_Al_Aqaba Report 10:05 PM Are you satisfied with the answer? Click here for Source Click here for Steps Click here for Tools Time taken: 129s, token: 0 tokens connected to Noor Al Aqaba LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`report_agent`
Tools used	`agent_external_report_tool`
API calls made	`4`
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	`0`
Network failures	`0`



4. Noor_Al_aqaba

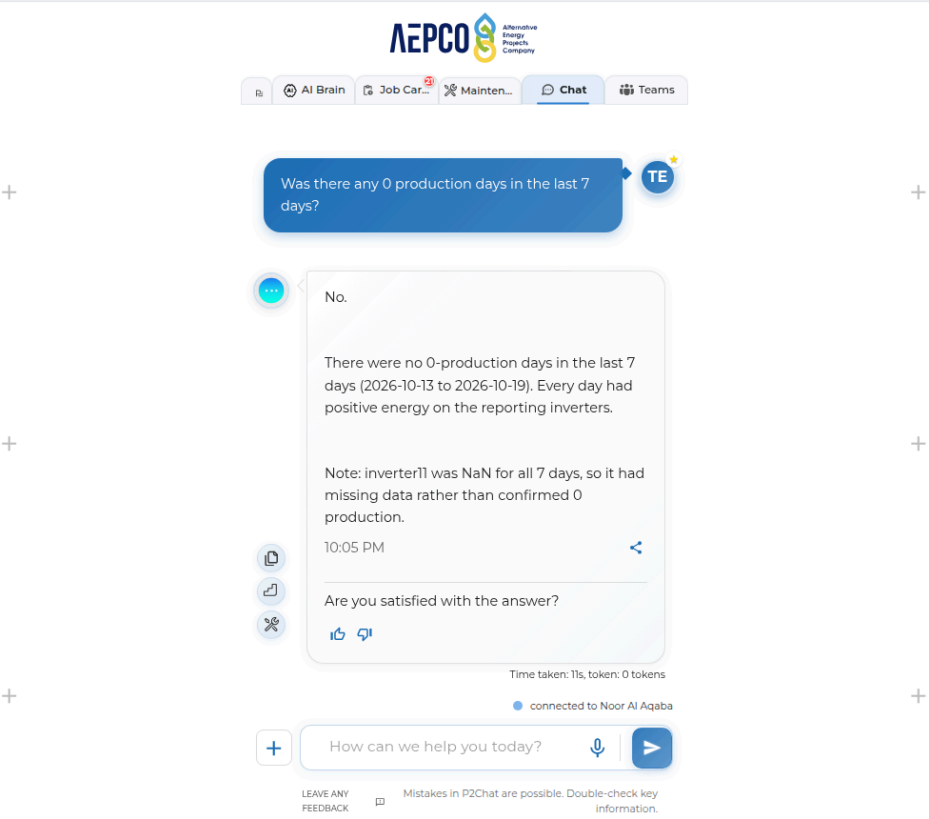
Production • 26.0s

Question: Was there any 0 production days in the last 7 days?

Response: No. There were no 0-production days in the last 7 days (2026-10-13 to 2026-10-19). Every day had positive energy on the reporting inverters. Note: inverter11 was NaN for all 7 days, so it had missing data rather than confirmed 0 production. 10:05 PM Are you satisfied with the answer? Time taken: 11s, token: 0 tokens connected to Noor Al Aqaba LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`functions.total_energy_func_ag`
API calls made	`1`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



5. Noor_Al_aqaba

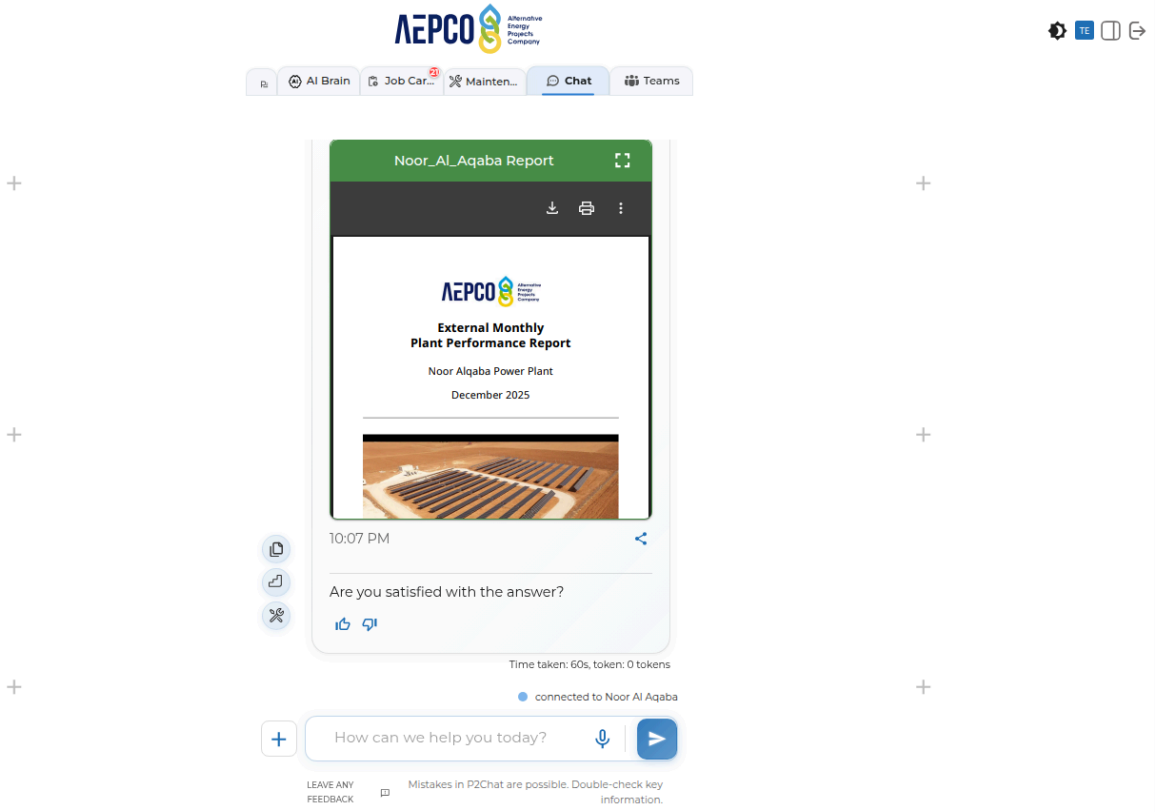
Production • 74.8s

Question: Please generate December 2025 performance report

Response: I've generated the external report for December 2025. You can view it here: 'https://report-aepco-sydney.s3.ap-southeast-2.amazonaws.com/Noor_Alqaba_Power_Plant_December_2025_external_report.pdf' Noor_Al_Aqaba Report 10:07 PM Are you satisfied with the answer? Time taken: 60s, token: 0 tokens connected to Noor Al Aqaba LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`report_agent`
Tools used	`agent_external_report_tool`
API calls made	`3`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



6. Noor_Al_aqaba

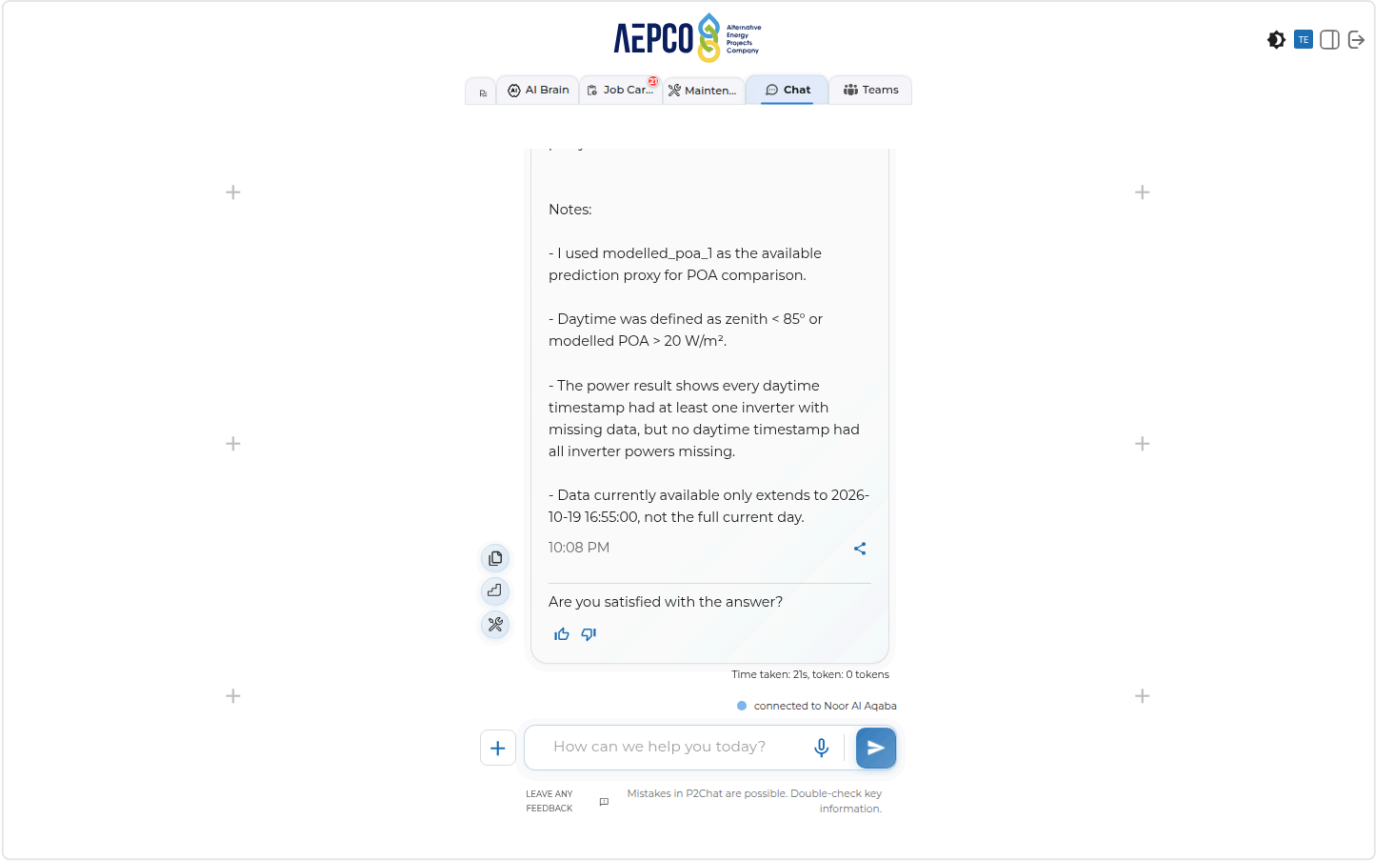
Production • 36.8s

Question: How is my data quality for POA and power? How many of them show missing data during daytime or mismatches with satellite prediction?

Response: For 2026-10-18 to 2026-10-19 16:55:00: - Daytime data points evaluated: 253 - Missing daytime POA points (pyranometer_1): 0 - Missing daytime measured power points: - Any inverter missing: 253 - All inverters missing: 0 - Mismatch instances: - POA vs prediction proxy (modelled_poa_1): 5 - Measured power vs irradiance prediction proxy: 6 Notes: - I used modelled_poa_1 as the available prediction proxy for POA comparison. - Daytime was defined as zenith < 85° or modelled POA > 20 W/m². - The power result shows every daytime timestamp had at least one inverter with missing data, but no daytime timestamp had all inverter powers missing. - Data currently available only extends to 2026-10-19 16:55:00, not the full current day. 10:08 PM Are you satisfied with the answer? Time taken: 21s, token: 0 tokens connected to Noor Al Aqaba LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`functions.extract_timeseries_data_ag, functions.execute_custom_code`
API calls made	2
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	0
Network failures	0



7. Noor_Al_aqaba

Production • 35.2s

Question: Please plot last 3 days hourly performance ratio for my MPPTs for inverter 2

Response: I've created the visualization for you. View Fullscreen 10:09 PM ★ Are you satisfied with the answer? Time taken: 21s, token: 0 tokens connected to Noor Al Aqaba LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.extract_timeseries_data_ag, functions.generate_graphs_via_code, functions.perf_ratio_ag`
API calls made	`2`
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	`0`
Network failures	`0`

8. Noor_Al_aqaba

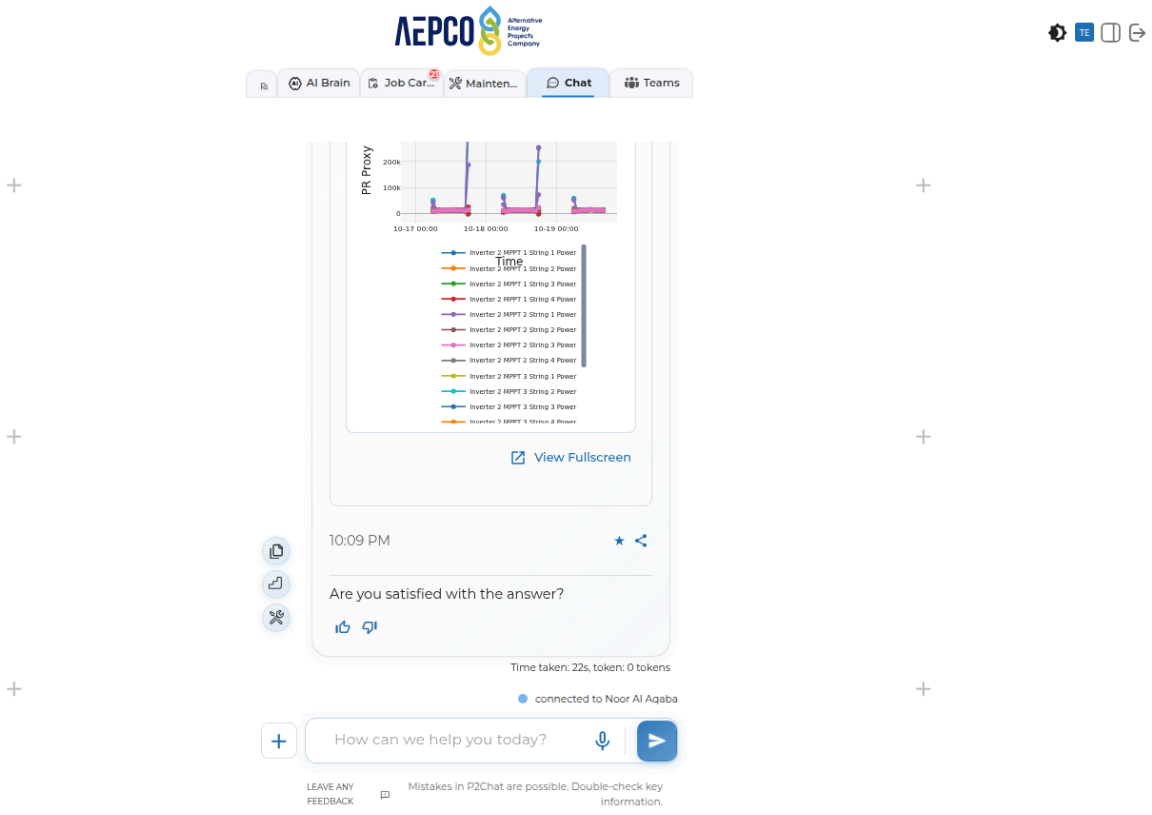
Production • 37.4s

Question: Please plot last 3 days hourly performance ratio for my Strings for inverter 2

Response: I've created the visualization for you. View Fullscreen 10:09 PM ★ Are you satisfied with the answer? Time taken: 22s, token: 0 tokens connected to Noor Al Aqaba LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.extract_timeseries_data_ag, functions.generate_graphs_via_code, functions.execute_custom_code, functions.perf_ratio_ag`
API calls made	`1`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



9. Noor_Al_aqaba

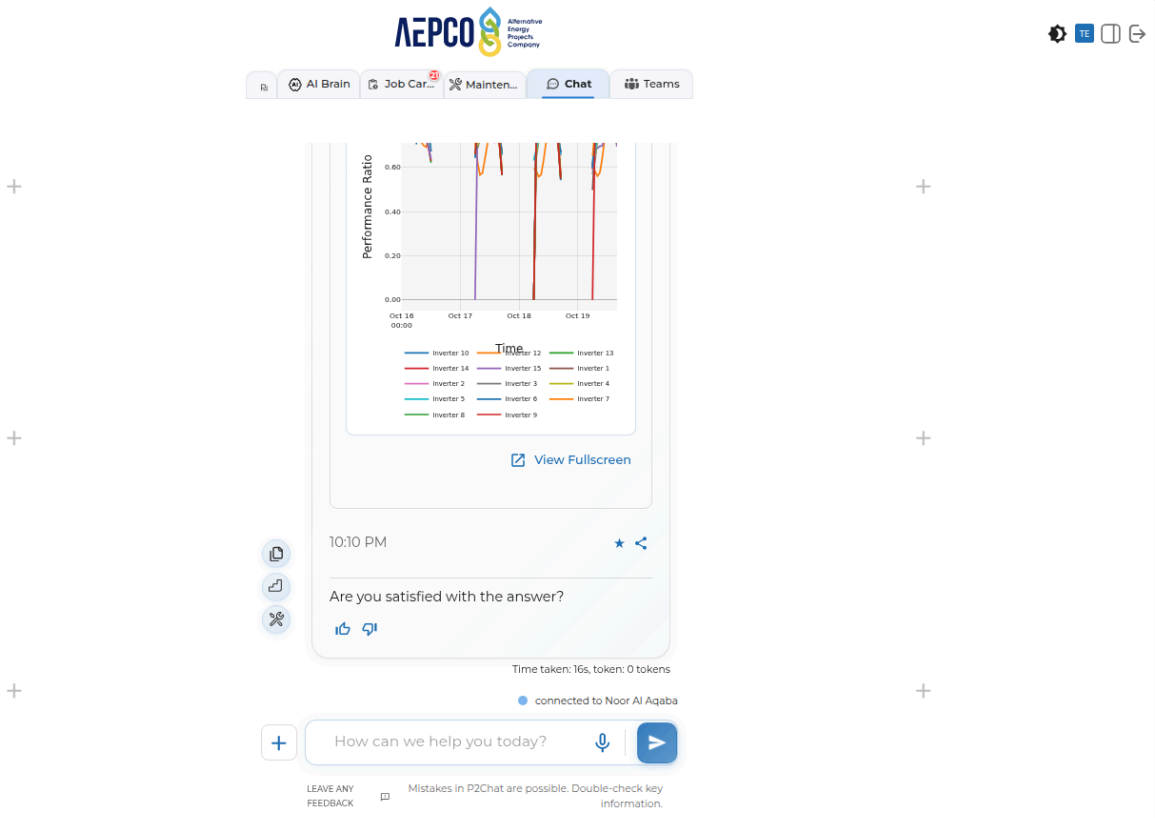
Production • 32.2s

Question: Please plot the last 4 days hourly performance ratio for all my inverters

Response: I've created the visualization for you. View Fullscreen 10:10 PM ★ Are you satisfied with the answer? Time taken: 16s, token: 0 tokens connected to Noor Al Aqaba LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.generate_graphs_via_code, functions.perf_ratio_ag`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



10. Noor_Al_aqaba

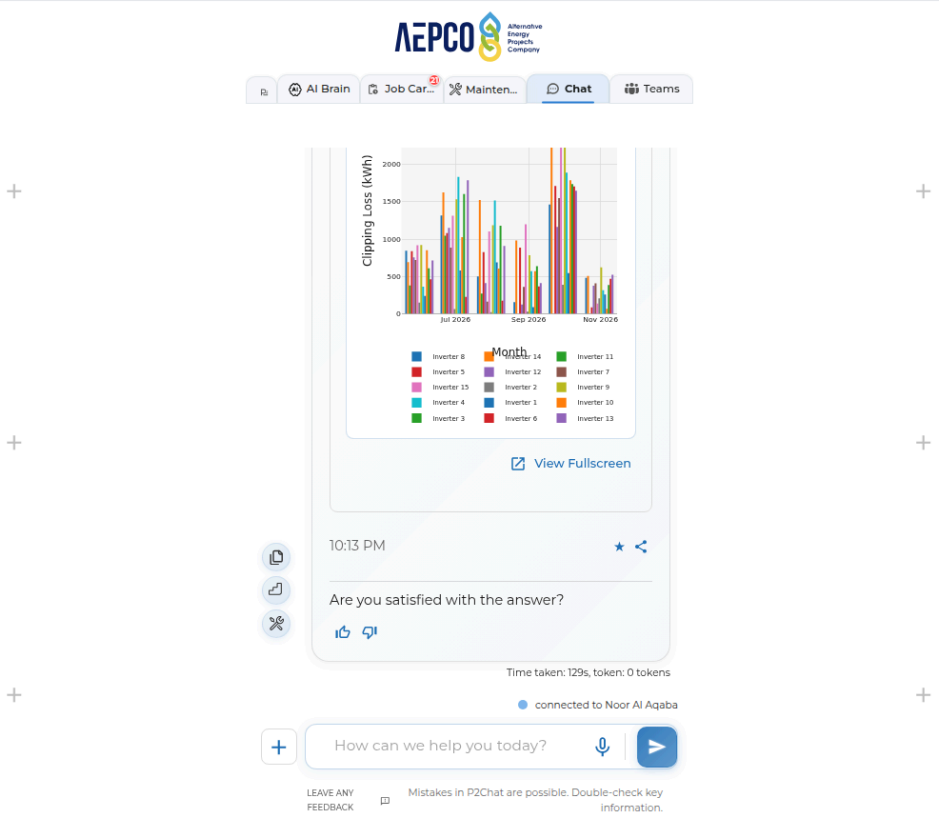
Production • 145.0s

Question: Please figure out how much clipping loss was found in the last 6 months broken by inverters, list them down and plot it as well

Response: Clipping Loss by Inverter (Last 6 Months: 2026-04-19 to 2026-10-19) - Inverter 1: 110,770.36 kWh - Inverter 2: 111,470.02 kWh - Inverter 3: 111,454.57 kWh - Inverter 4: 111,168.84 kWh - Inverter 5: 93,314.64 kWh - Inverter 6: 111,464.80 kWh - Inverter 7: 111,477.72 kWh - Inverter 8: 111,464.80 kWh - Inverter 9: 110,355.94 kWh - Inverter 10: 93,565.93 kWh - Inverter 11: 50,538.22 kWh - Inverter 12: 111,657.03 kWh - Inverter 13: 111,657.03 kWh - Inverter 14: 111,657.03 kWh - Inverter 15: 92,631.43 kWh Note: The clipping-loss API returned a partial result. Data from 2026-09-10 to 2026-10-19 failed due to rate limits, so actual values may be higher for all inverters. Visualization: - The bar chart of monthly clipping loss by inverter over the last six months has been created. Clipping Loss by Inverter (Last 6 Months: 2026-04-19 to 2026-10-19) - Inverter 1: 110,770.36 kWh - Inverter 2: 111,470.02 kWh - Inverter 3: 111,454.57 kWh - Inverter 4

TECHNICAL EVIDENCE

Agents used	`data_agent, graph_agent`
Tools used	`functions.generate_graphs_via_code, functions.clipping_loss_via_requests, functions.calculate_clipping_or_curtailment_losses_ag`
API calls made	`4`
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	`0`
Network failures	`0`



11. Noor_Al_aqaba

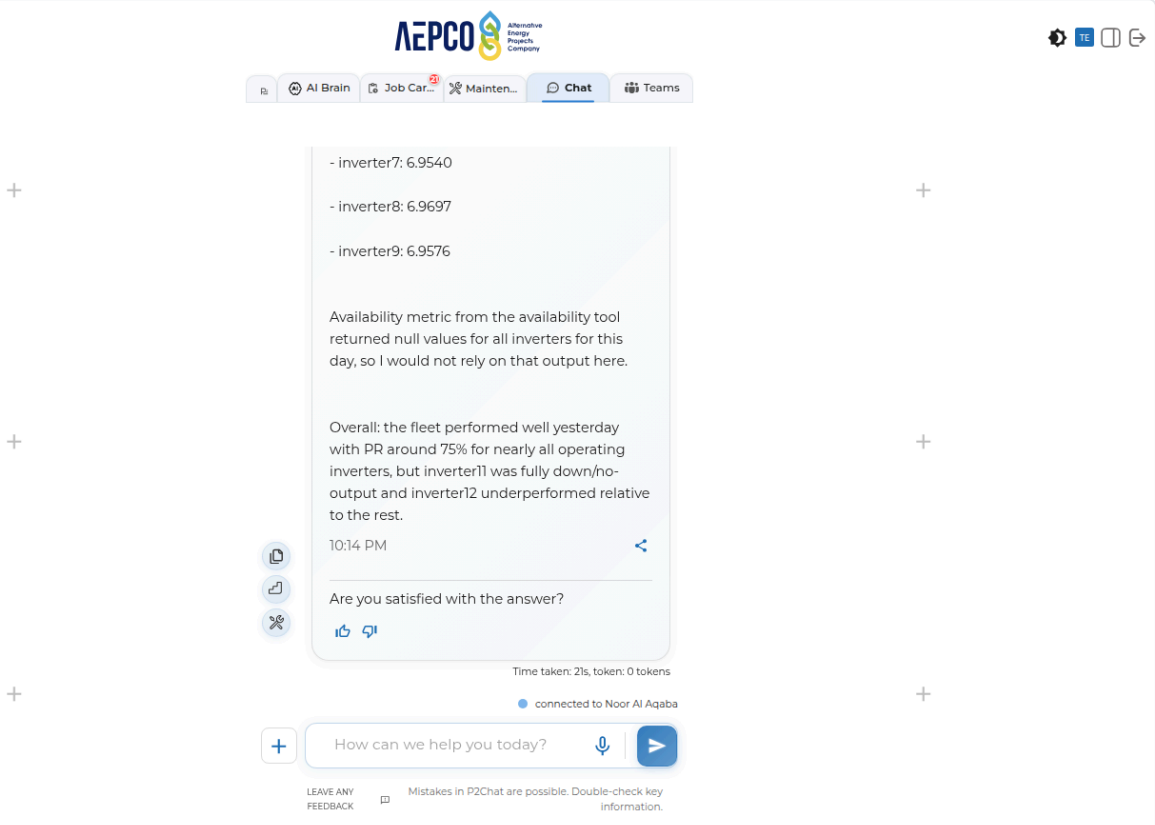
Production • 36.9s

Question: How did my plant perform yesterday?

Response: Yesterday (2026-10-18), the plant's most top-level available view is inverter-level. Key performance metrics: - Total energy generated: 35,610.7 kWh - Reference yield: 7.493 kWh/kWp - Specific yield: mostly ~6.23 to 7.05 kWh/kWp across inverters - Performance ratio (PR): mostly ~0.75 across operating inverters Highlights: - Best PR: inverter15 = 0.7646 - Lowest operating PR: inverter12 = 0.6724 - inverter11 produced 0.0 kWh and its PR/specific yield were null, indicating it was not operating or had no valid production data yesterday. Per-inverter energy (sorted): - inverter1: 2496.8 kWh - inverter10: 2413.0 kWh - inverter11: 0.0 kWh - inverter12: 2258.6 kWh - inverter13: 2515.2 kWh - inverter14: 2533.2 kWh - inverter15: 2443.4 kWh - inverter2: 2480.6 kWh - inverter3: 2487.5 kWh - inverter4: 2495.8 kWh - inverter5: 2395.3 kWh - inverter6: 2526.1 kWh - inverter7: 2519.1 kWh - inverter8: 2524.8 kWh - inverter9: 2520.4 kWh Per-inverter

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`multi_tool_use.parallel`
API calls made	`2`
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	`0`
Network failures	`0`



12. Noor_Al_aqaba

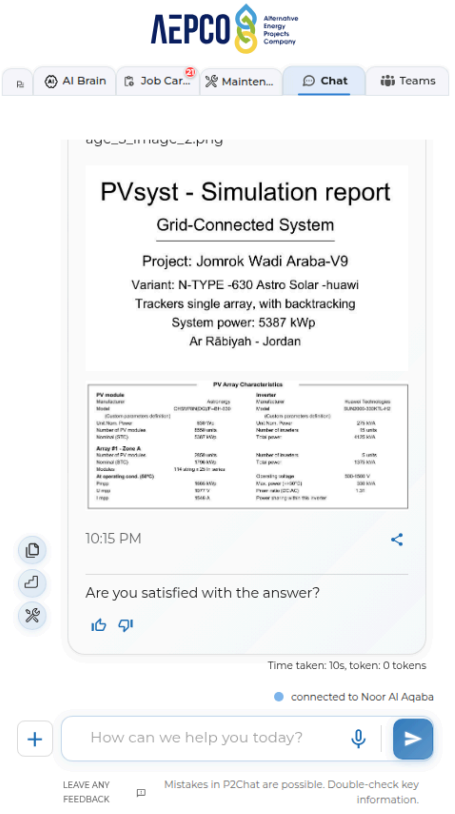
Production • 26.2s

Question: Based on pvsyst report, what is the rated power of this plant and array configuration

Response: Found 2 relevant PVsyst report sections: 1. [Noor_Al_Aqaba] Grid Connected System – Rated Power (Page 1, Table 1) 2. [Noor_Al_Aqaba] PV Array Characteristics (Page 3, Table 2) Please access the URLs provided to view the detailed information. Here is the reference: https://image-pvsyst-aepco-sydney.s3.ap-southeast-2.amazonaws.com/testuser1/Noor_Al_Aqaba/page_1_image_1.png https://image-pvsyst-aepco-sydney.s3.ap-southeast-2.amazonaws.com/testuser1/Noor_Al_Aqaba/page_3_image_2.png 10:15 PM Are you satisfied with the answer? Time taken: 10s, token: 0 tokens connected to Noor Al Aqaba LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`pvsyst_agent`
Tools used	`pvsyst_tool`
API calls made	3
Last API call	`GET /testuser2/chat/ → 404`
Console errors	3
Network failures	1



13. Noor_Al_aqaba

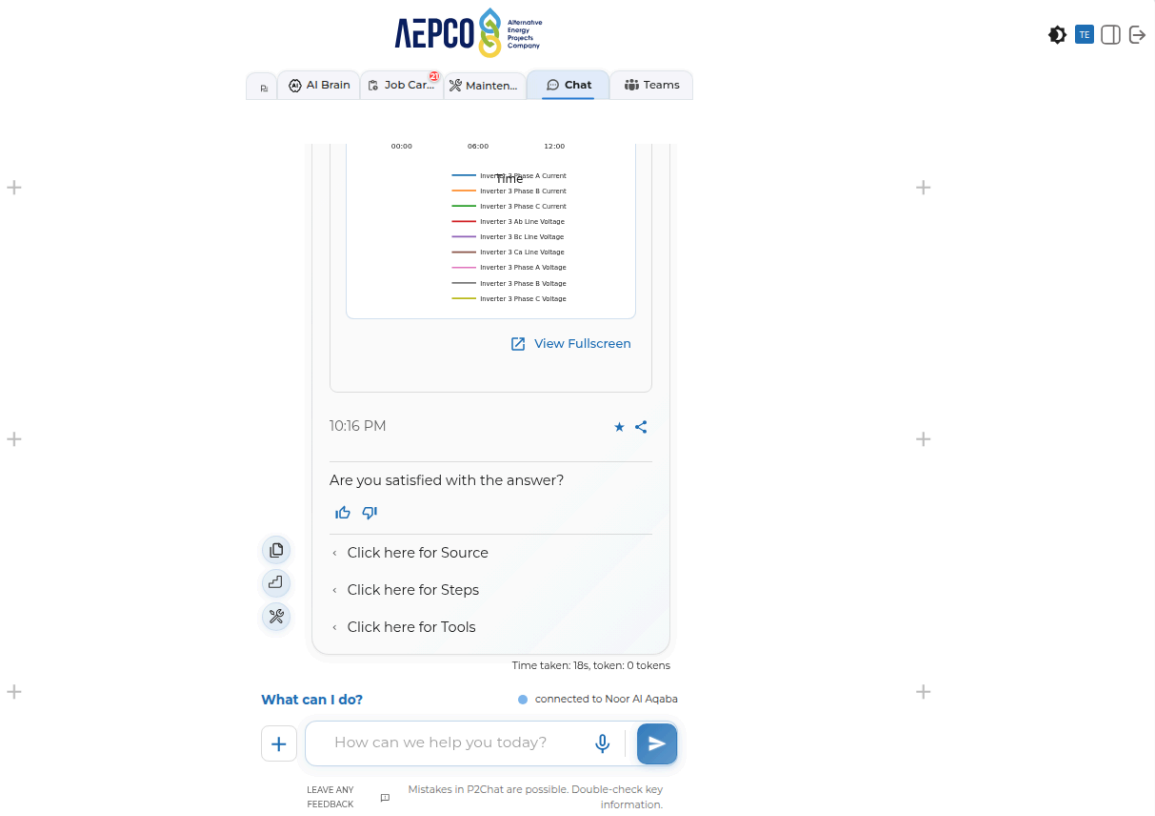
Production • 34.4s

Question: Please plot my current and voltage in a dual axes plot on the deepest possible granularity level for inverter3

Response: I've created the visualization for you. View Fullscreen 10:16 PM ★ Are you satisfied with the answer? Click here for Source Click here for Steps Click here for Tools Time taken: 18s, token: 0 tokens What can I do? connected to Noor Al Aqaba LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.extract_timeseries_data_ag, functions.execute_custom_code, functions.generate_graphs_via_code`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	2
Network failures	0



14. Noor_Al_aqaba

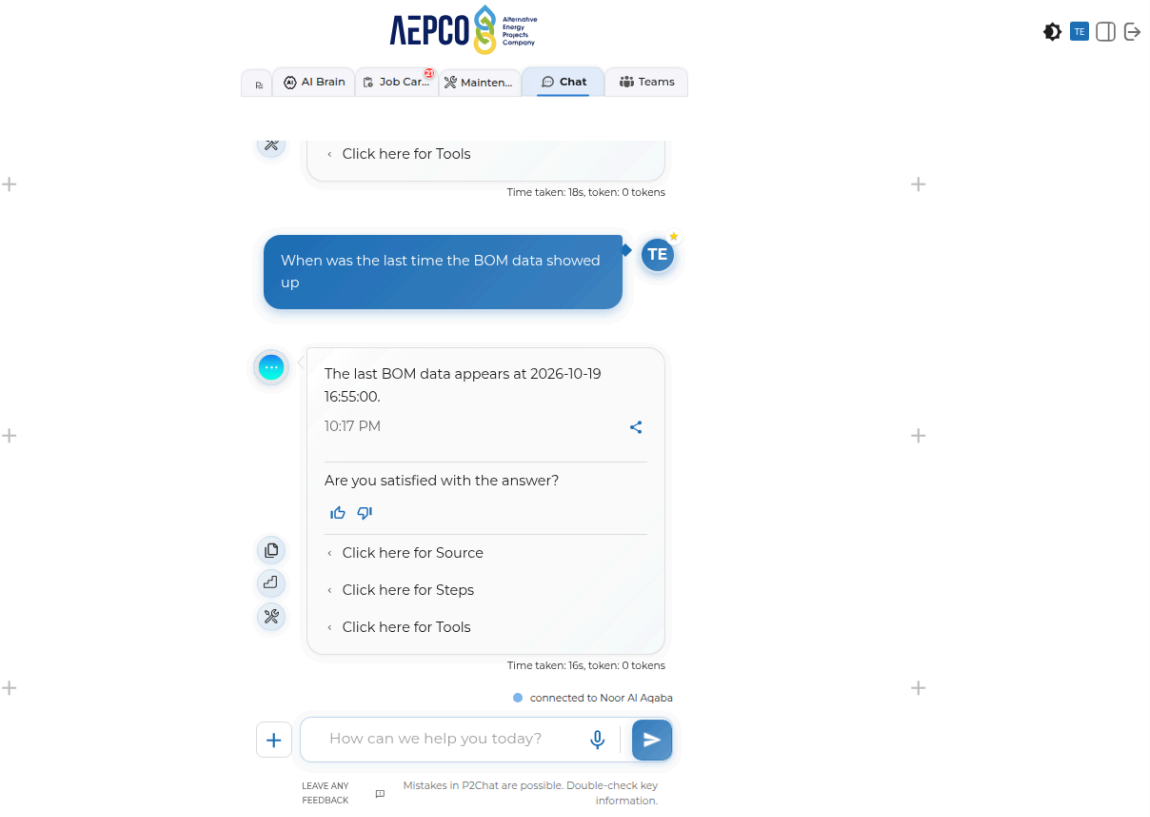
Production • 30.6s

Question: When was the last time the BOM data showed up

Response: The last BOM data appears at 2026-10-19 16:55:00. 10:17 PM Are you satisfied with the answer? Click here for Source Click here for Steps Click here for Tools Time taken: 16s, token: 0 tokens connected to Noor Al Aqaba LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`extract_timeseries_data_ag`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



15. Noor_Al_aqaba

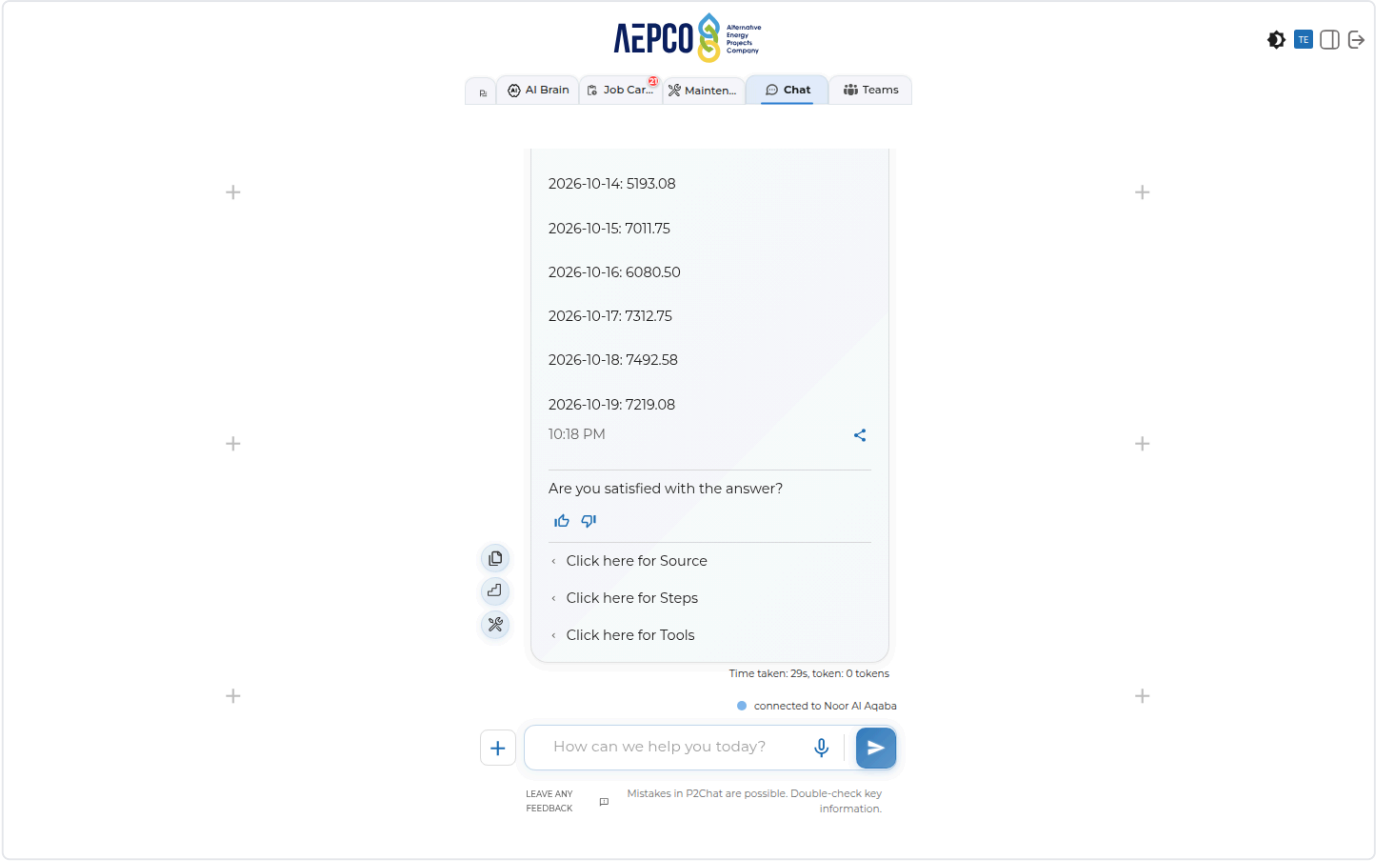
Production • 44.3s

Question: Please find out the daily POA poa summed up as a Wh/m^2, daily for the last 12 months and then calculate the median, mean and kurt

Response: Daily POA sums (Wh/m²) over the requested last 12 months were only available for an incomplete period: 2026-05-09 to 2026-10-19. Summary stats for available daily POA sums: - Median: 10082.33 Wh/m² - Mean: 9559.61 Wh/m² - Kurtosis: 1.3299 - Days with data: 164 Data coverage note: - Requested: 2025-10-19 to 2026-10-19 - Available sensor data: 2026-05-09 10:45 to 2026-10-19 16:55 Daily sums (Wh/m²): 2026-05-09: 5513.67 2026-05-10: 8749.50 2026-05-11: 10391.67 2026-05-12: 9233.67 2026-05-13: 10041.08 2026-05-14: 10073.25 2026-05-15: 6948.17 2026-05-16: 9748.33 2026-05-17: 9177.75 2026-05-18: 8215.33 2026-05-19: 9160.42 2026-05-20: 7635.75 2026-05-21: 10173.92 2026-05-22: 10097.75 2026-05-23: 10162.50 2026-05-24: 10796.58 2026-05-25: 10817.67 2026-05-26: 8859.33 2026-05-27: 10436.67 2026-05-28: 10813.42 2026-05-29: 10558.42 2026-05-30: 10942.00 2026-05-31: 10403.67 2026-06-01: 10978.42 2026-06-02: 10489.17 2026-06-03:

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`extract_timeseries_data_ag, execute_custom_code`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



16. Noor_Al_aqaba

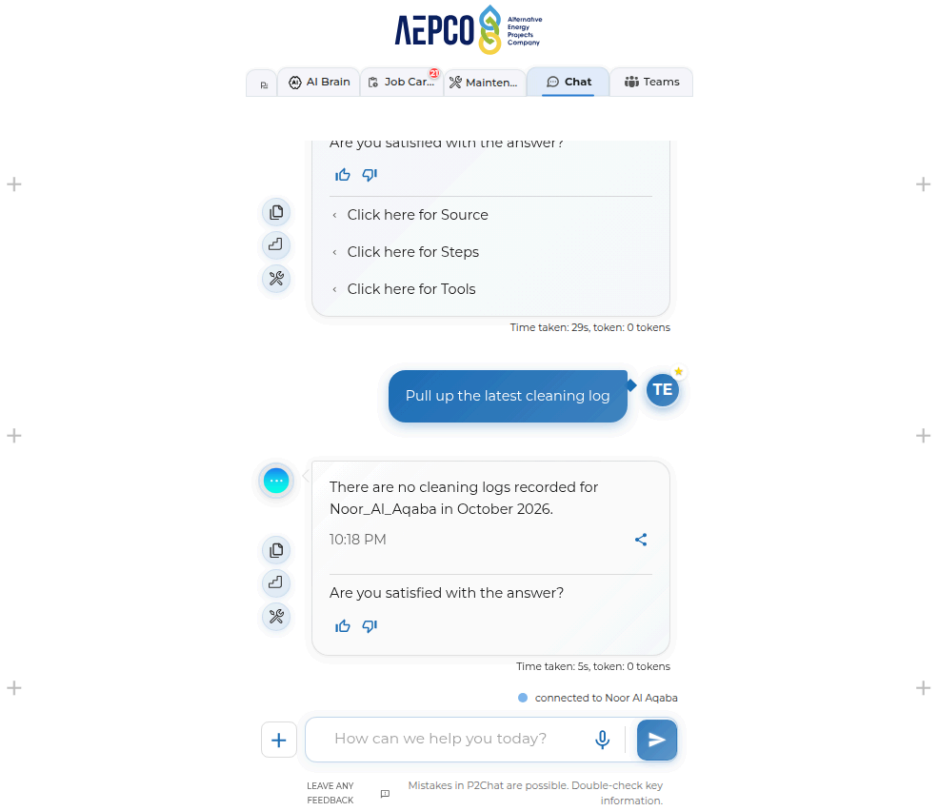
Production • 20.0s

Question: Pull up the latest cleaning log

Response: There are no cleaning logs recorded for Noor_Al_Aqaba in October 2026. 10:18 PM Are you satisfied with the answer? Time taken: 5s, token: 0 tokens connected to Noor Al Aqaba LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_entry_agent`
API calls made	2
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	0
Network failures	0



17. Noor_Al_aqaba

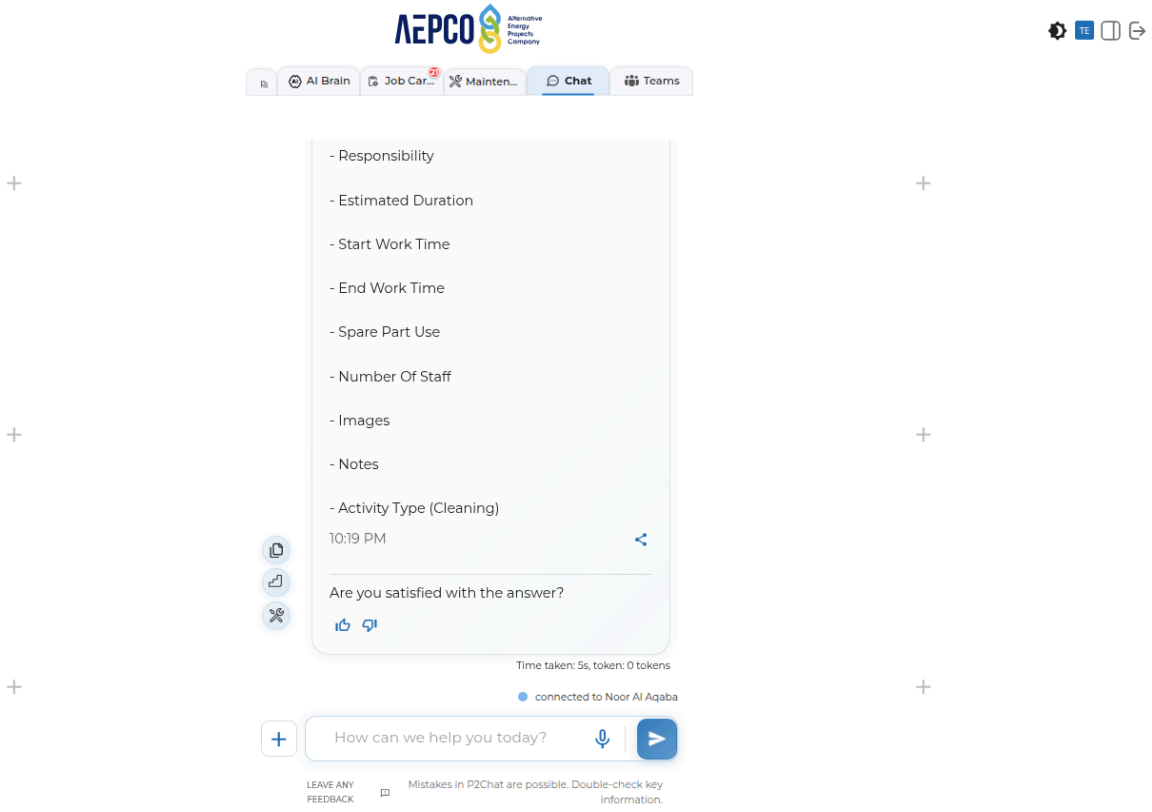
Production • 20.0s

Question: Jim cleaned the pyranometer today, can you add it to the cleaning log?

Response: To add a cleaning activity log, please provide the following required details: - Date - Fault Cause - Action Taken - Status Site will be set automatically to Noor_Al_Aqaba. Optional fields you can also provide, but they are not required: - Priority - Responsibility - Estimated Duration - Start Work Time - End Work Time - Spare Part Use - Number Of Staff - Images - Notes - Activity Type (Cleaning) 10:19 PM Are you satisfied with the answer? Time taken: 5s, token: 0 tokens connected to Noor Al Aqaba LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_entry_agent`
API calls made	`1`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



18. Noor_Al_aqaba

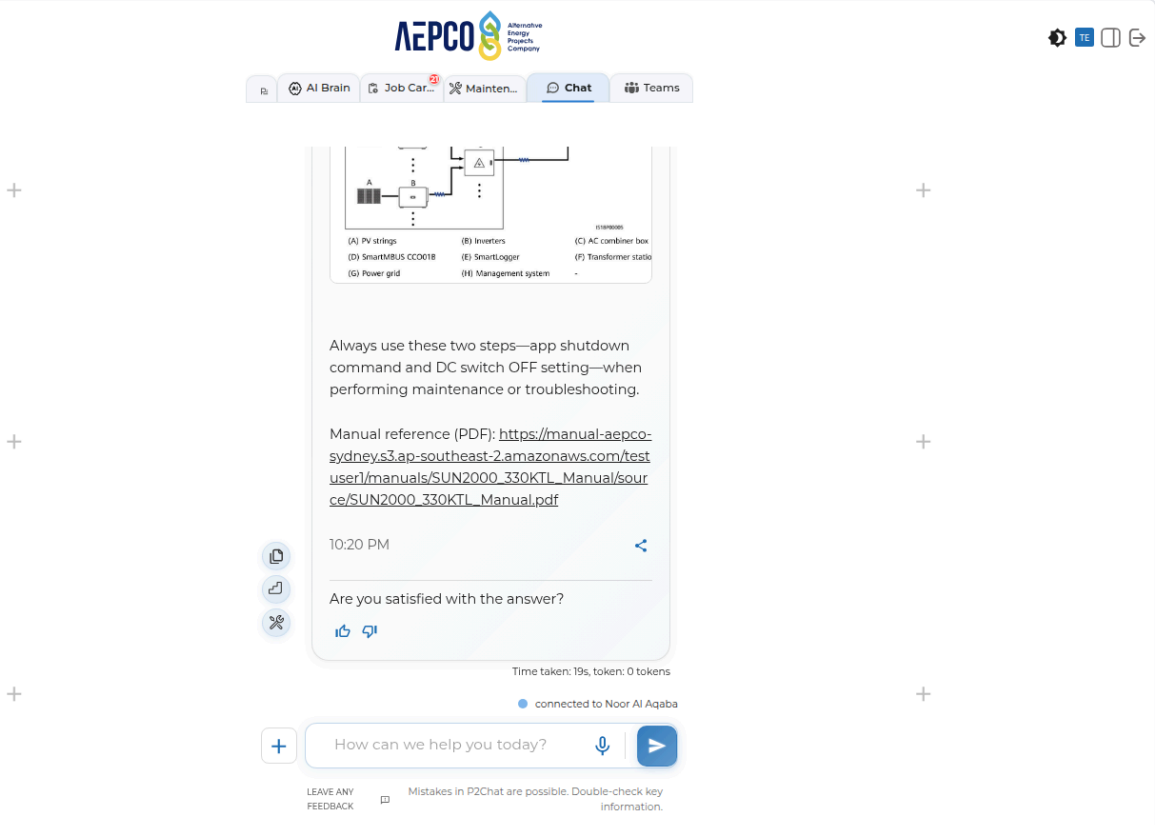
Production • 33.7s

Question: How can I shut down my inverters?

Response: To shut down your inverters (SUN2000 Series): 1. Open the inverter monitoring app and log in. Go to Maintenance > Inverter ON/OFF, and send a shutdown command to the inverter. 2. Physically set the DC SWITCH on the inverter to OFF. This isolates the inverter from the PV strings and AC connection. Refer to the system layout diagram to identify inverters (B), DC switches, and related components. - System layout diagram: - Inverter and connection diagram: Always use these two steps—app shutdown command and DC switch OFF setting—when performing maintenance or troubleshooting. Manual reference (PDF): https://manual-aepco-sydney.s3.ap-southeast-2.amazonaws.com/testuser1/manuals/SUN2000_330KTL_Manual/source/SUN2000_330KTL_Manual.pdf 10:20 PM Are you satisfied with the answer? Time taken: 19s, token: 0 tokens connected to Noor Al Aqaba LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`manual_agent`
Tools used	`functions.manual_rag_tool`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



19. Noor_Al_aqaba

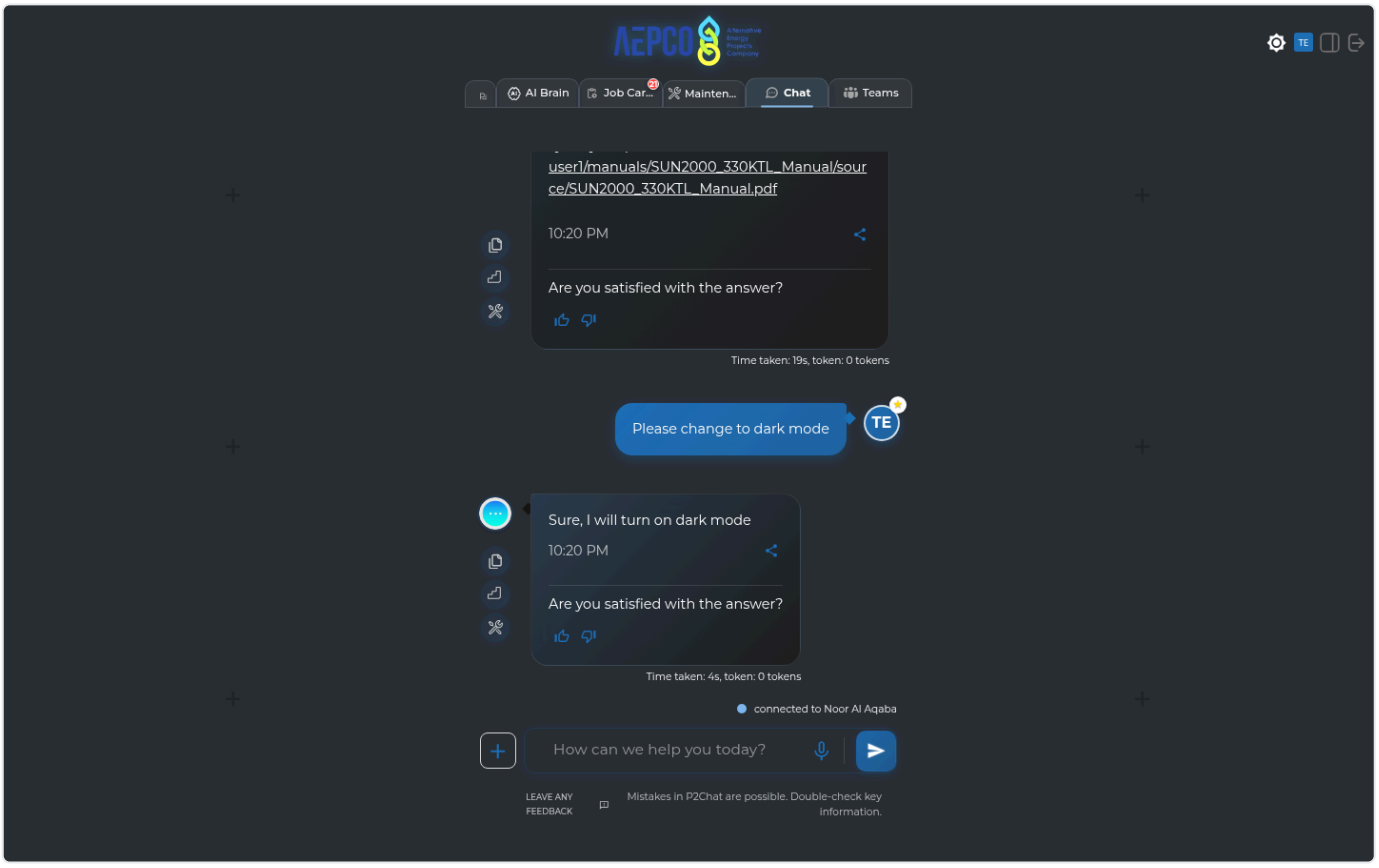
Production • 20.0s

Question: Please change to dark mode

Response: Sure, I will turn on dark mode 10:20 PM Are you satisfied with the answer? Time taken: 4s, token: 0 tokens connected to Noor Al Aqaba LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`ui_agent`
API calls made	2
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	0
Network failures	0



20. Noor_Al_aqaba

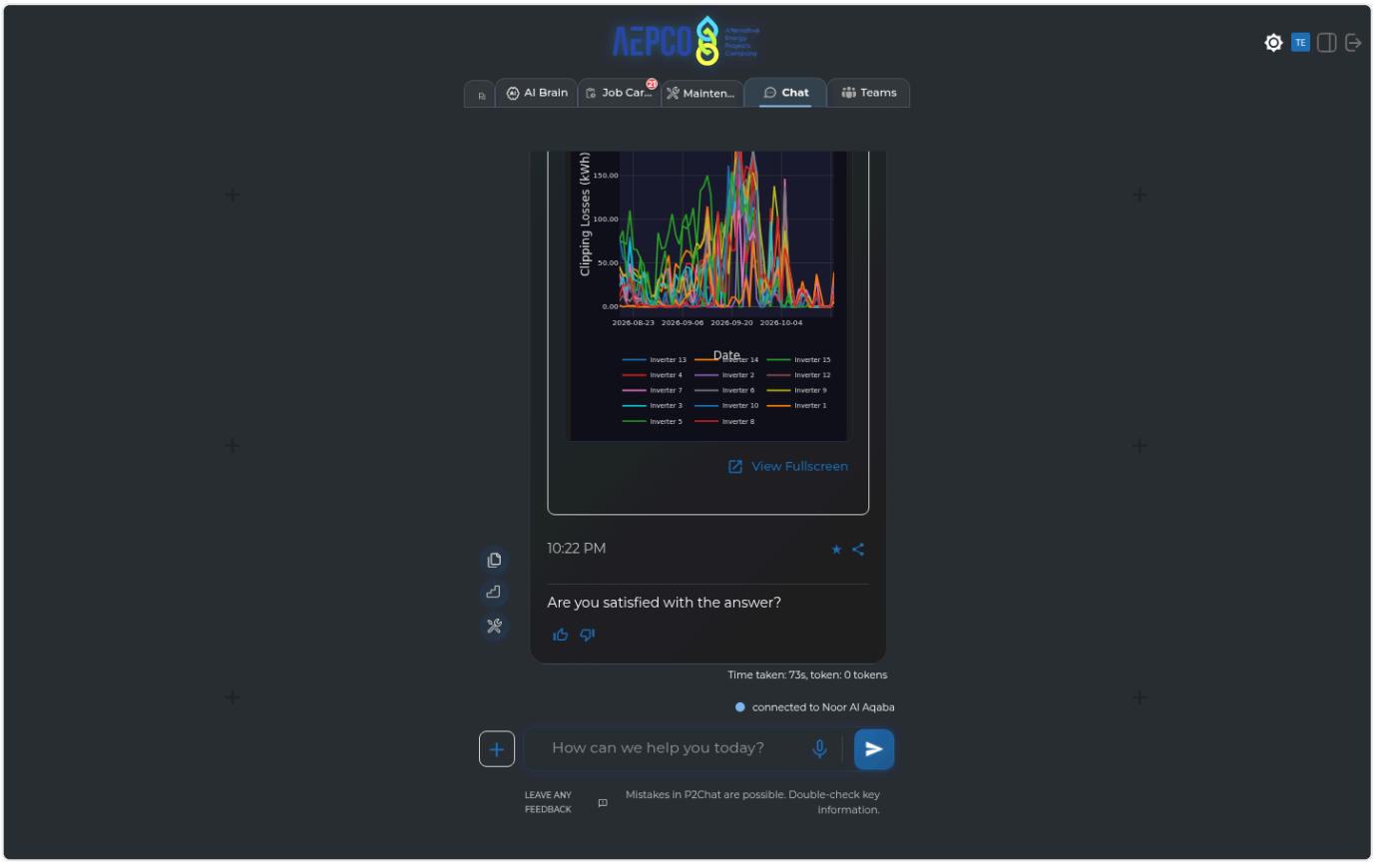
Production • 88.6s

Question: Please figure out how much clipping loss was found in the last 2 months broken by inverters, list them down and plot it as well

Response: Clipping Losses by Inverter (2026-08-19 to 2026-10-19): - Inverter 1: 33,912.17 kWh - Inverter 2: 34,348.01 kWh - Inverter 3: 34,344.96 kWh - Inverter 4: 34,344.96 kWh - Inverter 5: 27,316.40 kWh - Inverter 6: 34,350.03 kWh - Inverter 7: 34,350.03 kWh - Inverter 8: 34,350.03 kWh - Inverter 9: 34,109.96 kWh - Inverter 10: 27,317.34 kWh - Inverter 11: 0.00 kWh - Inverter 12: 34,348.01 kWh - Inverter 13: 34,348.01 kWh - Inverter 14: 34,348.01 kWh - Inverter 15: 27,317.34 kWh Data Quality Status: Good I've created a line chart showing daily clipping losses by inverter for the last two months. Clipping Losses by Inverter (2026-08-19 to 2026-10-19): - Inverter 1: 33,912.17 kWh - Inverter 2: 34,348.01 kWh - Inverter 3: 34,344.96 kWh - Inverter 4: 34,344.96 kWh - Inverter 5: 27,316.40 kWh - Inverter 6: 34,350.03 kWh - Inverter 7: 34,350.03 kWh - Inverter 8: 34,350.03 kWh - Inverter 9: 34,109.96 kWh - Inverter 10: 27,317.34 kWh - Inver

TECHNICAL EVIDENCE

Agents used	`data_agent, graph_agent`
Tools used	`functions.clipping_loss_via_requests, functions.generate_graphs_via_code, functions.calculate_clipping_or_curtailment_losses_ag`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



21. Noor_Al_aqaba

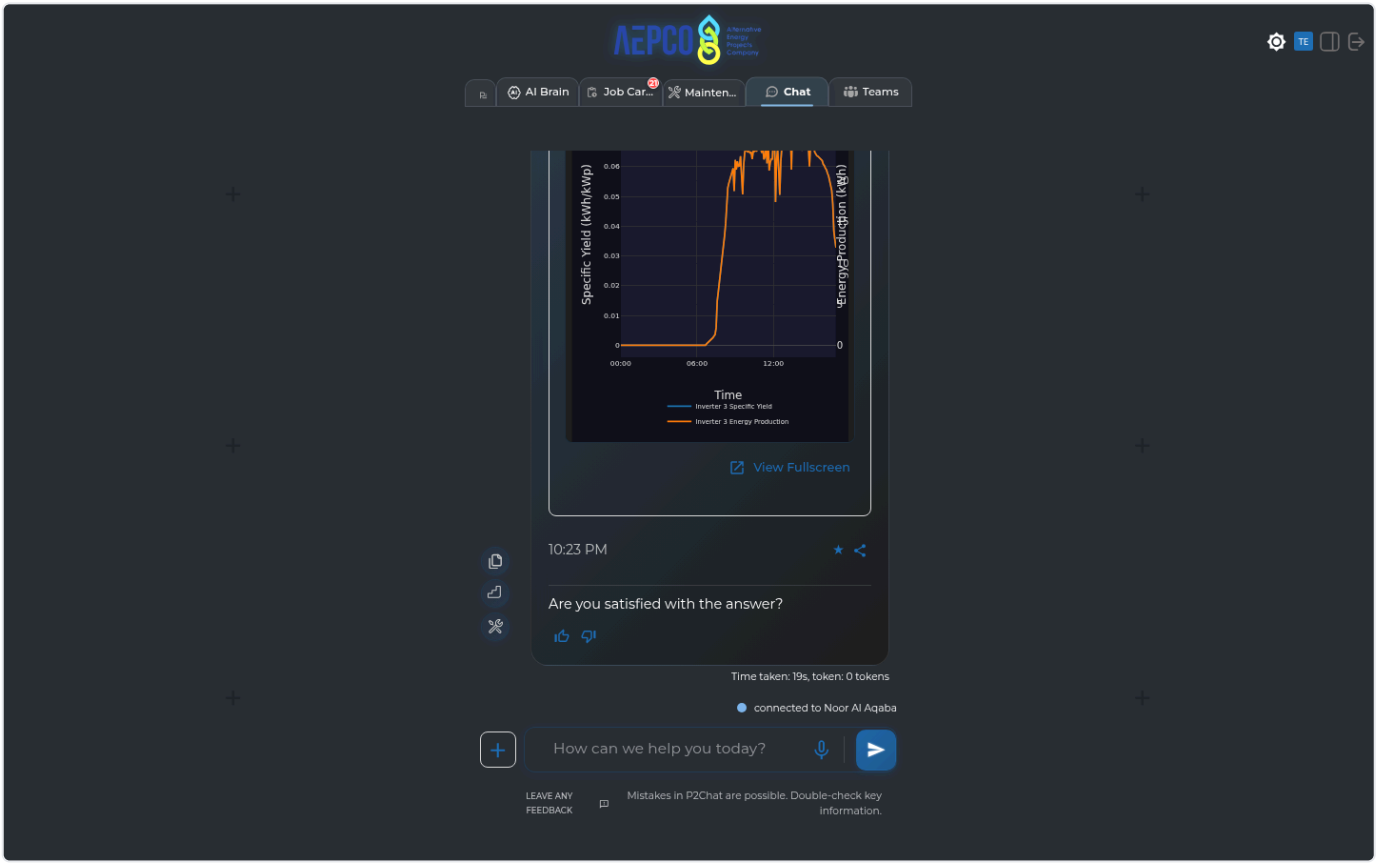
Production • 33.9s

Question: Please repeat the first plotting question, but interchange it with Specific yield on 1 axes, and energy production on the other axes on a single dual axes plot (Memory Test)

Response: I've created the visualization for you. View Fullscreen 10:23 PM ★ Are you satisfied with the answer? Time taken: 19s, token: 0 tokens connected to Noor Al Aqaba LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.specific_yield_ag, functions.generate_graphs_via_code, functions.total_energy_func_ag`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



22. MOF

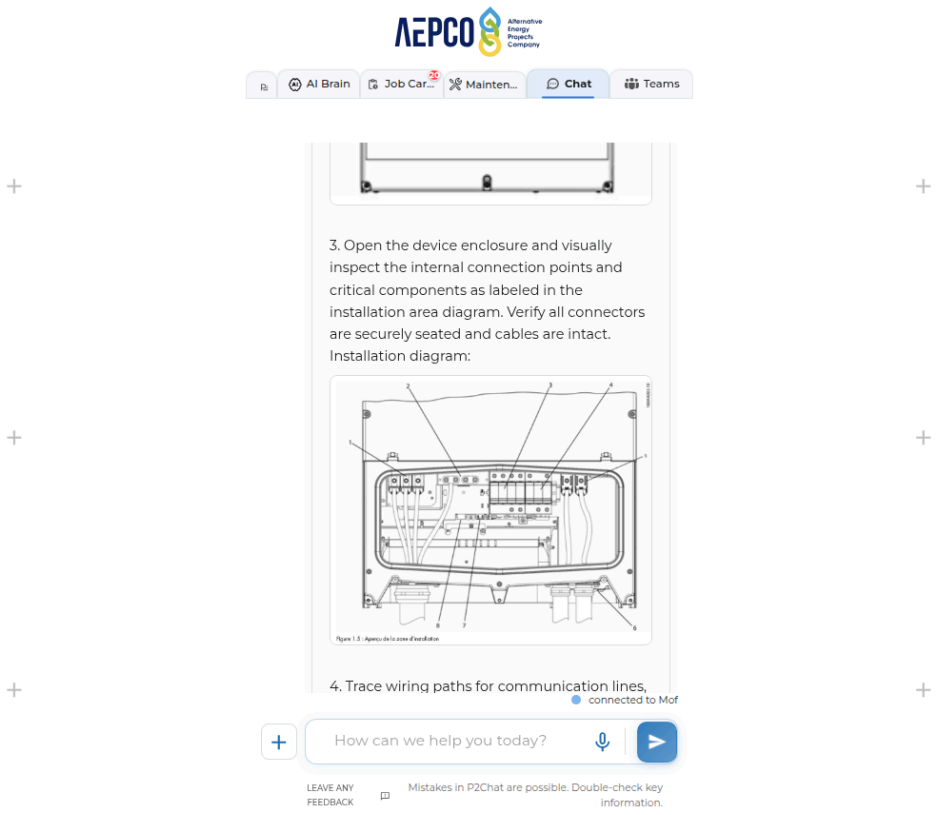
Production • 32.1s

Question: I have a COMMUNICATION_FAULT what do I do to my inverters?

Response: To troubleshoot the COMMUNICATION_FAULT error on inverters, follow these steps referencing the specific diagrams for each component: 1. Inspect the pathways for electrical flow and communication within the device to identify possible issues in system components and connections. View the pathway diagram here: 2. Check the front keypad interface on the enclosure for proper command input and system status. Refer to this diagram: 3. Open the device enclosure and visually inspect the internal connection points and critical components as labeled in the installation area diagram. Verify all connectors are securely seated and cables are intact. Installation diagram: 4. Trace wiring paths for communication lines, paying close attention to labeled cables/connectors. Ensure all are attached correctly and undamaged, referencing this overview: 5. If the fault persists, consult the internal layout to check labeled components such as electrical connections, fans, and communication interfaces

TECHNICAL EVIDENCE

Agents used	`manual_agent`
Tools used	`functions.manual_rag_tool`
API calls made	`12`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`1`
Network failures	`0`



23. MOF

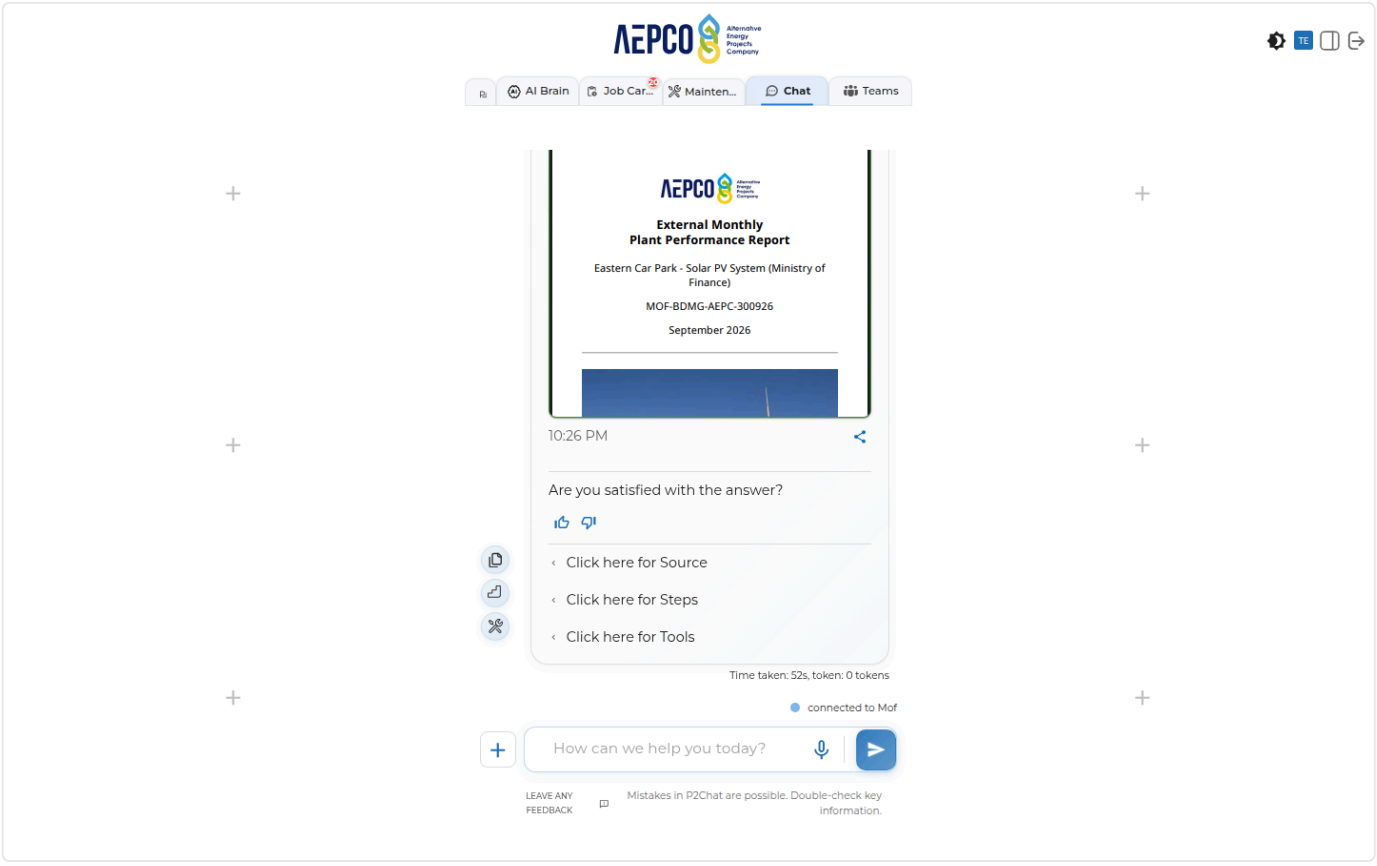
Production • 67.4s

Question: Please generate last months report

Response: I've generated the external report for September 2026. You can view it here: 'https://report-aepco-sydney.s3.ap-southeast-2.amazonaws.com/MOF_power_plant_September_2026_external_report.pdf' MOF Report 10:26 PM Are you satisfied with the answer? Click here for Source Click here for Steps Click here for Tools Time taken: 52s, token: 0 tokens connected to Mof LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`report_agent`
Tools used	`agent_external_report_tool`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



24. MOF

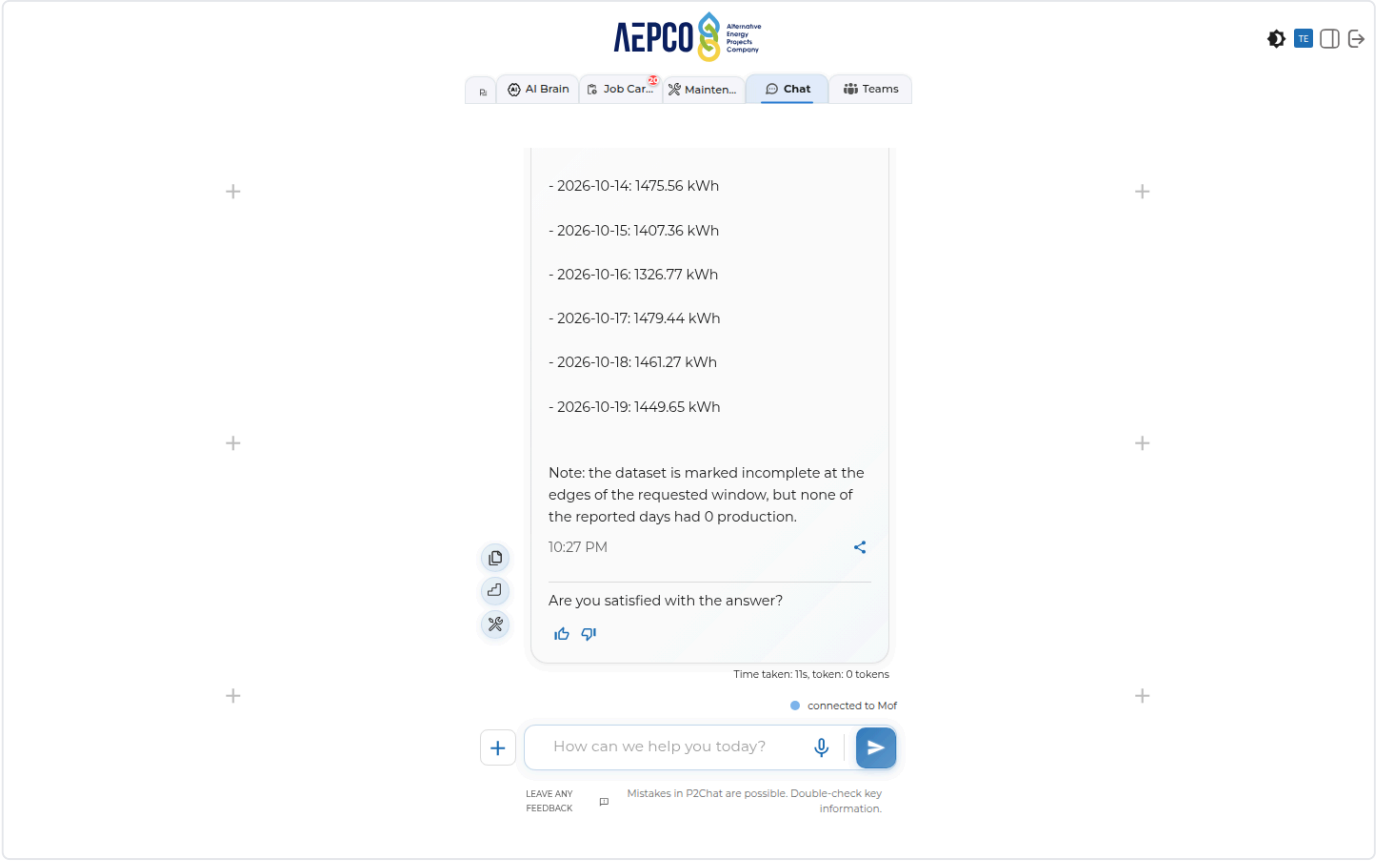
Production • 26.1s

Question: Was there any 0 production days in the last 7 days?

Response: No. There were no 0-production days in the last 7 days. Daily plant energy: - 2026-10-13: 1400.01 kWh - 2026-10-14: 1475.56 kWh - 2026-10-15: 1407.36 kWh - 2026-10-16: 1326.77 kWh - 2026-10-17: 1479.44 kWh - 2026-10-18: 1461.27 kWh - 2026-10-19: 1449.65 kWh Note: the dataset is marked incomplete at the edges of the requested window, but none of the reported days had 0 production. 10:27 PM Are you satisfied with the answer? Time taken: 11s, token: 0 tokens connected to Mof LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`functions.total_energy_func_ag`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



25. MOF

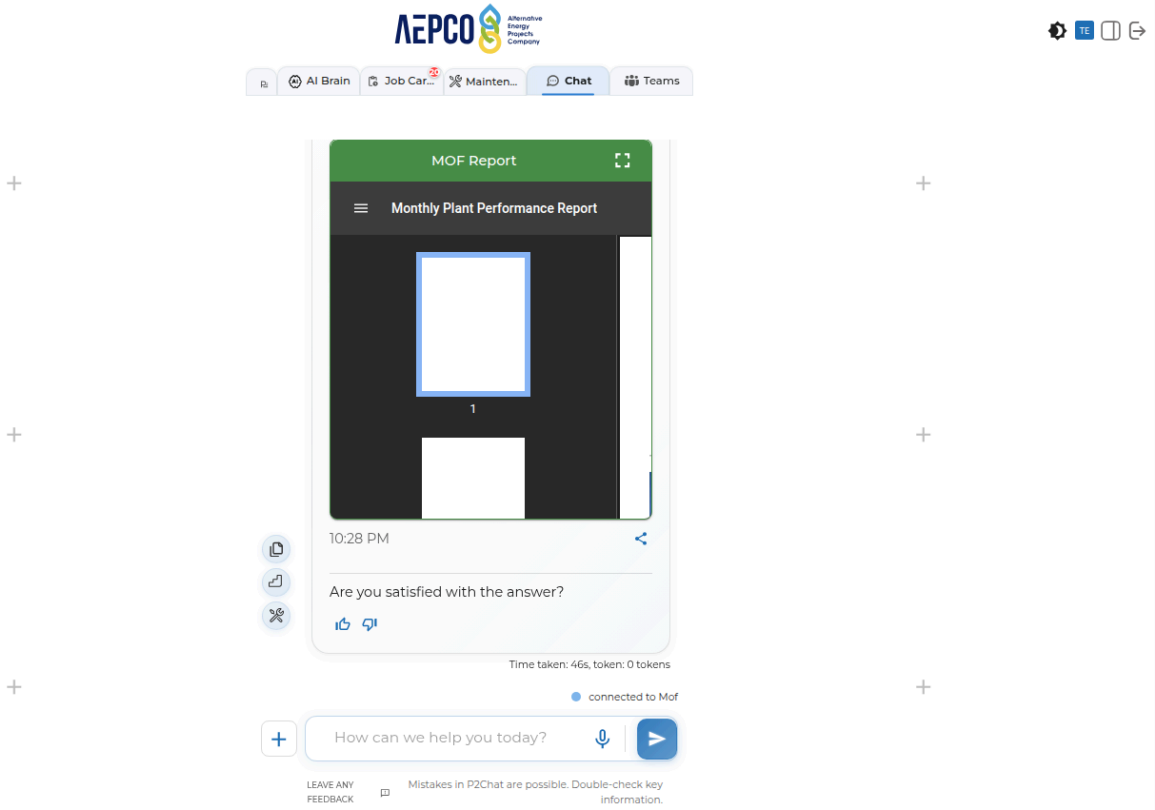
Production • 61.1s

Question: Please generate December 2025 performance report

Response: I've generated the performance report for December 2025. You can view it here: 'https://report-aepco-sydney.s3.ap-southeast-2.amazonaws.com/MOF_power_plant_December_2025_external_report.pdf' MOF Report 10:28 PM Are you satisfied with the answer? Time taken: 46s, token: 0 tokens connected to Mof LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`report_agent`
Tools used	`agent_external_report_tool`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



26. MOF

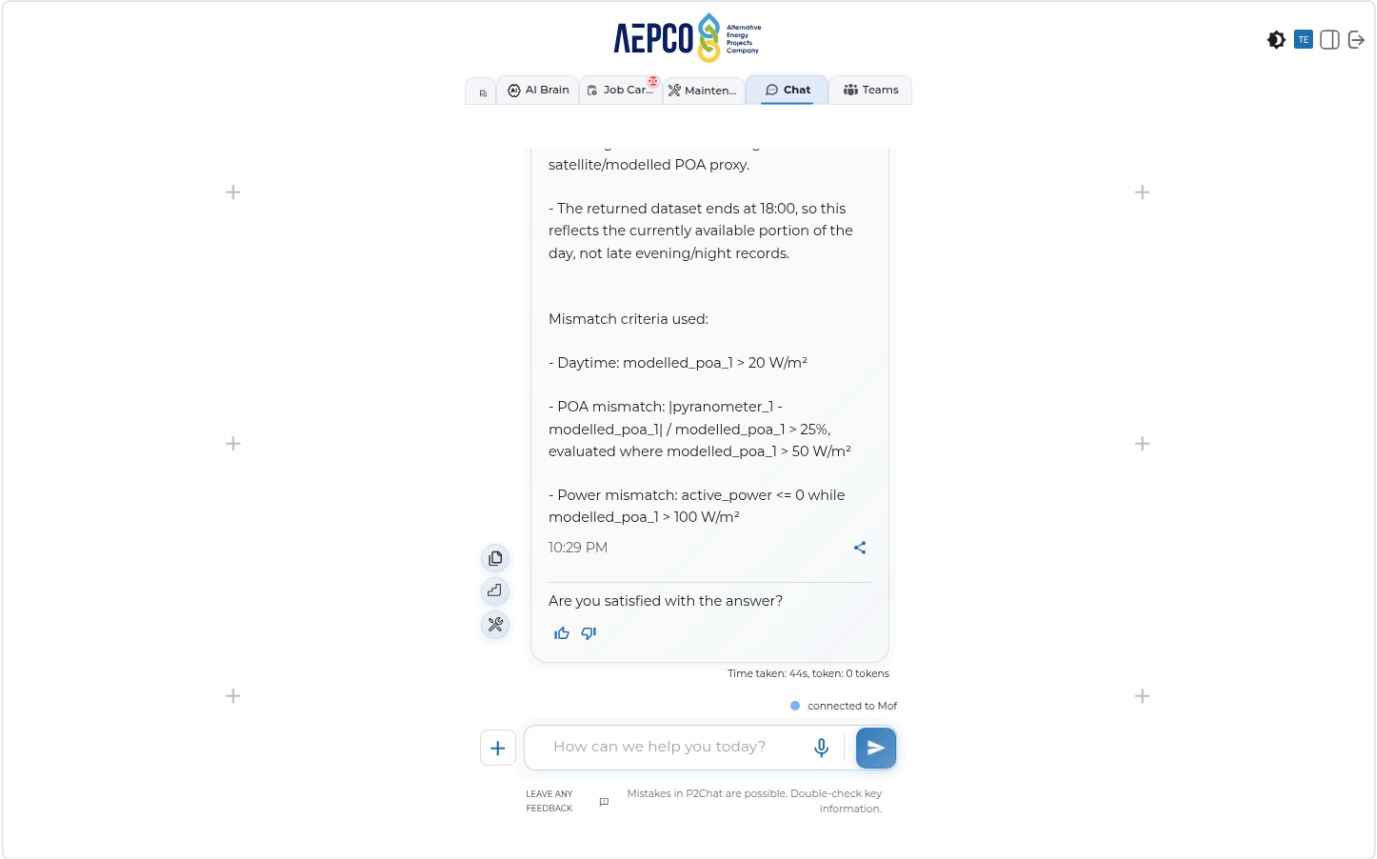
Production • 59.6s

Question: How is my data quality for POA and power? How many of them show missing data during daytime or mismatches with satellite prediction?

Response: For 2026-10-19, using daytime defined as modelled POA > 20 W/m²: - Daytime samples: 44 - Missing POA samples during daytime: 8 - Missing plant power samples during daytime: 0 - POA mismatches vs satellite/modelled POA proxy: 35 - Power mismatches vs satellite/modelled POA proxy: 0 Assessment: - Power data quality looks good for daytime completeness on the available data window. - POA data quality is weaker: 8 daytime gaps and a high mismatch count against the satellite/modelled POA proxy. - The returned dataset ends at 18:00, so this reflects the currently available portion of the day, not late evening/night records. Mismatch criteria used: - Daytime: modelled_poa_1 > 20 W/m² - POA mismatch: |pyranometer_1 - modelled_poa_1| / modelled_poa_1 > 25%, evaluated where modelled_poa_1 > 50 W/m² - Power mismatch: active_power <= 0 while modelled_poa_1 > 100 W/m² 10:29 PM Are you satisfied with the answer? Time taken: 44s, token: 0 tokens connected to Mof LEAVE

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`execute_custom_code, extract_timeseries_data_ag`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



27. MOF

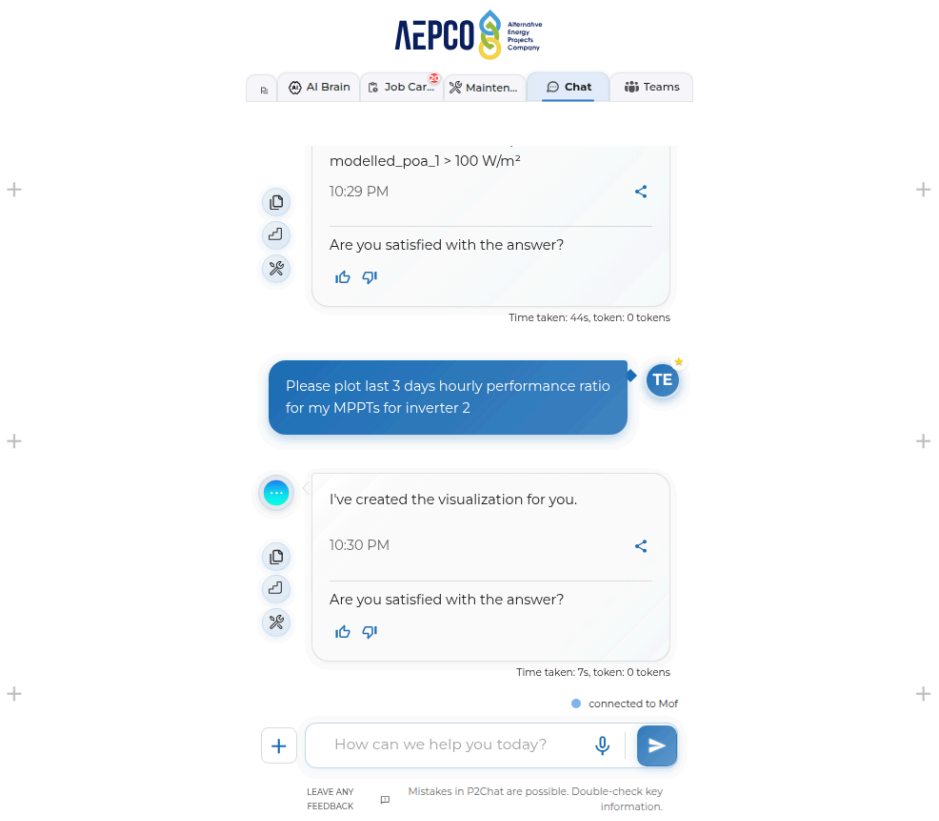
Production • 21.5s

Question: Please plot last 3 days hourly performance ratio for my MPPTs for inverter 2

Response: I've created the visualization for you. 10:30 PM Are you satisfied with the answer? Time taken: 7s, token: 0 tokens connected to Mof LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.perf_ratio_ag`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



28. MOF

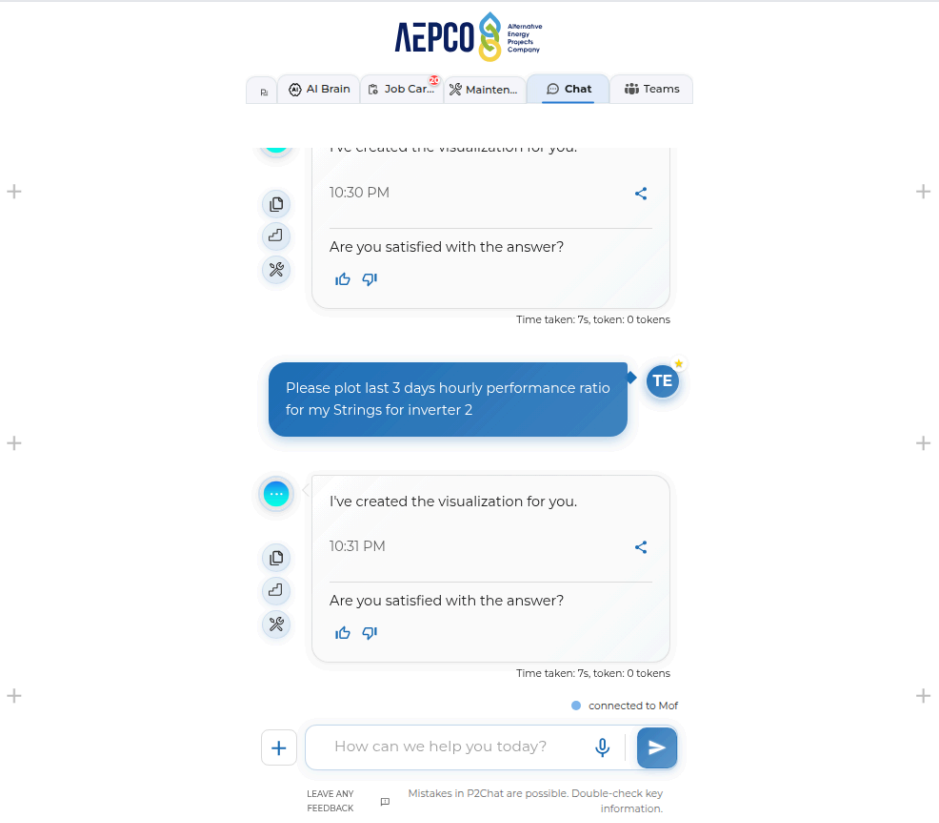
Production • 23.1s

Question: Please plot last 3 days hourly performance ratio for my Strings for inverter 2

Response: I've created the visualization for you. 10:31 PM Are you satisfied with the answer? Time taken: 7s, token: 0 tokens connected to Mof LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.perf_ratio_ag`
API calls made	2
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	0
Network failures	0



29. MOF

Production • 30.9s

Question: Please plot the last 4 days hourly performance ratio for all my inverters

Response: I've created the visualization for you. View Fullscreen 10:31 PM ★ Are you satisfied with the answer? Time taken: 15s, token: 0 tokens connected to Mof LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.perf_ratio_ag, functions.generate_graphs_via_code`
API calls made	`1`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`

30. MOF

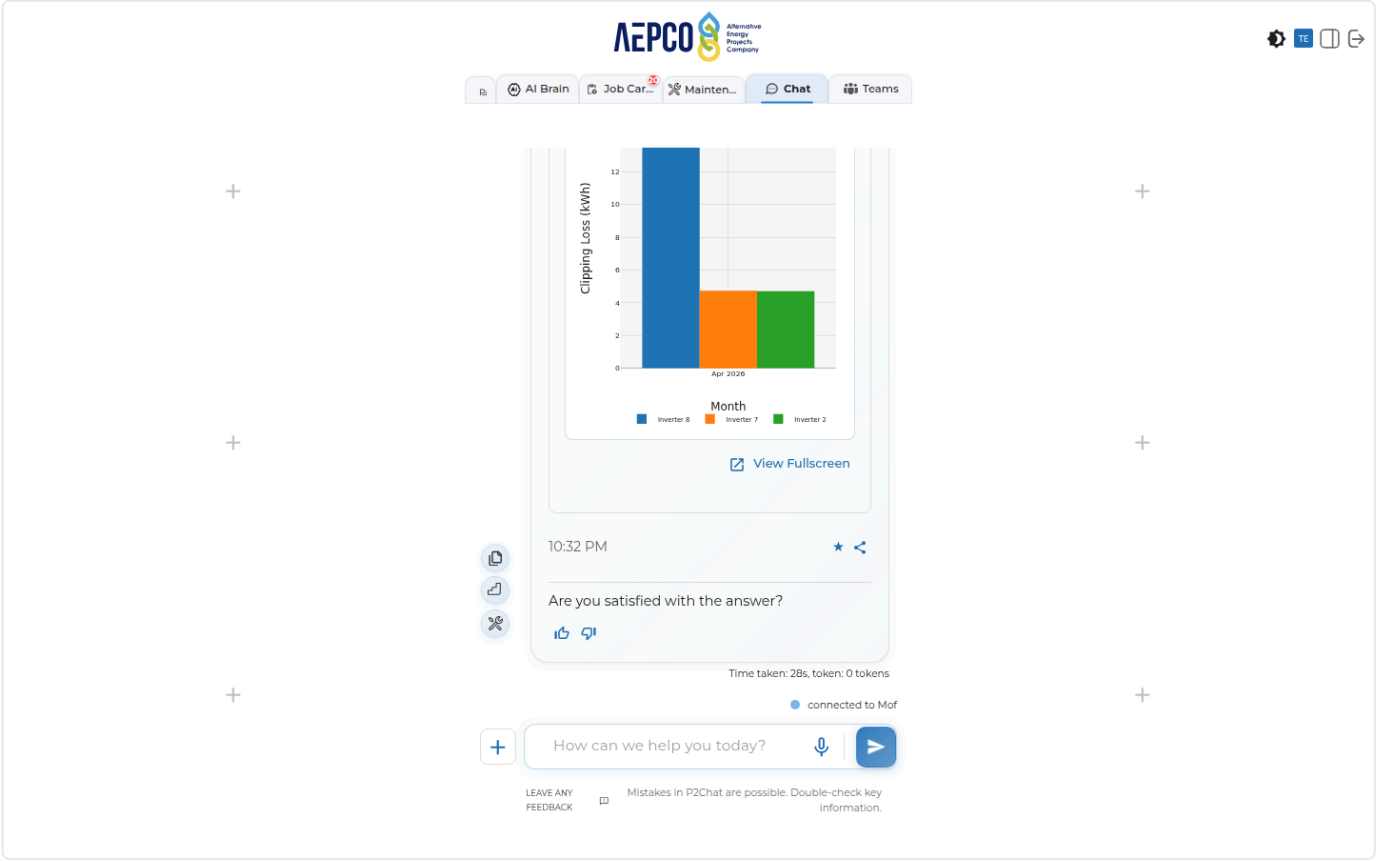
Production • 43.8s

Question: Please figure out how much clipping loss was found in the last 6 months broken by inverters, list them down and plot it as well

Response: Clipping Loss Summary (Last 6 Months: 2026-04-19 to 2026-10-19) - inverter1: 0.00 kWh - inverter2: 4.70 kWh - inverter3: 0.00 kWh - inverter4: 0.00 kWh - inverter5: 0.00 kWh - inverter6: 0.00 kWh - inverter7: 4.74 kWh - inverter8: 15.08 kWh Data Quality Status: Good The visualization has been prepared: a bar chart showing monthly clipping loss per inverter for the last 6 months. Clipping Loss Summary (Last 6 Months: 2026-04-19 to 2026-10-19) - inverter1: 0.00 kWh - inverter2: 4.70 kWh - inverter3: 0.00 kWh - inverter4: 0.00 kWh - inverter5: 0.00 kWh - inverter6: 0.00 kWh - inverter7: 4.74 kWh - inverter8: 15.08 kWh Data Quality Status: Good The visualization has been prepared: a bar chart showing monthly clipping loss per inverter for the last 6 months. View Fullscreen 10:32 PM ★ Are you satisfied with the answer? Time taken: 28s, token: 0 tokens connected to Mof LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key in

TECHNICAL EVIDENCE

Agents used	`data_agent, graph_agent`
Tools used	`functions.generate_graphs_via_code, functions.clipping_loss_via_requests, functions.calculate_clipping_or_curtailment_losses_ag`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



31. MOF

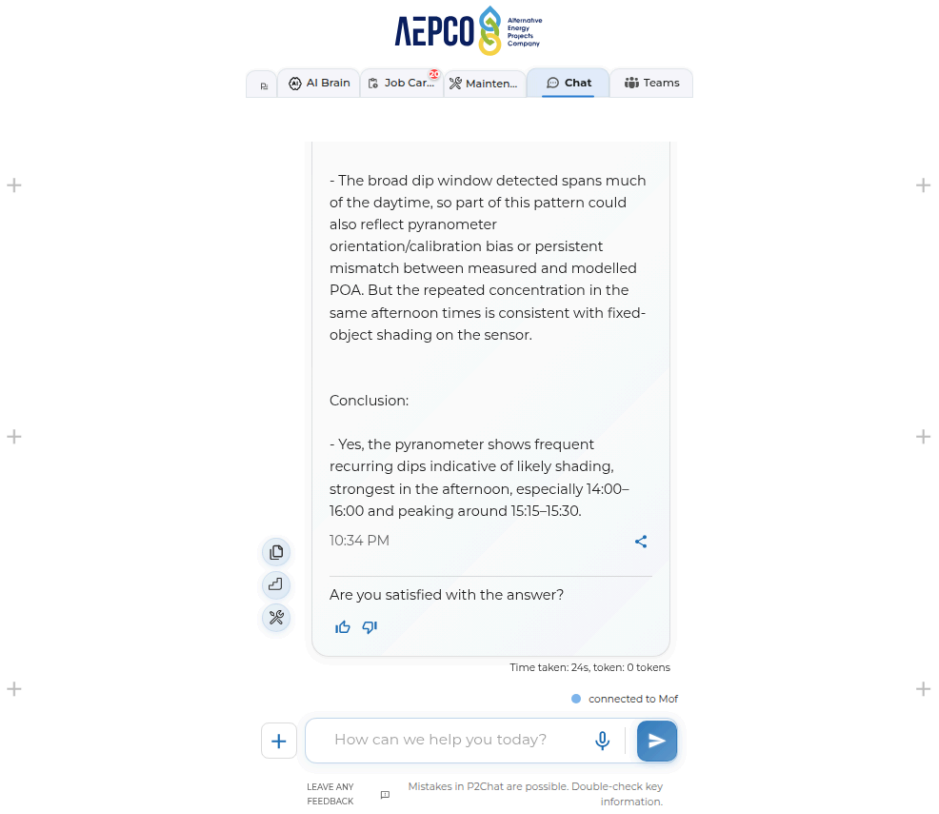
Production • 38.4s

Question: Is there shading happening with my plant? Please analyze the pyranometer and check for frequent dips at certain times

Response: I analyzed the last month of pyranometer data (2026-09-19 to 2026-10-19, available through 2026-10-19 18:00) against modelled POA irradiance and looked for repeated low measured/modelled irradiance ratios that are consistent with shading-type dips. Findings: - There is a strong, repeatable afternoon dip pattern. - The most frequent dip times are: - 15:30: dip frequency 100%, median measured/modelled ratio 0.620 - 15:15: 96.4%, ratio 0.635 - 15:00: 96.0%, ratio 0.630 - 16:00: 95.5%, ratio 0.593 - 14:00: 92.3%, ratio 0.696 - 15:45: 92.0%, ratio 0.608 - 14:45: 89.3%, ratio 0.670 - 14:30: 89.3%, ratio 0.678 - 13:15: 89.3%, ratio 0.713 Hourly pattern: - 12:00 hour: dip frequency 60.4% - 13:00 hour: 86.2% - 14:00 hour: 89.4% - 15:00 hour: 96.2% - 16:00 hour: 96.8% Interpretation: - The clearest shading signature is from roughly 13:00 to 16:00, intensifying through the afternoon. - The deepest and most repeatable dip is centered a

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`functions.extract_timeseries_data_ag, functions.execute_custom_code`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



32. MOF

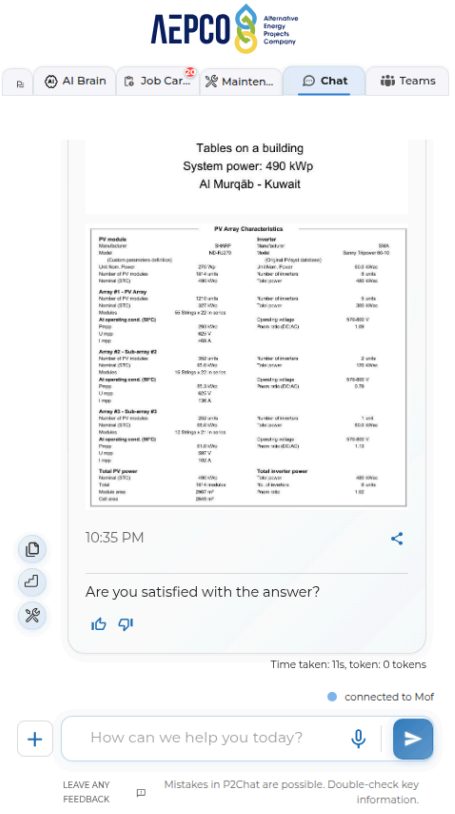
Production • 26.2s

Question: Based on pvsyst report, what is the rated power of this plant and array configuration

Response: Found 2 relevant PVsyst report sections: 1. [MOF] Front Page – System Rated Power (Page 1, Table 1) 2. [MOF] PV Array Characteristics – Array Configuration and Inverter Details (Page 3, Table 2) Please access the URLs provided to view the detailed information. Here is the reference: https://image-pvsyst-aepco-sydney.s3.ap-southeast-2.amazonaws.com/testuser1/MOF/page_1_image_1.jpg https://image-pvsyst-aepco-sydney.s3.ap-southeast-2.amazonaws.com/testuser1/MOF/page_3_image_2.jpg 10:35 PM Are you satisfied with the answer? Time taken: 11s, token: 0 tokens connected to Mof LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`pvsyst_agent`
Tools used	`pvsyst_tool`
API calls made	`4`
Last API call	`GET /testuser2/chat/ → 404`
Console errors	`4`
Network failures	`1`



33. MOF

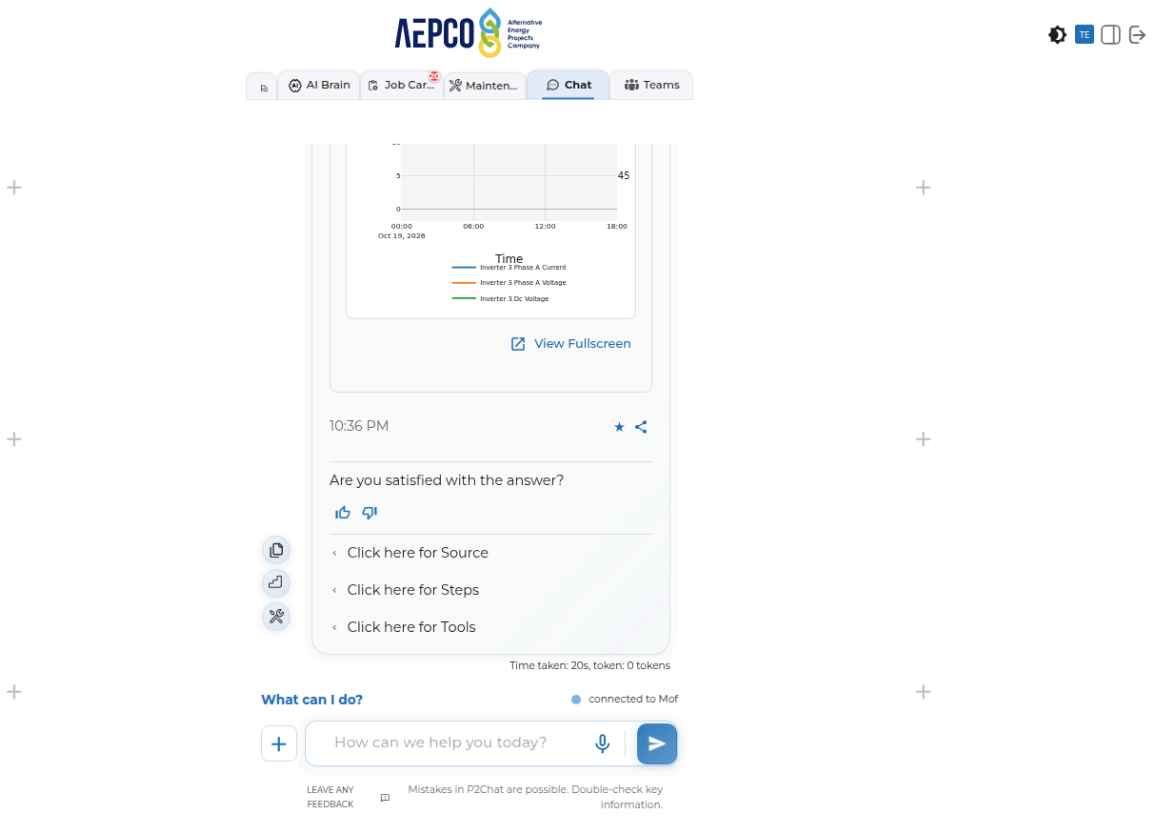
Production • 35.2s

Question: Please plot my current and voltage in a dual axes plot on the deepest possible granularity level for inverter3

Response: I've created the visualization for you. View Fullscreen 10:36 PM ★ Are you satisfied with the answer? Click here for Source Click here for Steps Click here for Tools Time taken: 20s, token: 0 tokens What can I do? connected to Mof LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.generate_graphs_via_code, functions.extract_timeseries_data_ag, functions.execute_custom_code`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`1`
Network failures	`0`



34. MOF

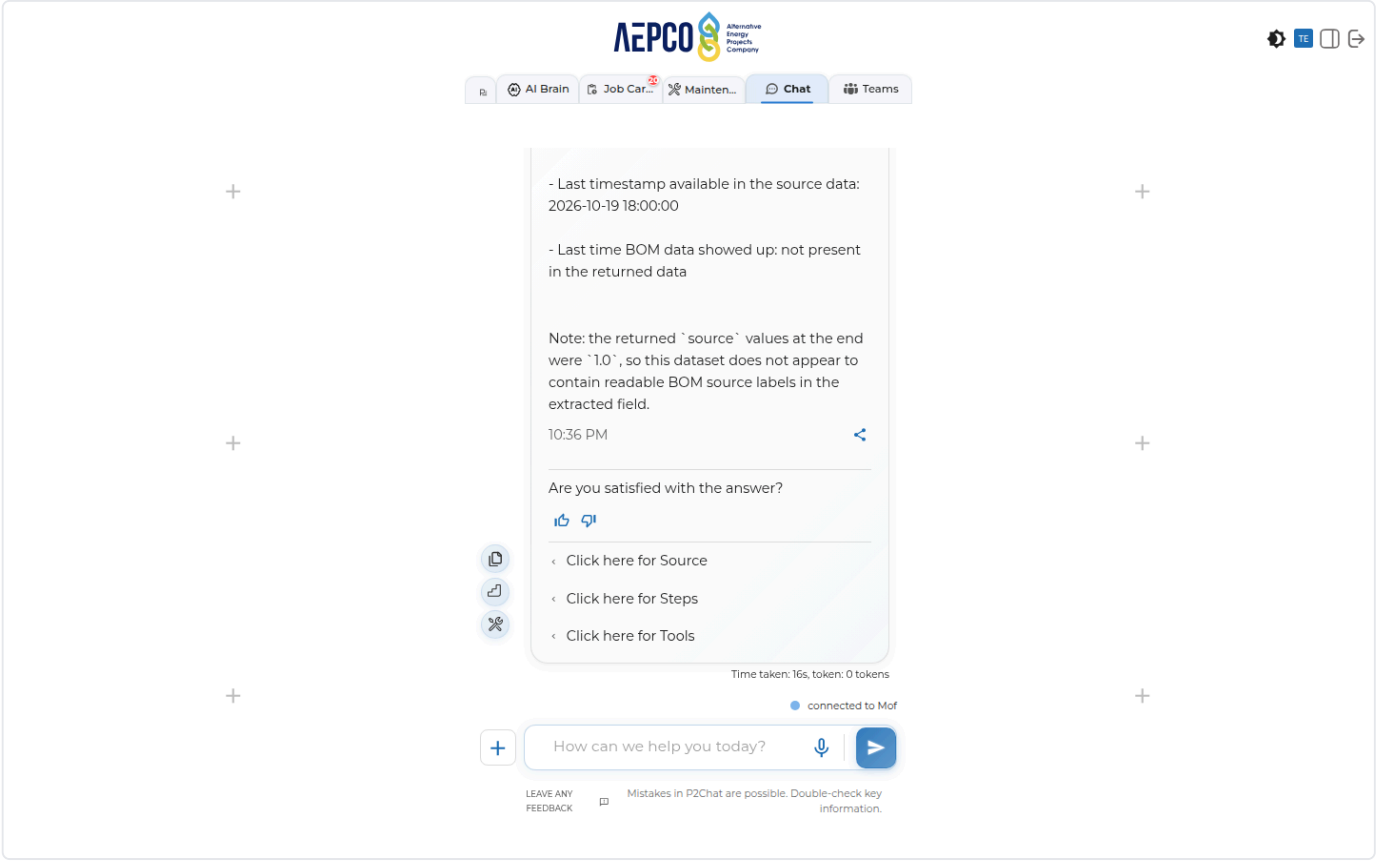
Production • 30.7s

Question: When was the last time the BOM data showed up

Response: I couldn't find any BOM-labelled records in the available `source` data. - Last timestamp available in the source data: 2026-10-19 18:00:00 - Last time BOM data showed up: not present in the returned data Note: the returned `source` values at the end were `1.0`, so this dataset does not appear to contain readable BOM source labels in the extracted field. 10:36 PM Are you satisfied with the answer? Click here for Source Click here for Steps Click here for Tools Time taken: 16s, token: 0 tokens connected to Mof LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`extract_timeseries_data_ag, execute_custom_code`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



35. MOF

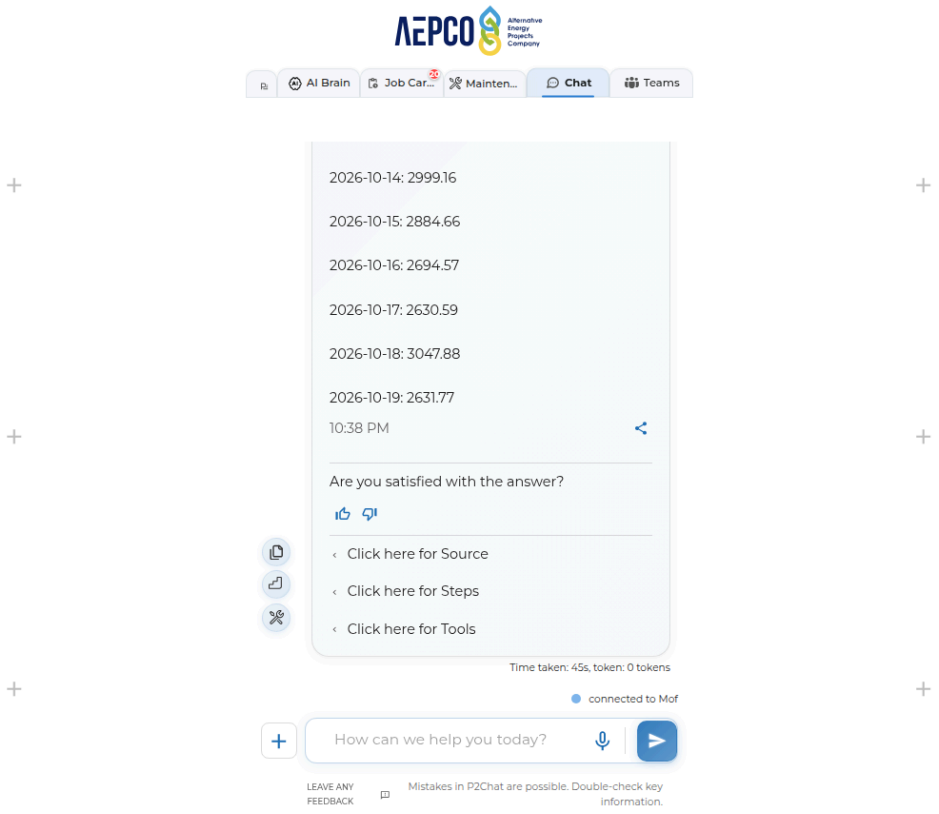
Production • 61.0s

Question: Please find out the daily POA poa summed up as a Wh/m^2, daily for the last 12 months and then calculate the median, mean and kurt

Response: Daily plane-of-array irradiance sums from `pyranometer_1` over the last 12 months were calculated in Wh/m² by summing 15-minute irradiance samples × 0.25 h for each day. Summary statistics for the 366 daily sums: - Median: 1449.68 Wh/m² - Mean: 2565.00 Wh/m² - Kurtosis: -1.6505 Note: the extracted data ends at 2026-10-19 18:00:00, so the final day (2026-10-19) is partial. Daily sums (Wh/m²): 2025-10-19: 885.85 2025-10-20: 904.79 2025-10-21: 889.05 2025-10-22: 878.82 2025-10-23: 858.98 2025-10-24: 840.18 2025-10-25: 827.76 2025-10-26: 794.18 2025-10-27: 840.57 2025-10-28: 827.31 2025-10-29: 825.42 2025-10-30: 787.88 2025-10-31: 765.16 2025-11-01: 784.85 2025-11-02: 857.27 2025-11-03: 835.38 2025-11-04: 818.88 2025-11-05: 786.32 2025-11-06: 699.71 2025-11-07: 648.38 2025-11-08: 626.68 2025-11-09: 667.28 2025-11-10: 666.11 2025-11-11: 672.94 2025-11-12: 592.17 2025-11-13: 547.42 2025-11-14: 495.79 2025-11-15: 254.63

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`extract_timeseries_data_ag, execute_custom_code`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



36. MOF

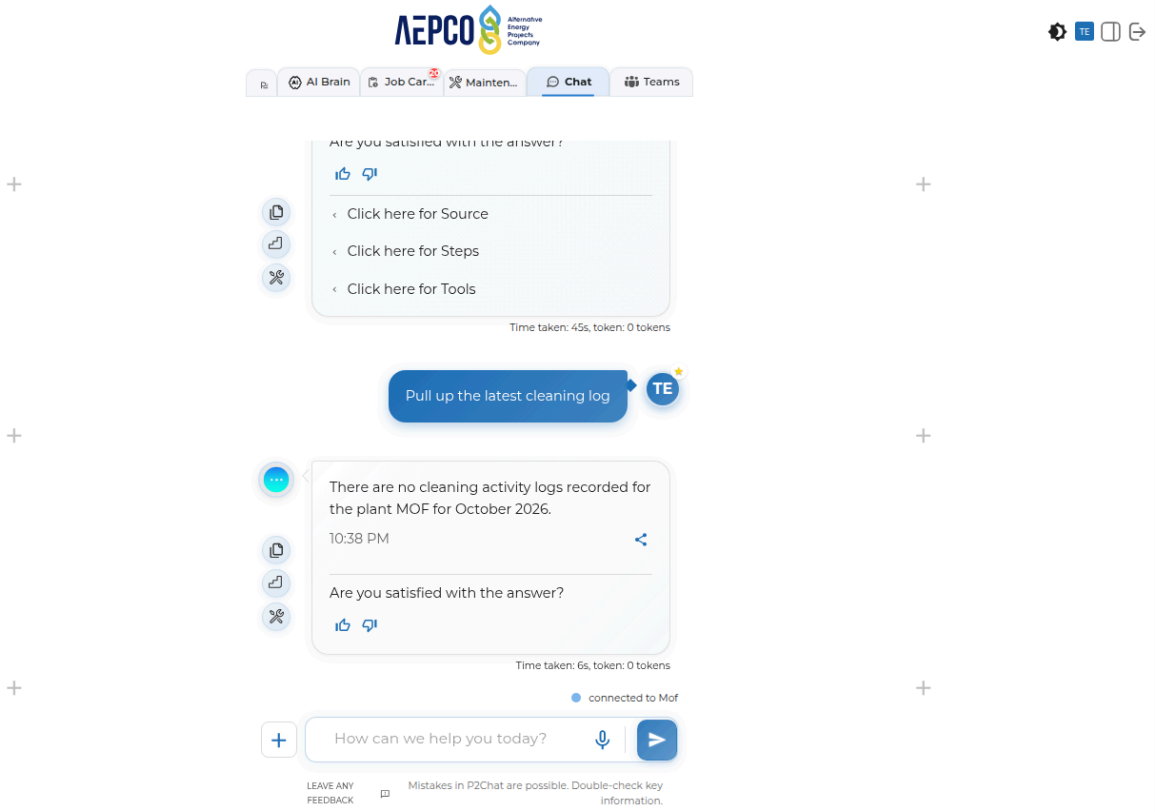
Production • 20.0s

Question: Pull up the latest cleaning log

Response: There are no cleaning activity logs recorded for the plant MOF for October 2026. 10:38 PM Are you satisfied with the answer? Time taken: 6s, token: 0 tokens connected to Mof LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_entry_agent`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



37. MOF

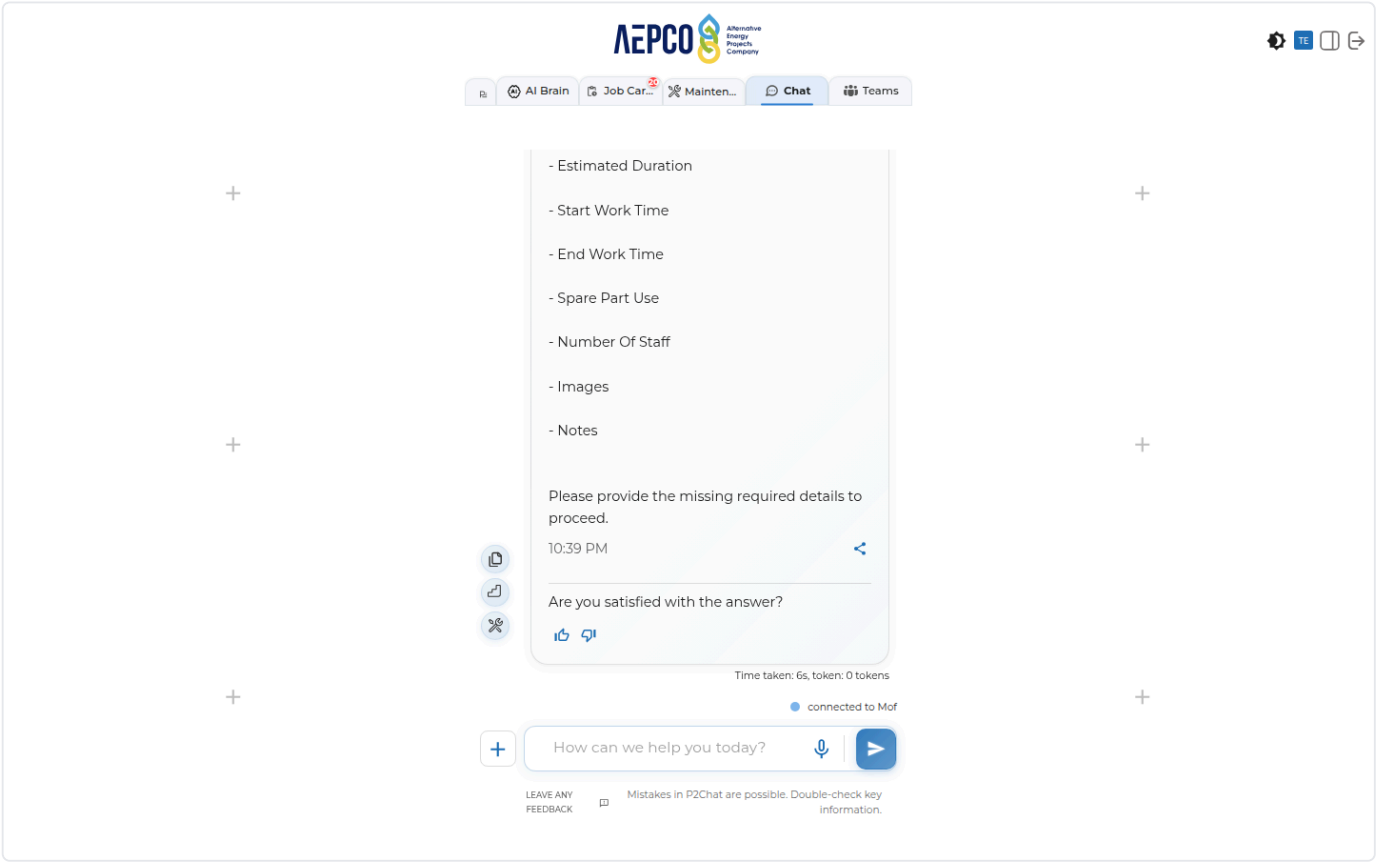
Production • 21.5s

Question: Jim cleaned the pyranometer today, can you add it to the cleaning log?

Response: To add this cleaning activity, please provide the following required details: - Date (it will be set to today: 2026-10-19, unless you'd like to change it) - Fault Cause - Action Taken (e.g., specify any special methods or notes about the cleaning) - Status (e.g., Completed, In Progress) Site will be set automatically as MOF. Optional fields you can also provide, but they are not required: - Priority - Responsibility (e.g., Jim or team) - Estimated Duration - Start Work Time - End Work Time - Spare Part Use - Number Of Staff - Images - Notes Please provide the missing required details to proceed. 10:39 PM Are you satisfied with the answer? Time taken: 6s, token: 0 tokens connected to Mof LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_entry_agent`
API calls made	`1`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



38. MOF

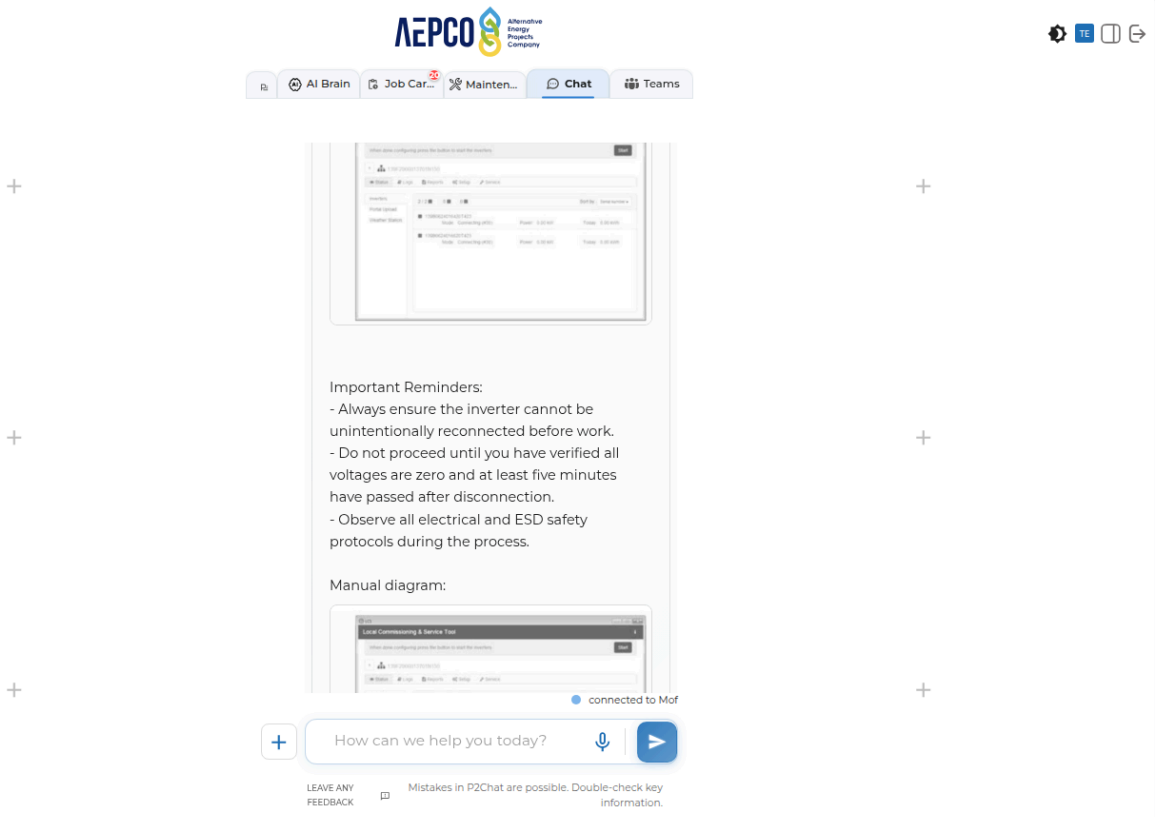
Production • 35.2s

Question: How can I shut down my inverters?

Response: To safely shut down your inverters, please follow these detailed steps as outlined in the manual: 1. Disconnect the inverter from the utility grid by switching off the main AC switch. This removes the AC supply to the inverter. 2. Turn off the PV (solar panel) side by using the integrated PV load-break (DC load-break) switch located on the inverter. This ensures the inverter is isolated from the PV array. 3. Make sure the inverter cannot be unintentionally reconnected. Lock out or tag out the switches if possible. 4. Confirm that no voltage is present: use a voltage detector to ensure the inverter and its related cables are voltage-free. 5. After disconnecting, WAIT AT LEAST FIVE MINUTES before opening or handling the inverter, as dangerous voltages may remain inside. 6. Before touching any internal components, discharge static by touching the grounded enclosure and observe all ESD (electrostatic discharge) safety rules. 7. If you need to open the inverter cover, use a TX 30 screwdriv

TECHNICAL EVIDENCE

Agents used	`manual_agent`
Tools used	`functions.manual_rag_tool`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



39. MOF

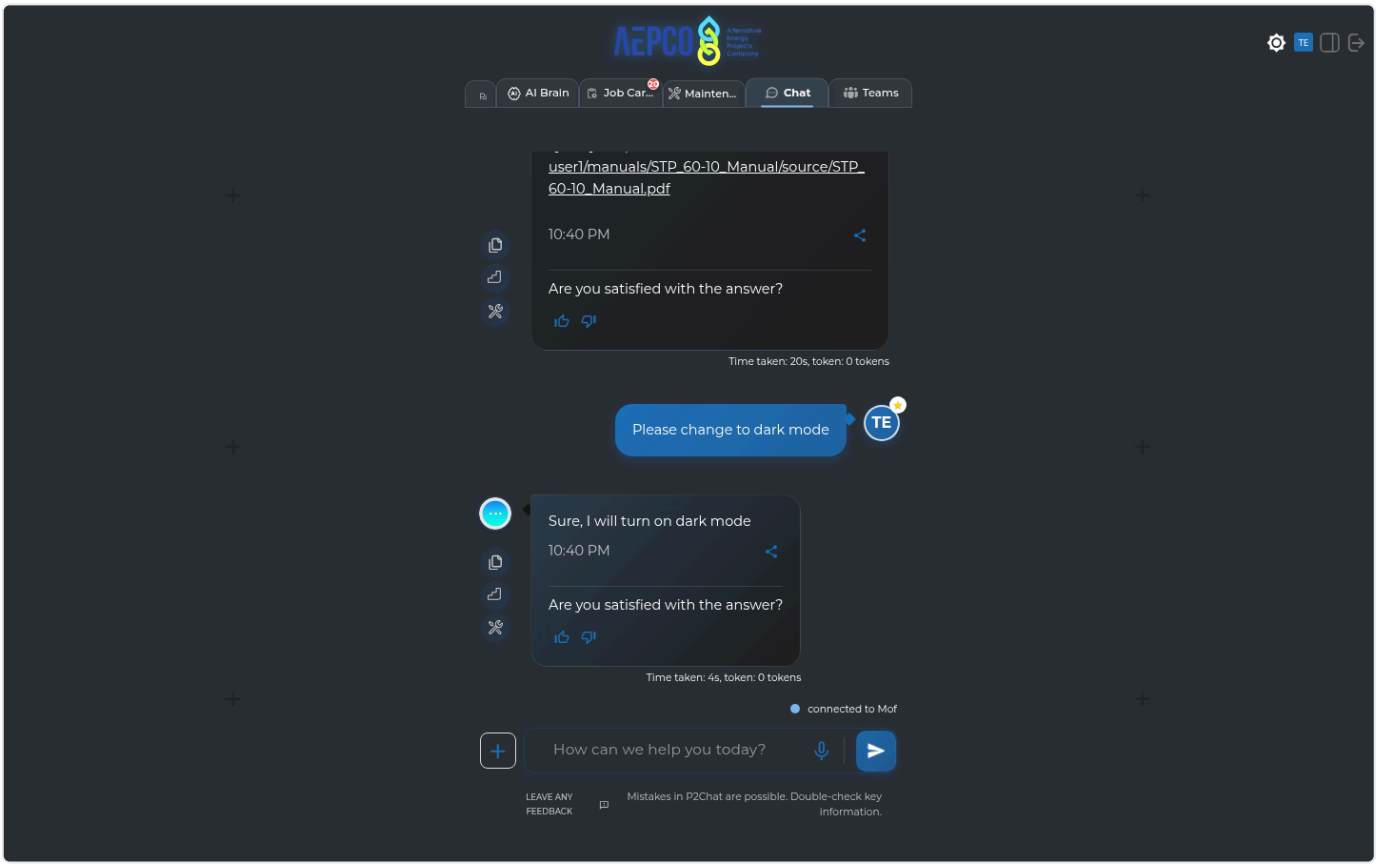
Production • 19.8s

Question: Please change to dark mode

Response: Sure, I will turn on dark mode 10:40 PM Are you satisfied with the answer? Time taken: 4s, token: 0 tokens connected to Mof LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`ui_agent`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



40. MOF

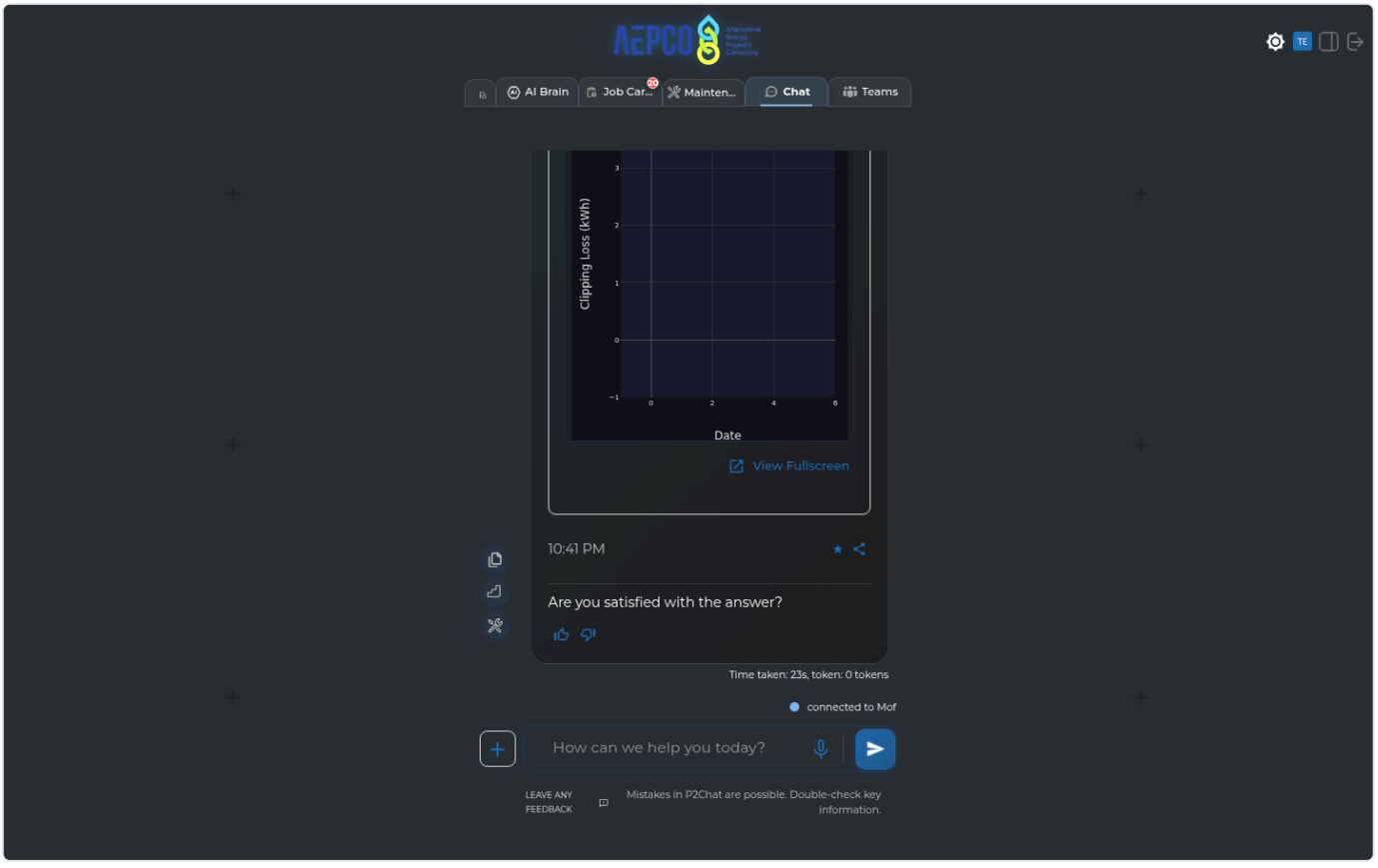
Production • 38.3s

Question: Please figure out how much clipping loss was found in the last 2 months broken by inverters, list them down and plot it as well

Response: Clipping Loss Summary (2026-08-19 to 2026-10-19) - inverter1: 0.0 kWh - inverter2: 0.0 kWh - inverter3: 0.0 kWh - inverter4: 0.0 kWh - inverter5: 0.0 kWh - inverter6: 0.0 kWh - inverter7: 0.0 kWh - inverter8: 0.0 kWh Note: The dataset ends at 2026-10-19 18:00:00, so the final requested day is covered only up to that timestamp. Visualization - A chart showing daily clipping loss by inverter for the last two months has been prepared. Clipping Loss Summary (2026-08-19 to 2026-10-19) - inverter1: 0.0 kWh - inverter2: 0.0 kWh - inverter3: 0.0 kWh - inverter4: 0.0 kWh - inverter5: 0.0 kWh - inverter6: 0.0 kWh - inverter7: 0.0 kWh - inverter8: 0.0 kWh Note: The dataset ends at 2026-10-19 18:00:00, so the final requested day is covered only up to that timestamp. Visualization - A chart showing daily clipping loss by inverter for the last two months has been prepared. View Fullscreen 10:41 PM ★ Are you satisfied with the answer? Time taken

TECHNICAL EVIDENCE

Agents used	`data_agent, graph_agent`
Tools used	`functions.clipping_loss_via_requests, functions.generate_graphs_via_code, functions.calculate_clipping_or_curtailment_losses_ag`
API calls made	`2`
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	`0`
Network failures	`0`



41. MOF

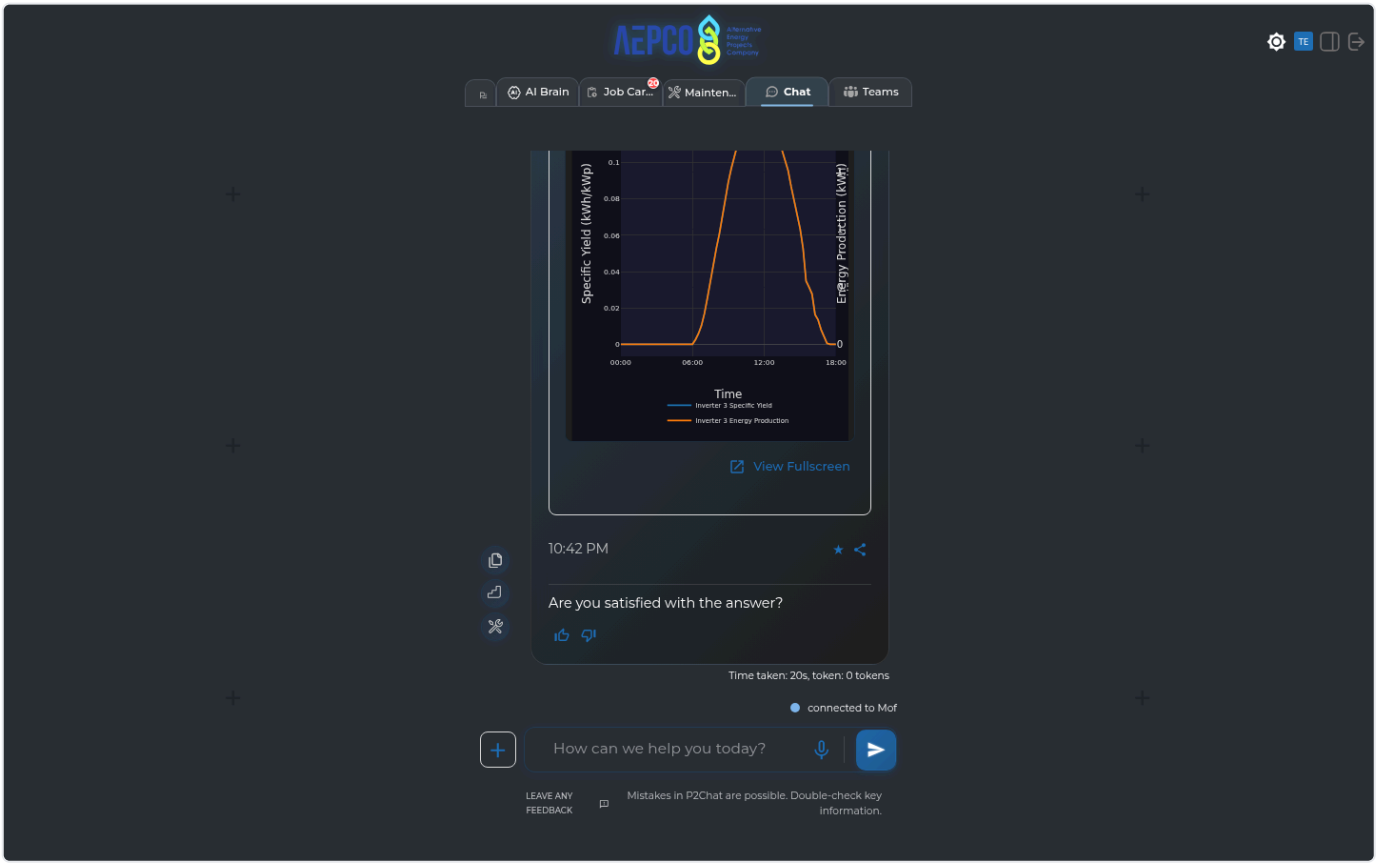
Production • 35.3s

Question: Please repeat the first plotting question, but interchange it with Specific yield on 1 axes, and energy production on the other axes on a single dual axes plot (Memory Test)

Response: I've created the visualization for you. View Fullscreen 10:42 PM ★ Are you satisfied with the answer? Time taken: 20s, token: 0 tokens connected to Mof LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`generate_graphs_via_code, total_energy_func_ag, specific_yield_ag`
API calls made	2
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	0
Network failures	0



42. KUPCO_HQ_DAJEEJ

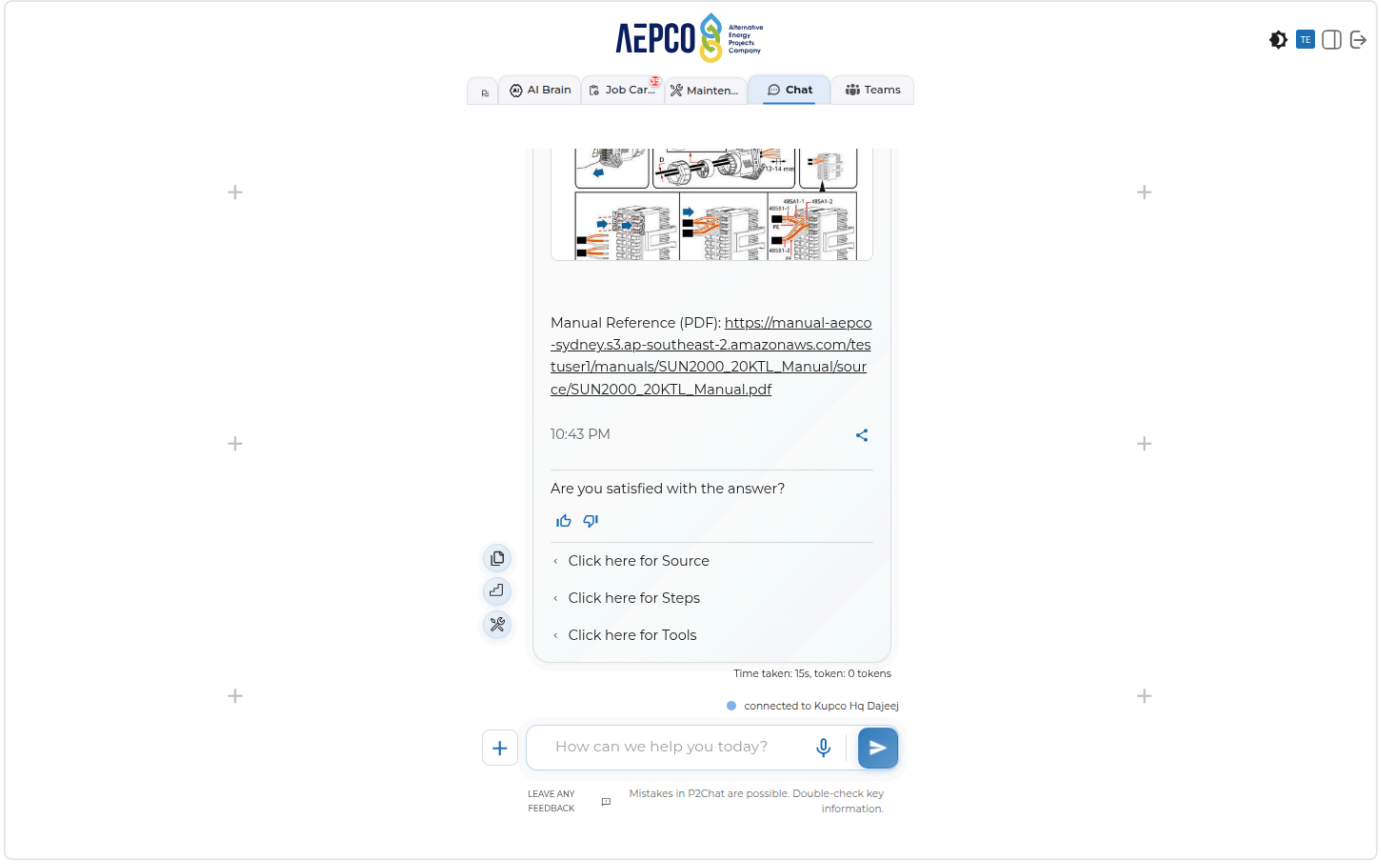
Production • 30.6s

Question: I have a COMMUNICATION_FAULT what do I do to my inverters?

Response: To troubleshoot a COMMUNICATION_FAULT on your inverters, follow these steps using the diagrams and guides from the manual: 1. Check all RS485A and RS485B communication cable connections between the SUN2000 inverters and the SmartLogger. Ensure each wire is inserted into the correct port and is secure. Reference network configuration diagram: 2. Verify the MBUS communication pathway (RS485 connections) from the inverters through the combiner box to the transformer station. Ensure all circuit breakers are closed and lines are terminated properly. Reference utility-scale communication diagram: 3. Confirm all physical connections using the full layout diagram. Pay special attention to the connection points between inverters, the SmartLogger, combiner box, and power meter. Diagram here: 4. Inspect cable installation. Check stripping dimensions and insertion sequence for each wire. Follow the guide for proper alignment and securing of connector nuts. See: 5. Use the pin assignment a

TECHNICAL EVIDENCE

Agents used	`manual_agent`
Tools used	`functions.manual_rag_tool`
API calls made	`13`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`1`
Network failures	`0`



43. KUPCO_HQ_DAJEEJ

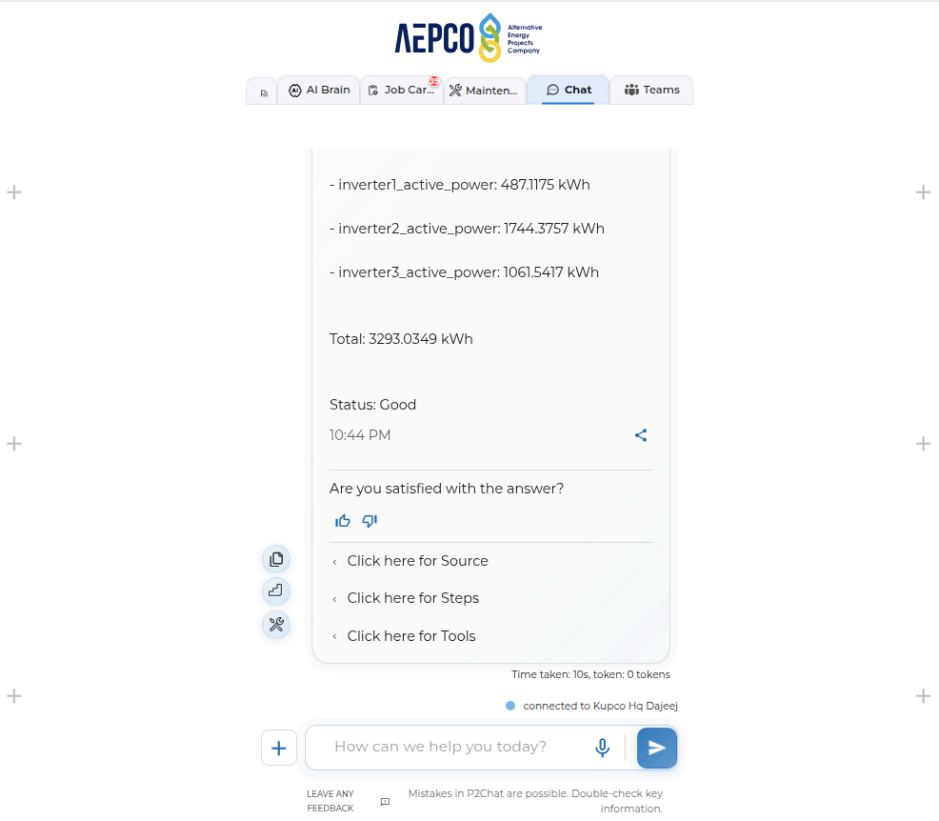
Production • 26.1s

Question: Use 'clipping_loss_via_requests' in order to calculate clipping loss if any for July 2026

Response: Clipping loss for July 2026 using `clipping_loss_via_requests`: - inverter1_active_power: 487.1175 kWh - inverter2_active_power: 1744.3757 kWh - inverter3_active_power: 1061.5417 kWh Total: 3293.0349 kWh Status: Good 10:44 PM Are you satisfied with the answer? Click here for Source Click here for Steps Click here for Tools Time taken: 10s, token: 0 tokens connected to Kupco Hq Dajeej LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`functions.clipping_loss_via_requests`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



44. KUPCO_HQ_DAJEEJ

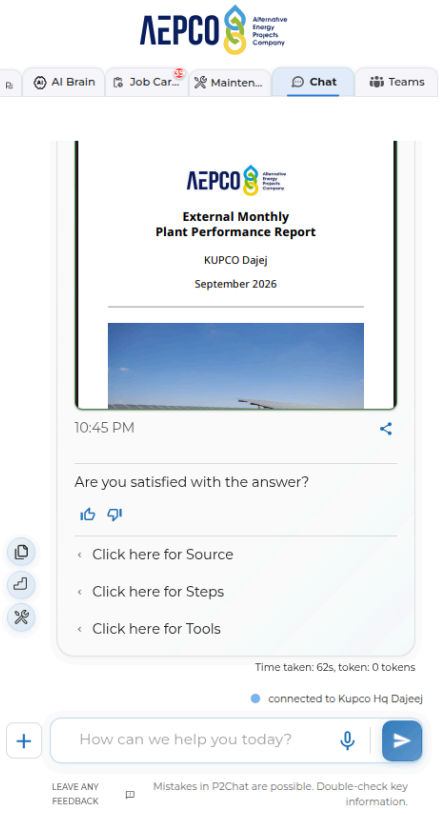
Production • 76.2s

Question: Please generate last months report

Response: I've generated the external report for September 2026. You can view it here: 'https://report-aepco-sydney.s3.ap-southeast-2.amazonaws.com/ABDM-L2_Project_September_2026_external_report.pdf' KUPCO_HQ_Dajeej Report 10:45 PM Are you satisfied with the answer? Click here for Source Click here for Steps Click here for Tools Time taken: 62s, token: 0 tokens connected to Kupco Hq Dajeej LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`report_agent`
Tools used	`agent_external_report_tool`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



45. KUPCO_HQ_DAJEEJ

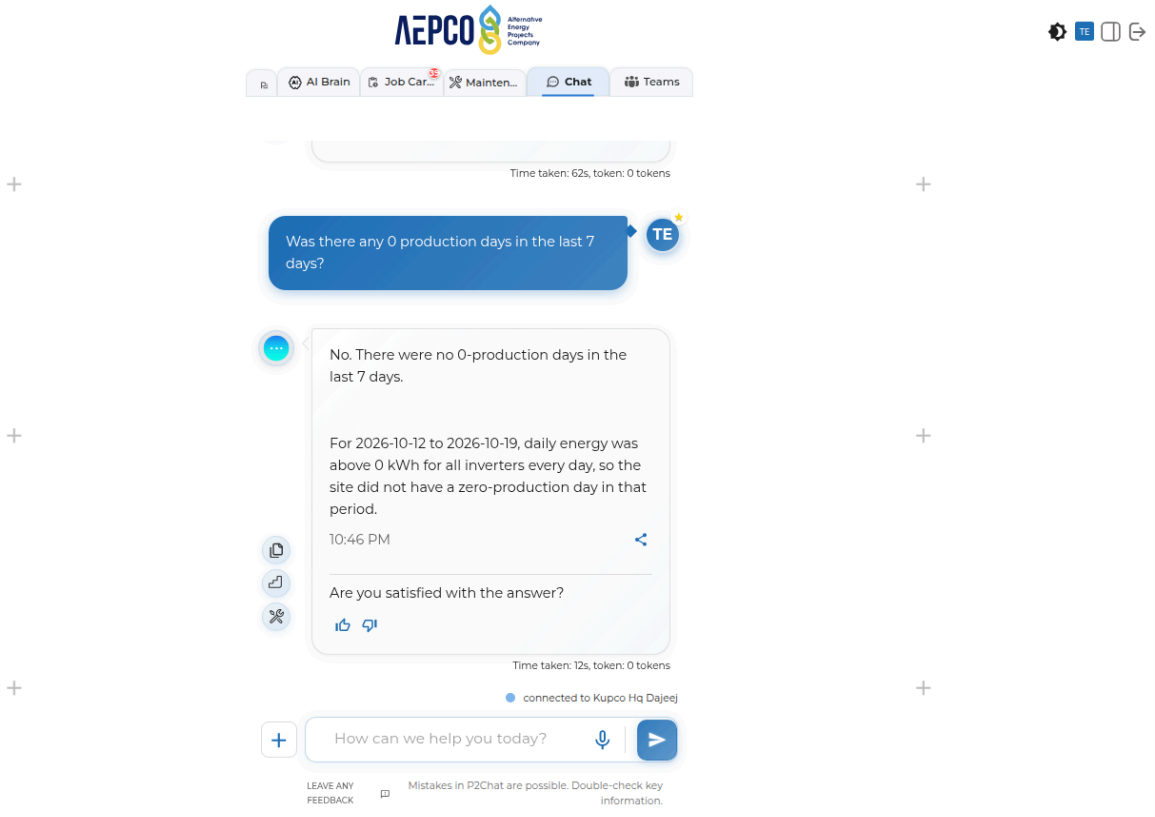
Production • 27.7s

Question: Was there any 0 production days in the last 7 days?

Response: No. There were no 0-production days in the last 7 days. For 2026-10-12 to 2026-10-19, daily energy was above 0 kWh for all inverters every day, so the site did not have a zero-production day in that period. 10:46 PM Are you satisfied with the answer? Time taken: 12s, token: 0 tokens connected to Kupco Hq Dajeej LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`functions.total_energy_func_ag`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



46. KUPCO_HQ_DAJEEJ

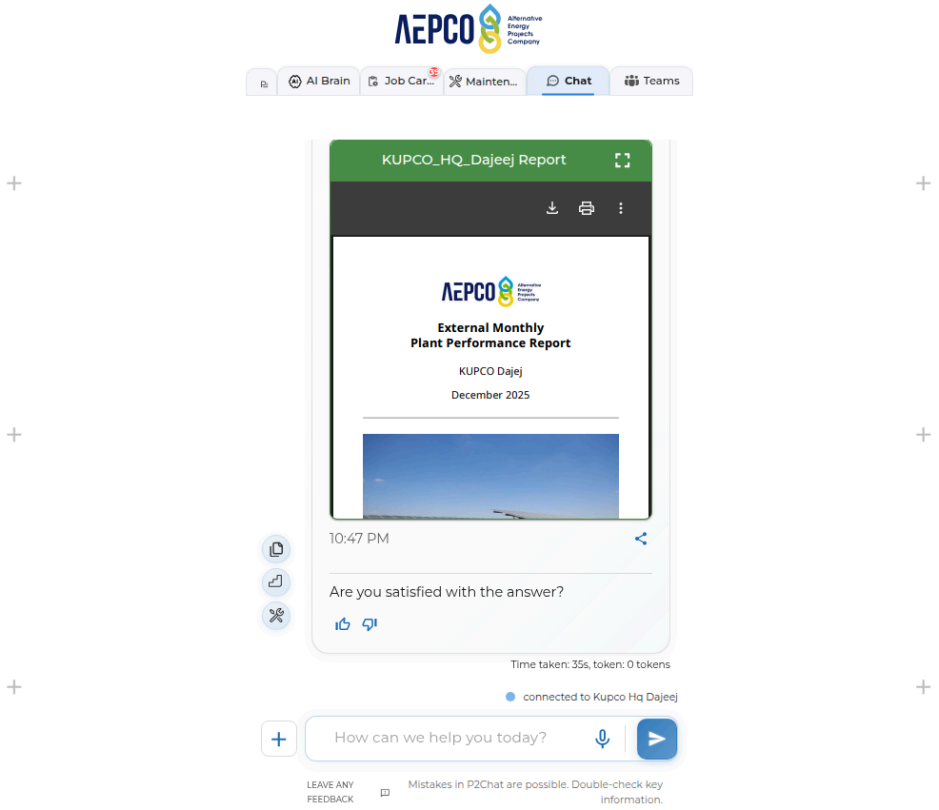
Production • 50.4s

Question: Please generate December 2025 performance report

Response: I've generated the performance report for December 2025. You can view it here: 'https://report-aepco-sydney.s3.ap-southeast-2.amazonaws.com/ABDM-L2_Project_December_2025_external_report.pdf' KUPCO_HQ_Dajeej Report 10:47 PM Are you satisfied with the answer? Time taken: 35s, token: 0 tokens connected to Kupco Hq Dajeej LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`report_agent`
Tools used	`agent_external_report_tool`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



47. KUPCO_HQ_DAJEEJ

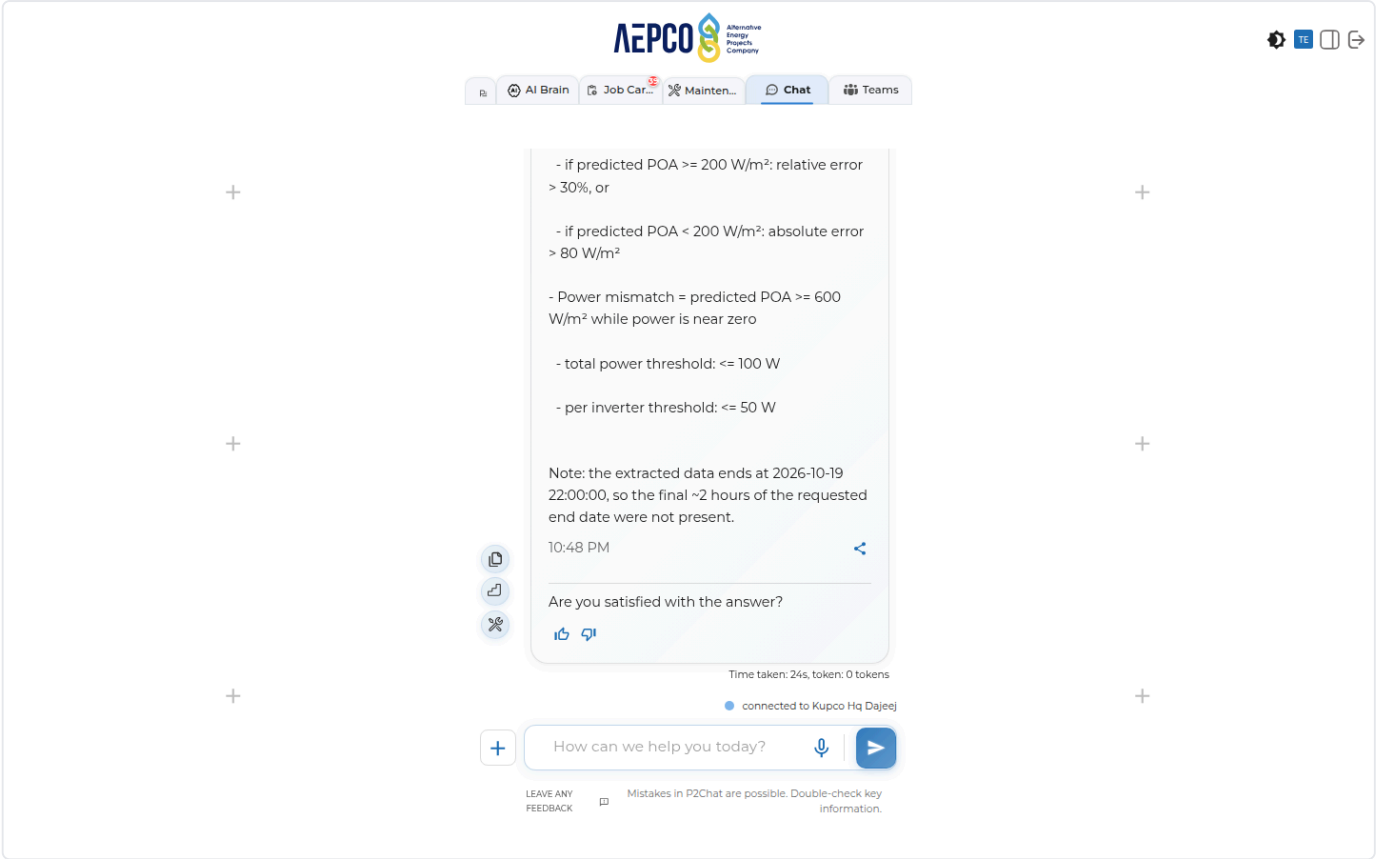
Production • 39.8s

Question: How is my data quality for POA and power? How many of them show missing data during daytime or mismatches with satellite prediction?

Response: Using 2026-10-12 to 2026-10-19 22:00 data only, the data quality counts are: - Daytime points assessed: 1013 - Missing POA during daytime (pyranometer_1): 7 - Missing power during daytime, summed across all 3 inverters: 0 - Missing power during daytime by inverter: - inverter1_active_power: 1 - inverter2_active_power: 6 - inverter3_active_power: 0 Mismatches with prediction: - POA mismatches vs predicted/modelled POA: 545 - Power mismatches vs prediction, total of all inverters: 0 - Power mismatches by inverter: - inverter1_active_power: 0 - inverter2_active_power: 1 - inverter3_active_power: 0 Rules used: - Daytime = zenith < 85° and modelled_poa_1 > 20 W/m² - POA mismatch = - if predicted POA >= 200 W/m²: relative error > 30%, or - if predicted POA < 200 W/m²: absolute error > 80 W/m² - Power mismatch = predicted POA >= 600 W/m² while power is near zero - total power threshold: <= 100 W - per inverter threshold: <= 50

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`functions.extract_timeseries_data_ag, functions.execute_custom_code`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



48. KUPCO_HQ_DAJEEJ

Production • 31.2s

Question: Please plot last 3 days hourly performance ratio for my MPPTs for inverter 2

Response: I've created the visualization for you. View Fullscreen 10:49 PM ★ Are you satisfied with the answer? Time taken: 15s, token: 0 tokens connected to Kupco Hq Dajeej LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.generate_graphs_via_code, functions.perf_ratio_ag`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0

49. KUPCO_HQ_DAJEEJ

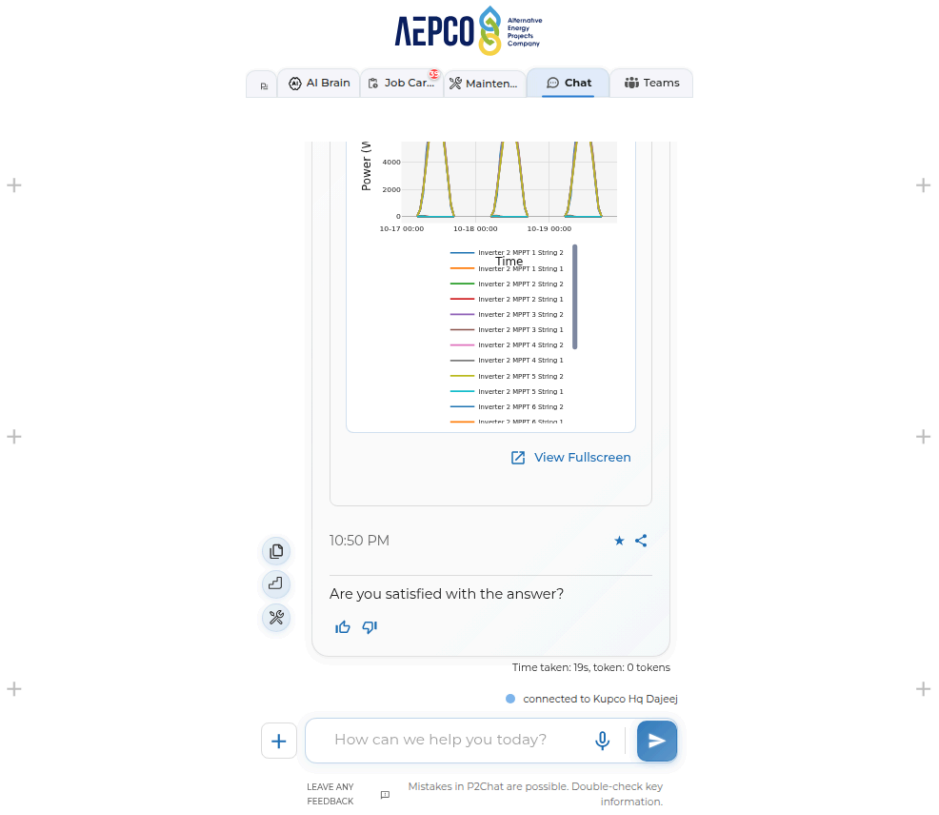
Production • 35.0s

Question: Please plot last 3 days hourly performance ratio for my Strings for inverter 2

Response: I've created the visualization for you. View Fullscreen 10:50 PM ★ Are you satisfied with the answer? Time taken: 19s, token: 0 tokens connected to Kupco Hq Dajeej LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.generate_graphs_via_code, functions.extract_timeseries_data_ag, functions.perf_ratio_ag, functions.execute_custom_code`
API calls made	`2`
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	`0`
Network failures	`0`



50. KUPCO_HQ_DAJEEJ

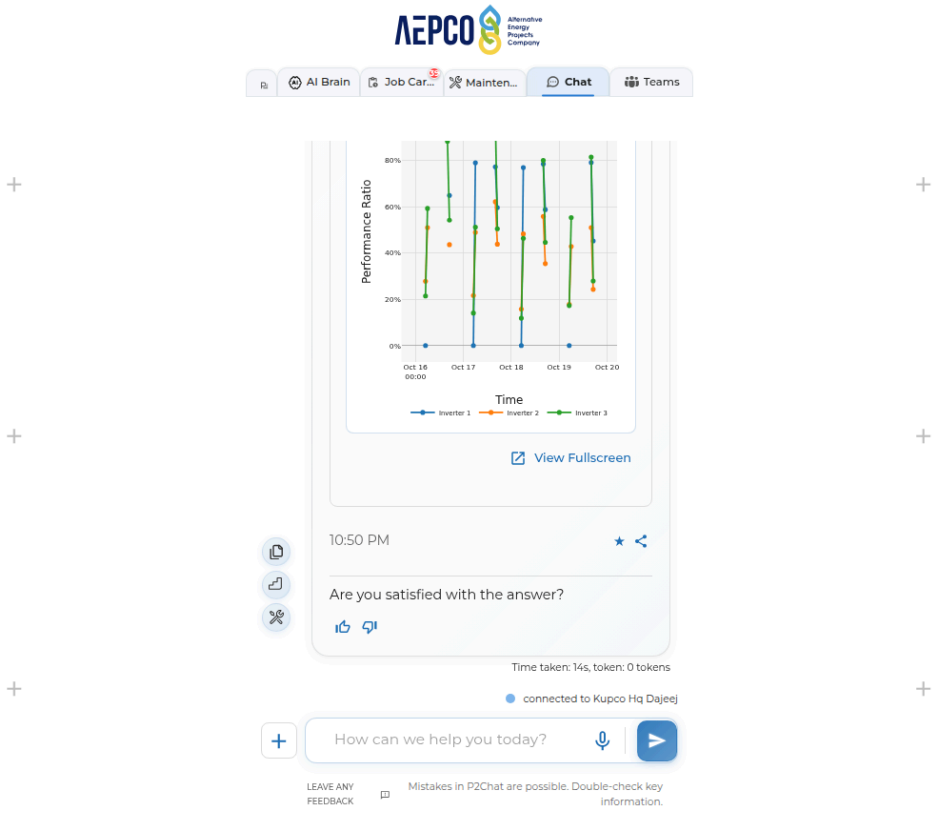
Production • 29.2s

Question: Please plot the last 4 days hourly performance ratio for all my inverters

Response: I've created the visualization for you. View Fullscreen 10:50 PM ★ Are you satisfied with the answer? Time taken: 14s, token: 0 tokens connected to Kupco Hq Dajeej LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.generate_graphs_via_code, functions.perf_ratio_ag`
API calls made	`1`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



51. KUPCO_HQ_DAJEEJ

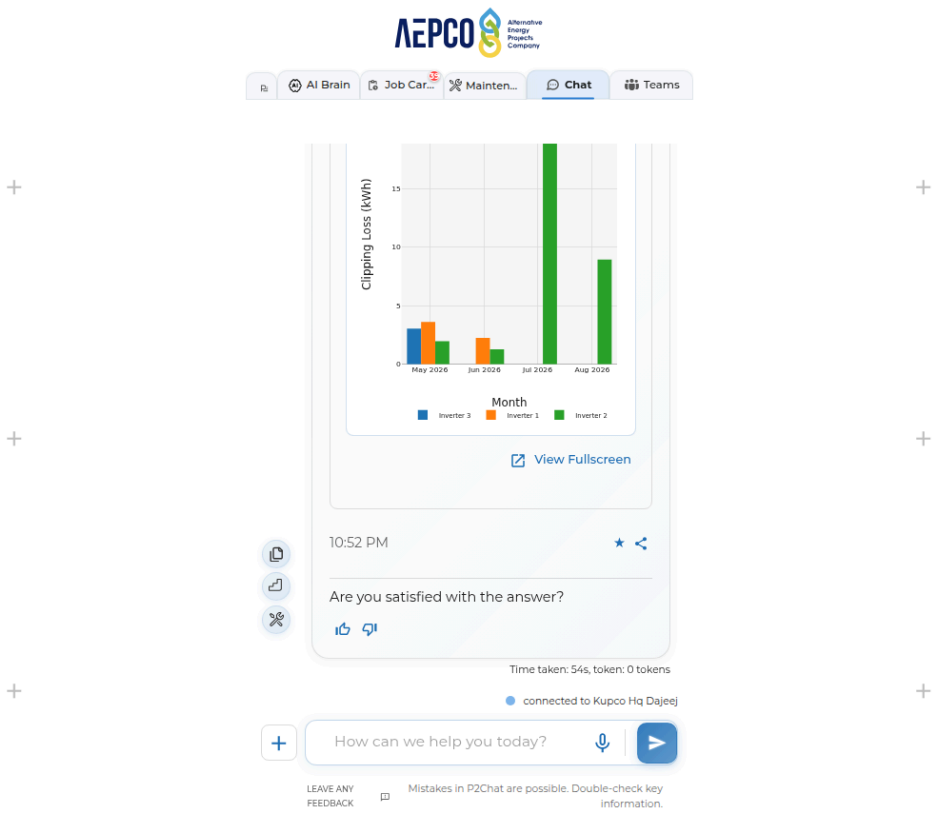
Production • 68.7s

Question: Please figure out how much clipping loss was found in the last 6 months broken by inverters, list them down and plot it as well

Response: Clipping Loss Summary (Last 6 Months: 2026-04-19 to 2026-10-19) - inverter1: 2,450.42 kWh - inverter2: 8,822.28 kWh - inverter3: 5,672.90 kWh Data Quality Status: Good The visualization has been prepared (Monthly Clipping Loss by Inverter for the last 6 months). Clipping Loss Summary (Last 6 Months: 2026-04-19 to 2026-10-19) - inverter1: 2,450.42 kWh - inverter2: 8,822.28 kWh - inverter3: 5,672.90 kWh Data Quality Status: Good The visualization has been prepared (Monthly Clipping Loss by Inverter for the last 6 months). View Fullscreen 10:52 PM ★ Are you satisfied with the answer? Time taken: 54s, token: 0 tokens connected to Kupco Hq Dajeej LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent, graph_agent`
Tools used	`functions.clipping_loss_via_requests, functions.generate_graphs_via_code, functions.calculate_clipping_or_curtailment_losses_ag`
API calls made	`3`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



52. KUPCO_HQ_DAJEEJ

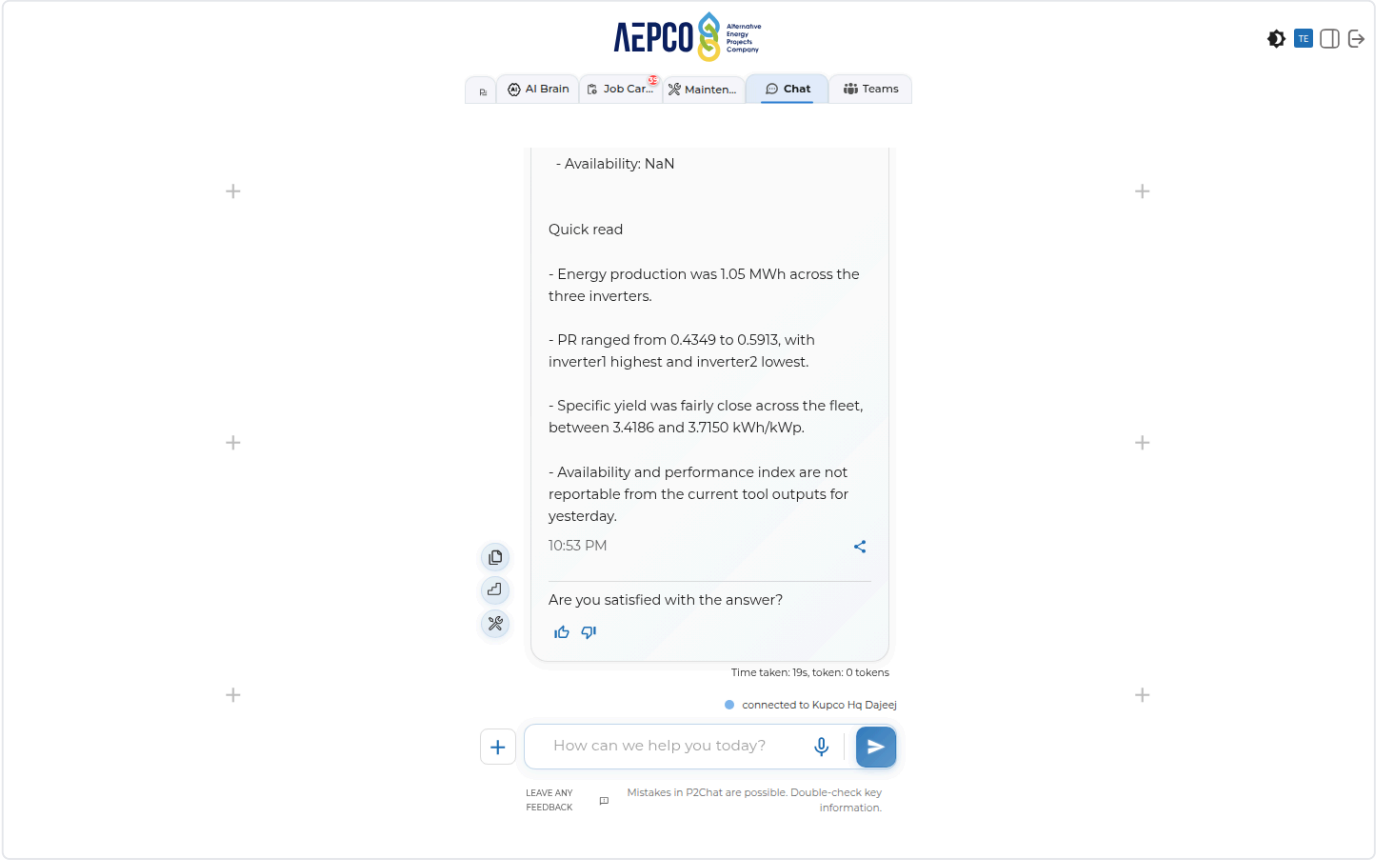
Production • 33.8s

Question: How did my plant perform yesterday?

Response: Yesterday (2026-10-18), the plant-level view isn't available in this dataset, so the most top-level view is the sum of the 3 inverters. Key KPIs - Total energy produced: 1,049.12 kWh - Reference yield (plant): 2.82 kWh/kWp - Inverter availability: all returned NaN, so availability could not be determined from the tool output - Performance index: unavailable due to a calculation error in the tool Per-inverter performance - inverter1 - Energy: 132.63 kWh - PR: 0.5913 - Specific yield: 3.7150 kWh/kWp - Availability: NaN - inverter2 - Energy: 473.78 kWh - PR: 0.4349 - Specific yield: 3.4888 kWh/kWp - Availability: NaN - inverter3 - Energy: 442.71 kWh - PR: 0.5058 - Specific yield: 3.4186 kWh/kWp - Availability: NaN Quick read - Energy production was 1.05 MWh across the three inverters. - PR ranged from 0.4349 to 0.5913, with inverter1 highest and inverter2 lowest. - Specific yield was fairly close across the fleet

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`multi_tool_use.parallel`
API calls made	2
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	0
Network failures	0



53. KUPCO_HQ_DAJEEJ

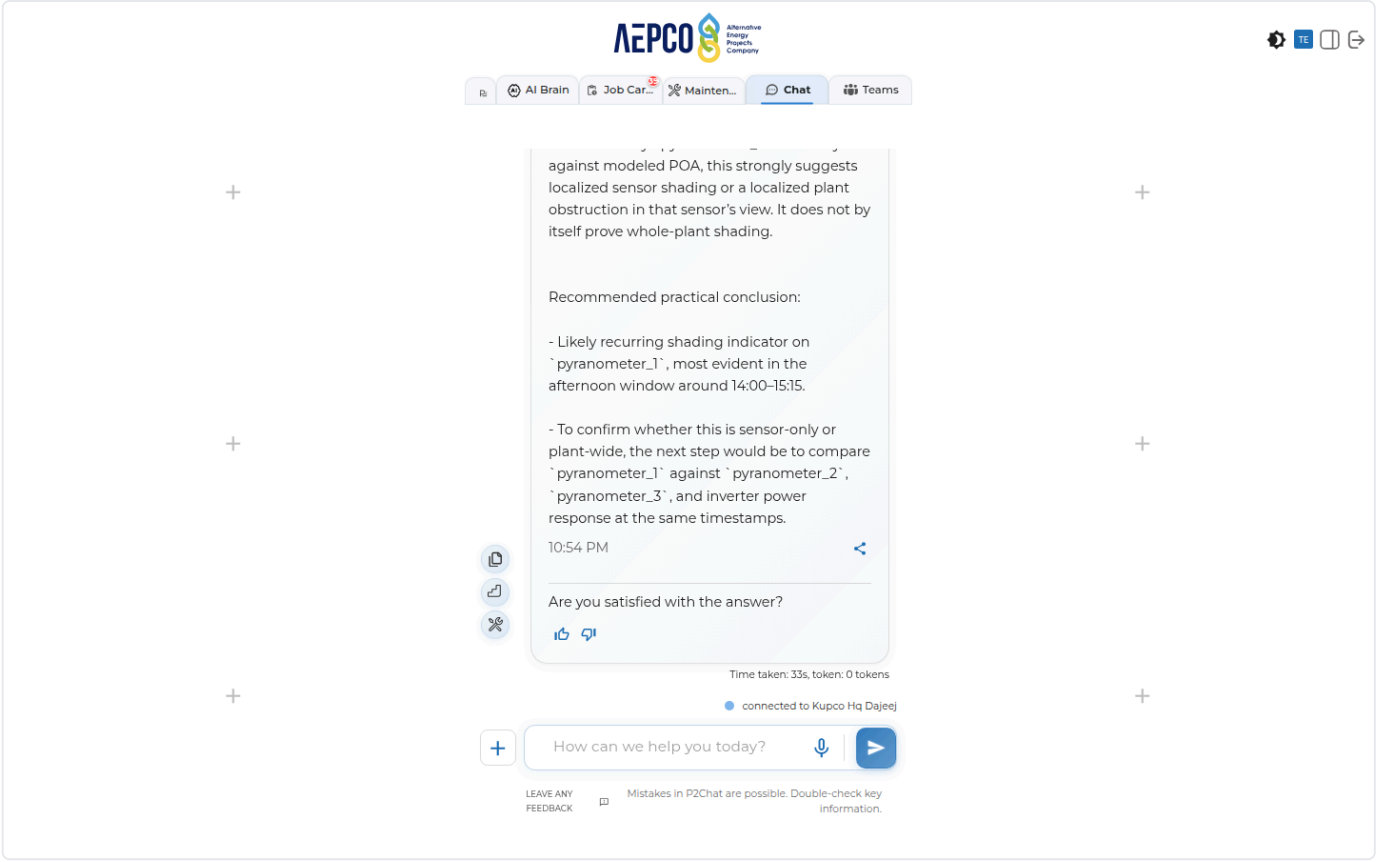
Production • 47.4s

Question: Is there shading happening with my plant? Please analyze the pyranometer and check for frequent dips at certain times

Response: I analyzed `pyranometer_1` over the last 30 days (2026-09-19 to 2026-10-19, daylight points only) against `modelled_poa_1`, using a time-of-day baseline to find recurring negative deviations that may indicate shading. Findings: - Overall daylight dip rate: 8.44% - Overall median `pyranometer_1 / modelled_poa_1` ratio: 0.737 - Data available through 2026-10-19 22:00, so the very end of the requested day was not yet present. Most frequent recurring dip windows: - 15:15–15:15: dip frequency 19.35%, average loss during dips 30.43%, average drop 195.5 W/m² - 14:20–14:20: dip frequency 19.35%, average loss during dips 24.29%, average drop 236.0 W/m² - 12:00–12:00: dip frequency 18.52%, average loss during dips 15.68%, average drop 214.4 W/m² - 14:55–15:00: dip frequency 17.74%, average loss during dips 28.04%, average drop 202.1 W/m² - 12:35–12:35: dip frequency 16.13%, average loss during dips 24.26%, average drop 262.8 W/m² - 14:35–14:35: dip frequency 16.13%, avera

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`functions.extract_timeseries_data_ag, functions.execute_custom_code`
API calls made	2
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	0
Network failures	0



54. KUPCO_HQ_DAJEEJ

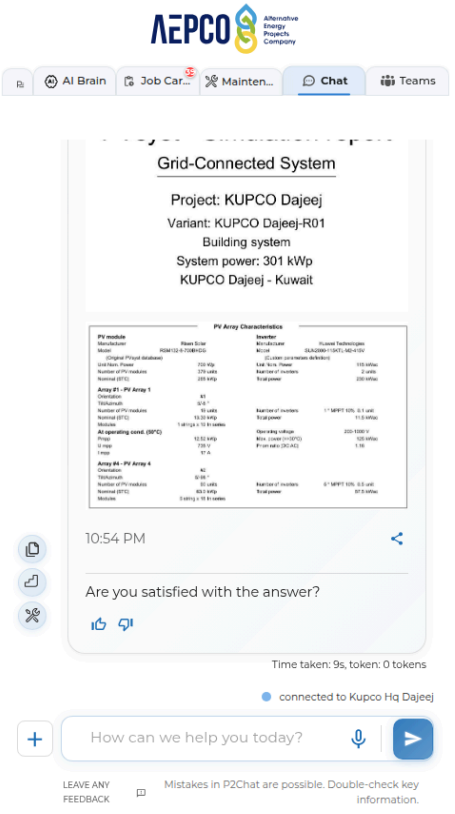
Production • 24.6s

Question: Based on pvsyst report, what is the rated power of this plant and array configuration

Response: Found 2 relevant PVsyst report sections: 1. [KUPCO_HQ_Dajeej] Front Page – Grid Connected System Overview (Page 1, Table 1) 2. [KUPCO_HQ_Dajeej] PV Array Characteristics – Array Configuration Details (Page 3, Table 2) Please access the URLs provided to view the detailed information. Here is the reference: https://image-pvsyst-aepco-sydney.s3.ap-southeast-2.amazonaws.com/testuser1/KUPCO_HQ_Dajeej/page_1_image_1.png https://image-pvsyst-aepco-sydney.s3.ap-southeast-2.amazonaws.com/testuser1/KUPCO_HQ_Dajeej/page_3_image_2.png 10:54 PM Are you satisfied with the answer? Time taken: 9s, token: 0 tokens connected to Kupco Hq Dajeej LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`pvsyst_agent`
Tools used	`pvsyst_tool`
API calls made	3
Last API call	`GET /testuser2/chat/ → 404`
Console errors	4
Network failures	1



55. KUPCO_HQ_DAJEEJ

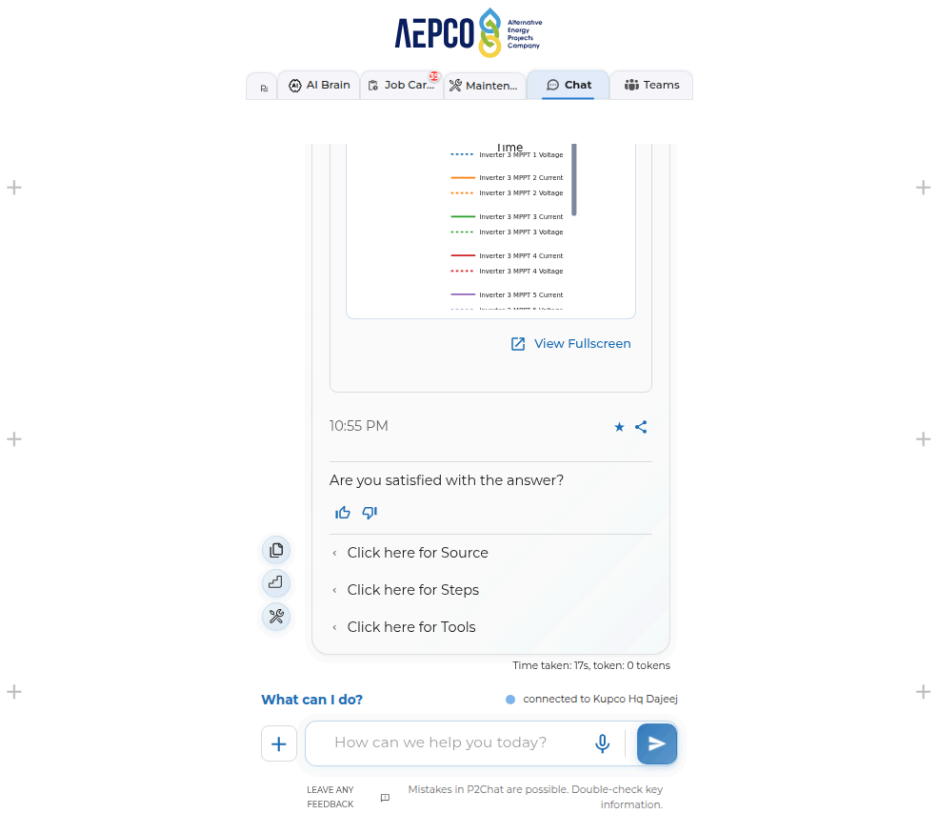
Production • 32.6s

Question: Please plot my current and voltage in a dual axes plot on the deepest possible granularity level for inverter3

Response: I've created the visualization for you. View Fullscreen 10:55 PM ★ Are you satisfied with the answer? Click here for Source Click here for Steps Click here for Tools Time taken: 17s, token: 0 tokens What can I do? connected to Kupco Hq Dajeej LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.extract_timeseries_data_ag, functions.generate_graphs_via_code, functions.execute_custom_code`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`1`
Network failures	`0`



56. KUPCO_HQ_DAJEEJ

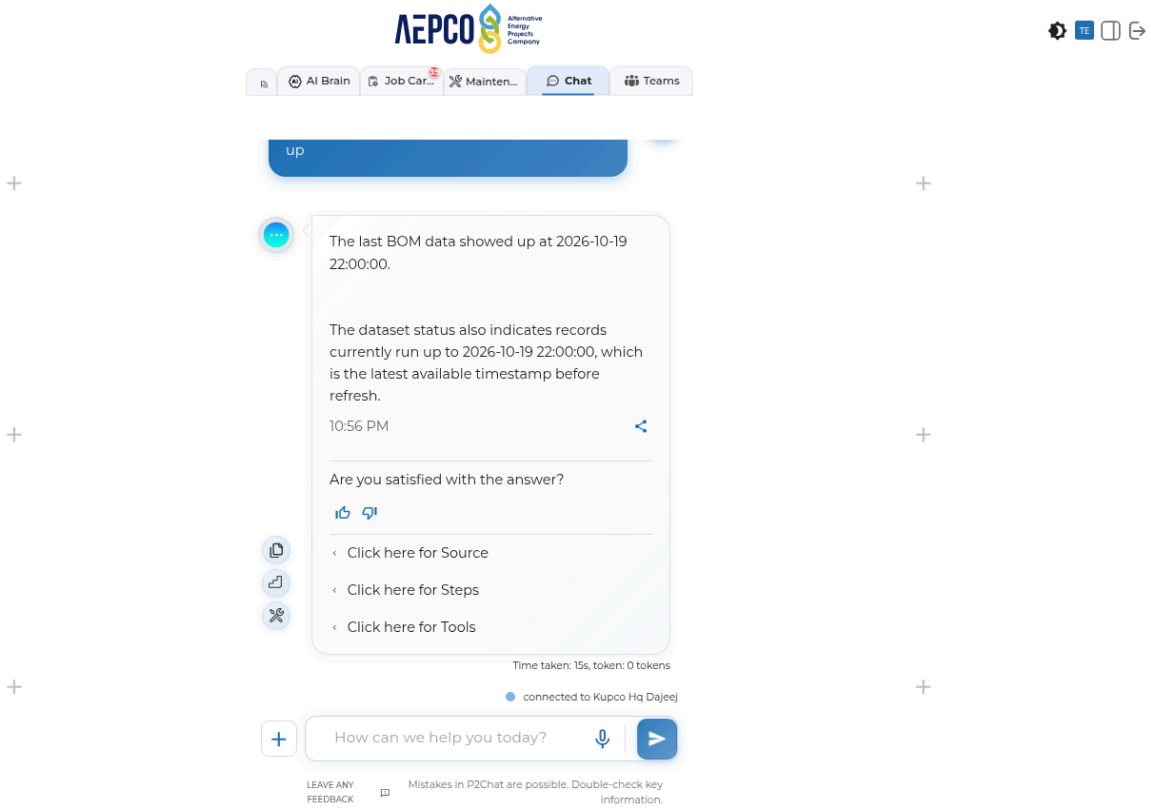
Production • 29.4s

Question: When was the last time the BOM data showed up

Response: The last BOM data showed up at 2026-10-19 22:00:00. The dataset status also indicates records currently run up to 2026-10-19 22:00:00, which is the latest available timestamp before refresh. 10:56 PM Are you satisfied with the answer? Click here for Source Click here for Steps Click here for Tools Time taken: 15s, token: 0 tokens connected to Kupco Hq Dajeej LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`functions.extract_timeseries_data_ag, functions.execute_custom_code`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



57. KUPCO_HQ_DAJEEJ

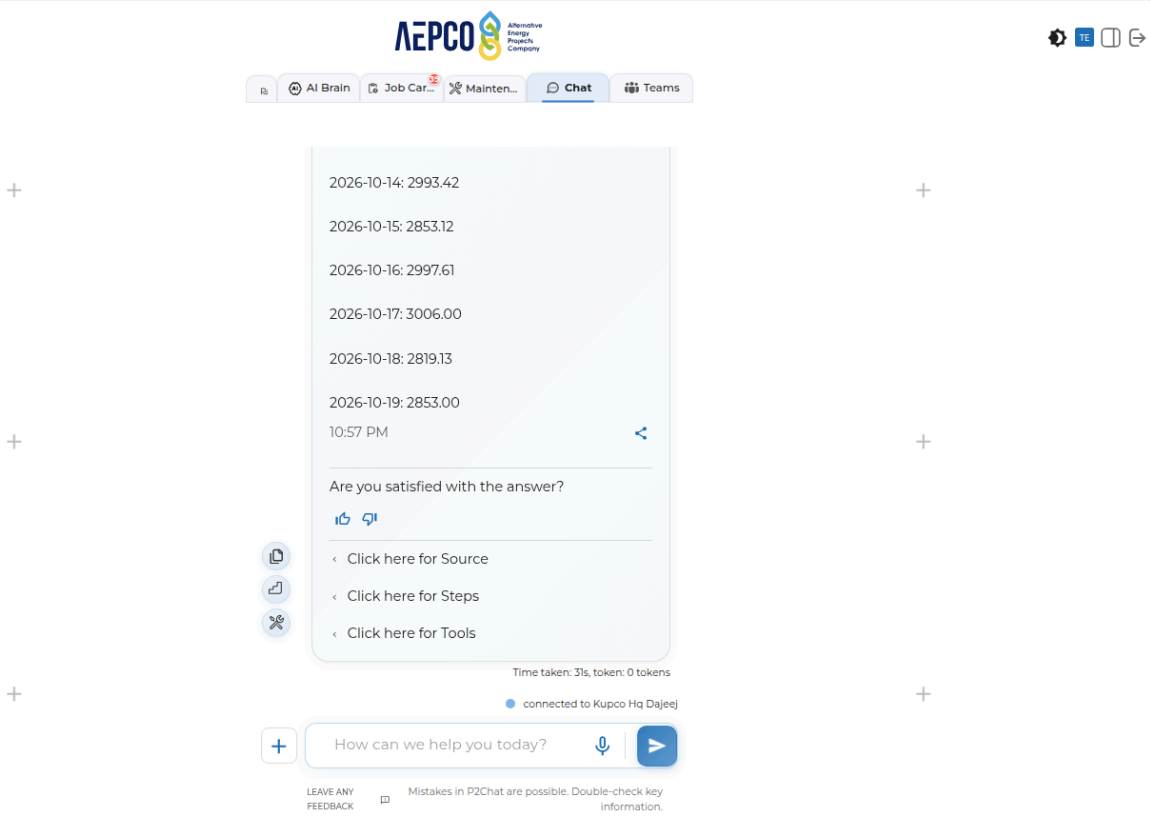
Production • 45.9s

Question: Please find out the daily POA poa summed up as a Wh/m^2, daily for the last 12 months and then calculate the median, mean and kurt

Response: Using `pyranometer_1`, I computed the daily POA sum in Wh/m² for the requested last 12 months window. Note: the database does not contain the full last 12 months. Available data in this extraction starts on 2026-04-12 and ends on 2026-10-19 22:00:00, so the statistics below are based on the available portion only. Summary statistics of daily POA sum: - Median: 6204.34 Wh/m² - Mean: 5722.39 Wh/m² - Kurtosis: 0.6274 - Number of days used: 191 Daily POA sums (Wh/m²): 2026-04-12: 4179.34 2026-04-13: 6496.27 2026-04-14: 6543.98 2026-04-15: 6702.76 2026-04-16: 5521.16 2026-04-17: 7207.08 2026-04-18: 6395.07 2026-04-19: 3950.52 2026-04-20: 1519.08 2026-04-21: 4832.97 2026-04-22: 6928.17 2026-04-23: 5062.22 2026-04-24: 5835.13 2026-04-25: 4872.32 2026-04-26: 5114.72 2026-04-27: 6485.17 2026-04-28: 6737.80 2026-04-29: 6865.51 2026-04-30: 6774.07 2026-05-01: 6749.66 2026-05-02: 6967.80 2026-05-03: 7135.47 2026-05-04: 6694.07 2026-0

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`functions.extract_timeseries_data_ag, functions.execute_custom_code`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



58. KUPCO_HQ_DAJEEJ

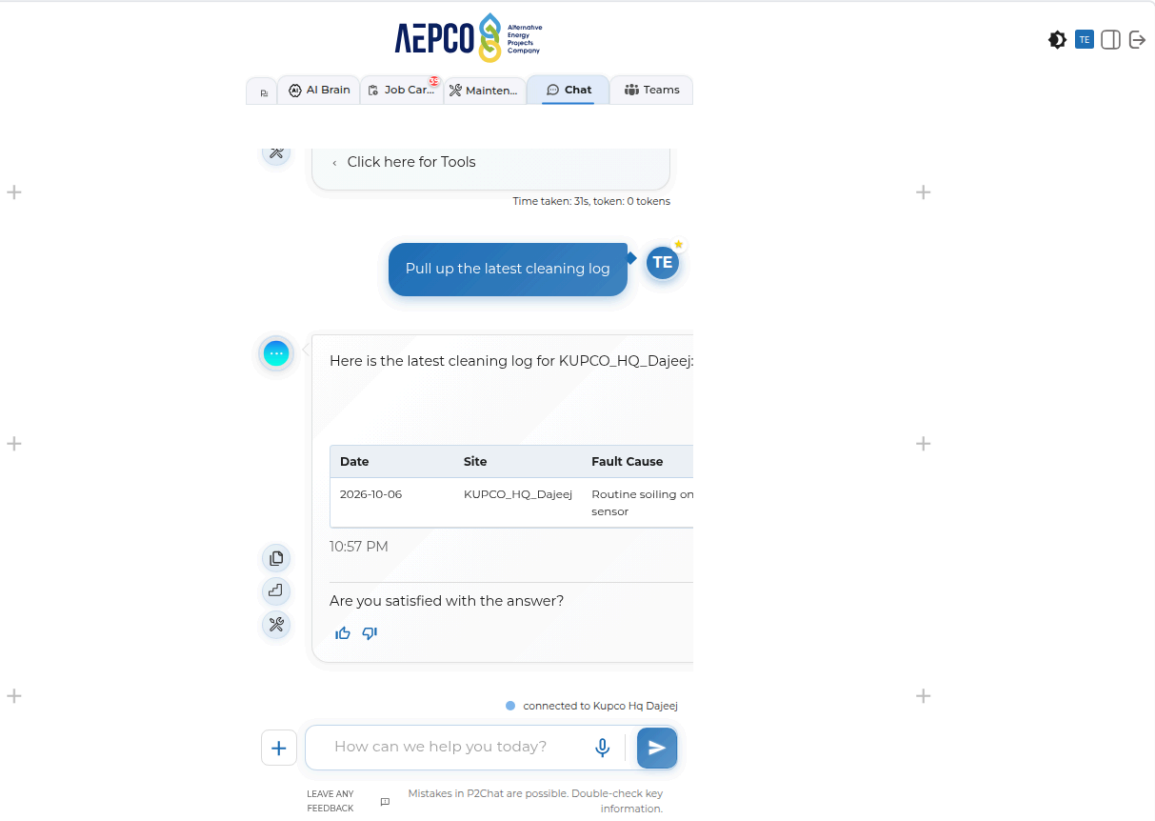
Production • 21.5s

Question: Pull up the latest cleaning log

Response: Here is the latest cleaning log for KUPCO_HQ_Dajeej: Edit Date Site Fault Cause Action Taken Status Priority Responsibility Estimated Duration Start Work Time End Work Time Spare Part Use Number Of Staff Images Notes 2026-10-06 KUPCO_HQ_Dajeej Routine soiling on pyranometer sensor Pyranometer cleaned by Jim Completed - - - - - - - - - 10:57 PM Are you satisfied with the answer? Time taken: 6s, token: 0 tokens connected to Kupco Hq Dajeej LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_entry_agent`
API calls made	2
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	0
Network failures	0



59. KUPCO_HQ_DAJEEJ

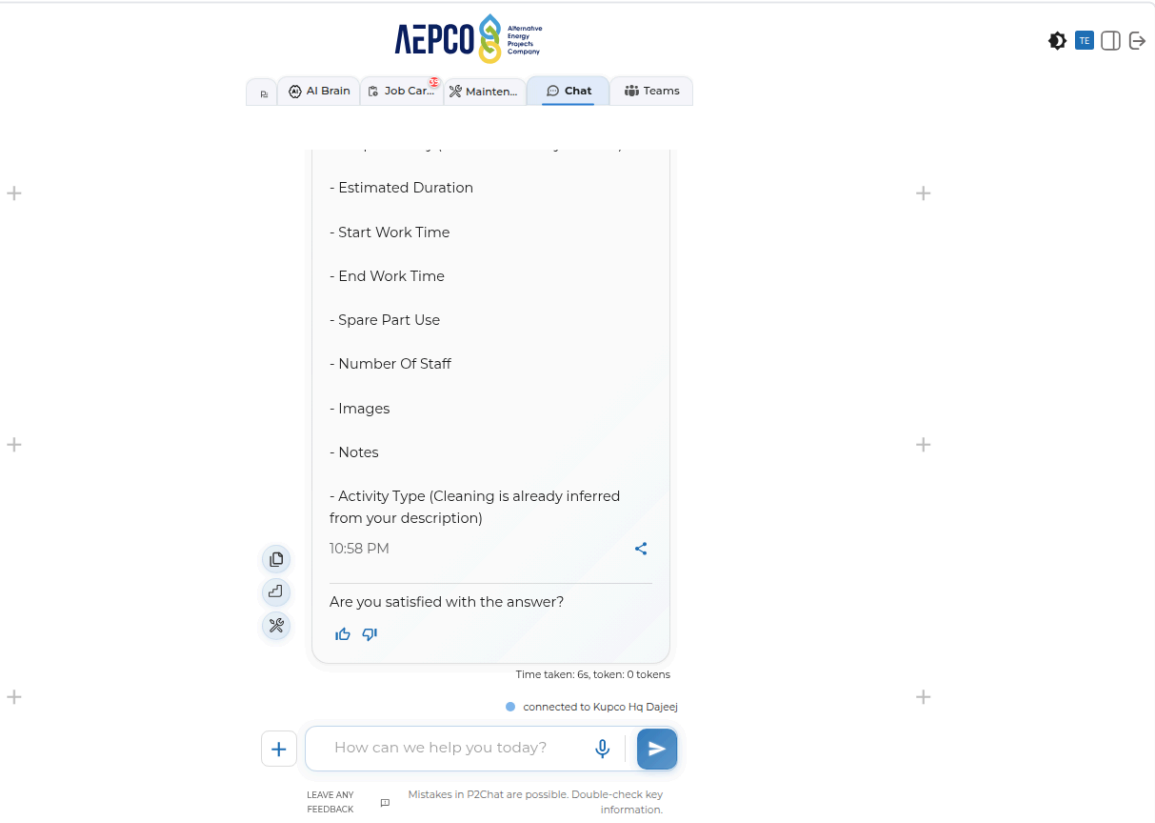
Production • 21.6s

Question: Jim cleaned the pyranometer today, can you add it to the cleaning log?

Response: To add a cleaning activity, please provide the following required details: - Date (you mentioned 'today', so I will use 2026-10-19 unless you want another date) - Fault Cause - Action Taken - Status Site will be set automatically as KUPCO_HQ_Dajeej. Optional fields you can also provide, but they are not required: - Priority - Responsibility (can set as Jim if you'd like) - Estimated Duration - Start Work Time - End Work Time - Spare Part Use - Number Of Staff - Images - Notes - Activity Type (Cleaning is already inferred from your description) 10:58 PM Are you satisfied with the answer? Time taken: 6s, token: 0 tokens connected to Kupco Hq Dajeej LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_entry_agent`
API calls made	`1`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



60. KUPCO_HQ_DAJEEJ

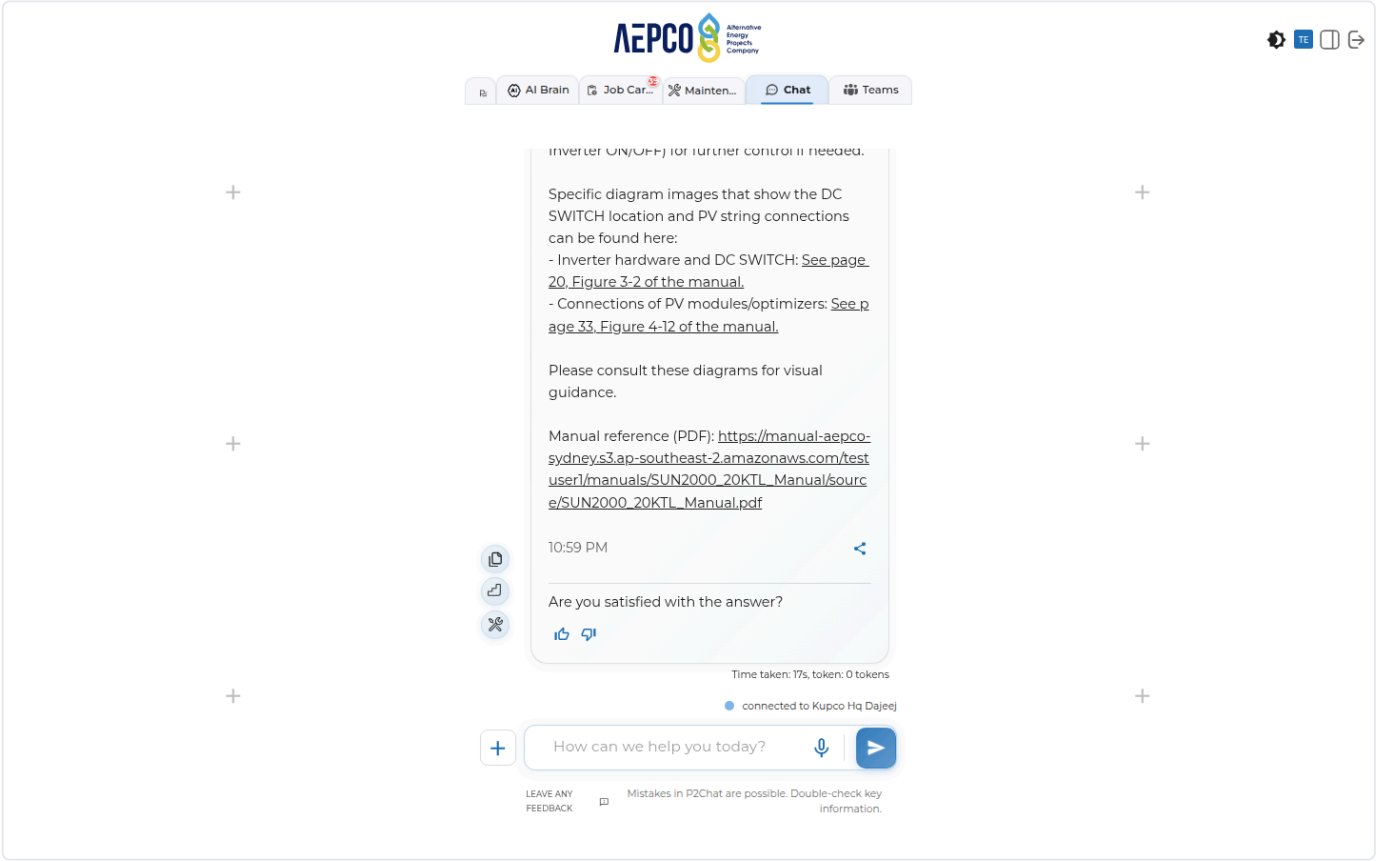
Production • 32.2s

Question: How can I shut down my inverters?

Response: To shut down your inverters, follow these steps, referencing your manual: 1. Log in to the app and select Maintenance > Inverter ON/OFF from the home screen to send a shutdown command to the inverter. 2. Set the DC SWITCH to the OFF position to fully disconnect the inverter from the DC power source. Refer to the labeled DC SWITCH in the inverter hardware diagrams for its exact location and appearance. 3. If you are troubleshooting specific PV strings or modules, also: - Disconnect the possibly faulty PV module and optimizer from the PV string, as shown in the wiring diagram. - Use a DC extension cable with an MC4 connector (as referenced in the diagrams) to bypass or connect modules as necessary. 4. After repairs or troubleshooting, to keep the inverter offline, maintain the DC SWITCH in the OFF position and confirm that the shutdown command has been sent via the app. If the inverter's display shows "Shutdown: Command", use the app (Maintenance > Inverter ON/OFF) for further con

TECHNICAL EVIDENCE

Agents used	`manual_agent`
Tools used	`functions.manual_rag_tool`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



61. KUPCO_HQ_DAJEEJ

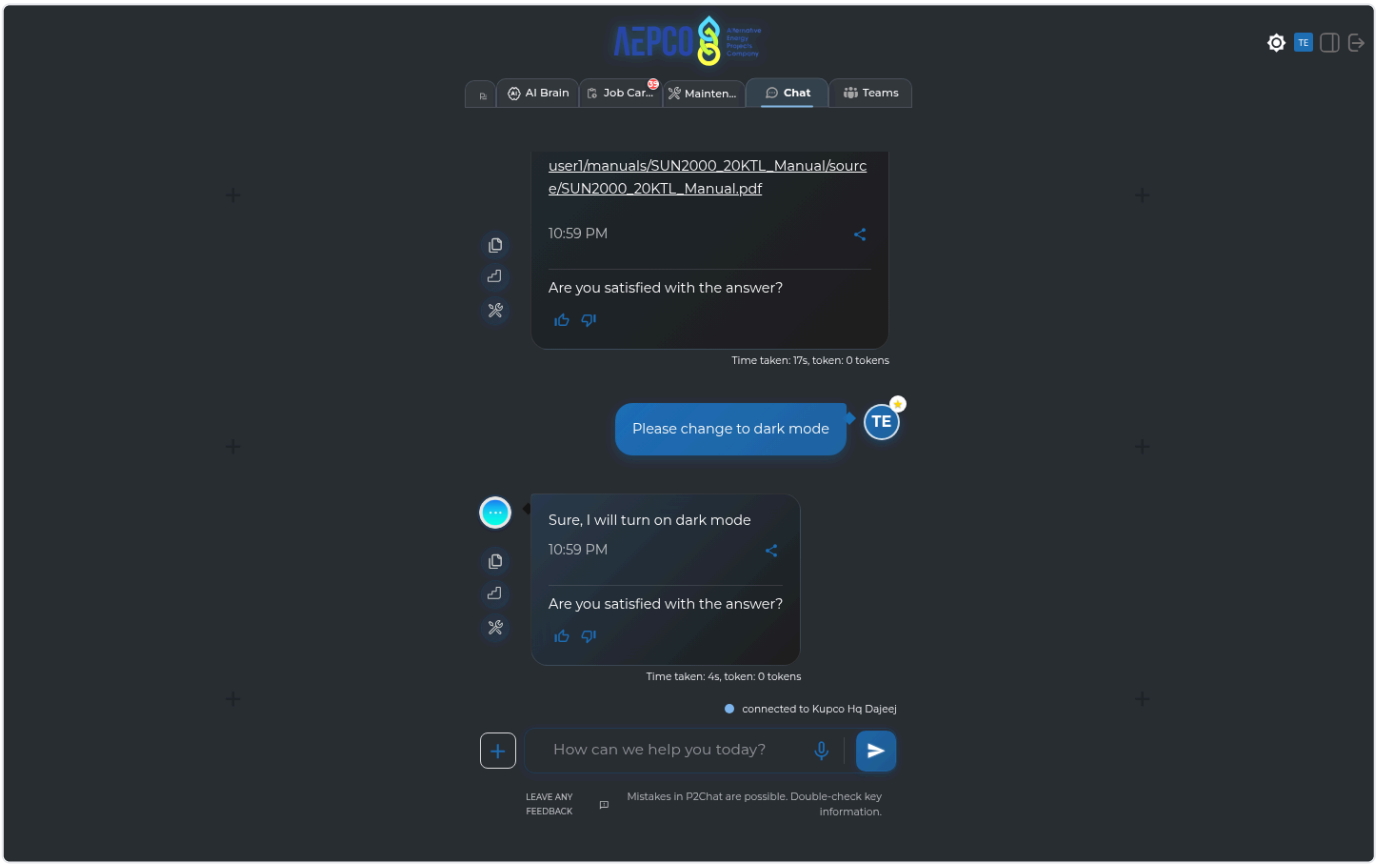
Production • 19.4s

Question: Please change to dark mode

Response: Sure, I will turn on dark mode 10:59 PM Are you satisfied with the answer? Time taken: 4s, token: 0 tokens connected to Kupco Hq Dajeej LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`ui_agent`
API calls made	2
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	0
Network failures	0



62. KUPCO_HQ_DAJEEJ

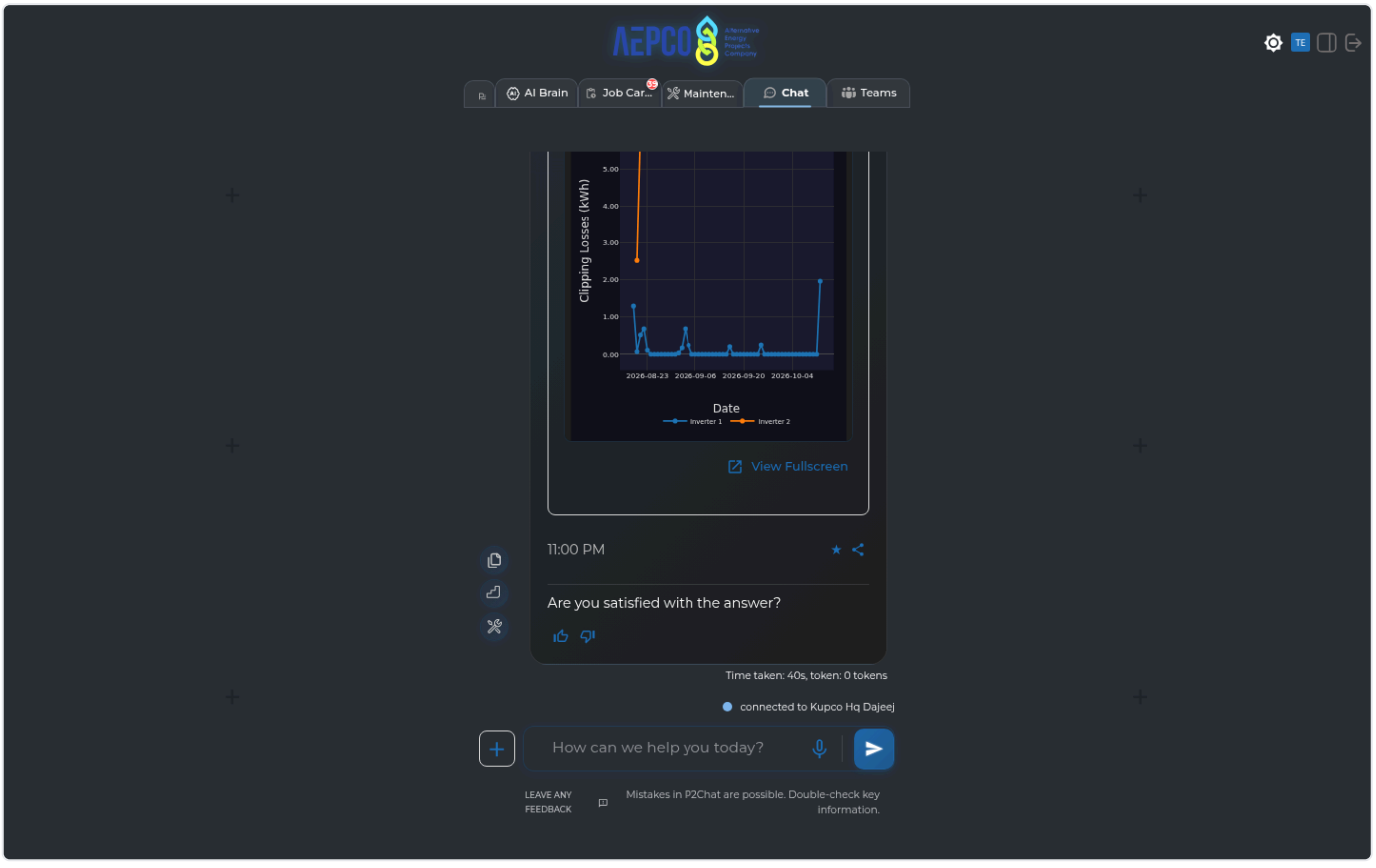
Production • 55.3s

Question: Please figure out how much clipping loss was found in the last 2 months broken by inverters, list them down and plot it as well

Response: Clipping Losses Summary (2026-08-19 to 2026-10-19): - inverter1: 237.0183 kWh - inverter2: 827.5514 kWh - inverter3: 421.2896 kWh Data Quality Status: - Good Visualization: - I've created a line chart showing daily clipping losses per inverter for the last 2 months ("Daily Clipping Losses per Inverter - Last 2 Months"). Clipping Losses Summary (2026-08-19 to 2026-10-19): - inverter1: 237.0183 kWh - inverter2: 827.5514 kWh - inverter3: 421.2896 kWh Data Quality Status: - Good Visualization: - I've created a line chart showing daily clipping losses per inverter for the last 2 months ("Daily Clipping Losses per Inverter - Last 2 Months"). View Fullscreen 11:00 PM ★ Are you satisfied with the answer? Time taken: 40s, token: 0 tokens connected to Kupco Hq Dajeej LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent, graph_agent`
Tools used	`functions.clipping_loss_via_requests, functions.calculate_clipping_or_curtailment_losses_ag, functions.generate_graphs_via_code`
API calls made	`2`
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	`0`
Network failures	`0`



63. KUPCO_HQ_DAJEEJ

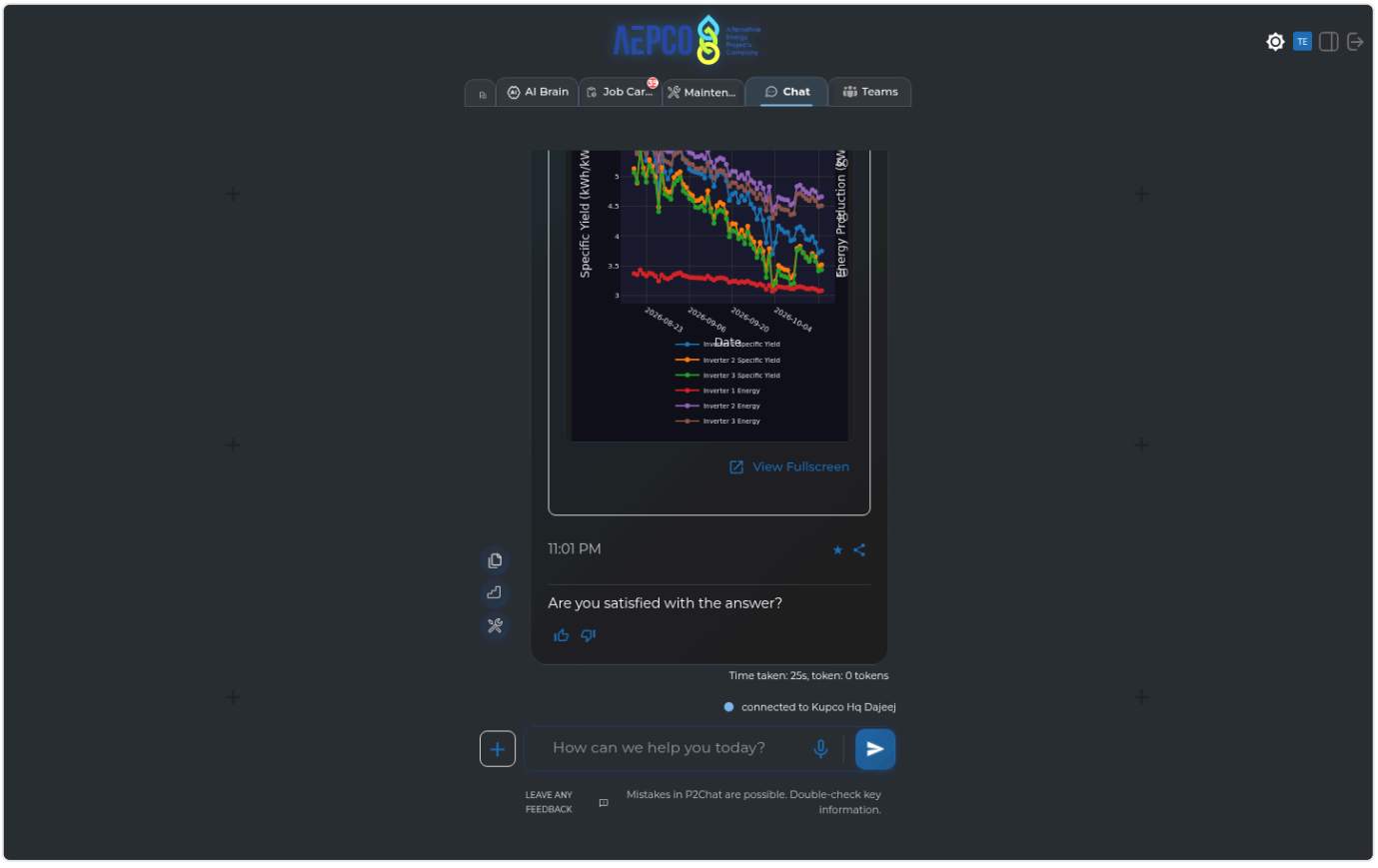
Production • 39.9s

Question: Please repeat the first plotting question, but interchange it with Specific yield on 1 axes, and energy production on the other axes on a single dual axes plot (Memory Test)

Response: Here is the summary for the inverter performance over the last two months and the requested visualization: - Specific yield (kWh/kWp) by inverter and month: - Inverter 1: 2026-08 = 68.99, 2026-09 = 147.04, 2026-10 = 75.66 - Inverter 2: 2026-08 = 64.98, 2026-09 = 132.09, 2026-10 = 67.05 - Inverter 3: 2026-08 = 63.92, 2026-09 = 128.88, 2026-10 = 65.61 - Energy production (kWh) by inverter and month: - Inverter 1: 2026-08 = 2 462.86, 2026-09 = 5 249.45, 2026-10 = 2 701.05 - Inverter 2: 2026-08 = 8 824.71, 2026-09 = 17 938.33, 2026-10 = 9 105.85 - Inverter 3: 2026-08 = 8 277.83, 2026-09 = 16 689.79, 2026-10 = 8 496.98 I've created a dual-axis line chart showing daily specific yield (left axis) and energy production (right axis) for each inverter over the requested period. Here is the summary for the inverter performance over the last two months and the requested visualization: - Specific yield (kWh/kWp) by inverter and month: - Inverter 1: 2026-08 =

TECHNICAL EVIDENCE

Agents used	`data_agent, graph_agent`
Tools used	`multi_tool_use.parallel(functions.specific_yield_ag, functions.total_energy_func_ag), multi_tool_use.parallel, functions.generate_graphs_via_code`
API calls made	`2`
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	`0`
Network failures	`0`



64. Galleria_Mall

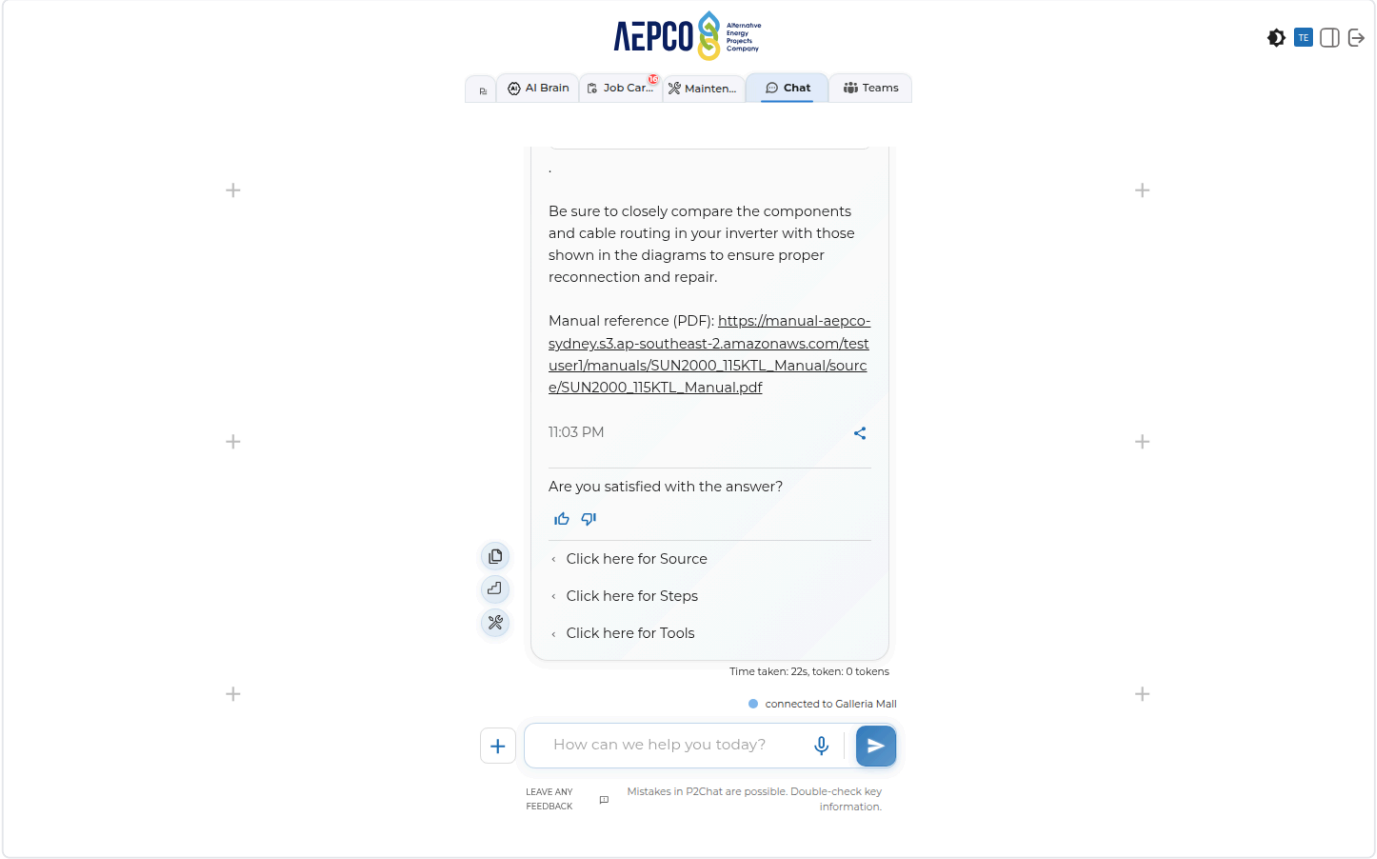
Production • 36.9s

Question: I have a COMMUNICATION_FAULT what do I do to my inverters?

Response: To troubleshoot a COMMUNICATION_FAULT on your inverters, follow these steps referencing the provided diagrams: 1. Visually inspect all key components and connections as shown in this diagram: . 2. Verify cable orientation and ensure all cables are securely connected according to the layout in this diagram: . 3. Check the internal component labeled "IS12H00012" for correct installation. Use this diagram to locate and verify it: . 4. If a fan (such as FAN 2 or FAN 3) is suspected to be faulty, refer to their labeled positions and inspect/replace as indicated in the following diagrams: , . 5. If you need to remove cable ties from FAN 3 before servicing, follow this visual guide: . Be sure to closely compare the components and cable routing in your inverter with those shown in the diagrams to ensure proper reconnection and repair. Manual reference (PDF): https://manual-aepco-sydney.s3.ap-southeast-2.amazonaws.com/testuser1/manuals/SUN2000_115KTL_Manual/source/SUN2000_115KTL_Manual

TECHNICAL EVIDENCE

Agents used	`manual_agent`
Tools used	`functions.manual_rag_tool`
API calls made	`12`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`2`
Network failures	`0`



65. Galleria_Mall

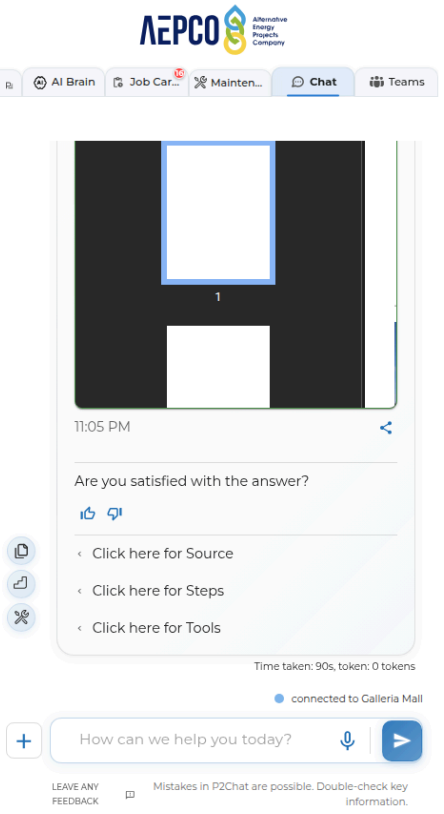
Production • 105.5s

Question: Please generate last months report

Response: I've generated the plant performance report for September 2026. You can view it here: 'https://report-aepco-sydney.s3.ap-southeast-2.amazonaws.com/Galleria_Mall_September_2026_external_report.pdf' Galleria_Mall Report 11:05 PM Are you satisfied with the answer? Click here for Source Click here for Steps Click here for Tools Time taken: 90s, token: 0 tokens connected to Galleria Mall LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`report_agent`
Tools used	`agent_external_report_tool`
API calls made	`3`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



66. Galleria_Mall

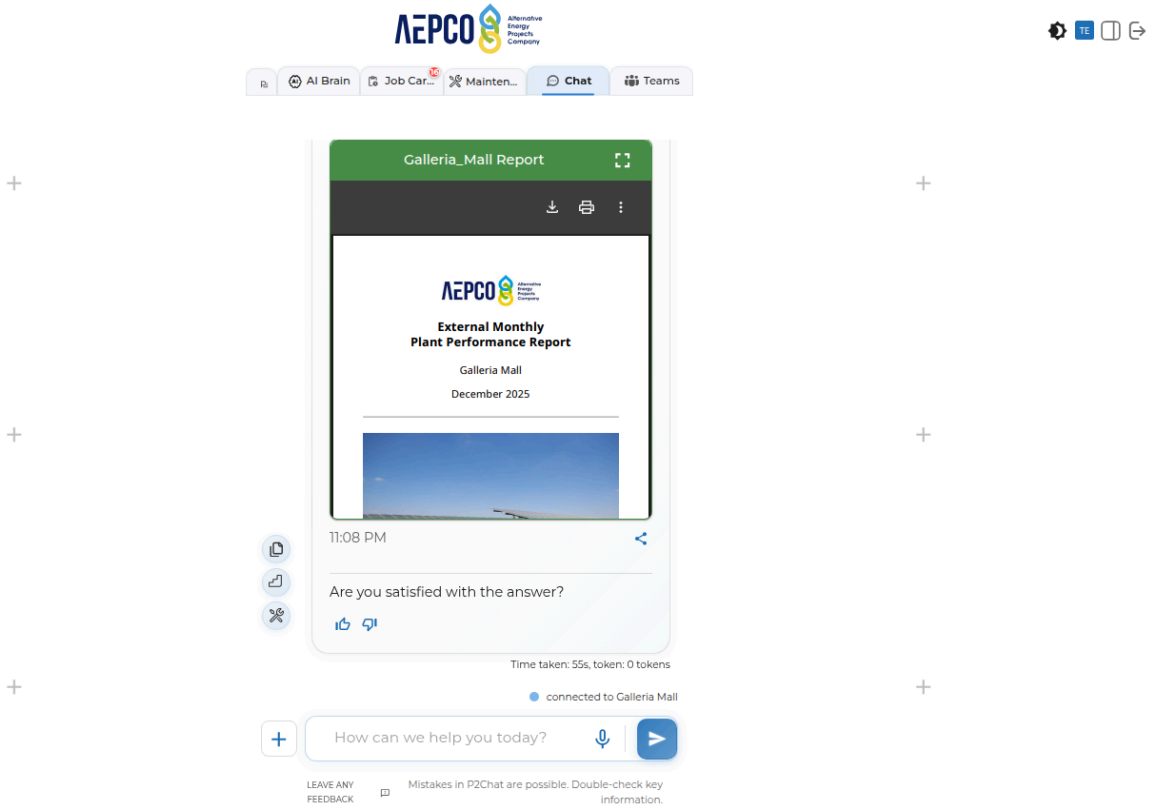
Production • 70.4s

Question: Please generate December 2025 performance report

Response: I've generated the performance report for December 2025. You can view it here: 'https://report-aepco-sydney.s3.ap-southeast-2.amazonaws.com/Galleria_Mall_December_2025_external_report.pdf' Galleria_Mall Report 11:08 PM Are you satisfied with the answer? Time taken: 55s, token: 0 tokens connected to Galleria Mall LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`report_agent`
Tools used	`agent_external_report_tool`
API calls made	`3`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



67. Galleria_Mall

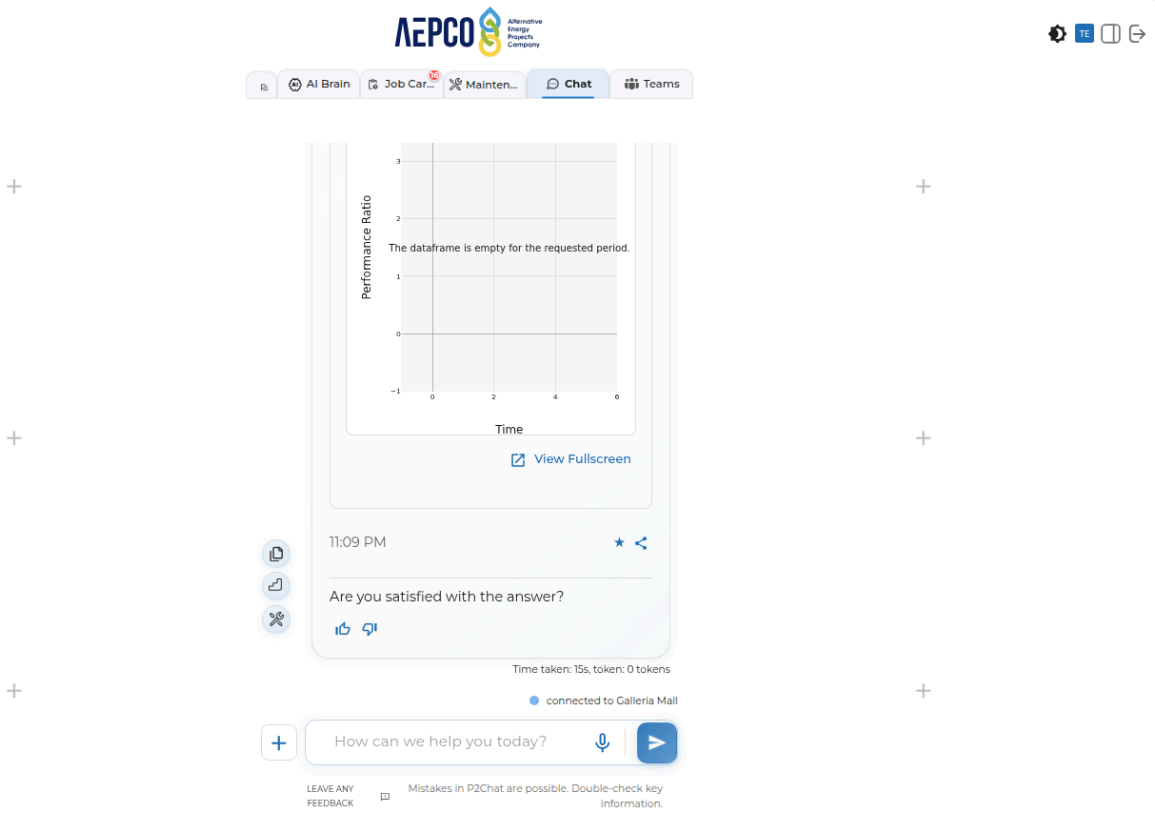
Production • 29.2s

Question: Please plot last 3 days hourly performance ratio for my MPPTs for inverter 2

Response: I've created the visualization for you. View Fullscreen 11:09 PM ★ Are you satisfied with the answer? Time taken: 15s, token: 0 tokens connected to Galleria Mall LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.generate_graphs_via_code, functions.perf_ratio_ag`
API calls made	`1`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



68. Galleria_Mall

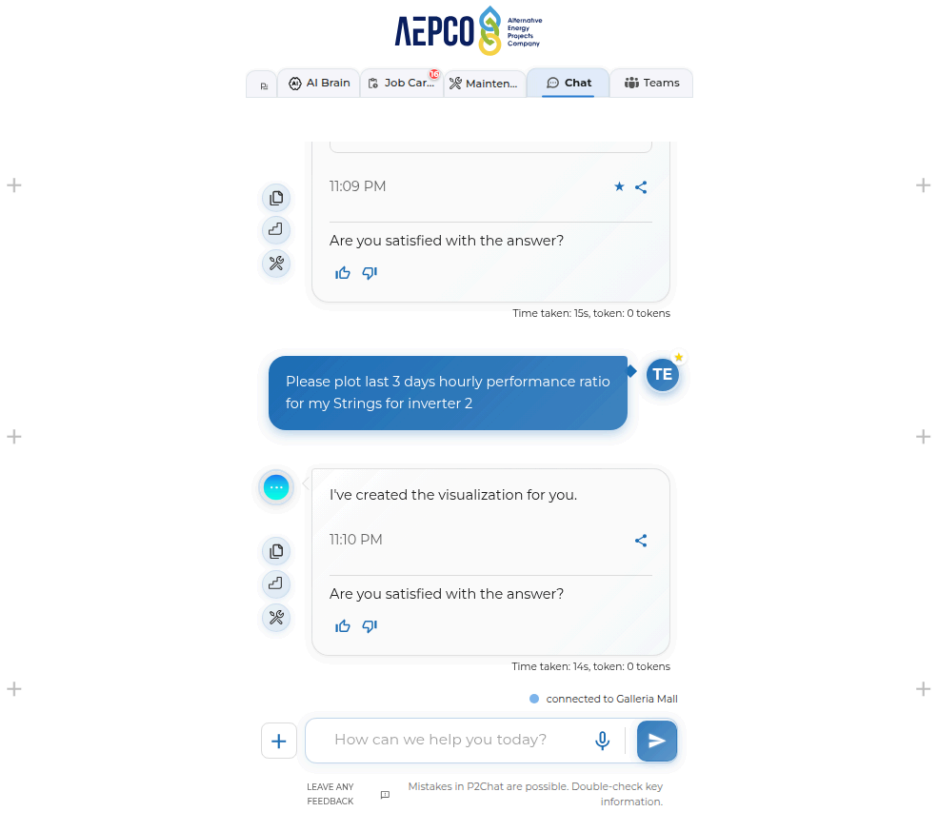
Production • 29.2s

Question: Please plot last 3 days hourly performance ratio for my Strings for inverter 2

Response: I've created the visualization for you. 11:10 PM Are you satisfied with the answer? Time taken: 14s, token: 0 tokens connected to Galleria Mall LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.extract_timeseries_data_ag, functions.perf_ratio_ag, functions.inspect_xpool`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



69. Galleria_Mall

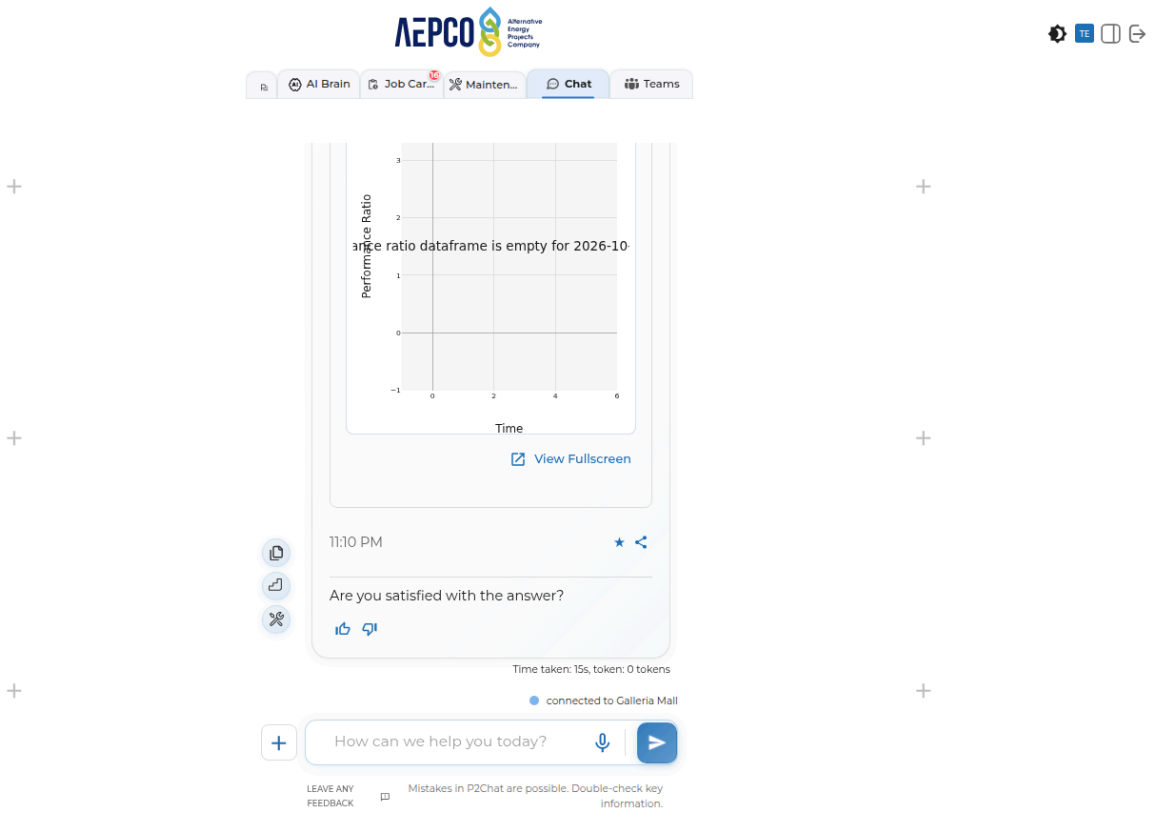
Production • 30.3s

Question: Please plot the last 4 days hourly performance ratio for all my inverters

Response: I've created the visualization for you. View Fullscreen 11:10 PM ★ Are you satisfied with the answer? Time taken: 15s, token: 0 tokens connected to Galleria Mall LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.generate_graphs_via_code, functions.perf_ratio_ag`
API calls made	2
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	0
Network failures	0



70. Galleria_Mall

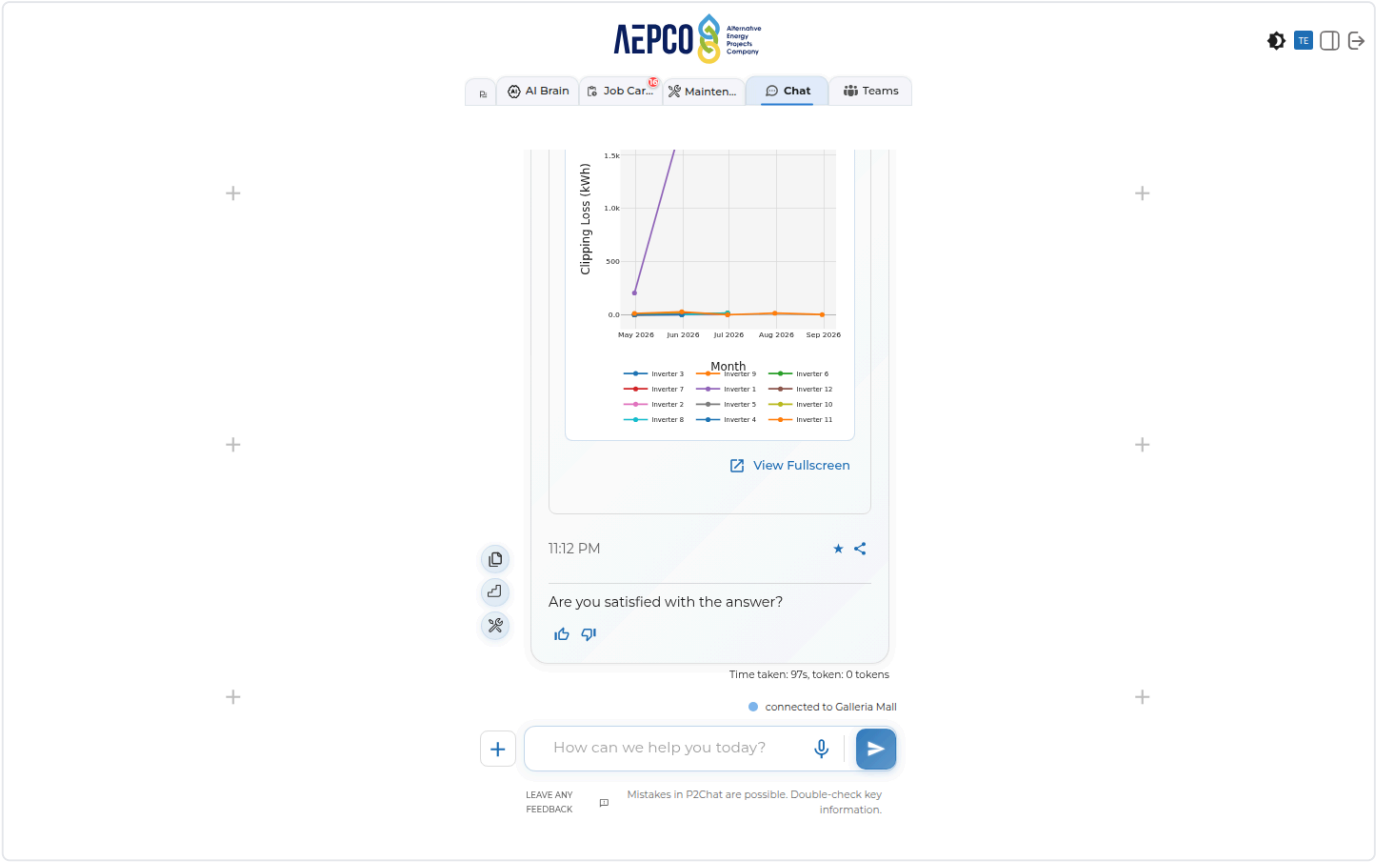
Production • 111.7s

Question: Please figure out how much clipping loss was found in the last 6 months broken by inverters, list them down and plot it as well

Response: Here is the summary of clipping losses over the last 6 months (2026-04-19 to 2026-10-19), broken down by inverter: - Inverter 1: 2004.248 kWh - Inverter 2: 5.731 kWh - Inverter 3: 0.663 kWh - Inverter 4: 2.032 kWh - Inverter 5: 20.916 kWh - Inverter 6: 21.356 kWh - Inverter 7: 21.506 kWh - Inverter 8: 16.872 kWh - Inverter 9: 12.485 kWh - Inverter 10: 25.981 kWh - Inverter 11: 59.408 kWh - Inverter 12: 10.624 kWh A line chart showing monthly clipping loss by inverter for the last 6 months has been created. Here is the summary of clipping losses over the last 6 months (2026-04-19 to 2026-10-19), broken down by inverter: - Inverter 1: 2004.248 kWh - Inverter 2: 5.731 kWh - Inverter 3: 0.663 kWh - Inverter 4: 2.032 kWh - Inverter 5: 20.916 kWh - Inverter 6: 21.356 kWh - Inverter 7: 21.506 kWh - Inverter 8: 16.872 kWh - Inverter 9: 12.485 kWh - Inverter 10: 25.981 kWh - Inverter 11: 59.408 kWh - Inverter 12: 10.624 kWh A line chart sh

TECHNICAL EVIDENCE

Agents used	`data_agent, graph_agent`
Tools used	`functions.generate_graphs_via_code, functions.calculate_clipping_or_curtailment_losses_ag, functions.clipping_loss_via_requests`
API calls made	`3`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



71. Galleria_Mall

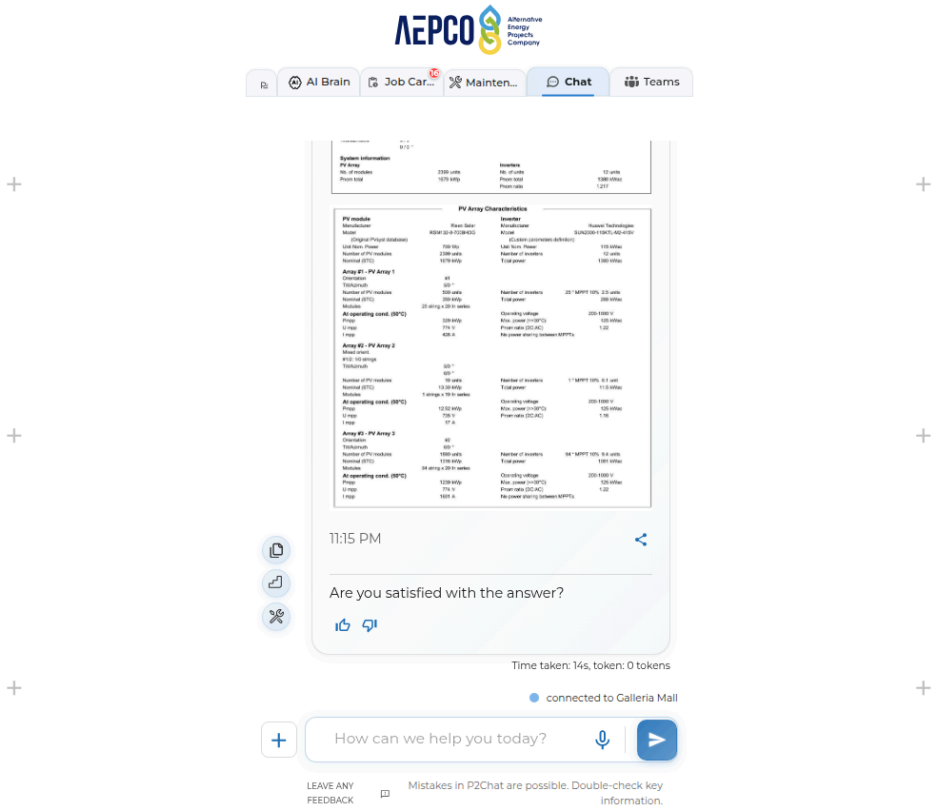
Production • 29.3s

Question: Based on pvsyst report, what is the rated power of this plant and array configuration

Response: Found 2 relevant PVsyst report sections: 1. [Galleria_Mall] System Summary (Page 2, Table 2) – includes rated power and general configuration overview. 2. [Galleria_Mall] PV Array Characteristics (Page 3, Table 2) – details of PV array orientations, module counts, and inverter setup. Please access the URLs provided to view the detailed information. Here is the reference: https://image-pvsyst-aepco-sydney.s3.ap-southeast-2.amazonaws.com/testuser1/Galleria_Mall/page_2_image_2.png https://image-pvsyst-aepco-sydney.s3.ap-southeast-2.amazonaws.com/testuser1/Galleria_Mall/page_3_image_2.png 11:15 PM Are you satisfied with the answer? Time taken: 14s, token: 0 tokens connected to Galleria Mall LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`pvsyst_agent`
Tools used	`pvsyst_tool`
API calls made	`4`
Last API call	`GET /testuser2/chat/ → 404`
Console errors	`3`
Network failures	`1`



72. Galleria_Mall

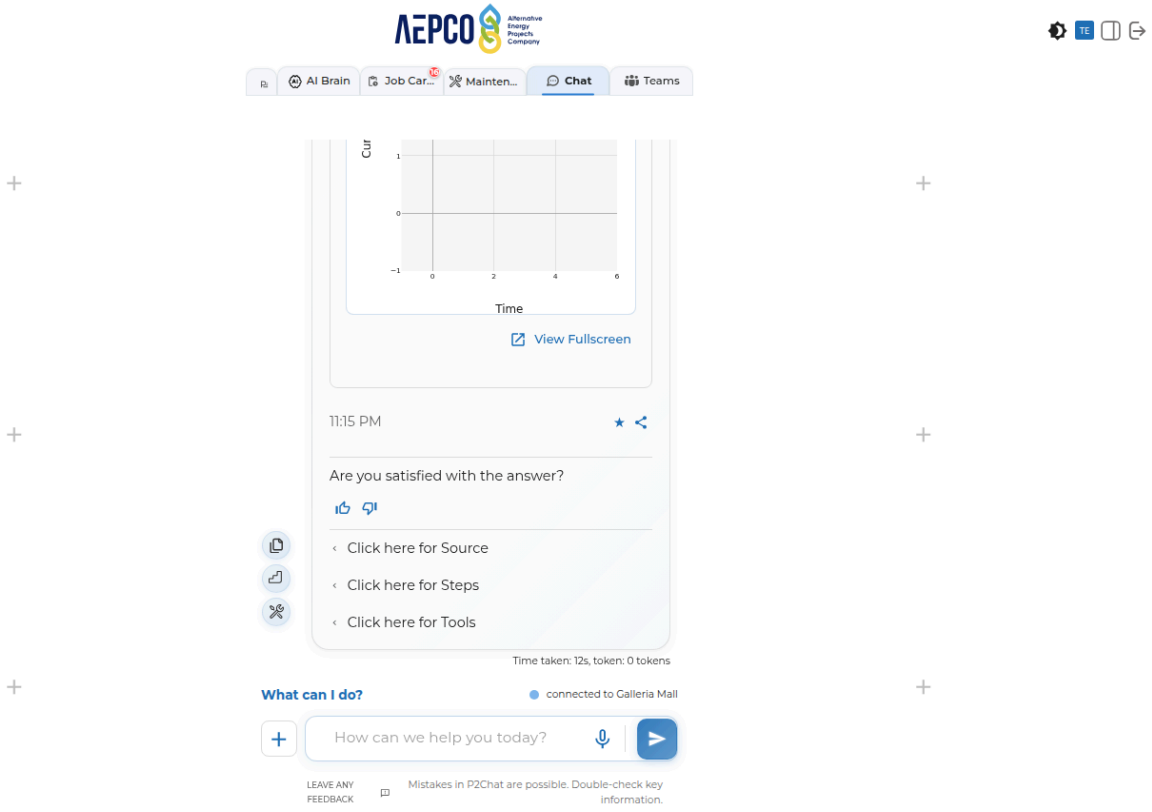
Production • 27.7s

Question: Please plot my current and voltage in a dual axes plot on the deepest possible granularity level for inverter3

Response: I've created the visualization for you. View Fullscreen 11:15 PM ★ Are you satisfied with the answer? Click here for Source Click here for Steps Click here for Tools Time taken: 12s, token: 0 tokens What can I do? connected to Galleria Mall LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.extract_timeseries_data_ag, functions.generate_graphs_via_code`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	3
Network failures	0



73. Galleria_Mall

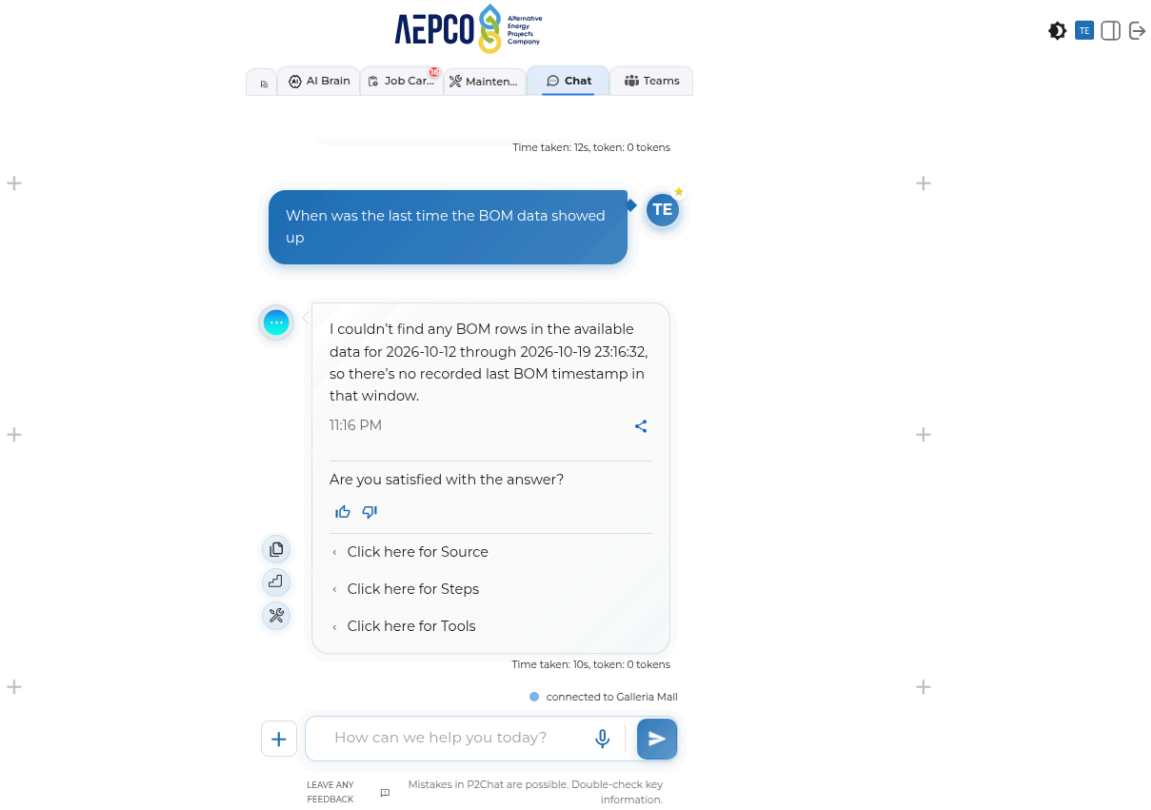
Production • 24.6s

Question: When was the last time the BOM data showed up

Response: I couldn't find any BOM rows in the available data for 2026-10-12 through 2026-10-19 23:16:32, so there's no recorded last BOM timestamp in that window. 11:16 PM Are you satisfied with the answer? Click here for Source Click here for Steps Click here for Tools Time taken: 10s, token: 0 tokens connected to Galleria Mall LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`functions.extract_timeseries_data_ag`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



74. Galleria_Mall

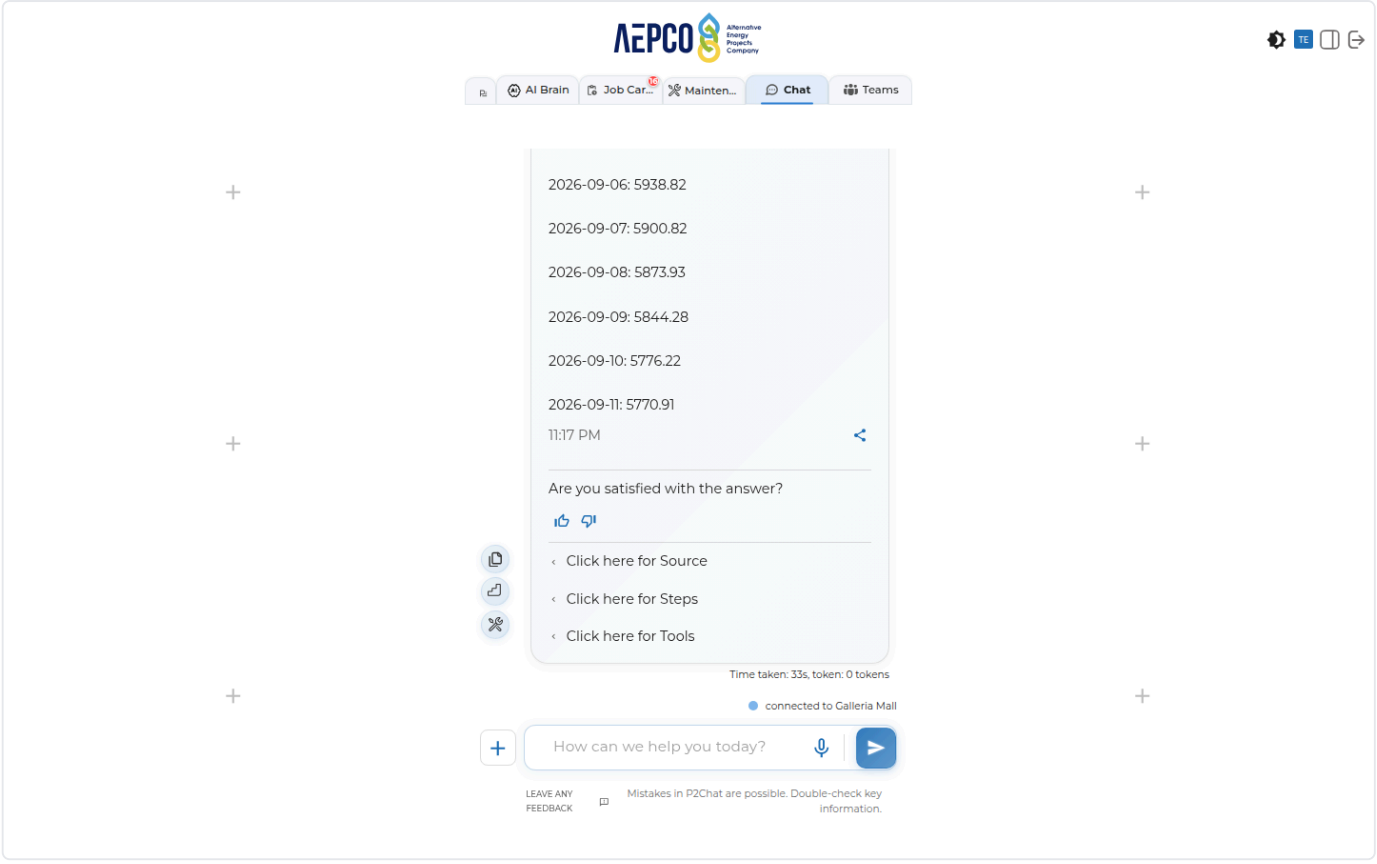
Production • 47.4s

Question: Please find out the daily POA poa summed up as a Wh/m^2, daily for the last 12 months and then calculate the median, mean and kurt

Response: Using `pyranometer_1`, I summed the 5-minute POA irradiance into daily irradiation totals as Wh/m² over the requested last 12 months window. Result: - Median daily POA: 6943.73 Wh/m² - Mean daily POA: 6714.55 Wh/m² - Kurtosis of daily POA values: 8.2525 Data coverage note: - The requested window was 2025-10-20 to 2026-10-19 - Available `pyranometer_1` data only spans 2026-04-05 to 2026-09-11 - So the statistics are based on 160 days with data, not a full 12 months Daily POA totals (Wh/m²): 2026-04-05: 6704.45 2026-04-06: 6793.26 2026-04-07: 6848.05 2026-04-08: 6760.17 2026-04-09: 4432.46 2026-04-10: 6582.76 2026-04-11: 3884.04 2026-04-12: 4312.05 2026-04-13: 6611.83 2026-04-14: 6873.08 2026-04-15: 7191.16 2026-04-16: 5991.03 2026-04-17: 7068.52 2026-04-18: 6275.19 2026-04-19: 3980.02 2026-04-20: 1524.48 2026-04-21: 4740.04 2026-04-22: 6620.00 2026-04-23: 4896.32 2026-04-24: 5638.57 2026-04-25: 4724.33 2026-04-26: 6091.48

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`functions.extract_timeseries_data_ag, functions.execute_custom_code`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



75. Galleria_Mall

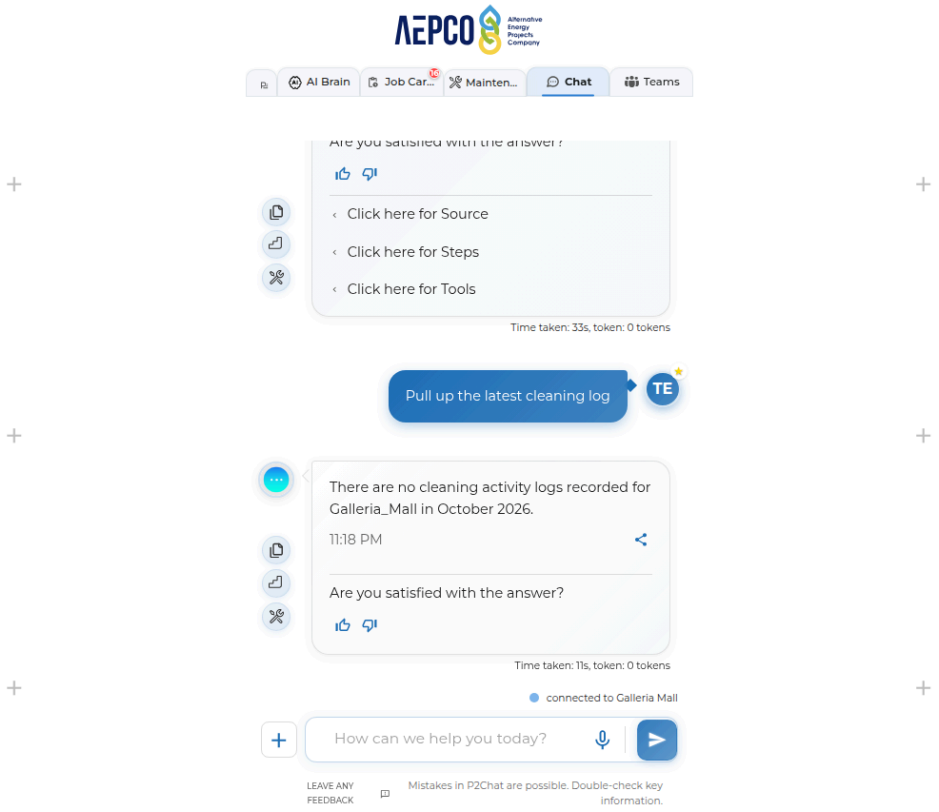
Production • 26.1s

Question: Pull up the latest cleaning log

Response: There are no cleaning activity logs recorded for Galleria_Mall in October 2026. 11:18 PM Are you satisfied with the answer? Time taken: 11s, token: 0 tokens connected to Galleria Mall LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_entry_agent`
API calls made	2
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	0
Network failures	0



76. Galleria_Mall

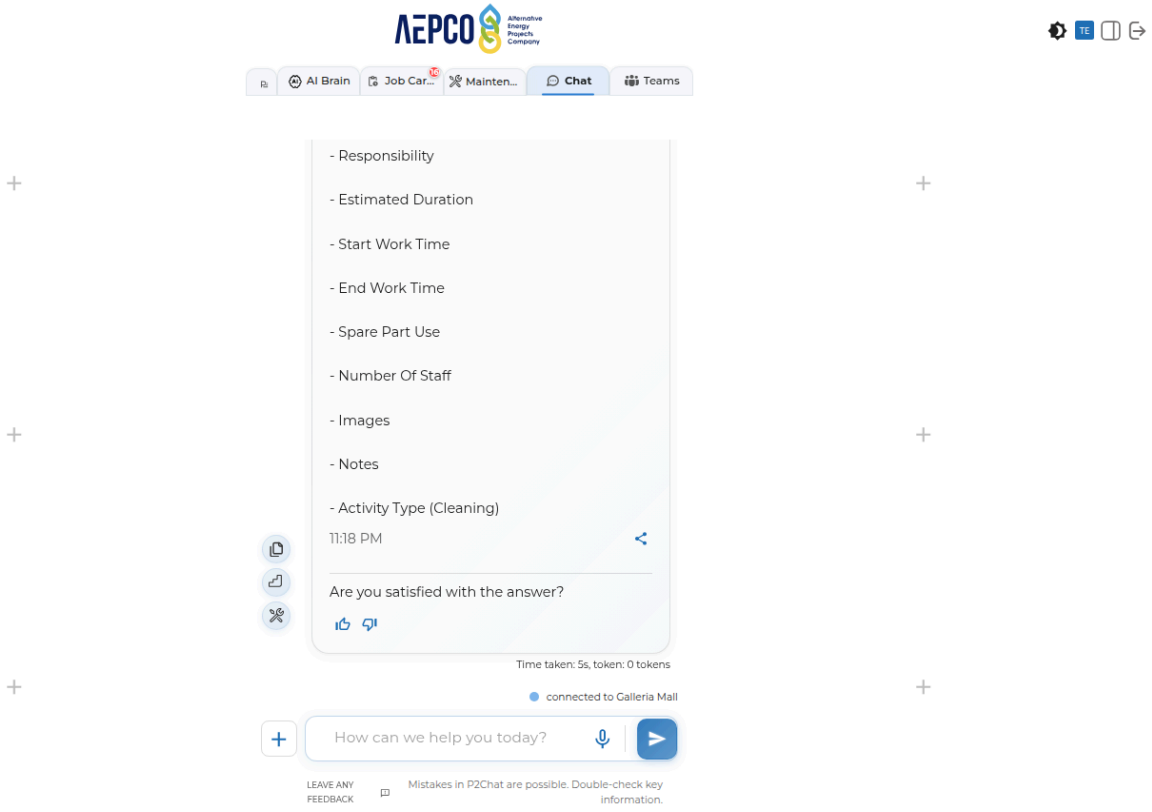
Production • 20.0s

Question: Jim cleaned the pyranometer today, can you add it to the cleaning log?

Response: To add a cleaning activity log, please provide the following required details: - Date - Fault Cause - Action Taken - Status Site will be set automatically from the connected plant. Optional fields you can also provide, but they are not required: - Priority - Responsibility - Estimated Duration - Start Work Time - End Work Time - Spare Part Use - Number Of Staff - Images - Notes - Activity Type (Cleaning) 11:18 PM Are you satisfied with the answer? Time taken: 5s, token: 0 tokens connected to Galleria Mall LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_entry_agent`
API calls made	`1`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



77. Galleria_Mall

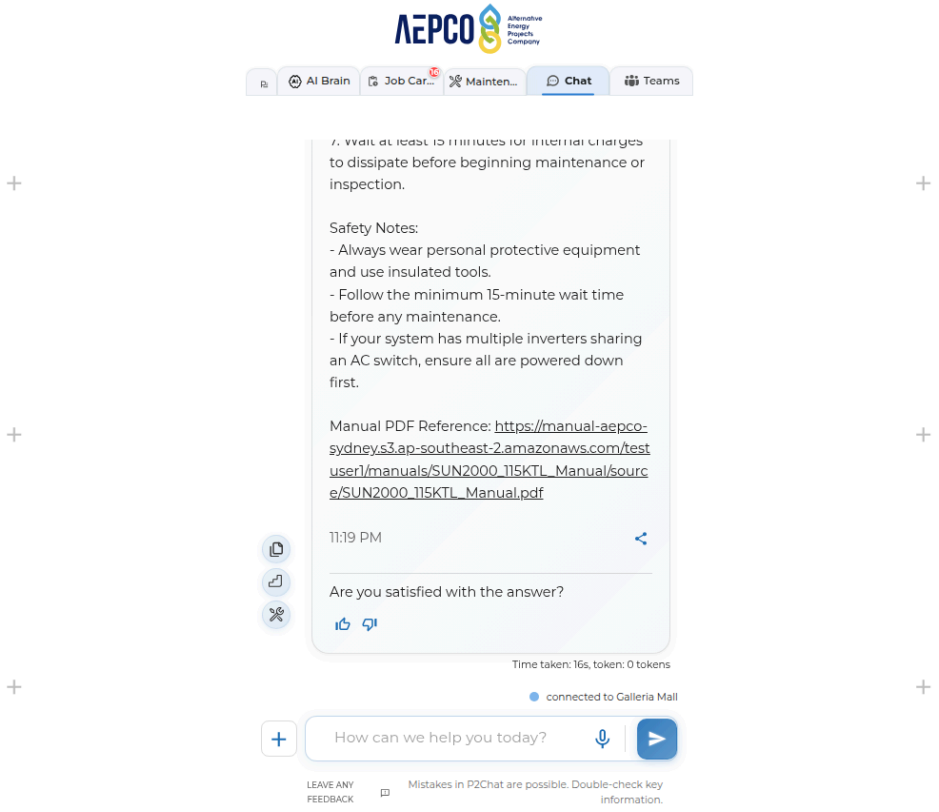
Production • 30.7s

Question: How can I shut down my inverters?

Response: To safely shut down your inverters, follow these steps (referencing the SUN2000_115KTL manual): 1. If the inverter is operating normally, send a shutdown command from the management app, SmartLogger, or your backend system. 2. If the inverter is already shut down due to a fault, proceed to the next step. 3. Turn off the AC switch located between the inverter and the power grid. You can identify the AC switch and its placement in your system using this diagram: 4. Use a clamp meter to measure the DC current on each PV string input to the inverter. Check this diagram for measurement points: - If all DC inputs show less than or equal to 0.01 A, continue. - Otherwise, wait until PV string current drops below 0.01 A (typically at night), then proceed. 5. Open the maintenance compartment, install the support strut if necessary, and use a multimeter to confirm there is no AC voltage on the AC terminal block before any further work. 6. Switch all DC switches of the inverter to the O

TECHNICAL EVIDENCE

Agents used	`manual_agent`
Tools used	`functions.manual_rag_tool`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



78. Galleria_Mall

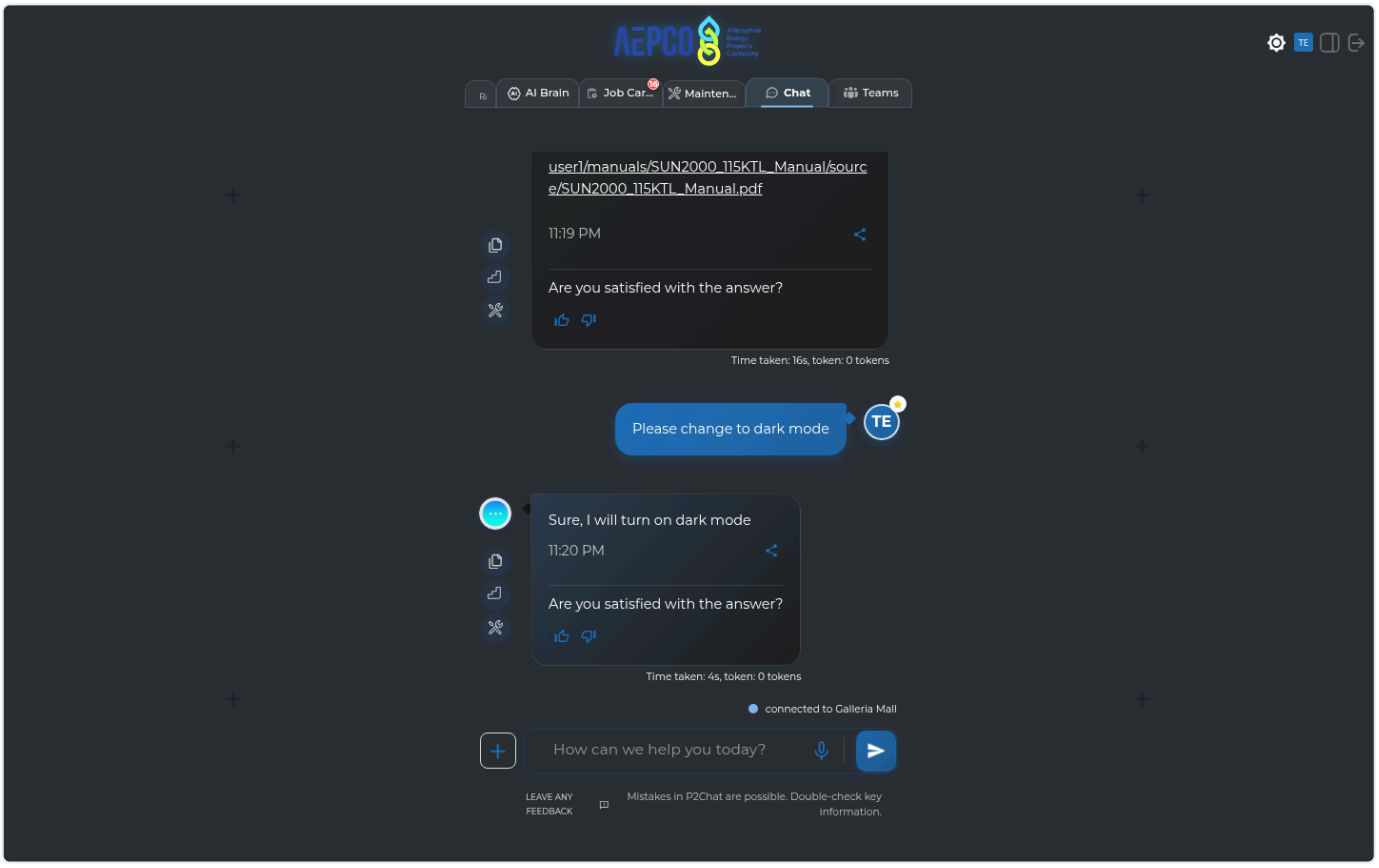
Production • 20.1s

Question: Please change to dark mode

Response: Sure, I will turn on dark mode 11:20 PM Are you satisfied with the answer? Time taken: 4s, token: 0 tokens connected to Galleria Mall LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`ui_agent`
API calls made	`1`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



79. Galleria_Mall

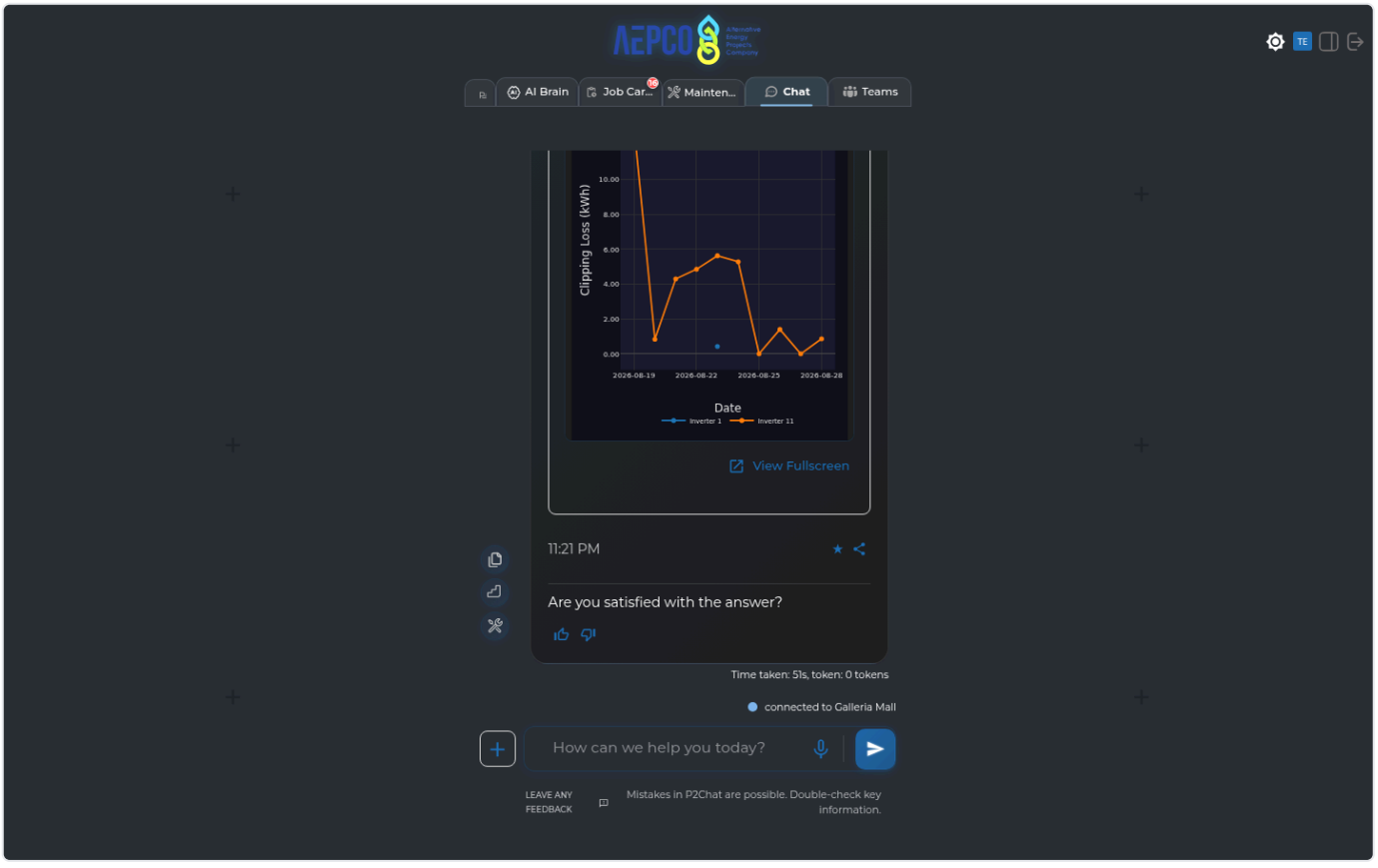
Production • 65.9s

Question: Please figure out how much clipping loss was found in the last 2 months broken by inverters, list them down and plot it as well

Response: Clipping Loss by Inverter (2026-08-19 to 2026-10-19) - Inverter 1: 0.430 kWh - Inverter 2: 0.000 kWh - Inverter 3: 0.000 kWh - Inverter 4: 0.000 kWh - Inverter 5: 0.000 kWh - Inverter 6: 0.000 kWh - Inverter 7: 0.000 kWh - Inverter 8: 0.000 kWh - Inverter 9: 0.000 kWh - Inverter 10: 0.000 kWh - Inverter 11: 36.261 kWh - Inverter 12: 0.000 kWh Total clipping loss across all inverters: 36.691 kWh Note: Available data only runs through 2026-09-11 19:00:00, so this result is based on the portion of the requested two-month window for which data exists. I've created a line chart showing daily clipping loss by inverter for the last two months. Clipping Loss by Inverter (2026-08-19 to 2026-10-19) - Inverter 1: 0.430 kWh - Inverter 2: 0.000 kWh - Inverter 3: 0.000 kWh - Inverter 4: 0.000 kWh - Inverter 5: 0.000 kWh - Inverter 6: 0.000 kWh - Inverter 7: 0.000 kWh - Inverter 8: 0.000 kWh - Inverter 9: 0.000 kWh - Inverter 10: 0.000 kWh - In

TECHNICAL EVIDENCE

Agents used	`data_agent, graph_agent`
Tools used	`functions.clipping_loss_via_requests, functions.generate_graphs_via_code, functions.calculate_clipping_or_curtailment_losses_ag`
API calls made	3
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	0
Network failures	0



80. Galleria_Mall

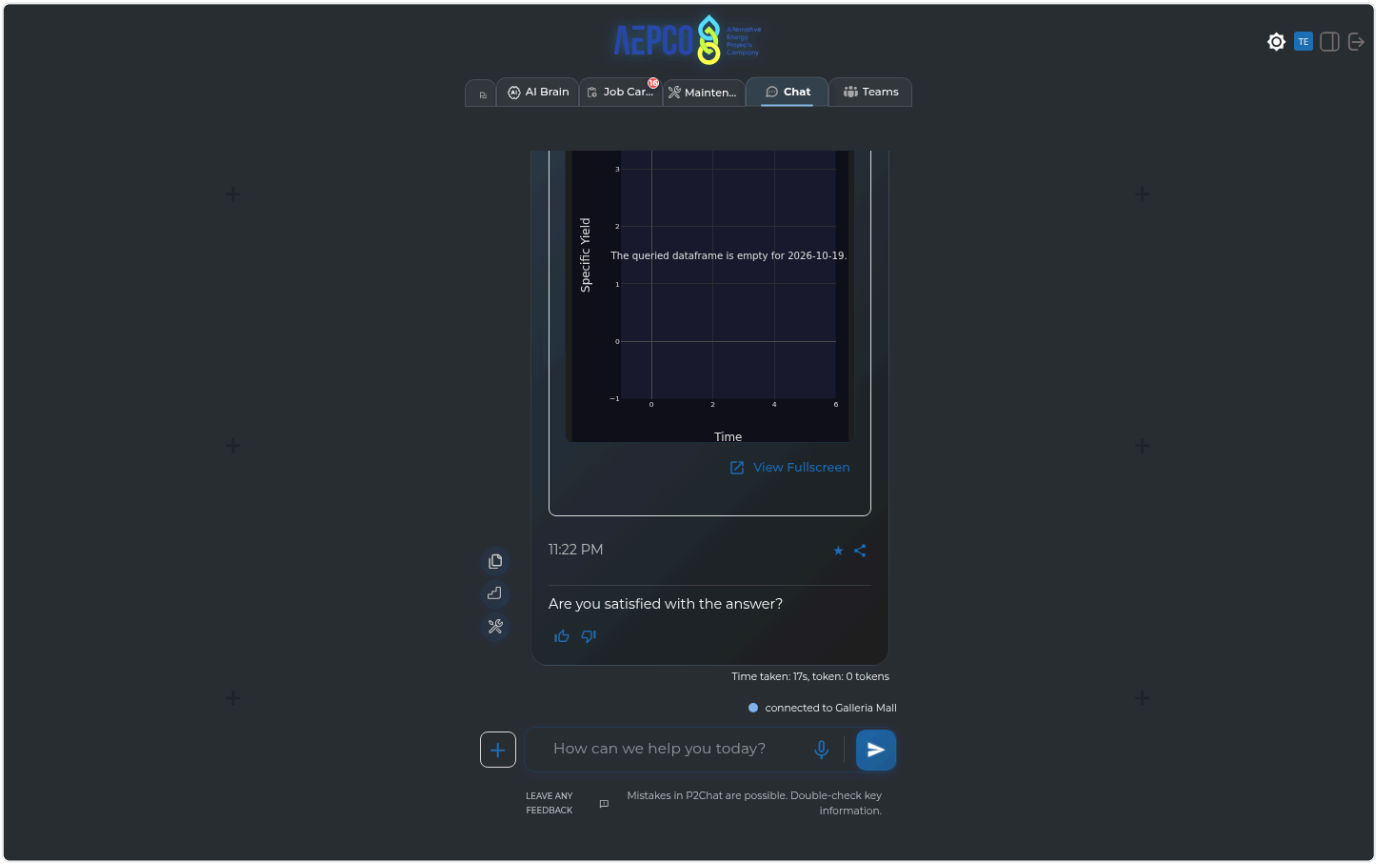
Production • 32.3s

Question: Please repeat the first plotting question, but interchange it with Specific yield on 1 axes, and energy production on the other axes on a single dual axes plot (Memory Test)

Response: I've created the visualization for you. View Fullscreen 11:22 PM ★ Are you satisfied with the answer? Time taken: 17s, token: 0 tokens connected to Galleria Mall LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.total_energy_func_ag, functions.generate_graphs_via_code, functions.specific_yield_ag`
API calls made	`2`
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	`0`
Network failures	`0`



81. GID

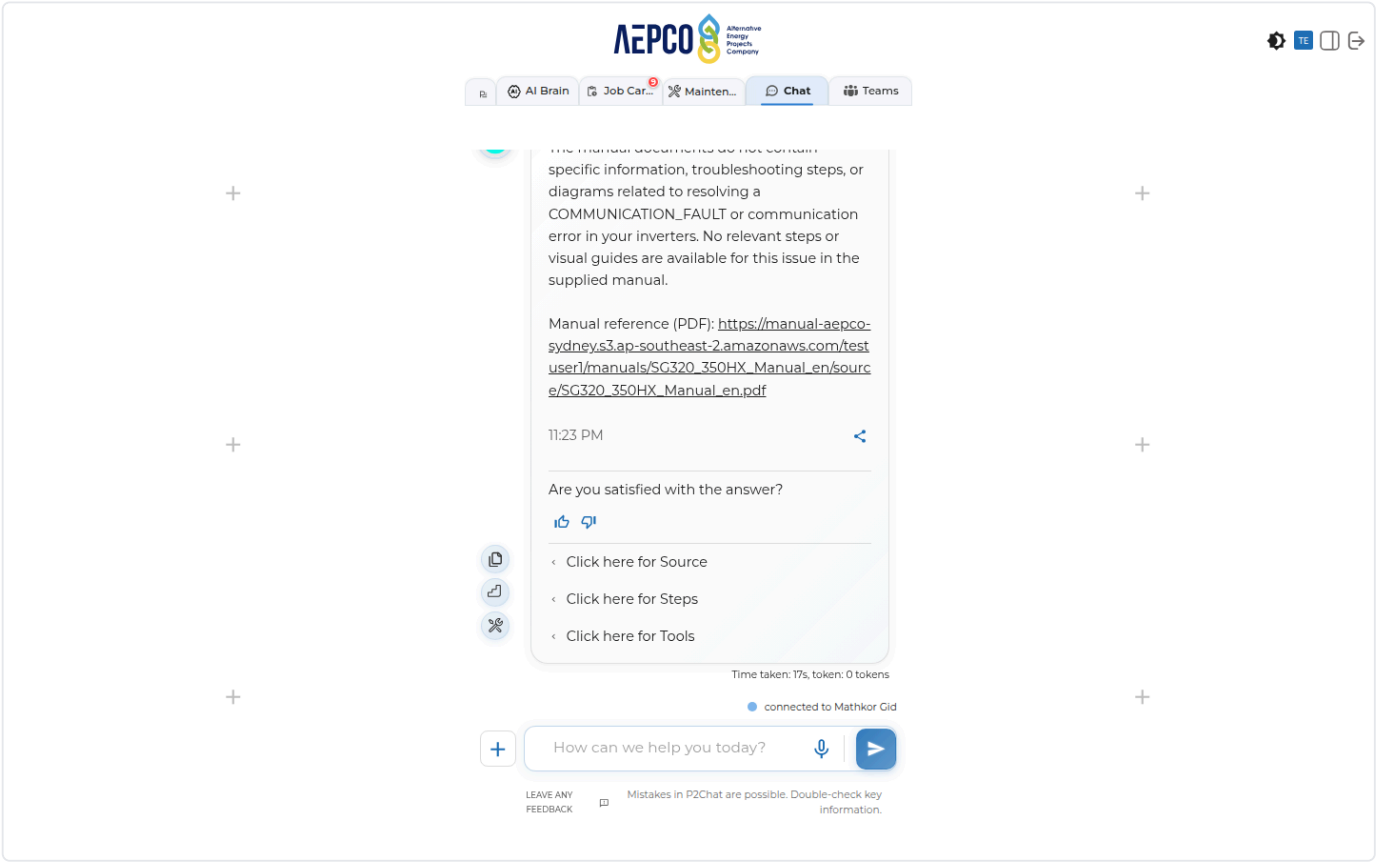
Production • 32.3s

Question: I have a COMMUNICATION_FAULT what do I do to my inverters?

Response: The manual documents do not contain specific information, troubleshooting steps, or diagrams related to resolving a COMMUNICATION_FAULT or communication error in your inverters. No relevant steps or visual guides are available for this issue in the supplied manual. Manual reference (PDF): https://manual-aepco-sydney.s3.ap-southeast-2.amazonaws.com/testuser1/manuals/SG320_350HX_Manual_en/source/SG320_350HX_Manual_en.pdf 11:23 PM Are you satisfied with the answer? Click here for Source Click here for Steps Click here for Tools Time taken: 17s, token: 0 tokens connected to Mathkor Gid LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`manual_agent`
API calls made	`1`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



82. GID

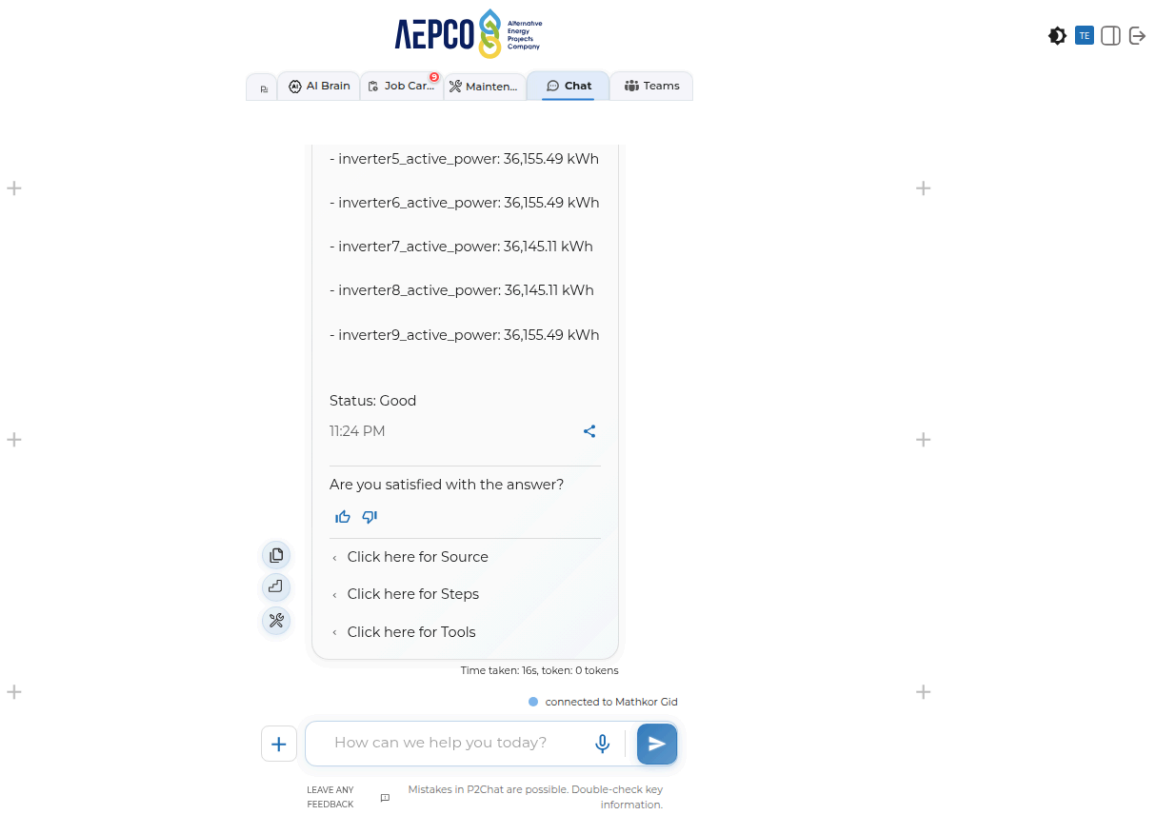
Production • 30.7s

Question: Use 'clipping_loss_via_requests' in order to calculate clipping loss if any for July 2026

Response: Clipping loss for July 2026 was: - inverter1_active_power: 31,189.95 kWh - inverter2_active_power: 31,079.45 kWh - inverter3_active_power: 31,194.90 kWh - inverter4_active_power: 31,190.07 kWh - inverter5_active_power: 36,155.49 kWh - inverter6_active_power: 36,155.49 kWh - inverter7_active_power: 36,145.11 kWh - inverter8_active_power: 36,145.11 kWh - inverter9_active_power: 36,155.49 kWh Status: Good 11:24 PM Are you satisfied with the answer? Click here for Source Click here for Steps Click here for Tools Time taken: 16s, token: 0 tokens connected to Mathkor Gid LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`functions.clipping_loss_via_requests`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



83. GID

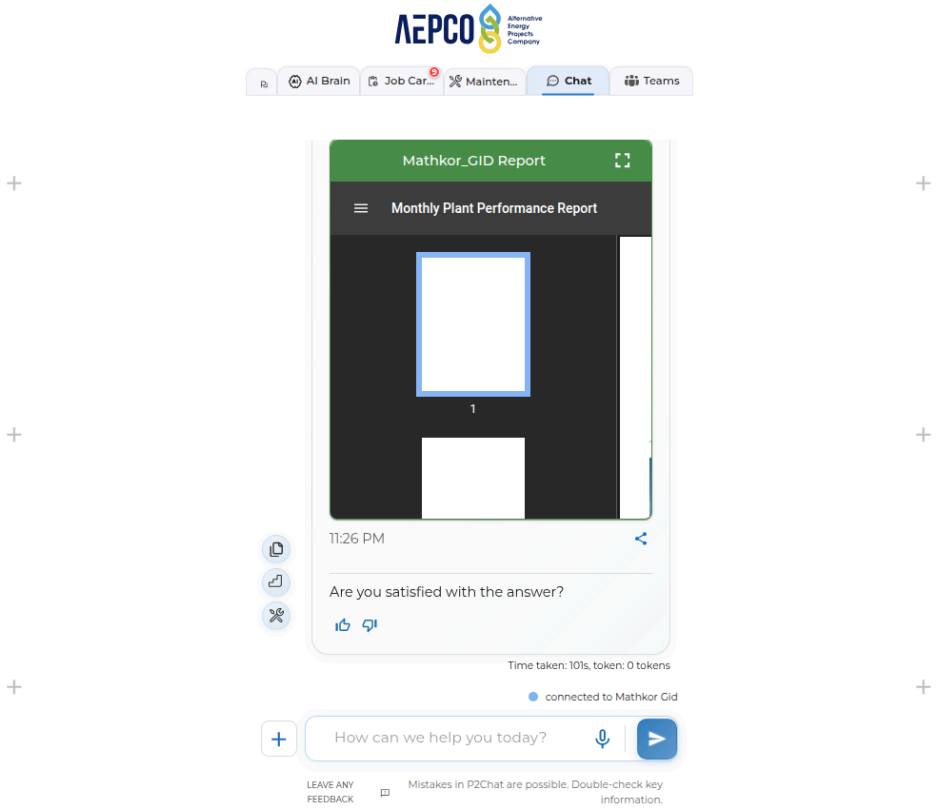
Production • 116.8s

Question: Please generate last months report

Response: I've generated the external performance report for September 2026. You can view it here: 'https://report-aepco-sydney.s3.ap-southeast-2.amazonaws.com/Mathkor_Solar_Plant_September_2026_external_report.pdf' Mathkor_GID Report 11:26 PM Are you satisfied with the answer? Time taken: 101s, token: 0 tokens connected to Mathkor Gid LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`report_agent`
Tools used	`agent_external_report_tool`
API calls made	`3`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



84. GID

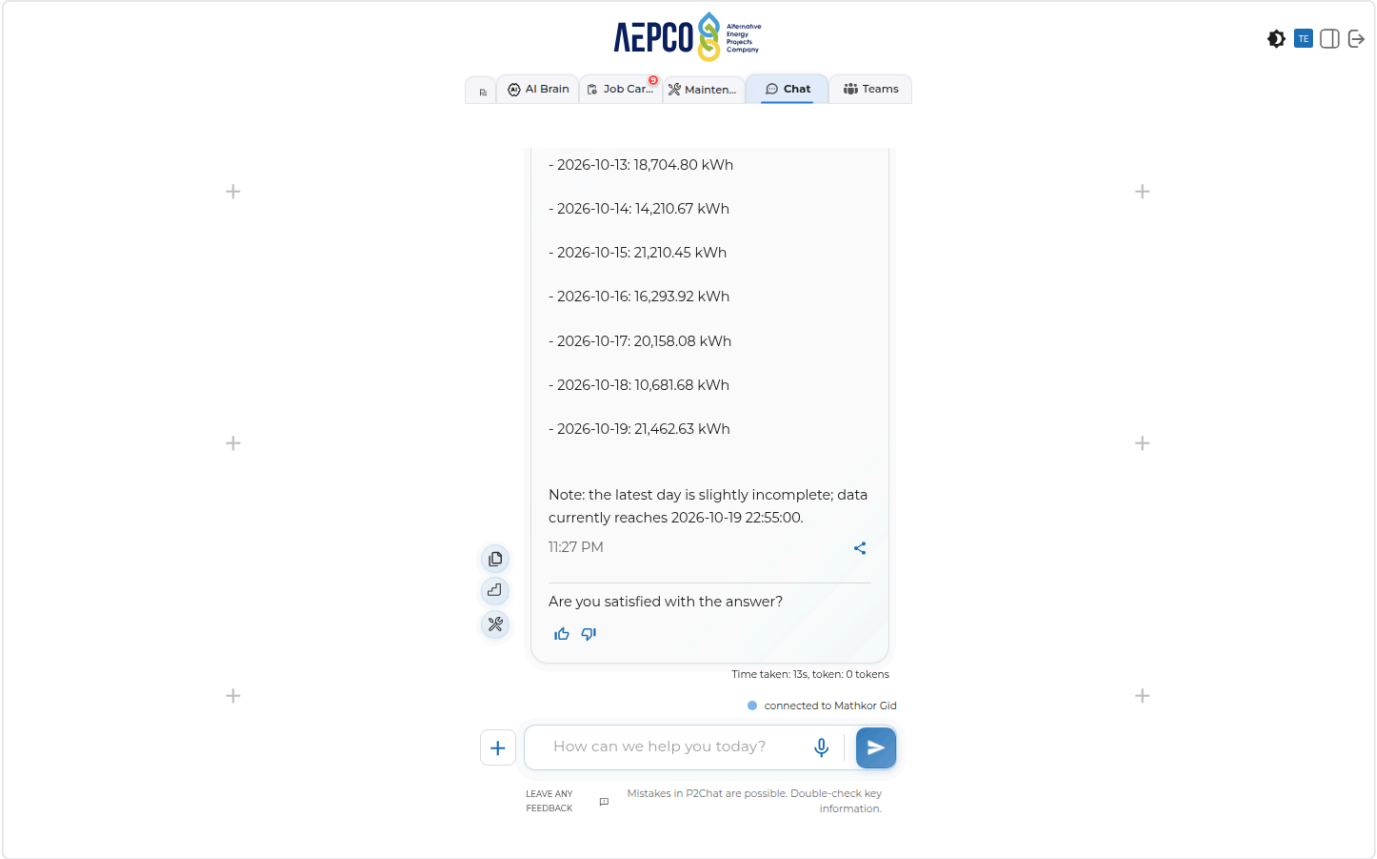
Production • 27.7s

Question: Was there any 0 production days in the last 7 days?

Response: There were no zero-production days in the last 7 days (2026-10-13 to 2026-10-19). Daily energy: - 2026-10-13: 18,704.80 kWh - 2026-10-14: 14,210.67 kWh - 2026-10-15: 21,210.45 kWh - 2026-10-16: 16,293.92 kWh - 2026-10-17: 20,158.08 kWh - 2026-10-18: 10,681.68 kWh - 2026-10-19: 21,462.63 kWh Note: the latest day is slightly incomplete; data currently reaches 2026-10-19 22:55:00. 11:27 PM Are you satisfied with the answer? Time taken: 13s, token: 0 tokens connected to Mathkor Gid LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`functions.total_energy_func_ag`
API calls made	`2`
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	`0`
Network failures	`0`



85. GID

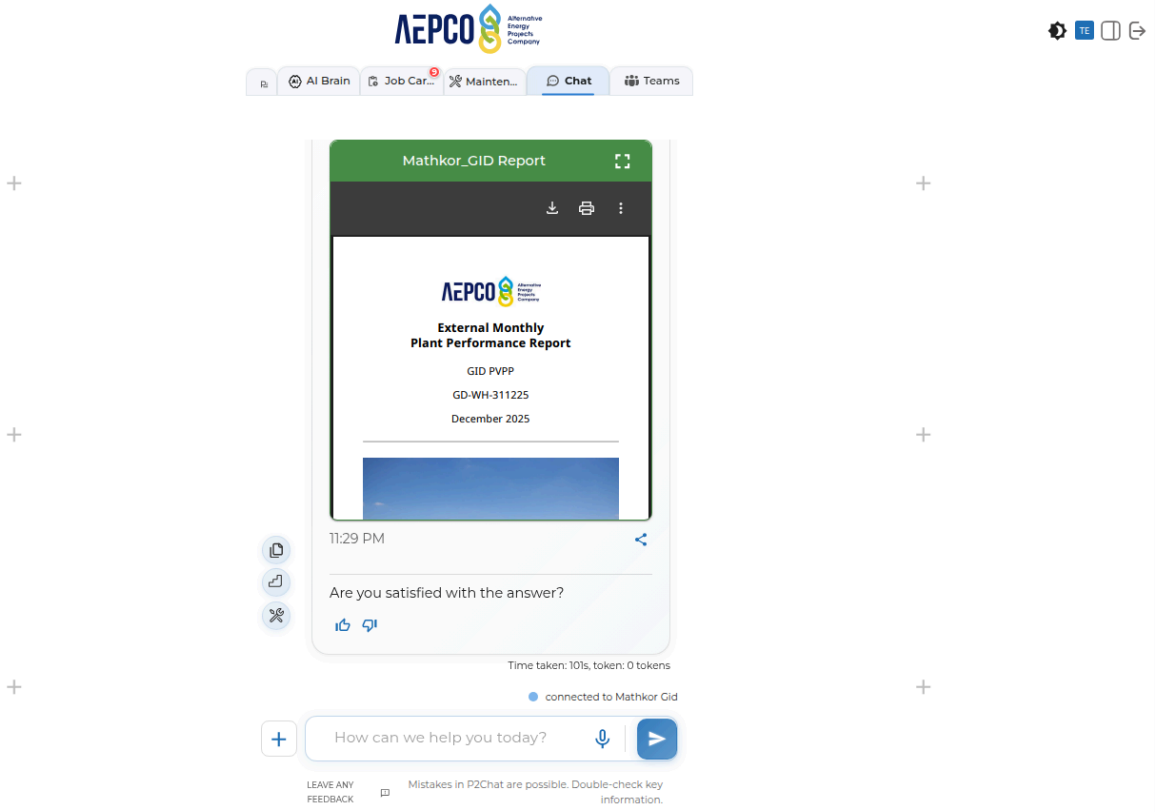
Production • 116.0s

Question: Please generate December 2025 performance report

Response: I've generated the external performance report for December 2025. You can view it here: 'https://report-aepco-sydney.s3.ap-southeast-2.amazonaws.com/Mathkor_Solar_Plant_December_2025_external_report.pdf' Mathkor_GID Report 11:29 PM Are you satisfied with the answer? Time taken: 101s, token: 0 tokens connected to Mathkor Gid LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`report_agent`
Tools used	`agent_external_report_tool`
API calls made	`3`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



86. GID

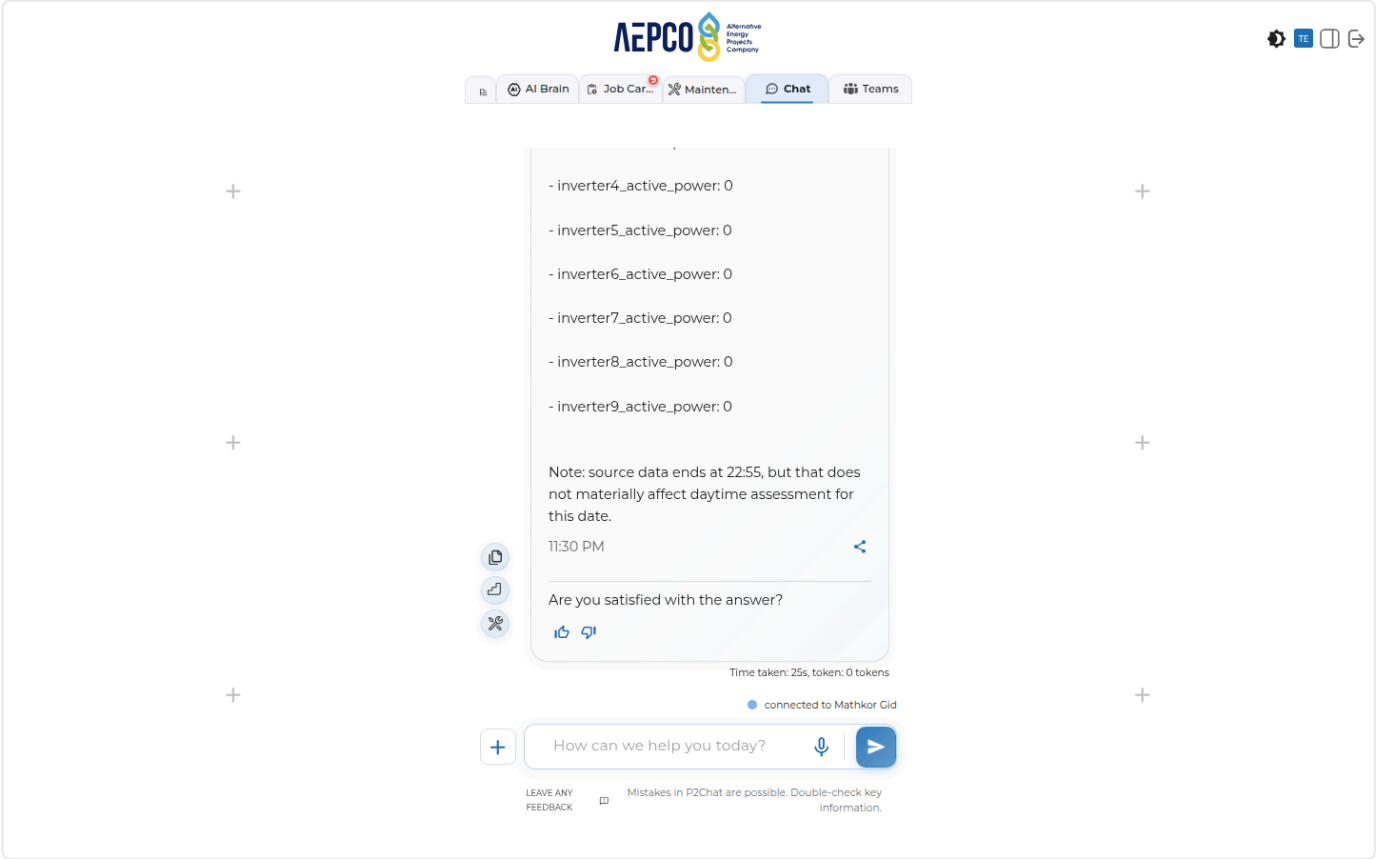
Production • 39.9s

Question: How is my data quality for POA and power? How many of them show missing data during daytime or mismatches with satellite prediction?

Response: For 2026-10-19 daytime only: - Daytime data points assessed: 130 - Missing POA points during daytime: 0 - Missing power points during daytime: 0 - POA mismatches vs modelled/satellite-predicted POA: 5 - Power mismatches between plant active power and summed inverter power: 1 Per-inverter missing daytime power points: - inverter1_active_power: 0 - inverter2_active_power: 0 - inverter3_active_power: 0 - inverter4_active_power: 0 - inverter5_active_power: 0 - inverter6_active_power: 0 - inverter7_active_power: 0 - inverter8_active_power: 0 - inverter9_active_power: 0 Note: source data ends at 22:55, but that does not materially affect daytime assessment for this date. 11:30 PM Are you satisfied with the answer? Time taken: 25s, token: 0 tokens connected to Mathkor Gid LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`multi_tool_use.parallel, functions.execute_custom_code, functions.final_result, functions.query_plant_details_ag, functions.extract_timeseries_data_ag`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



87. GID

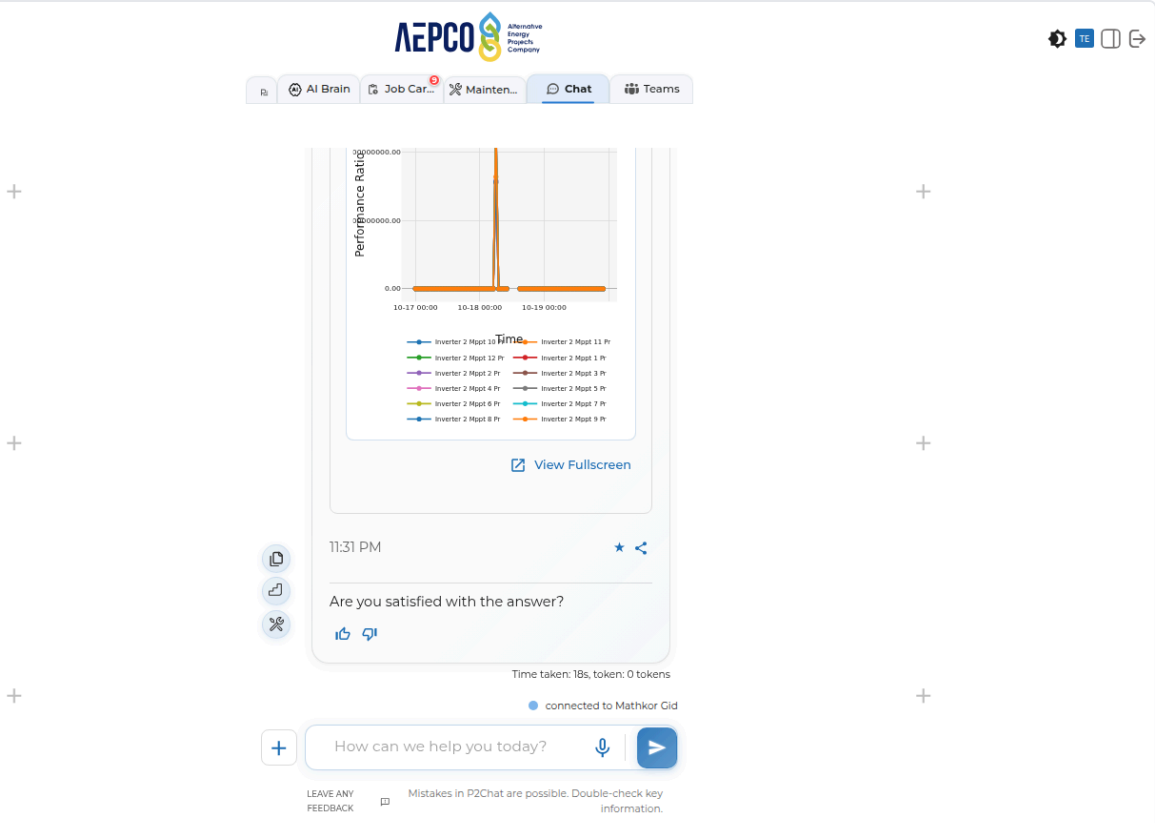
Production • 34.2s

Question: Please plot last 3 days hourly performance ratio for my MPPTs for inverter 2

Response: I've created the visualization for you. View Fullscreen 11:31 PM ★ Are you satisfied with the answer? Time taken: 18s, token: 0 tokens connected to Mathkor Gid LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.generate_graphs_via_code, functions.perf_ratio_ag`
API calls made	2
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	0
Network failures	0



88. GID

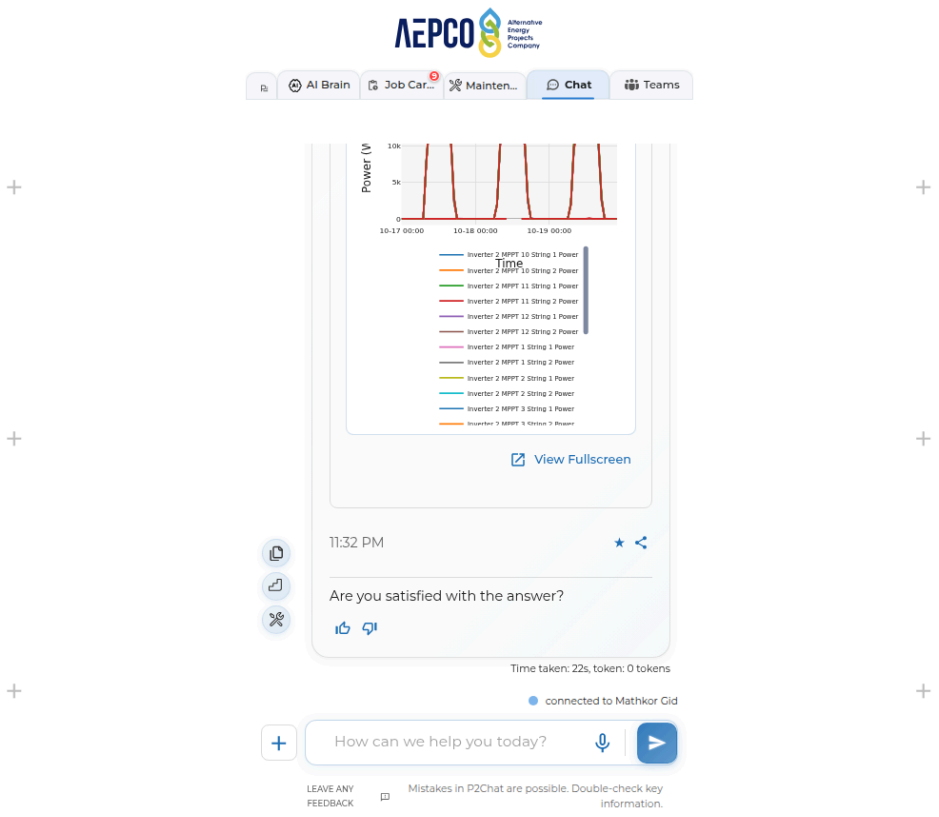
Production • 38.4s

Question: Please plot last 3 days hourly performance ratio for my Strings for inverter 2

Response: I've created the visualization for you. View Fullscreen 11:32 PM ★ Are you satisfied with the answer? Time taken: 22s, token: 0 tokens connected to Mathkor Gid LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.extract_timeseries_data_ag, functions.execute_custom_code, functions.generate_graphs_via_code, functions.perf_ratio_ag`
API calls made	`2`
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	`0`
Network failures	`0`



89. GID

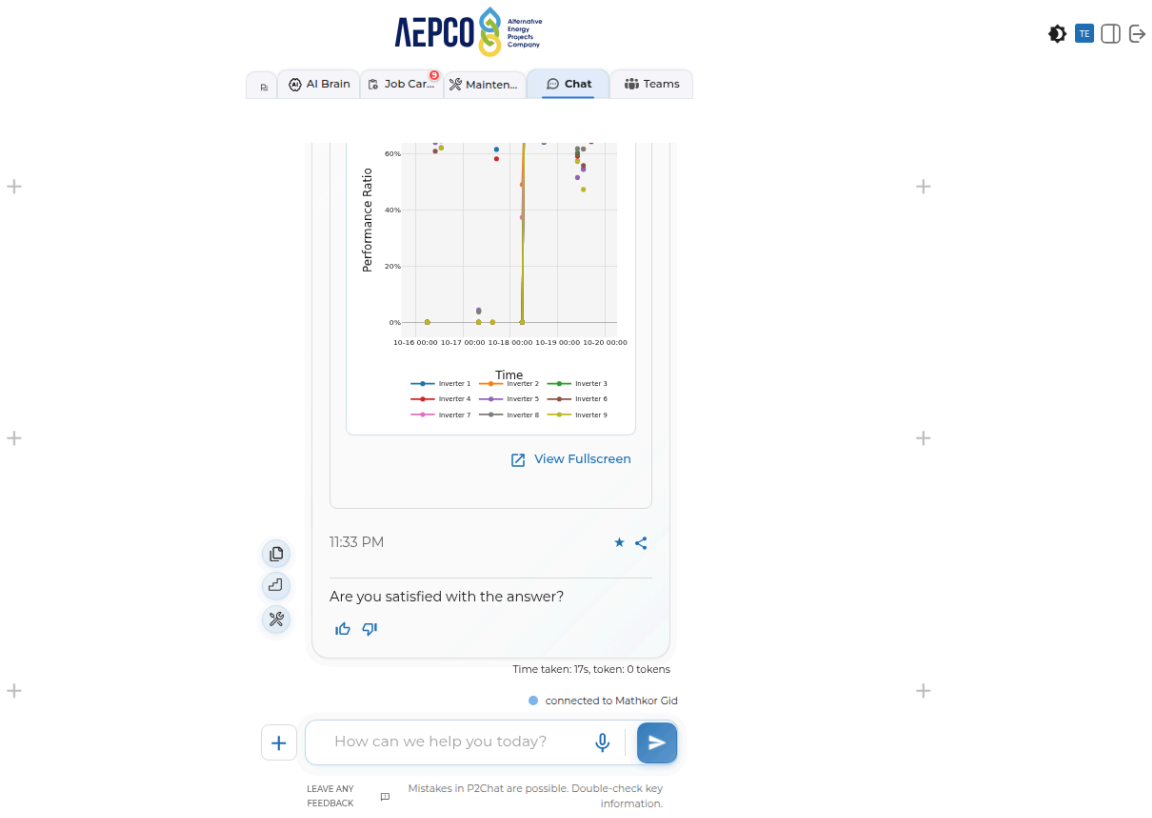
Production • 32.3s

Question: Please plot the last 4 days hourly performance ratio for all my inverters

Response: I've created the visualization for you. View Fullscreen 11:33 PM ★ Are you satisfied with the answer? Time taken: 17s, token: 0 tokens connected to Mathkor Gid LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.generate_graphs_via_code, functions.perf_ratio_ag`
API calls made	`1`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



90. GID

Production • 128.6s

Question: Please figure out how much clipping loss was found in the last 6 months broken by inverters, list them down and plot it as well

Response: Monthly Clipping Loss by Inverter (Last 6 Months) - inverter1: 130,674.33 kWh - inverter2: 130,294.98 kWh - inverter3: 130,420.02 kWh - inverter4: 130,370.76 kWh - inverter5: 153,838.81 kWh - inverter6: 153,834.51 kWh - inverter7: 154,136.24 kWh - inverter8: 154,147.28 kWh - inverter9: 153,848.52 kWh Data Quality Status: Good The visualization has been prepared. Monthly Clipping Loss by Inverter (Last 6 Months) - inverter1: 130,674.33 kWh - inverter2: 130,294.98 kWh - inverter3: 130,420.02 kWh - inverter4: 130,370.76 kWh - inverter5: 153,838.81 kWh - inverter6: 153,834.51 kWh - inverter7: 154,136.24 kWh - inverter8: 154,147.28 kWh - inverter9: 153,848.52 kWh Data Quality Status: Good The visualization has been prepared. View Fullscreen 11:35 PM ★ Are you satisfied with the answer? Time taken: 113s, token: 0 tokens connected to Mathkor Gid LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent, graph_agent`
Tools used	`functions.generate_graphs_via_code, functions.calculate_clipping_or_curtailment_losses_ag, functions.clipping_loss_via_requests`
API calls made	`4`
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	`0`
Network failures	`0`

91. GID

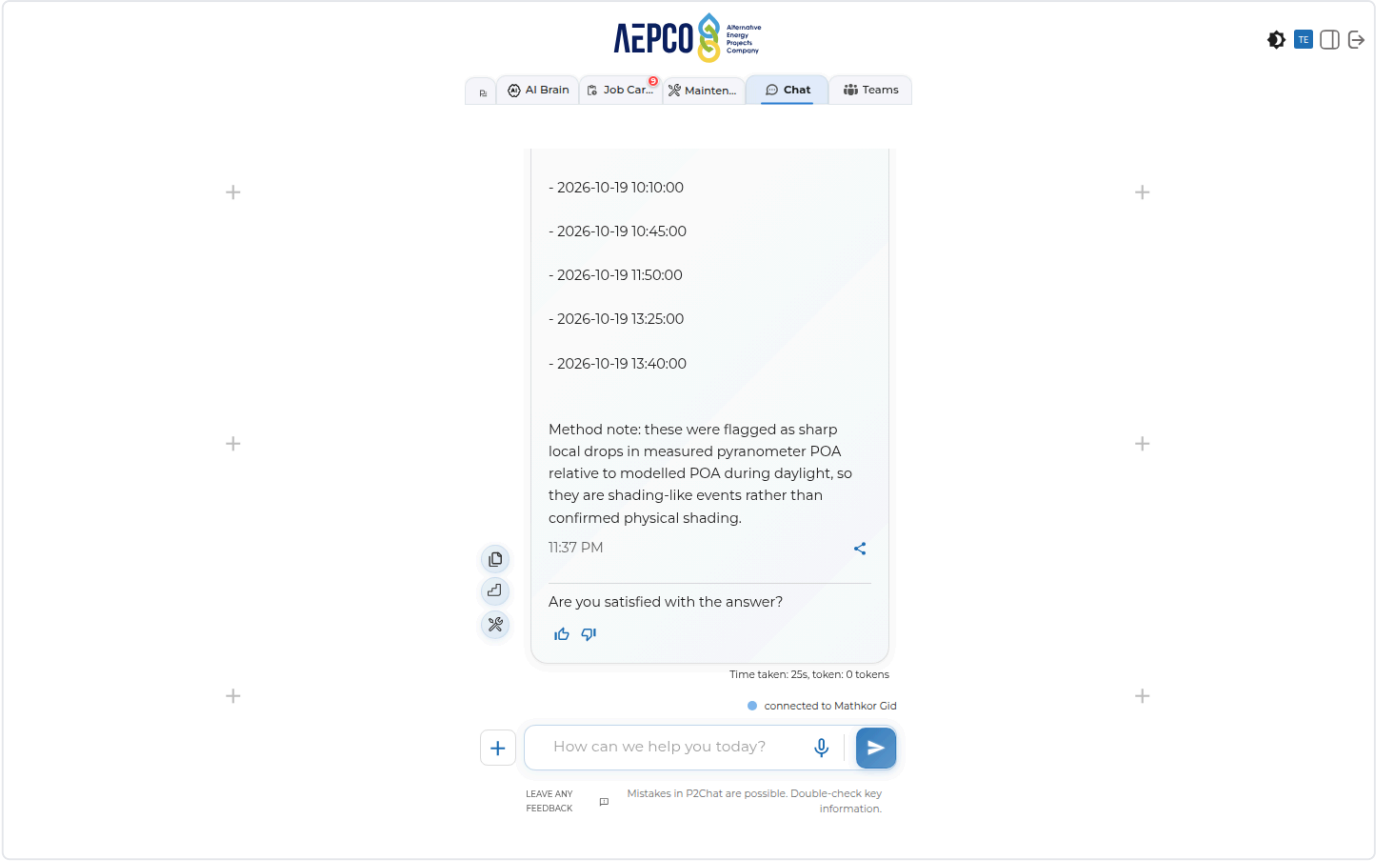
Production • 40.0s

Question: Is there shading happening with my plant? Please analyze the pyranometer and check for frequent dips at certain times

Response: I analyzed the last 8 days of pyranometer data against modelled POA and detected 48 shading-like irradiance dips. Most frequent dip times of day: - 09:10 — 3 times - 07:30 — 2 times - 09:30 — 2 times - 09:55 — 2 times - 10:15 — 2 times - 13:00 — 2 times - 13:55 — 2 times - 15:25 — 2 times - 15:40 — 2 times Detected dip timestamps: - 2026-10-12 09:10:00 - 2026-10-12 10:15:00 - 2026-10-12 12:40:00 - 2026-10-12 13:00:00 - 2026-10-12 13:20:00 - 2026-10-12 13:55:00 - 2026-10-12 14:25:00 - 2026-10-12 14:45:00 - 2026-10-12 15:00:00 - 2026-10-12 15:20:00 - 2026-10-12 15:40:00 - 2026-10-13 07:25:00 - 2026-10-13 09:30:00 - 2026-10-13 09:45:00 - 2026-10-13 10:30:00 - 2026-10-13 11:20:00 - 2026-10-13 11:45:00 - 2026-10-14 09:10:00 - 2026-10-14 10:15:00 - 2026-10-14 11:55:00 - 2026-10-14 13:00:00 - 2026-10-14 13:35:00 - 2026-10-14 13:55:00 - 2026-10-14 15:25:00 - 2026-10-14 15:40:00 - 2026-10-14 16:05:00 - 2026-10-14 16:4

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`functions.extract_timeseries_data_ag, functions.execute_custom_code`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



92. GID

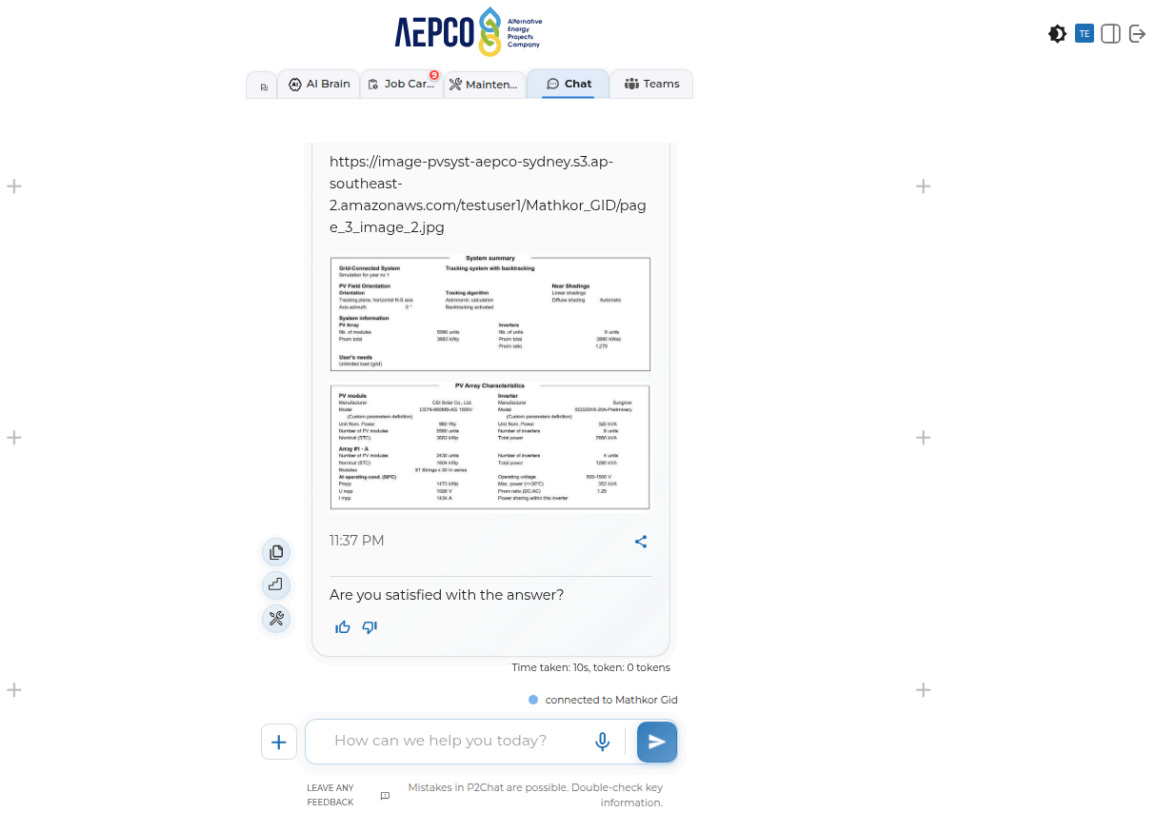
Production • 26.2s

Question: Based on pvsyst report, what is the rated power of this plant and array configuration

Response: Found 2 relevant PVsyst report sections: 1. [Mathkor_GID] System summary (Page 2, Table 2) 2. [Mathkor_GID] PV Array Characteristics (Page 3, Table 2) Please access the URLs provided to view the detailed rated power information. Here is the reference: https://image-pvsyst-aepco-sydney.s3.ap-southeast-2.amazonaws.com/testuser1/Mathkor_GID/page_2_image_2.jpg https://image-pvsyst-aepco-sydney.s3.ap-southeast-2.amazonaws.com/testuser1/Mathkor_GID/page_3_image_2.jpg 11:37 PM Are you satisfied with the answer? Time taken: 10s, token: 0 tokens connected to Mathkor Gid LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`pvsyst_agent`
Tools used	`pvsyst_tool`
API calls made	`4`
Last API call	`GET /testuser2/chat/ → 404`
Console errors	`3`
Network failures	`1`



93. GID

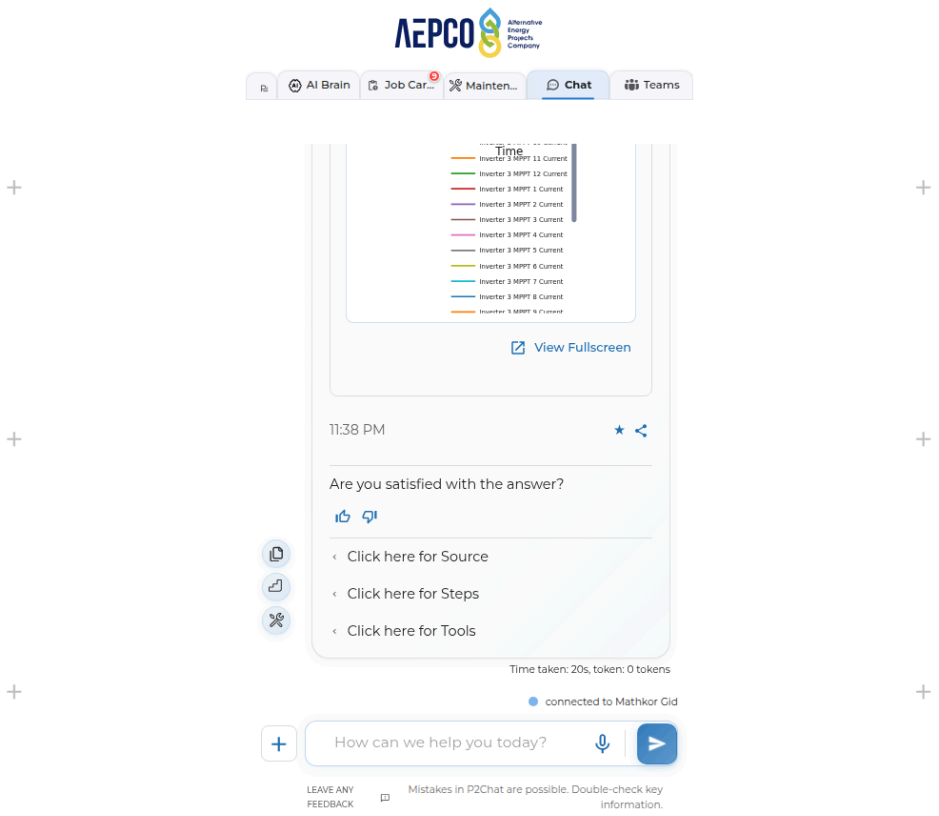
Production • 35.7s

Question: Please plot my current and voltage in a dual axes plot on the deepest possible granularity level for inverter3

Response: I've created the visualization for you. View Fullscreen 11:38 PM ★ Are you satisfied with the answer? Click here for Source Click here for Steps Click here for Tools Time taken: 20s, token: 0 tokens connected to Mathkor Gid LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.execute_custom_code, functions.extract_timeseries_data_ag, functions.generate_graphs_via_code`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`1`
Network failures	`0`



94. GID

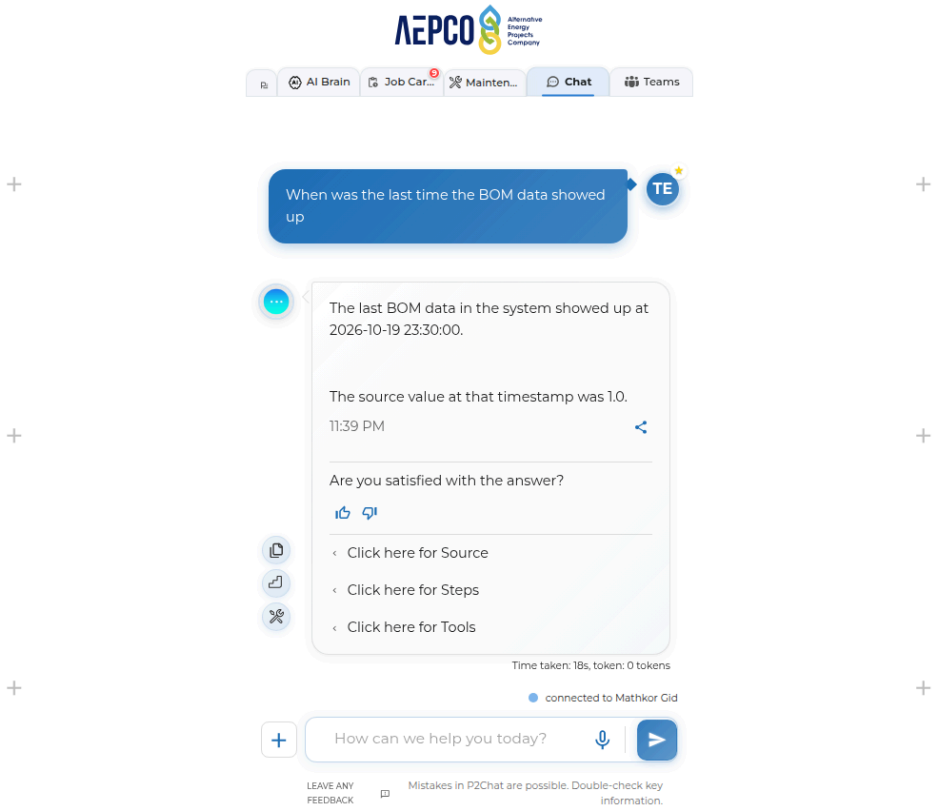
Production • 33.7s

Question: When was the last time the BOM data showed up

Response: The last BOM data in the system showed up at 2026-10-19 23:30:00. The source value at that timestamp was 1.0. 11:39 PM Are you satisfied with the answer? Click here for Source Click here for Steps Click here for Tools Time taken: 18s, token: 0 tokens connected to Mathkor Gid LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`execute_custom_code, extract_timeseries_data_ag`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



95. GID

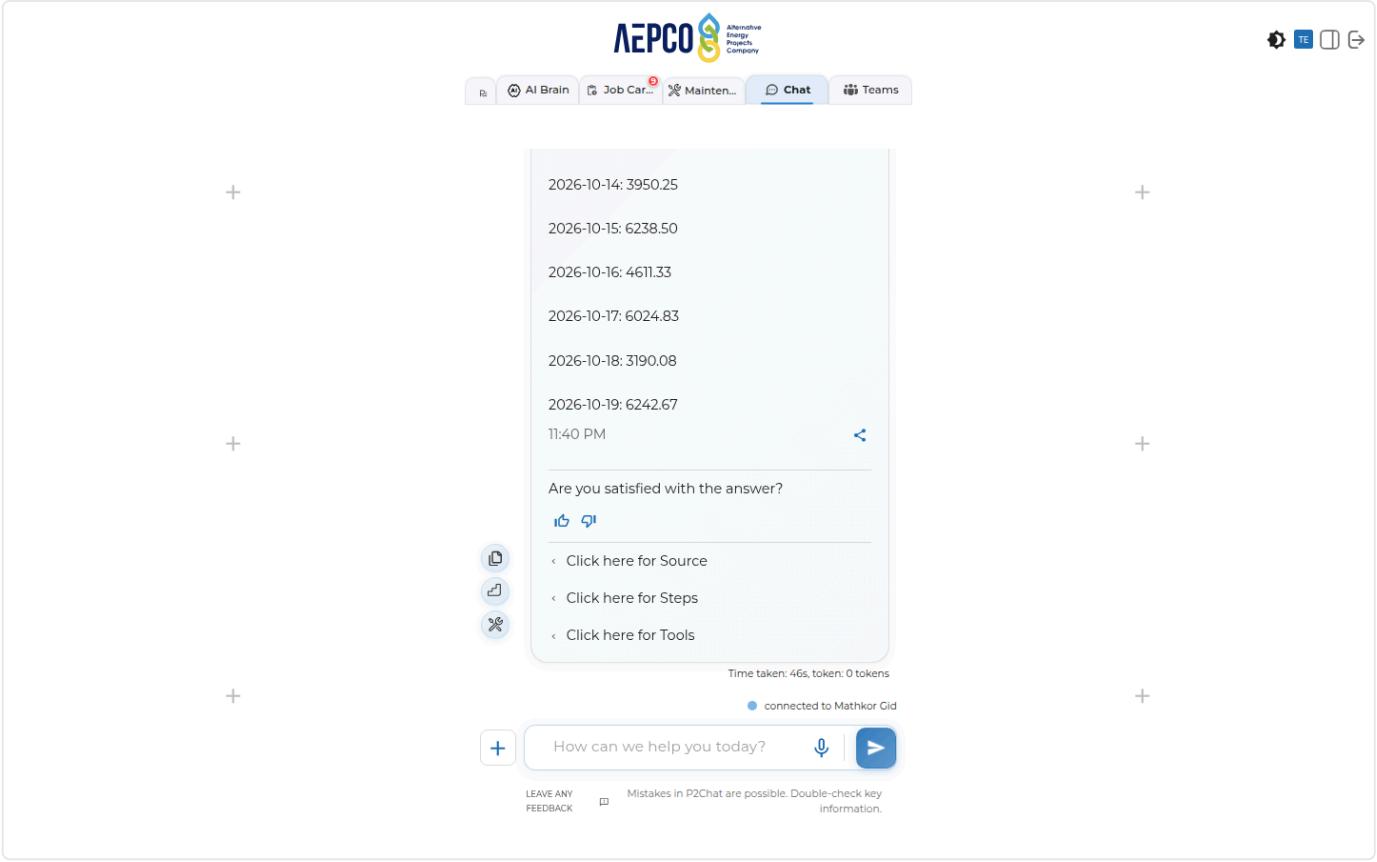
Production • 61.1s

Question: Please find out the daily POA poa summed up as a Wh/m^2, daily for the last 12 months and then calculate the median, mean and kurt

Response: Daily summed POA irradiance for the last 12 months was calculated from `pyranometer_1` by converting each 5-minute POA value from W/m² to Wh/m² using × 5/60, then summing by day. Summary statistics: - Median daily POA: 7277.00 Wh/m² - Mean daily POA: 6934.54 Wh/m² - Kurtosis: -0.9771 Notes: - The extracted data runs from 2025-10-19 to 2026-10-19. - Latest available timestamp is 2026-10-19 23:30:00, slightly short of the requested end-of-day refresh. - One daily value is NaN: 2025-12-12. Daily summed POA irradiance (Wh/m²): 2025-10-19: 6524.17 2025-10-20: 6575.17 2025-10-21: 6271.92 2025-10-22: 6186.58 2025-10-23: 6054.08 2025-10-24: 5254.33 2025-10-25: 6428.67 2025-10-26: 6299.33 2025-10-27: 3025.50 2025-10-28: 5656.25 2025-10-29: 5747.33 2025-10-30: 5582.83 2025-10-31: 5411.08 2025-11-01: 5513.33 2025-11-02: 4835.33 2025-11-03: 5081.83 2025-11-04: 4107.58 2025-11-05: 4937.67 2025-11-06: 5212.42 2025-11-07: 4733.25 2025-11-

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`execute_custom_code, extract_timeseries_data_ag`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



96. GID

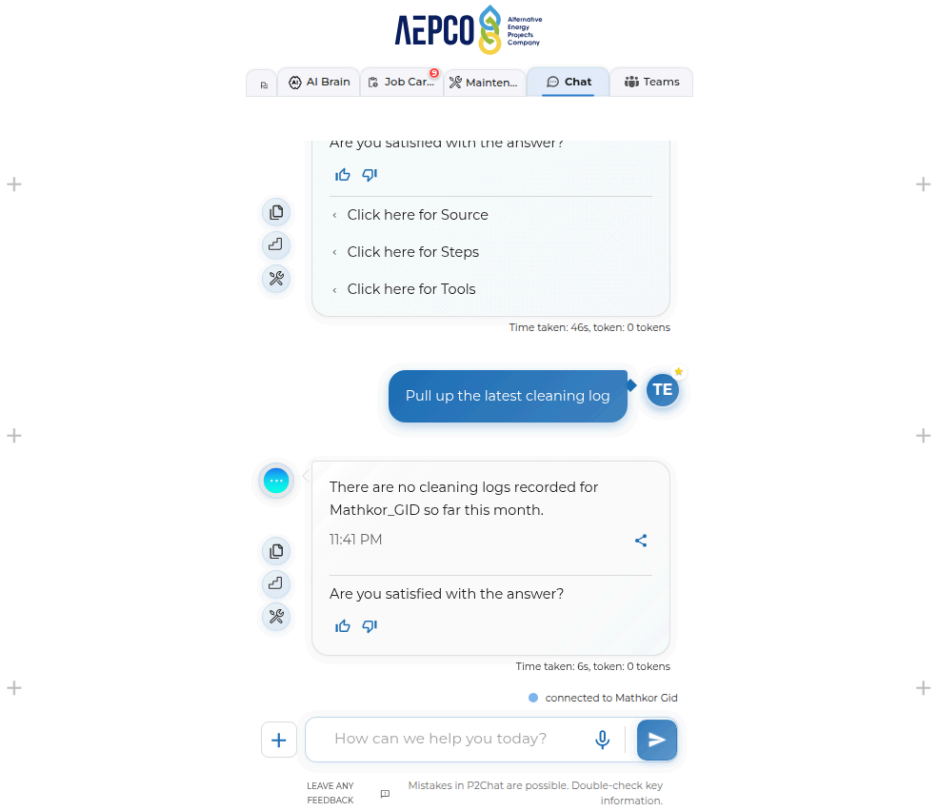
Production • 21.6s

Question: Pull up the latest cleaning log

Response: There are no cleaning logs recorded for Mathkor_GID so far this month. 11:41 PM Are you satisfied with the answer? Time taken: 6s, token: 0 tokens connected to Mathkor Gid LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_entry_agent`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



97. GID

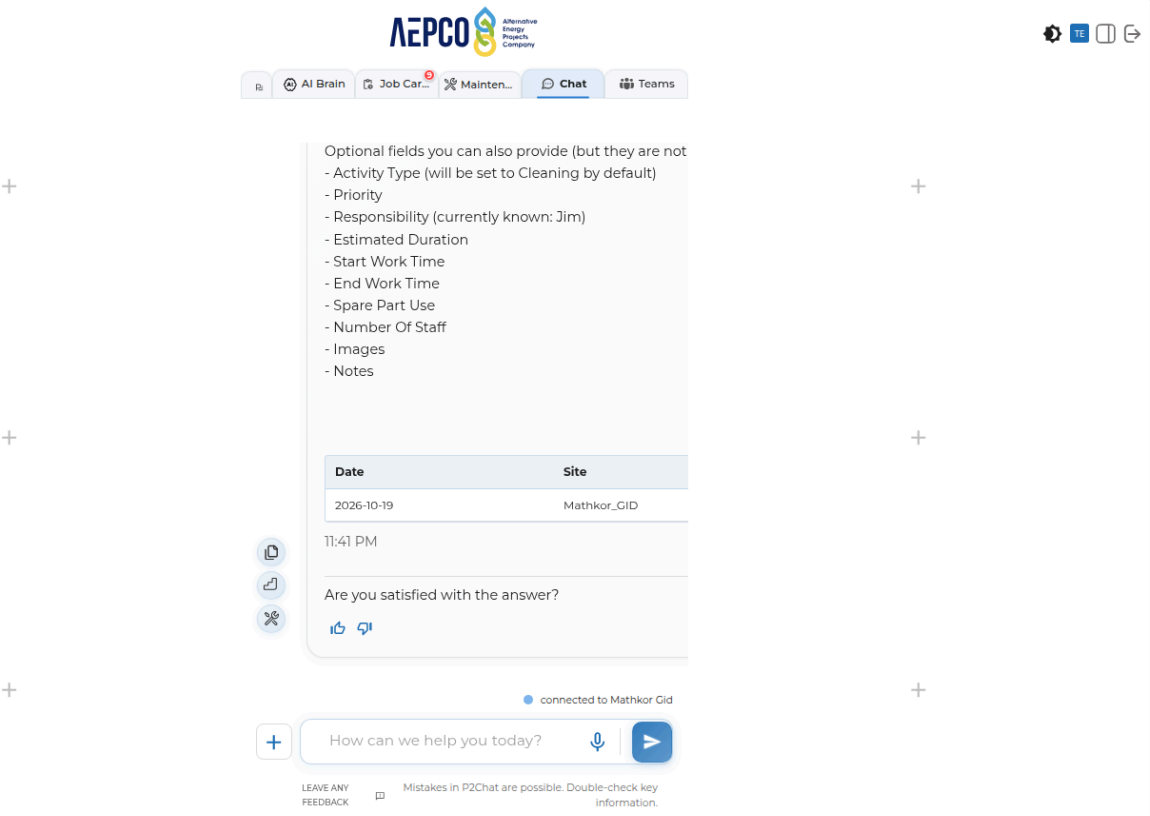
Production • 21.6s

Question: Jim cleaned the pyranometer today, can you add it to the cleaning log?

Response: To add a cleaning activity log, please provide the following required details: - Date (today is 2026-10-19; should I use this date?) - Fault Cause - Action Taken - Status Site will be set automatically as Mathkor_GID. Optional fields you can also provide (but they are not required): - Activity Type (will be set to Cleaning by default) - Priority - Responsibility (currently known: Jim) - Estimated Duration - Start Work Time - End Work Time - Spare Part Use - Number Of Staff - Images - Notes Edit Date Site Responsibility 2026-10-19 Mathkor_GID Jim 11:41 PM Are you satisfied with the answer? Time taken: 6s, token: 0 tokens connected to Mathkor Gid LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_entry_agent`
API calls made	`2`
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	`0`
Network failures	`0`



98. GID

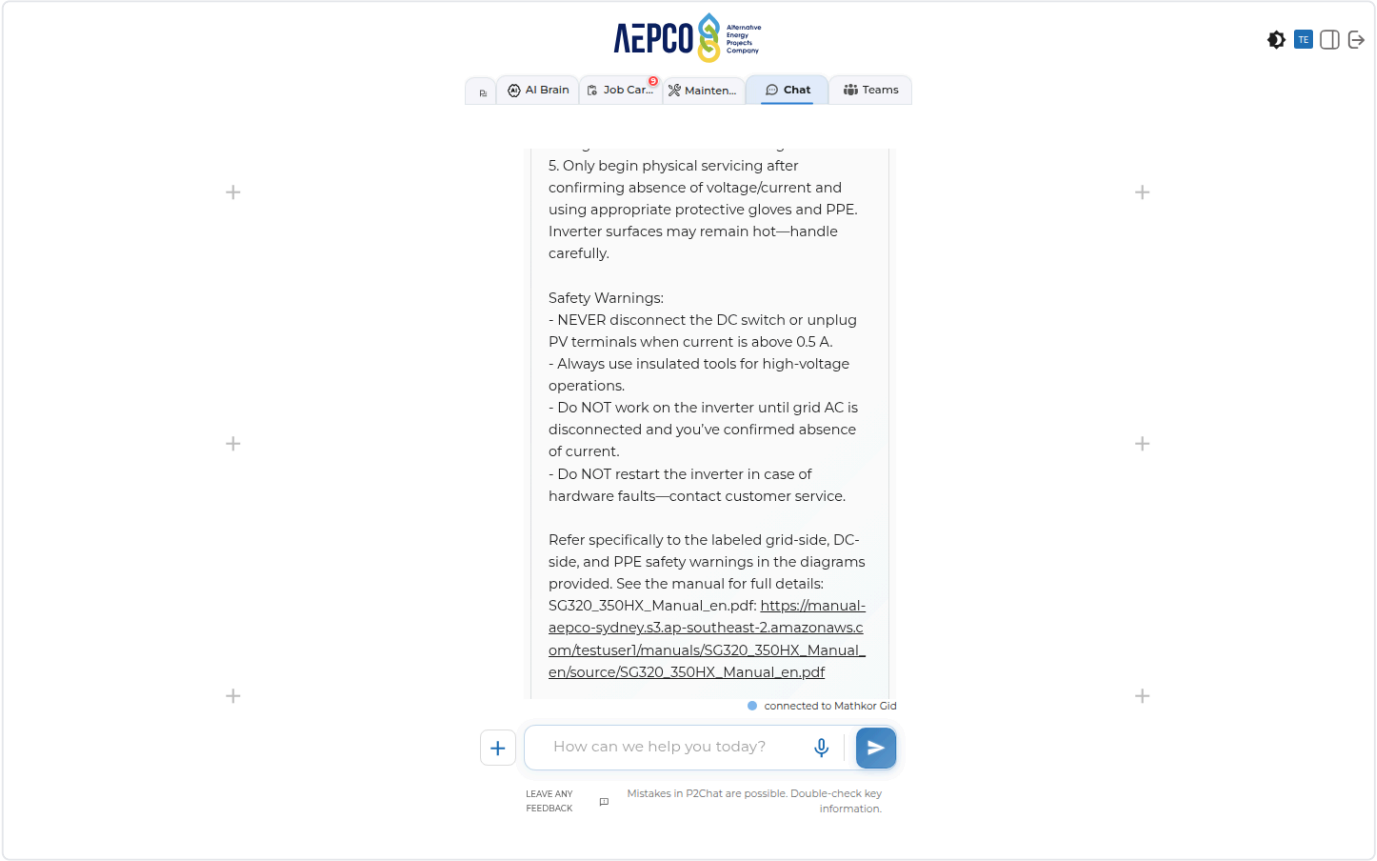
Production • 35.3s

Question: How can I shut down my inverters?

Response: To safely shut down your inverters, follow these steps exactly as outlined in the manufacturer manual: 1. Disconnect the grid-side AC circuit breaker before performing any service. This breaker is labeled as the grid-side input in your system schematic. See the system flow diagram to identify this connection: - System connection diagram: 2. Check the indicator on the inverter: - If the indicator is OFF: Wait until nighttime (when PV modules aren't generating current) before proceeding. - If the indicator is ON: You may proceed directly to the next step. 3. When it is safe (after confirming the current is below 0.5 A), disconnect the DC switch. This switch is typically between the PV array and inverter; refer to this diagram for the connection location: - PV to inverter connection: 4. Wait at least 25 minutes for all internal components to discharge after shutoff. Use a certified instrument to verify there is NO voltage or current before servicing. 5. Only begin physica

TECHNICAL EVIDENCE

Agents used	`manual_agent`
Tools used	`functions.manual_rag_tool`
API calls made	`1`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



99. GID

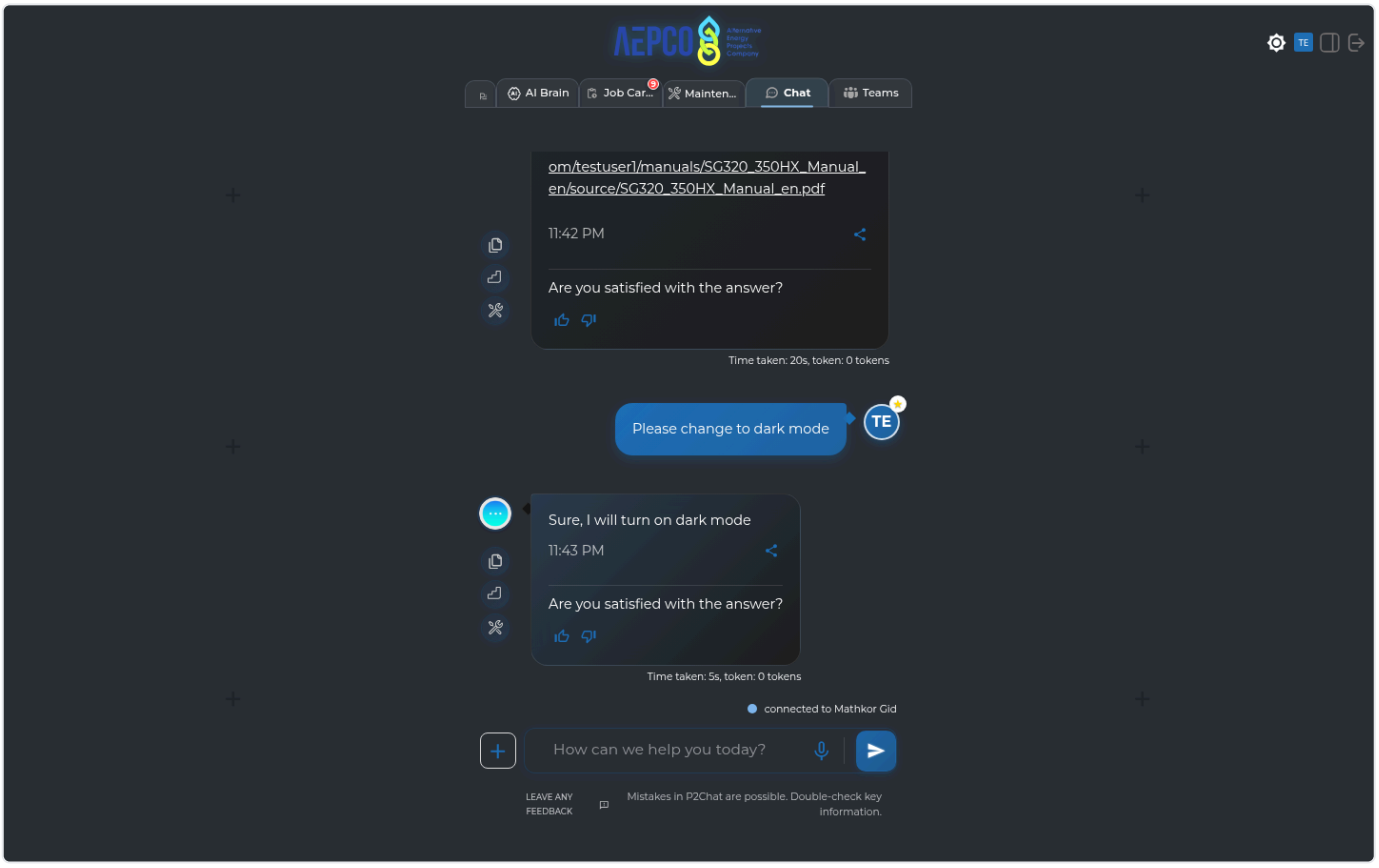
Production • 21.1s

Question: Please change to dark mode

Response: Sure, I will turn on dark mode 11:43 PM Are you satisfied with the answer? Time taken: 5s, token: 0 tokens connected to Mathkor Gid LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`ui_agent`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



100. GID

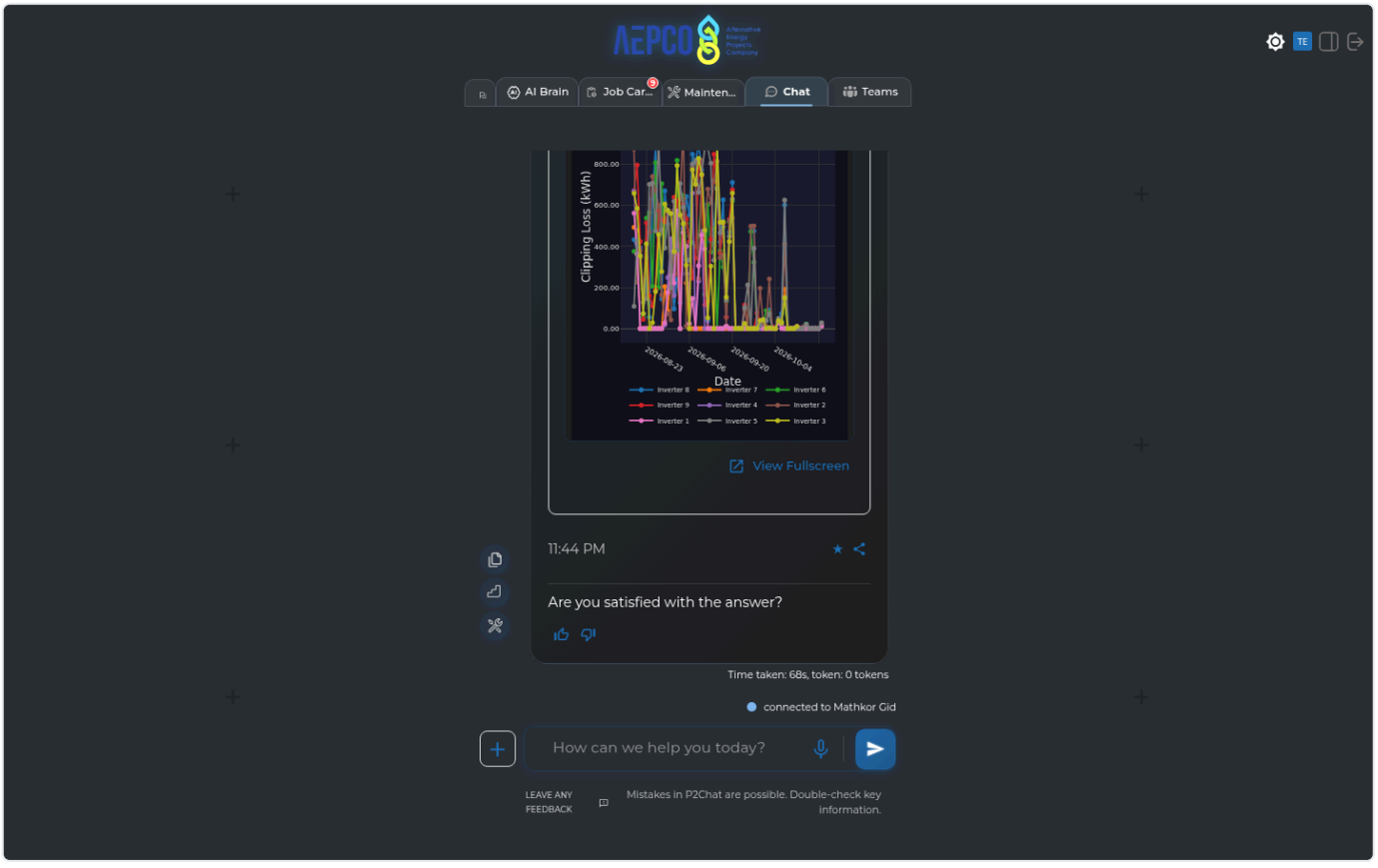
Production • 82.8s

Question: Please figure out how much clipping loss was found in the last 2 months broken by inverters, list them down and plot it as well

Response: Here are the total clipping losses by inverter for the last two months (2026-08-19 to 2026-10-19): - Inverter 1: 27,083.78 kWh - Inverter 2: 27,086.98 kWh - Inverter 3: 27,086.98 kWh - Inverter 4: 26,824.47 kWh - Inverter 5: 33,663.96 kWh - Inverter 6: 33,665.38 kWh - Inverter 7: 33,663.96 kWh - Inverter 8: 33,663.96 kWh - Inverter 9: 33,668.36 kWh Data quality status: Good I've created a line chart showing daily clipping loss for each inverter over the requested period. Here are the total clipping losses by inverter for the last two months (2026-08-19 to 2026-10-19): - Inverter 1: 27,083.78 kWh - Inverter 2: 27,086.98 kWh - Inverter 3: 27,086.98 kWh - Inverter 4: 26,824.47 kWh - Inverter 5: 33,663.96 kWh - Inverter 6: 33,665.38 kWh - Inverter 7: 33,663.96 kWh - Inverter 8: 33,663.96 kWh - Inverter 9: 33,668.36 kWh Data quality status: Good I've created a line chart showing daily clipping loss for each inverter over the requested period

TECHNICAL EVIDENCE

Agents used	`data_agent, graph_agent`
Tools used	`functions.calculate_clipping_or_curtailment_losses_ag, functions.generate_graphs_via_code, functions.clipping_loss_via_requests`
API calls made	`3`
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	`0`
Network failures	`0`



101. GID

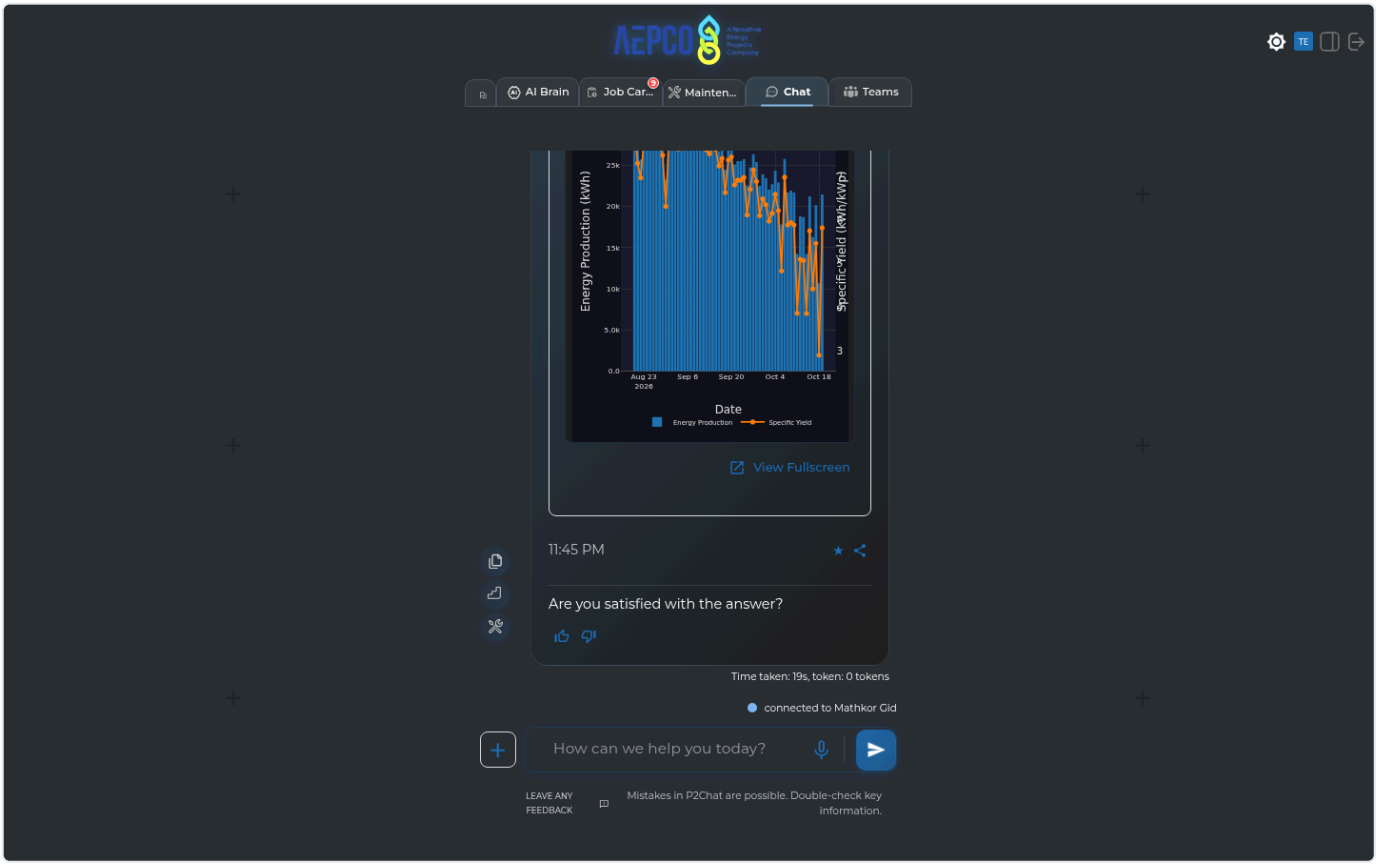
Production • 33.9s

Question: Please repeat the first plotting question, but interchange it with Specific yield on 1 axes, and energy production on the other axes on a single dual axes plot (Memory Test)

Response: I've created the visualization for you. View Fullscreen 11:45 PM ★ Are you satisfied with the answer? Time taken: 19s, token: 0 tokens connected to Mathkor Gid LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.generate_graphs_via_code, multi_tool_use.parallel`
API calls made	`1`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



102. Abdali_Mall_Netmetering

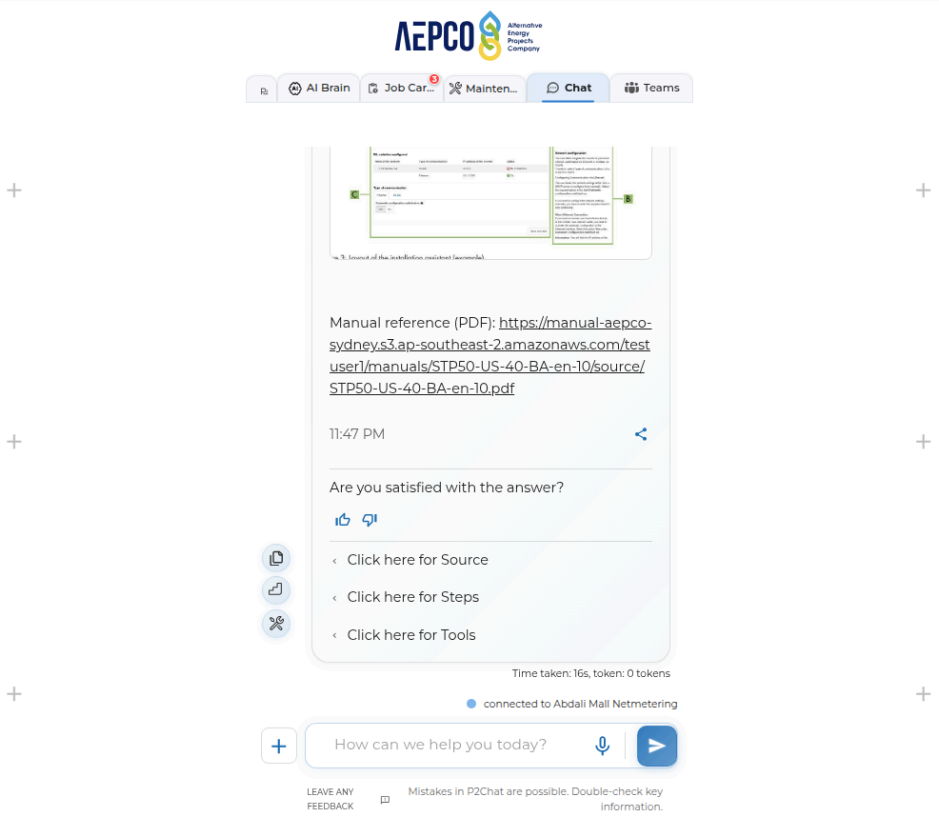
Production • 30.8s

Question: I have a COMMUNICATION_FAULT what do I do to my inverters?

Response: To troubleshoot a COMMUNICATION_FAULT on your inverters, follow these detailed steps using the manual: 1. Open the device user interface and go to the network configuration overview. See the network configuration diagram here: 2. Check the status indicators for both WLAN and Ethernet as shown in the diagram. If either indicates "No connection," do the following: - For WLAN issues, ensure the device is connecting to the correct Wi-Fi network and the WLAN settings match those needed. Confirm the configured IP address or set it to automatic configuration, following the overview in the diagram. - For Ethernet issues, ensure the Ethernet cable is securely and properly connected to both the inverter and the network switch or router. Confirm the assigned IP address matches your network. Try bypassing network devices by connecting directly to the router/switch as shown in the diagram. 3. Use the installation assistant if necessary to redo or adjust network settings—the interface for t

TECHNICAL EVIDENCE

Agents used	`manual_agent`
Tools used	`functions.manual_rag_tool`
API calls made	`11`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



103. Abdali_Mall_Netmetering

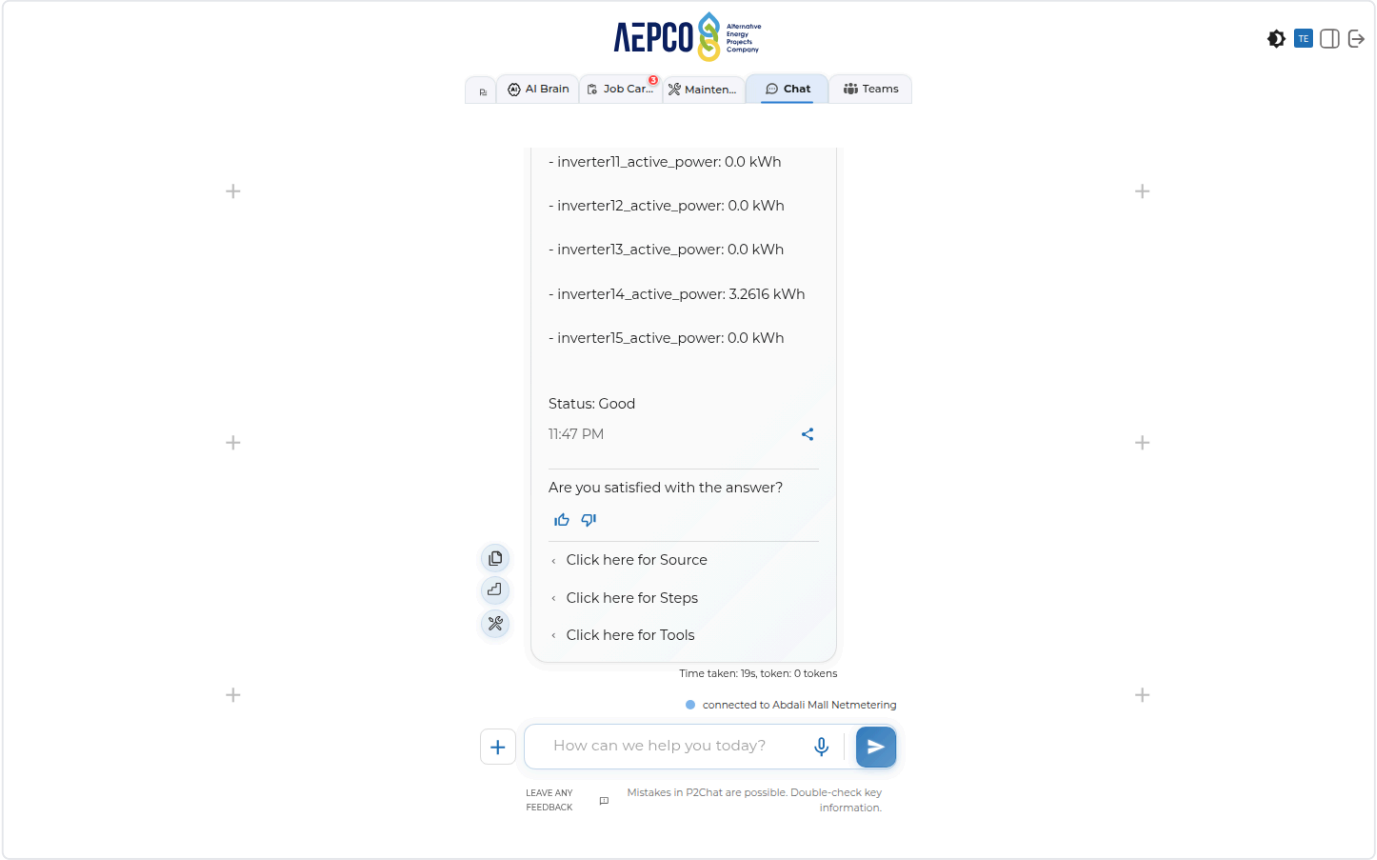
Production • 33.7s

Question: Use 'clipping_loss_via_requests' in order to calculate clipping loss if any for July 2026

Response: Clipping loss for July 2026: - inverter1_active_power: 218.943 kWh - inverter2_active_power: 131.9139 kWh - inverter3_active_power: 188.2499 kWh - inverter4_active_power: 318.8087 kWh - inverter5_active_power: 467.7012 kWh - inverter6_active_power: 0.0 kWh - inverter7_active_power: 0.0 kWh - inverter8_active_power: 13.2321 kWh - inverter9_active_power: 0.0 kWh - inverter10_active_power: 0.0 kWh - inverter11_active_power: 0.0 kWh - inverter12_active_power: 0.0 kWh - inverter13_active_power: 0.0 kWh - inverter14_active_power: 3.2616 kWh - inverter15_active_power: 0.0 kWh Status: Good 11:47 PM Are you satisfied with the answer? Click here for Source Click here for Steps Click here for Tools Time taken: 19s, token: 0 tokens connected to Abdali Mall Netmetering LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`functions.clipping_loss_via_requests`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



104. Abdali_Mall_Netmetering

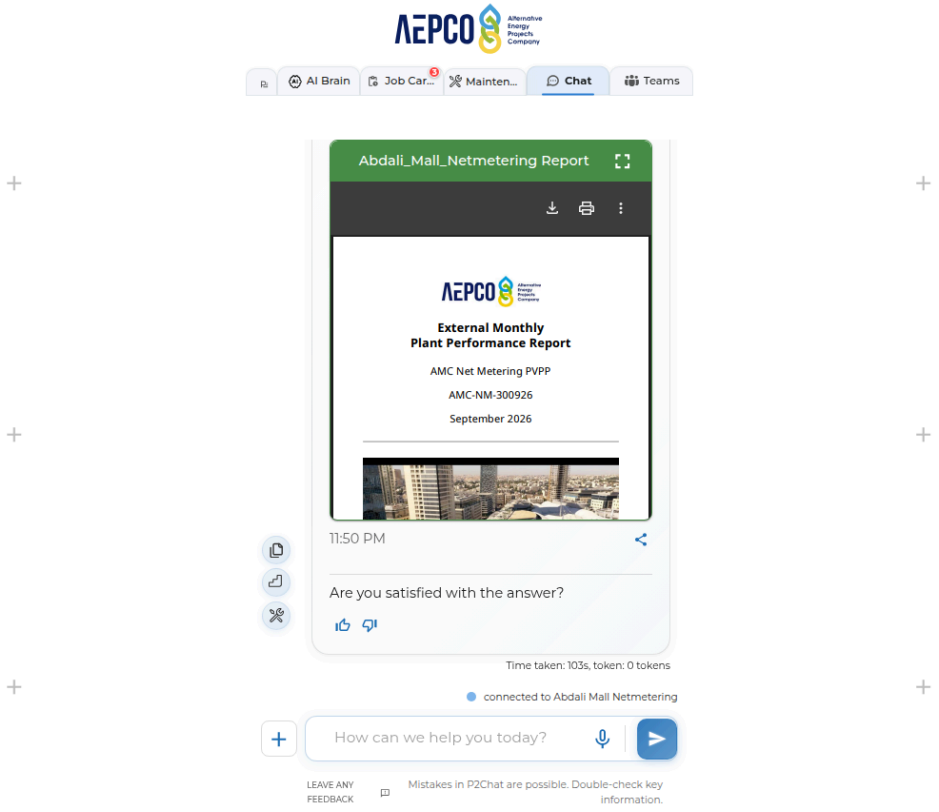
Production • 119.1s

Question: Please generate last months report

Response: I've generated the external plant performance report for September 2026. You can view it here: 'https://report-aepco-sydney.s3.ap-southeast-2.amazonaws.com/ABDM-L2_Project_September_2026_external_report.pdf' Abdali_Mall_Netmetering Report 11:50 PM Are you satisfied with the answer? Time taken: 103s, token: 0 tokens connected to Abdali Mall Netmetering LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`report_agent`
Tools used	`agent_external_report_tool`
API calls made	`3`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



105. Abdali_Mall_Netmetering

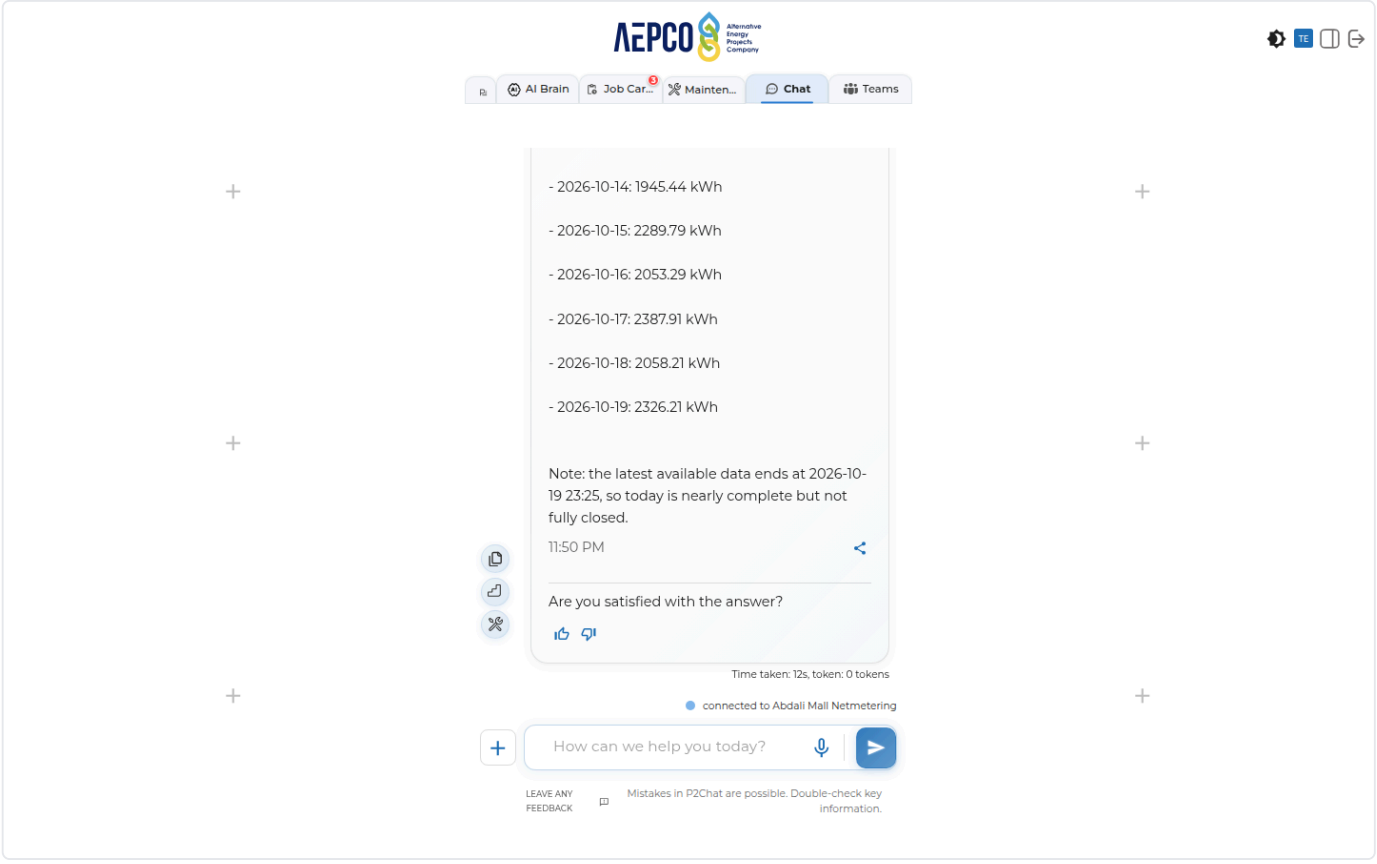
Production • 27.7s

Question: Was there any 0 production days in the last 7 days?

Response: No. There were no 0-production days in the last 7 days. Daily plant energy: - 2026-10-13: 2270.19 kWh - 2026-10-14: 1945.44 kWh - 2026-10-15: 2289.79 kWh - 2026-10-16: 2053.29 kWh - 2026-10-17: 2387.91 kWh - 2026-10-18: 2058.21 kWh - 2026-10-19: 2326.21 kWh Note: the latest available data ends at 2026-10-19 23:25, so today is nearly complete but not fully closed. 11:50 PM Are you satisfied with the answer? Time taken: 12s, token: 0 tokens connected to Abdali Mall Netmetering LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`functions.total_energy_func_ag`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



106. Abdali_Mall_Netmetering

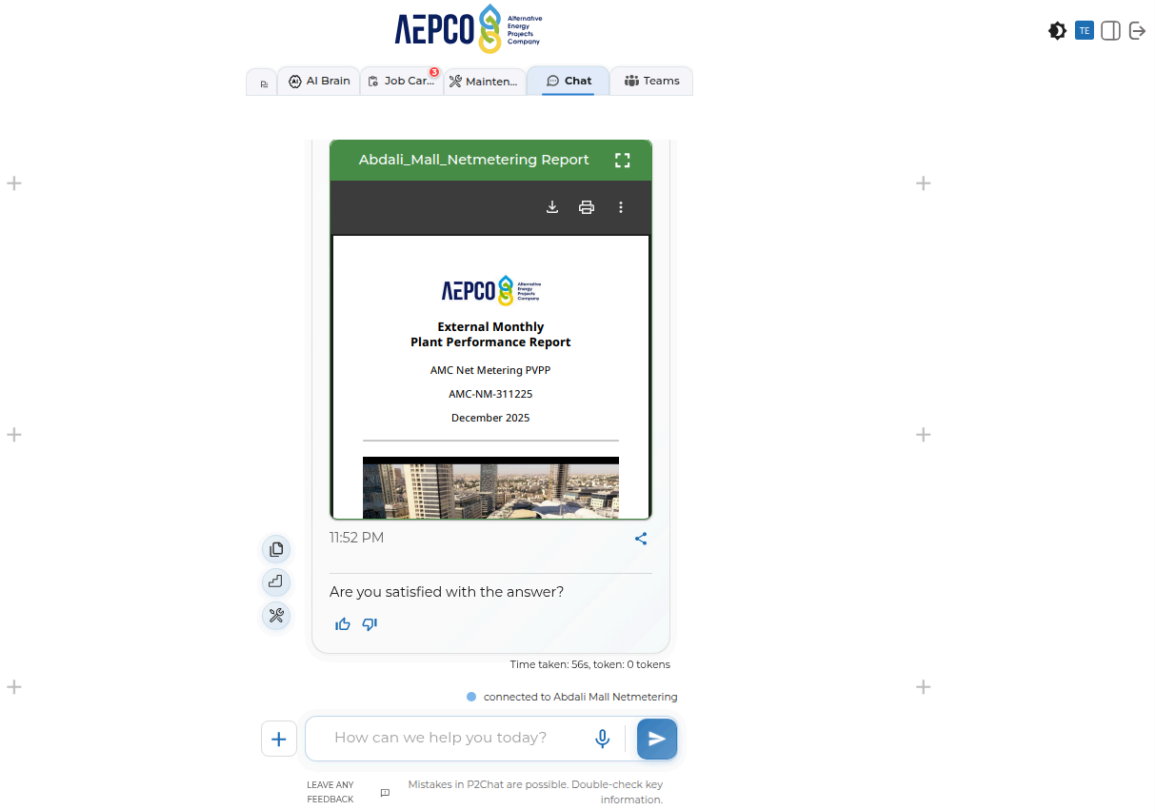
Production • 70.4s

Question: Please generate December 2025 performance report

Response: I've generated the external performance report for December 2025. You can view it here: 'https://report-aepco-sydney.s3.ap-southeast-2.amazonaws.com/ABDM-L2_Project_December_2025_external_report.pdf' Abdali_Mall_Netmetering Report 11:52 PM Are you satisfied with the answer? Time taken: 56s, token: 0 tokens connected to Abdali Mall Netmetering LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`report_agent`
Tools used	`agent_external_report_tool`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



107. Abdali_Mall_Netmetering

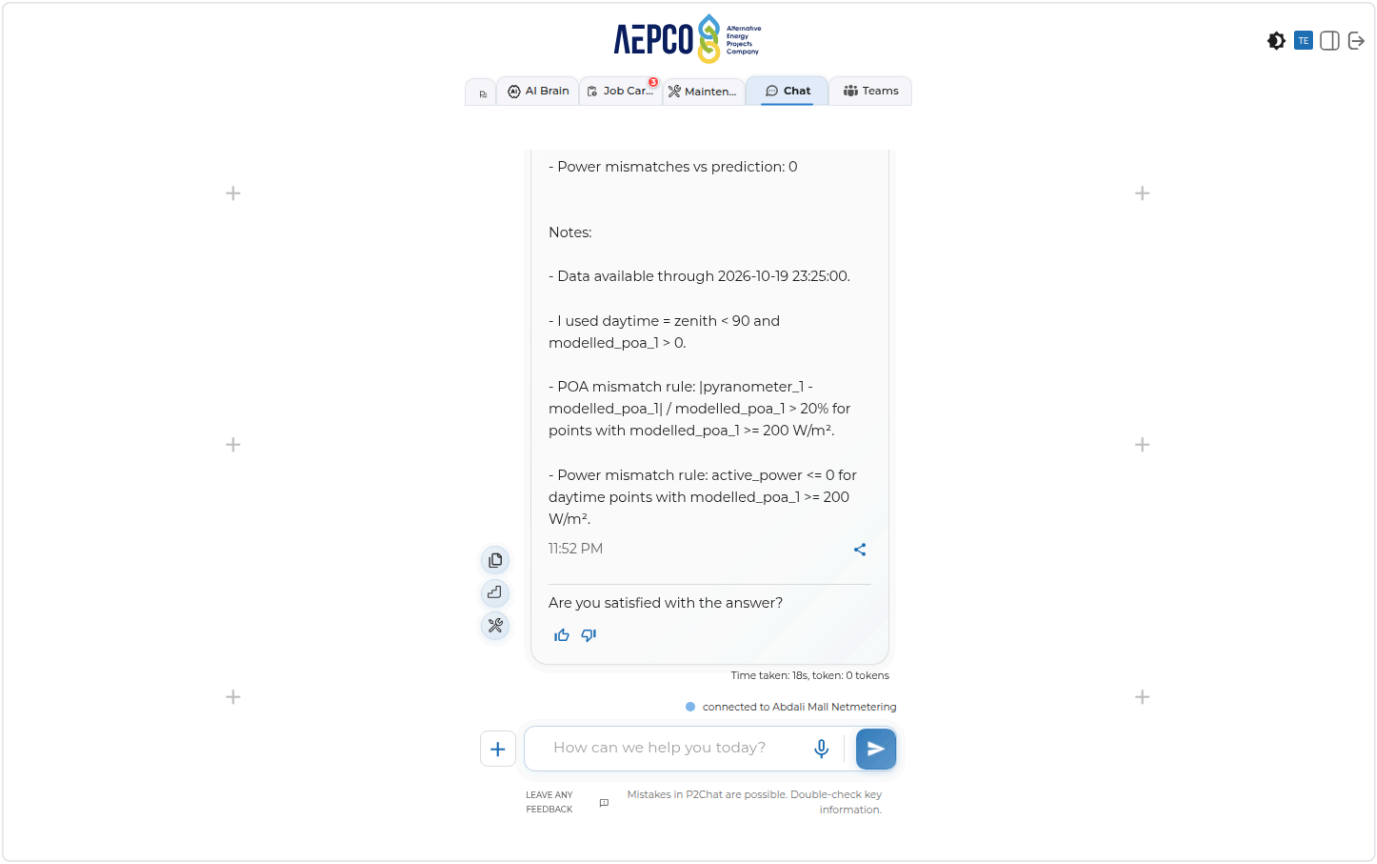
Production • 32.3s

Question: How is my data quality for POA and power? How many of them show missing data during daytime or mismatches with satellite prediction?

Response: For 2026-10-19: - Daytime points assessed: 132 - Missing POA daytime points: 0 - Missing power daytime points: 0 - POA mismatches vs prediction: 61 - Power mismatches vs prediction: 0 Notes: - Data available through 2026-10-19 23:25:00. - I used daytime = zenith < 90 and modelled_poa_1 > 0. - POA mismatch rule: |pyranometer_1 - modelled_poa_1| / modelled_poa_1 > 20% for points with modelled_poa_1 >= 200 W/m². - Power mismatch rule: active_power <= 0 for daytime points with modelled_poa_1 >= 200 W/m². 11:52 PM Are you satisfied with the answer? Time taken: 18s, token: 0 tokens connected to Abdali Mall Netmetering LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`functions.execute_custom_code, functions.extract_timeseries_data_ag`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



108. Abdali_Mall_Netmetering

Production • 30.7s

Question: Please plot last 3 days hourly performance ratio for my MPPTs for inverter 2

Response: I've created the visualization for you. View Fullscreen 11:53 PM ★ Are you satisfied with the answer? Time taken: 16s, token: 0 tokens connected to Abdali Mall Netmetering LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.generate_graphs_via_code, functions.perf_ratio_ag`
API calls made	2
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	0
Network failures	0

109. Abdali_Mall_Netmetering

Production • 32.3s

Question: Please plot last 3 days hourly performance ratio for my Strings for inverter 2

Response: I've created the visualization for you. View Fullscreen 11:54 PM ★ Are you satisfied with the answer? Time taken: 17s, token: 0 tokens connected to Abdali Mall Netmetering LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.generate_graphs_via_code, functions.perf_ratio_ag`
API calls made	2
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	0
Network failures	0

110. Abdali_Mall_Netmetering

Production • 32.3s

Question: Please plot the last 4 days hourly performance ratio for all my inverters

Response: I've created the visualization for you. View Fullscreen 11:55 PM ★ Are you satisfied with the answer? Time taken: 17s, token: 0 tokens connected to Abdali Mall Netmetering LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.generate_graphs_via_code, functions.perf_ratio_ag`
API calls made	`1`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`

111. Abdali_Mall_Netmetering

Production • 84.8s

Question: Please figure out how much clipping loss was found in the last 6 months broken by inverters, list them down and plot it as well

Response: Monthly Clipping Loss per Inverter (Last 6 Months) Clipping losses (2026-04-19 to 2026-10-19), by inverter: - inverter1: 859.5414 kWh - inverter2: 563.7428 kWh - inverter3: 755.2135 kWh - inverter4: 1205.8126 kWh - inverter5: 1741.4402 kWh - inverter6: 0.0410 kWh - inverter7: 0.0000 kWh - inverter8: 87.6991 kWh - inverter9: 0.0000 kWh - inverter10: 0.9826 kWh - inverter11: 0.0000 kWh - inverter12: 0.0000 kWh - inverter13: 0.0000 kWh - inverter14: 50.5393 kWh - inverter15: 2.8942 kWh Note: Values may be understated for part of the period due to API rate-limit errors (HTTP 429) during data retrieval. The visualization has been prepared. Monthly Clipping Loss per Inverter (Last 6 Months) Clipping losses (2026-04-19 to 2026-10-19), by inverter: - inverter1: 859.5414 kWh - inverter2: 563.7428 kWh - inverter3: 755.2135 kWh - inverter4: 1205.8126 kWh - inverter5: 1741.4402 kWh - inverter6: 0.0410 kWh - inverter7: 0.0000 kWh - inverter8

TECHNICAL EVIDENCE

Agents used	`data_agent, graph_agent`
Tools used	`functions.clipping_loss_via_requests, functions.calculate_clipping_or_curtailment_losses_ag, functions.generate_graphs_via_code`
API calls made	`3`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`

112. Abdali_Mall_Netmetering

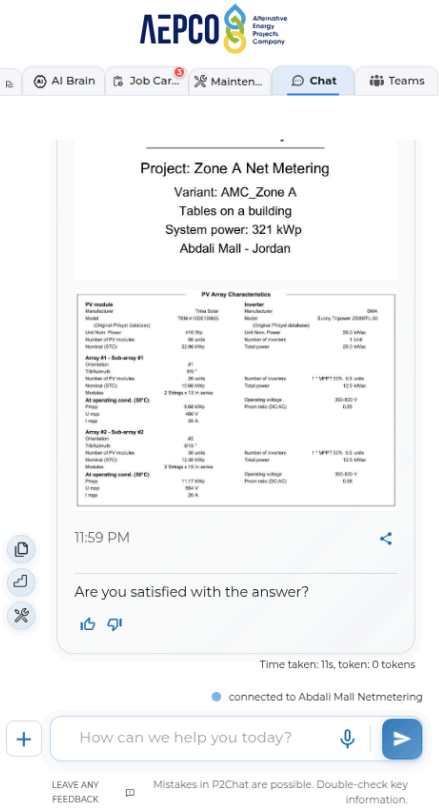
Production • 26.2s

Question: Based on pvsyst report, what is the rated power of this plant and array configuration

Response: Found 2 relevant PVsyst report sections: 1. [AMC_Zone_A] Front Page - Rated Power of Plant (Page: 1, Table: 1) 2. [AMC_Zone_F&G] PV Array Characteristics - Array Configuration (Page: 3, Table: 2) Please access the URLs provided to view the detailed information. Here is the reference: https://image-pvsyst-aepco-sydney.s3.ap-southeast-2.amazonaws.com/testuser1/Abdali_Mall_Netmetering/AMC_Zone_A/page_1_image_1.png https://image-pvsyst-aepco-sydney.s3.ap-southeast-2.amazonaws.com/testuser1/Abdali_Mall_Netmetering/AMC_Zone_F&G/page_3_image_2.png 11:59 PM Are you satisfied with the answer? Time taken: 11s, token: 0 tokens connected to Abdali Mall Netmetering LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`pvsyst_agent`
Tools used	`pvsyst_tool`
API calls made	3
Last API call	`GET /testuser2/chat/ → 404`
Console errors	3
Network failures	0



113. Abdali_Mall_Netmetering

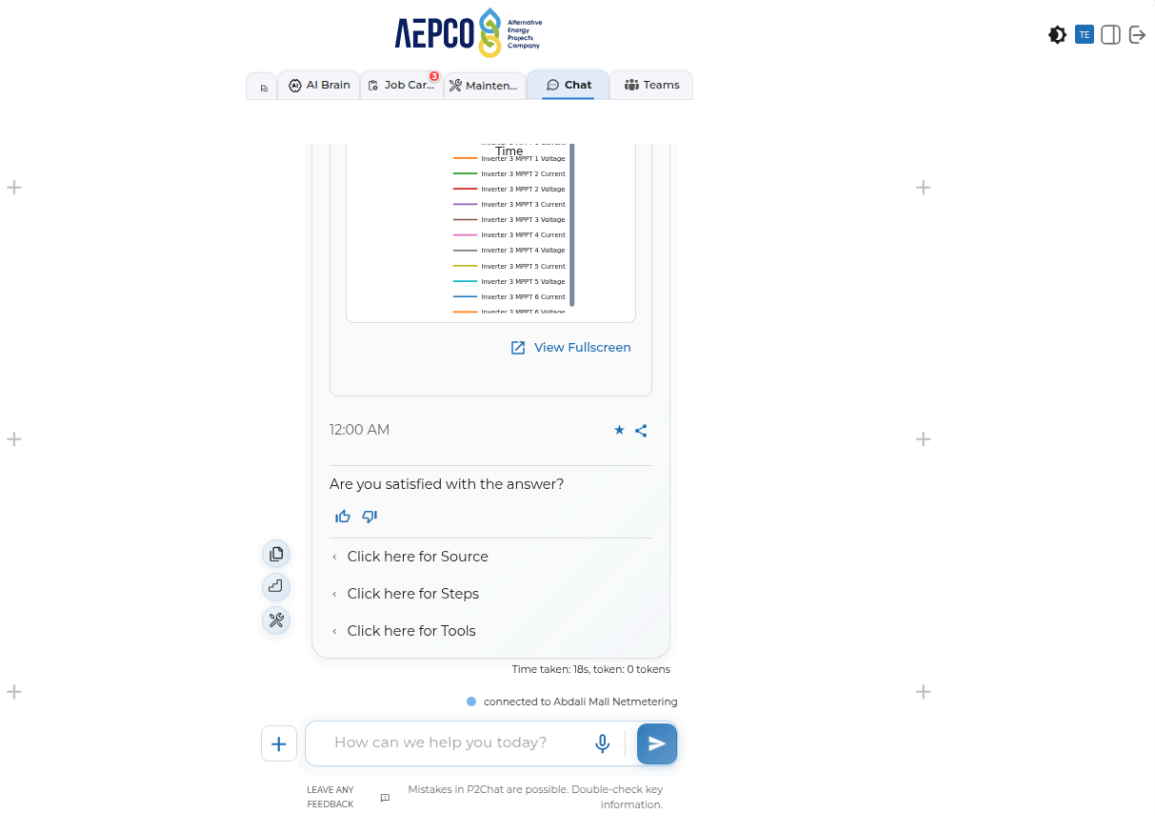
Production • 33.7s

Question: Please plot my current and voltage in a dual axes plot on the deepest possible granularity level for inverter3

Response: I've created the visualization for you. View Fullscreen 12:00 AM ★ Are you satisfied with the answer? Click here for Source Click here for Steps Click here for Tools Time taken: 18s, token: 0 tokens connected to Abdali Mall Netmetering LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.extract_timeseries_data_ag, functions.generate_graphs_via_code, functions.execute_custom_code`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`1`
Network failures	`0`



114. Abdali_Mall_Netmetering

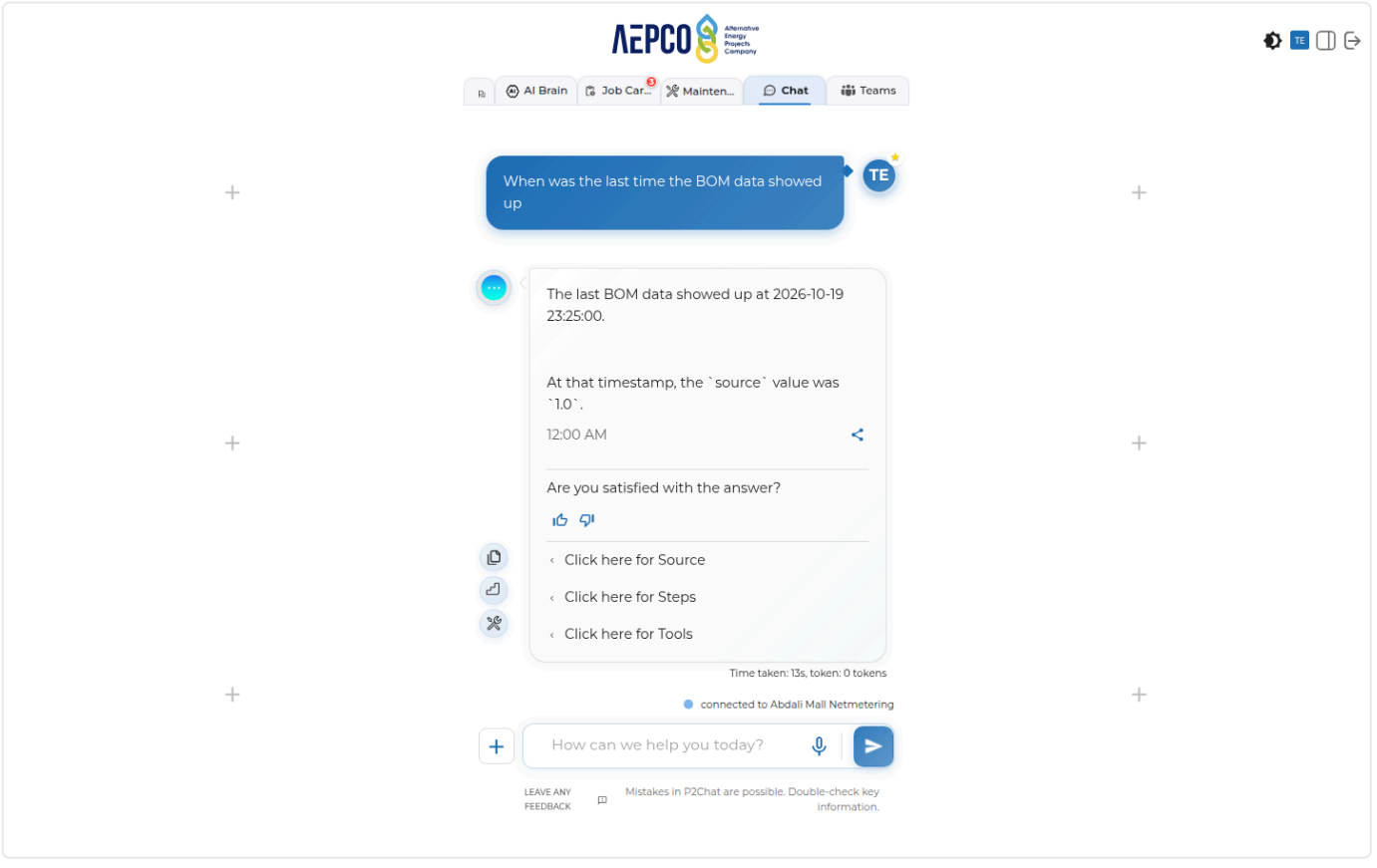
Production • 27.6s

Question: When was the last time the BOM data showed up

Response: The last BOM data showed up at 2026-10-19 23:25:00. At that timestamp, the `source` value was `1.0`. 12:00 AM Are you satisfied with the answer? Click here for Source Click here for Steps Click here for Tools Time taken: 13s, token: 0 tokens connected to Abdali Mall Netmetering LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`execute_custom_code, extract_timeseries_data_ag`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



115. Abdali_Mall_Netmetering

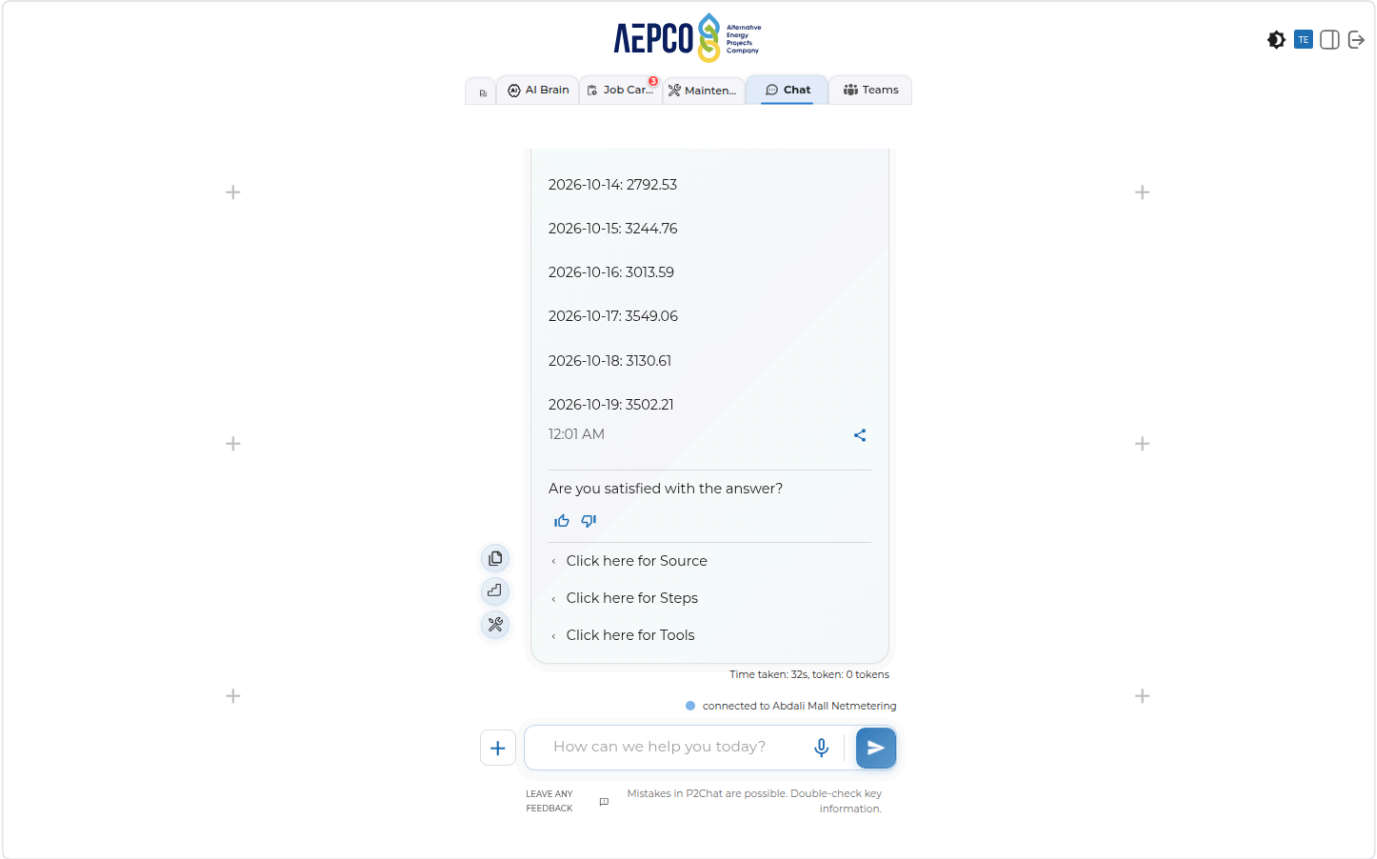
Production • 47.4s

Question: Please find out the daily POA poa summed up as a Wh/m^2, daily for the last 12 months and then calculate the median, mean and kurt

Response: Using `pyranometer_1`, I summed the 5-minute POA irradiance into daily irradiation as Wh/m² over the requested last 12 months window. However, the dataset is incomplete for that window: - Requested: 2025-10-20 to 2026-10-19 - Available data: 2026-04-01 00:05 to 2026-10-19 23:25 So the statistics below are for the available days only. Summary: - Days with data: 197 - Median daily POA: 5689.77 Wh/m² - Mean daily POA: 5400.47 Wh/m² - Kurtosis: 2.1938 Daily POA sums (Wh/m²): 2026-04-01: 1336.30 2026-04-02: 3648.51 2026-04-03: 4473.41 2026-04-04: 5132.93 2026-04-05: 4362.45 2026-04-06: 4437.39 2026-04-07: 4125.43 2026-04-08: 4305.31 2026-04-09: 5567.46 2026-04-10: 5637.33 2026-04-11: 4500.74 2026-04-12: 5393.14 2026-04-13: 5683.54 2026-04-14: 5687.77 2026-04-15: 4390.14 2026-04-16: 5551.00 2026-04-17: 5130.53 2026-04-18: 5338.41 2026-04-19: 4489.48 2026-04-20: 4899.28 2026-04-21: 6117.98 2026-04-22: 6301.33 2026-04-23: 63

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`execute_custom_code, extract_timeseries_data_ag`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



116. Abdali_Mall_Netmetering

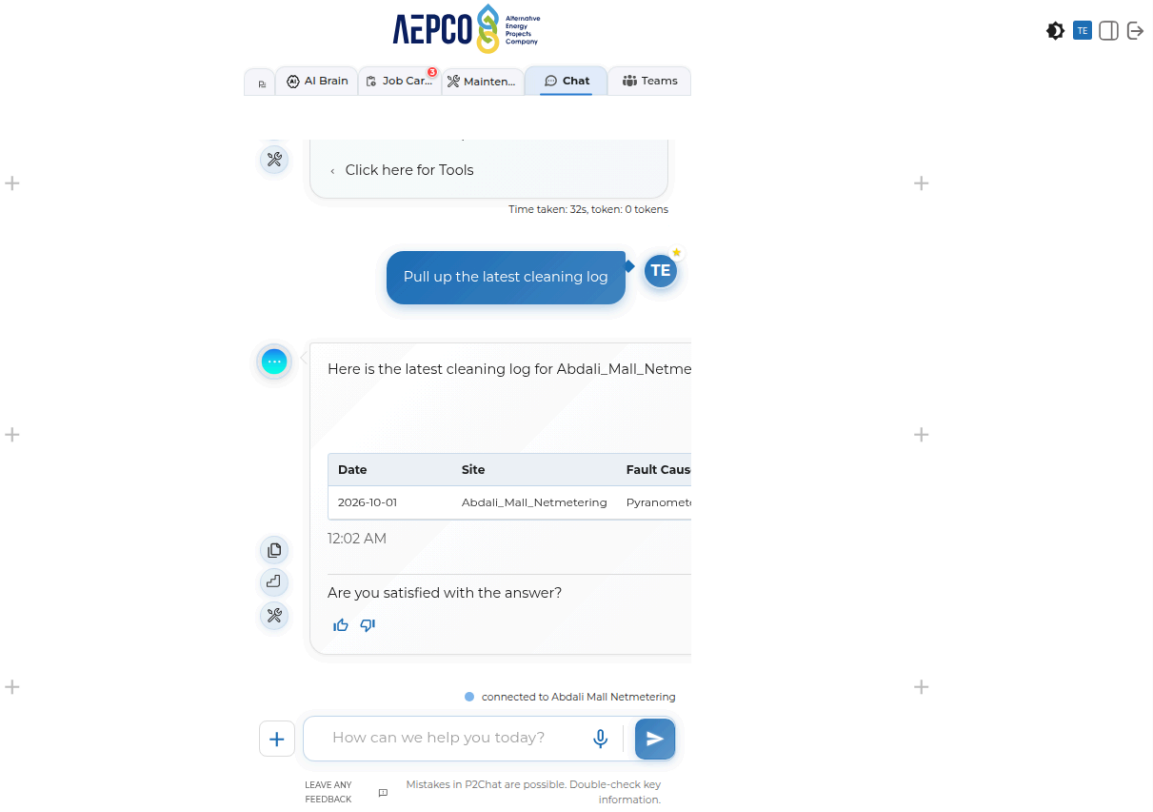
Production • 21.5s

Question: Pull up the latest cleaning log

Response: Here is the latest cleaning log for Abdali_Mall_Netmetering: Edit Date Site Fault Cause Action Taken Status Priority Responsibility Estimated Duration Start Work Time End Work Time Spare Part Use Number Of Staff Images Notes 2026-10-01 Abdali_Mall_Netmetering Pyranometer cleaning required Jim cleaned the pyranometer Completed - - - - - - - - - 12:02 AM Are you satisfied with the answer? Time taken: 6s, token: 0 tokens connected to Abdali Mall Netmetering LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_entry_agent`
API calls made	2
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	0
Network failures	0



117. Abdali_Mall_Netmetering

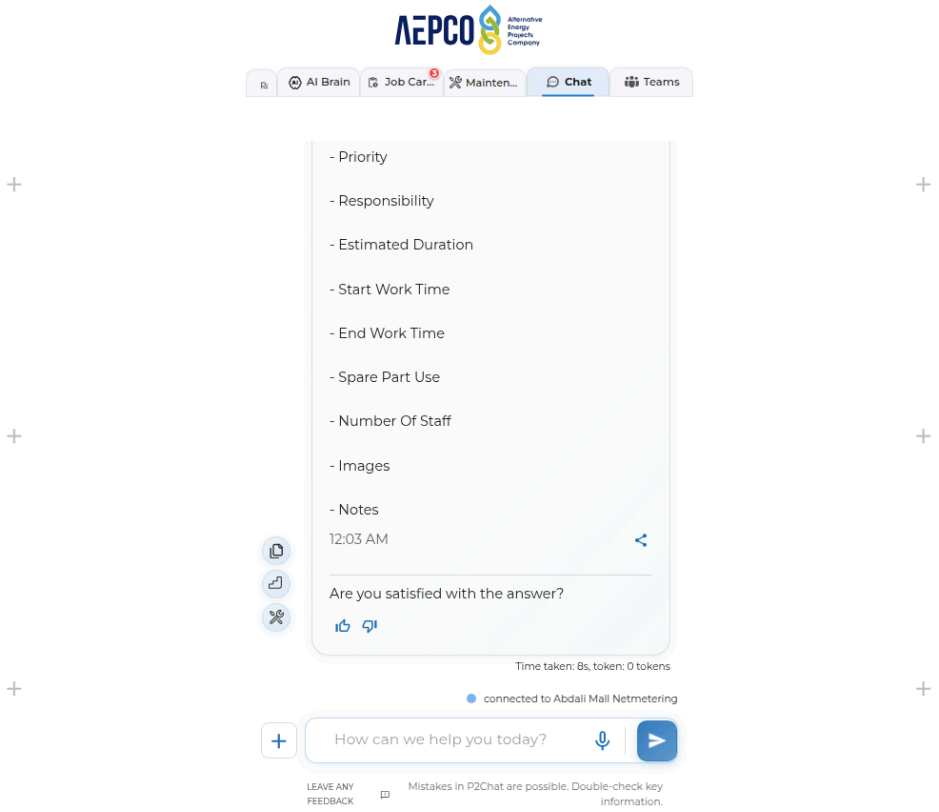
Production • 23.1s

Question: Jim cleaned the pyranometer today, can you add it to the cleaning log?

Response: To add a cleaning activity, please provide the following required details: - Date - Fault Cause - Action Taken - Status Site will be set automatically from the connected plant. Optional fields you can also provide, but they are not required: - Activity Type (Inspection, Cleaning, HSE Incident, Spare Parts Usage, General Site Activity) - Priority - Responsibility - Estimated Duration - Start Work Time - End Work Time - Spare Part Use - Number Of Staff - Images - Notes 12:03 AM Are you satisfied with the answer? Time taken: 8s, token: 0 tokens connected to Abdali Mall Netmetering LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_entry_agent`
API calls made	`1`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



118. Abdali_Mall_Netmetering

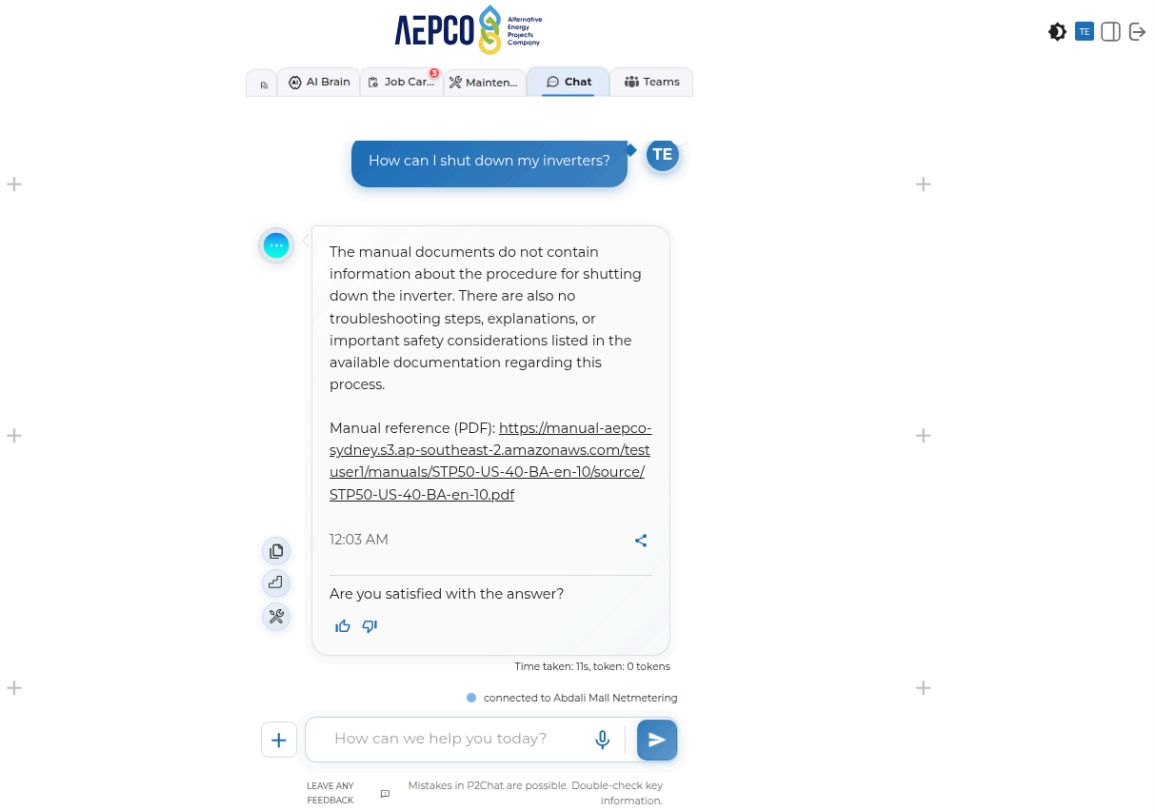
Production • 26.2s

Question: How can I shut down my inverters?

Response: The manual documents do not contain information about the procedure for shutting down the inverter. There are also no troubleshooting steps, explanations, or important safety considerations listed in the available documentation regarding this process. Manual reference (PDF): https://manual-aepco-sydney.s3.ap-southeast-2.amazonaws.com/testuser1/manuals/STP50-US-40-BA-en-10/source/STP50-US-40-BA-en-10.pdf 12:03 AM Are you satisfied with the answer? Time taken: 11s, token: 0 tokens connected to Abdali Mall Netmetering LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`manual_agent`
Tools used	`functions.manual_rag_tool`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



119. Abdali_Mall_Netmetering

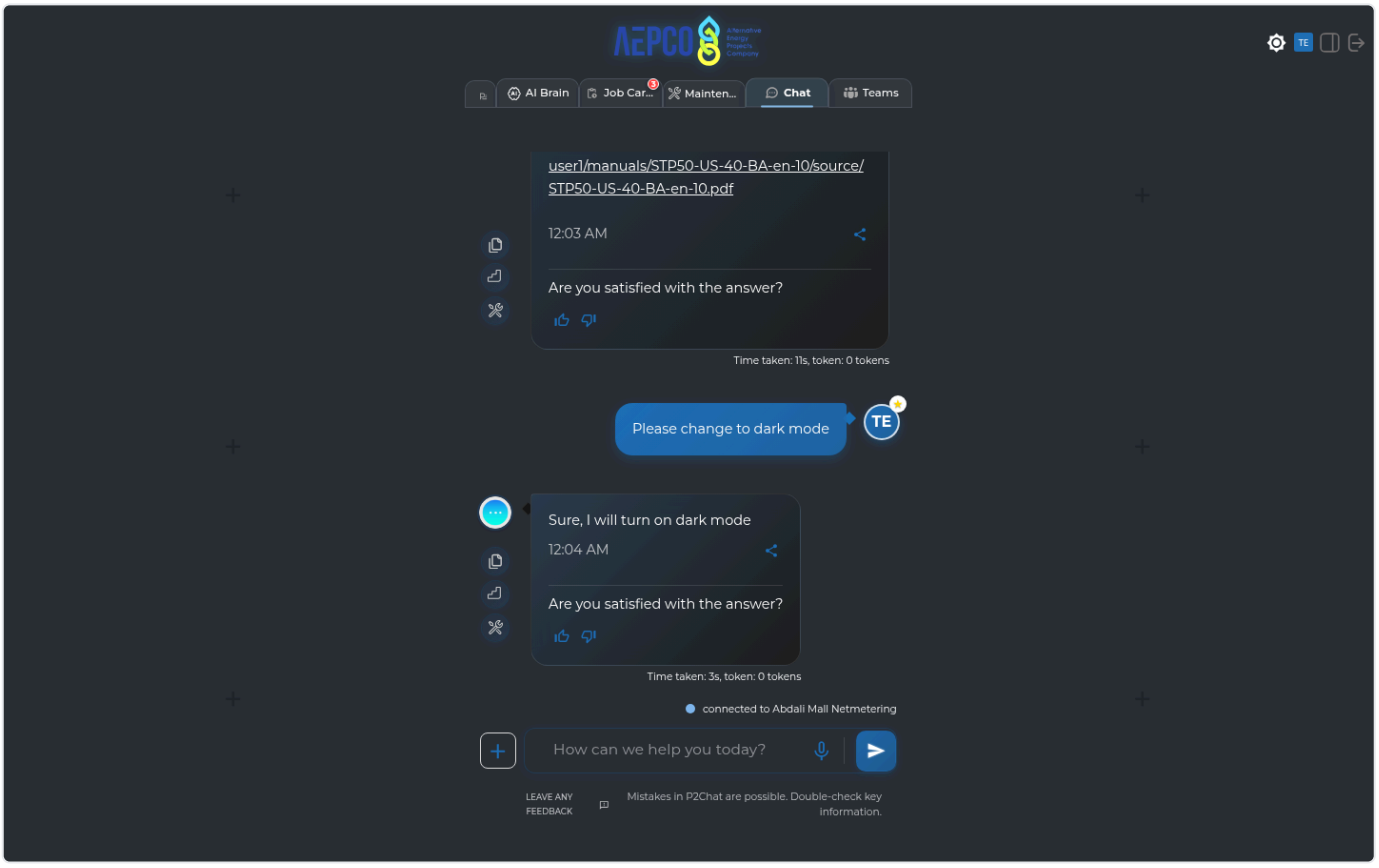
Production • 19.1s

Question: Please change to dark mode

Response: Sure, I will turn on dark mode 12:04 AM Are you satisfied with the answer? Time taken: 3s, token: 0 tokens connected to Abdali Mall Netmetering LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`ui_agent`
API calls made	`1`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



120. Abdali_Mall_Netmetering

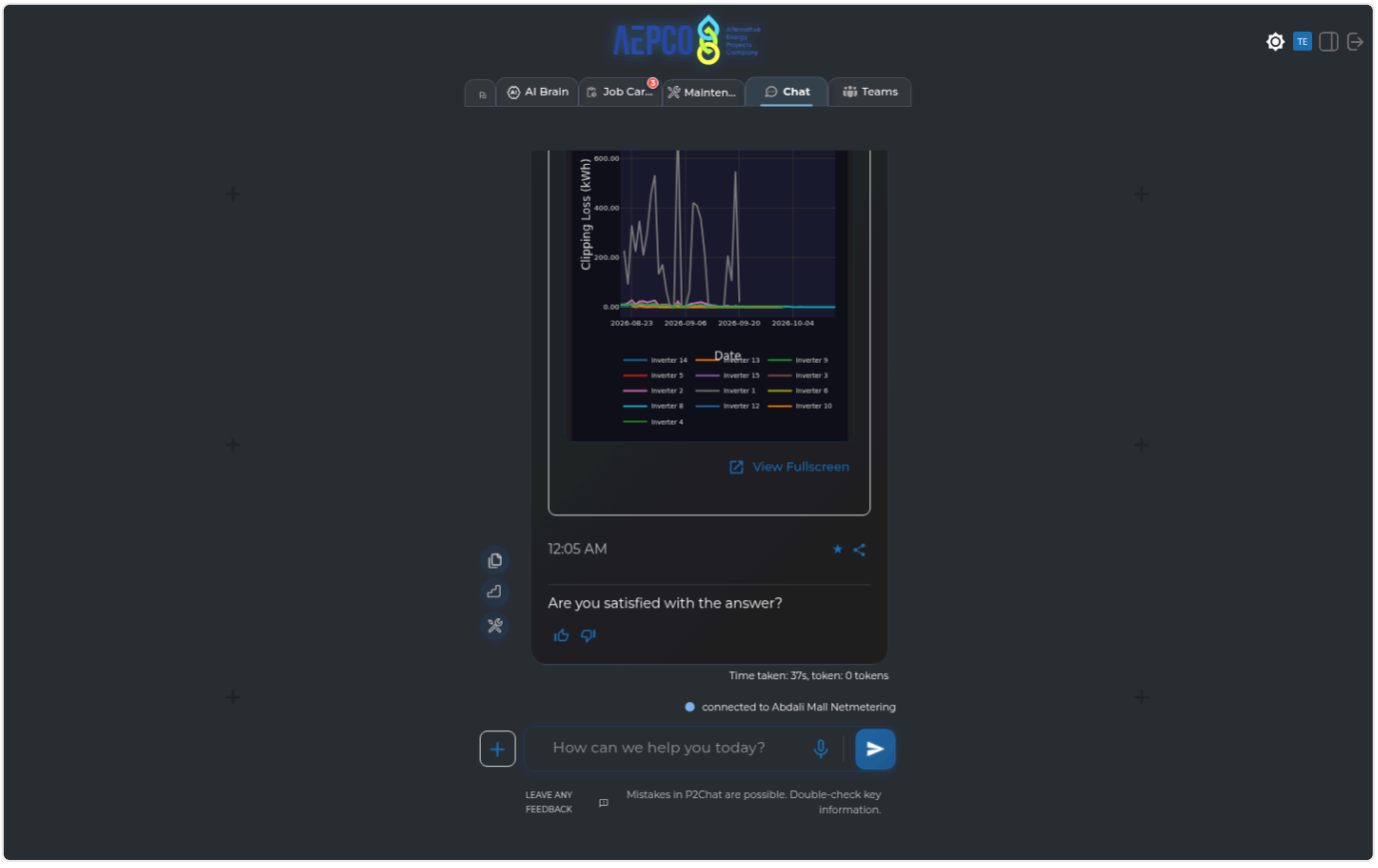
Production • 52.1s

Question: Please figure out how much clipping loss was found in the last 2 months broken by inverters, list them down and plot it as well

Response: Clipping Loss for the Last 2 Months (2026-08-20 to 2026-10-20), by Inverter: - Inverter 1: 0.9575 kWh - Inverter 2: 0.4991 kWh - Inverter 3: 0.7735 kWh - Inverter 4: 1.7151 kWh - Inverter 5: 8.5769 kWh - Inverter 6: 0.0000 kWh - Inverter 7: 0.0000 kWh - Inverter 8: 0.0435 kWh - Inverter 9: 0.0000 kWh - Inverter 10: 0.0000 kWh - Inverter 11: 0.0000 kWh - Inverter 12: 0.0000 kWh - Inverter 13: 0.0000 kWh - Inverter 14: 0.0000 kWh - Inverter 15: 0.0000 kWh Total Clipping Loss Across All Inverters: 12.5656 kWh Data Quality Status: Good I've created a line chart showing daily clipping loss per inverter over the last two months. Clipping Loss for the Last 2 Months (2026-08-20 to 2026-10-20), by Inverter: - Inverter 1: 0.9575 kWh - Inverter 2: 0.4991 kWh - Inverter 3: 0.7735 kWh - Inverter 4: 1.7151 kWh - Inverter 5: 8.5769 kWh - Inverter 6: 0.0000 kWh - Inverter 7: 0.0000 kWh - Inverter 8: 0.0435 kWh - Inverter 9: 0.0000 kWh - Inve

TECHNICAL EVIDENCE

Agents used	`data_agent, graph_agent`
Tools used	`functions.clipping_loss_via_requests, functions.generate_graphs_via_code, functions.calculate_clipping_or_curtailment_losses_ag`
API calls made	3
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	0
Network failures	0



121. Abdali_Mall_Netmetering

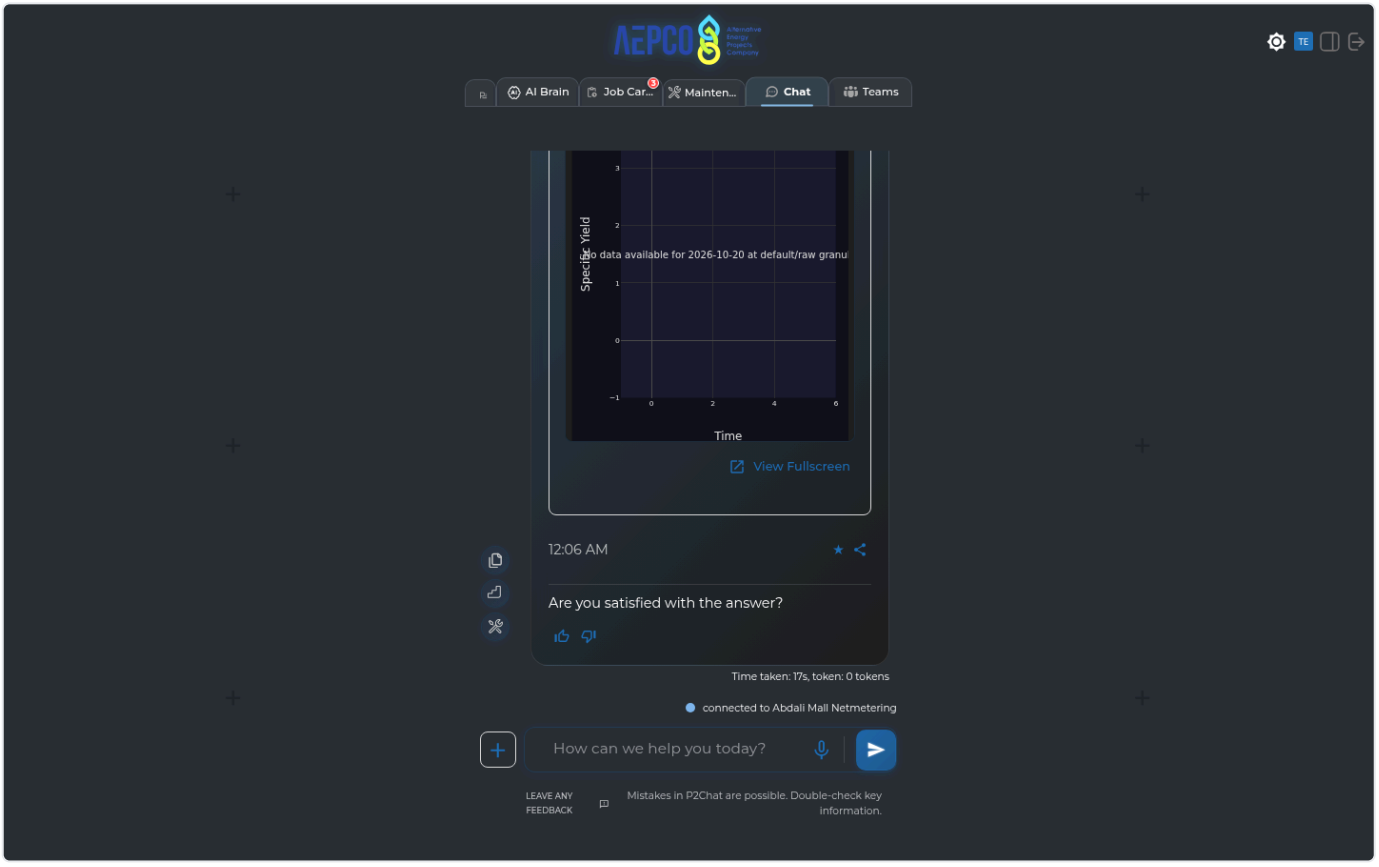
Production • 32.3s

Question: Please repeat the first plotting question, but interchange it with Specific yield on 1 axes, and energy production on the other axes on a single dual axes plot (Memory Test)

Response: I've created the visualization for you. View Fullscreen 12:06 AM ★ Are you satisfied with the answer? Time taken: 17s, token: 0 tokens connected to Abdali Mall Netmetering LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.specific_yield_ag, functions.generate_graphs_via_code, functions.total_energy_func_ag`
API calls made	`1`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



122. BF_Wheeling

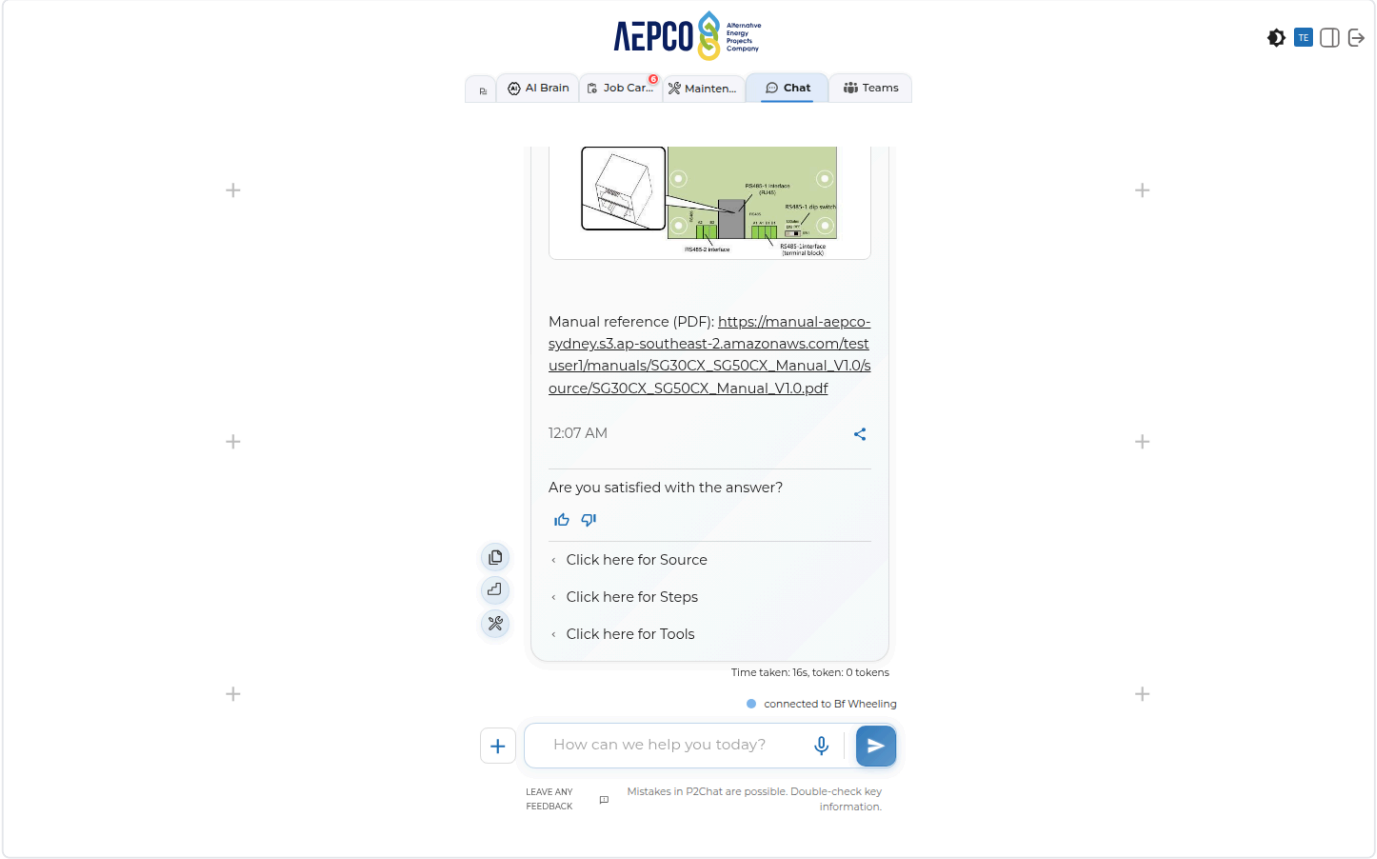
Production • 30.7s

Question: I have a COMMUNICATION_FAULT what do I do to my inverters?

Response: To troubleshoot a COMMUNICATION_FAULT on your inverters, please follow these steps using the referenced diagrams for guidance: 1. Check the RS485-1 interface (RJ45) and ensure the cable is securely connected as shown in the upper section of this diagram: 2. Set the dip switches for RS485-1 according to your communication requirements. Refer to dip switch settings in this diagram: 3. Inspect the RS485-2 interface (terminal block) and ensure all wires are securely attached to the correct terminals as shown in the lower part of this diagram: 4. Confirm the emergency stop dry contacts and fault output dry contacts wiring are consistent with the wiring shown. See diagram: 5. If all physical connections are correct but the communication fault persists, double-check the configuration of all dip switches to ensure they match your system's requirements. Reference: Manual reference (PDF): https://manual-aepco-sydney.s3.ap-southeast-2.amazonaws.com/testuser1/manuals/SG30CX_SG50CX_Manu

TECHNICAL EVIDENCE

Agents used	`manual_agent`
Tools used	`functions.manual_rag_tool`
API calls made	`12`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



123. BF_Wheeling

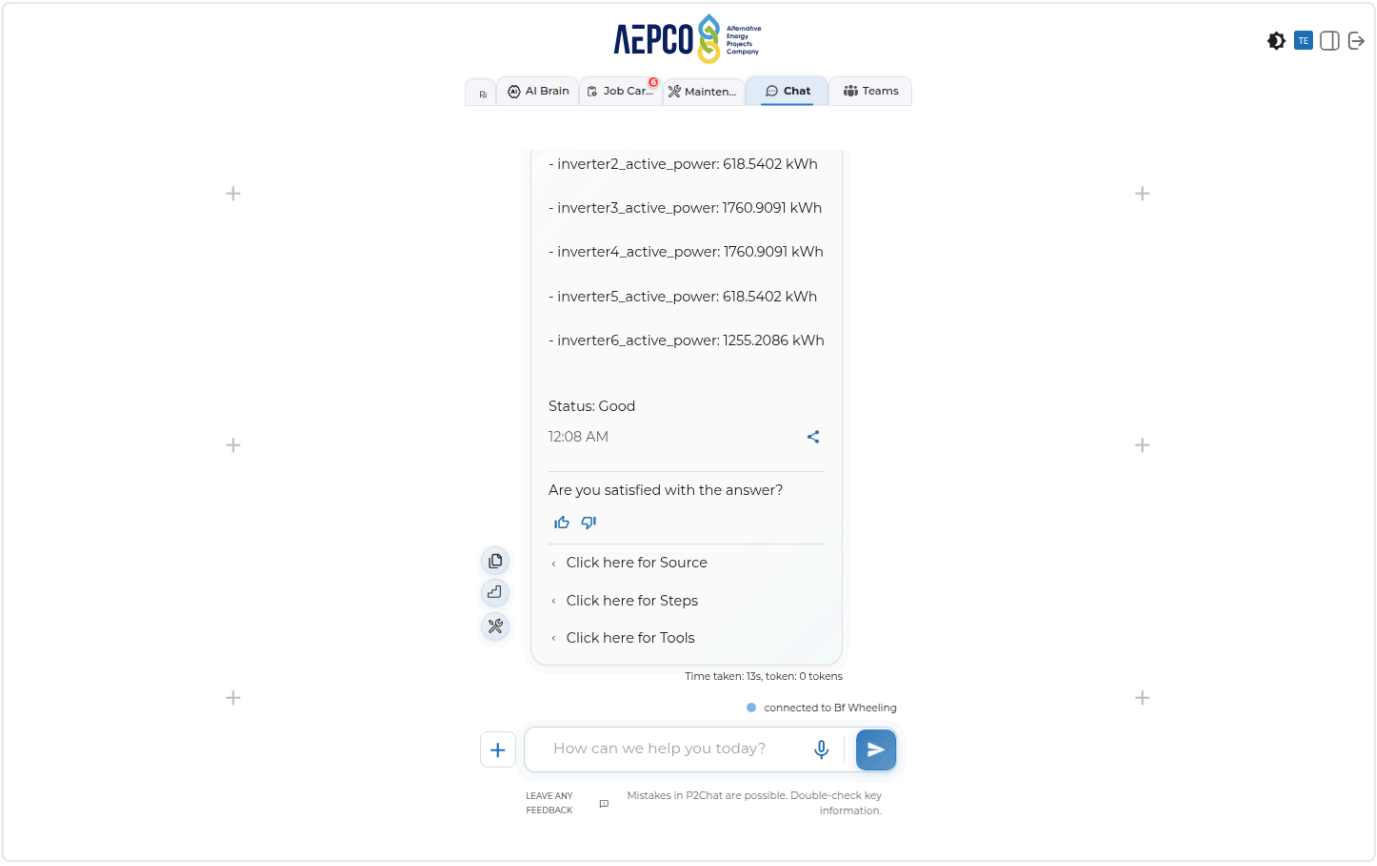
Production • 29.1s

Question: Use 'clipping_loss_via_requests' in order to calculate clipping loss if any for July 2026

Response: Clipping loss via requests for July 2026: - inverter1_active_power: 1760.9091 kWh - inverter2_active_power: 618.5402 kWh - inverter3_active_power: 1760.9091 kWh - inverter4_active_power: 1760.9091 kWh - inverter5_active_power: 618.5402 kWh - inverter6_active_power: 1255.2086 kWh Status: Good 12:08 AM Are you satisfied with the answer? Click here for Source Click here for Steps Click here for Tools Time taken: 13s, token: 0 tokens connected to Bf Wheeling LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`functions.clipping_loss_via_requests`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



124. BF_Wheeling

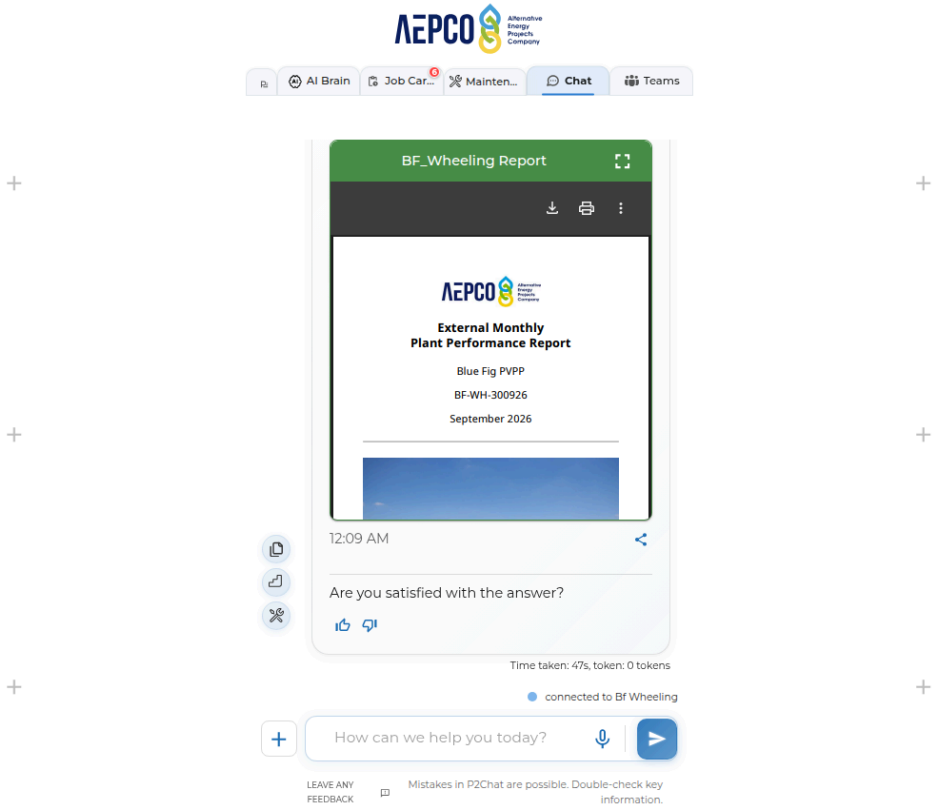
Production • 62.6s

Question: Please generate last months report

Response: I've generated the external report for September 2026. You can view it here: 'https://report-aepco-sydney.s3.ap-southeast-2.amazonaws.com/BF_Wheeling_Solar_Plant_September_2026_external_report.pdf' BF_Wheeling Report 12:09 AM Are you satisfied with the answer? Time taken: 47s, token: 0 tokens connected to Bf Wheeling LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`report_agent`
Tools used	`agent_external_report_tool`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



125. BF_Wheeling

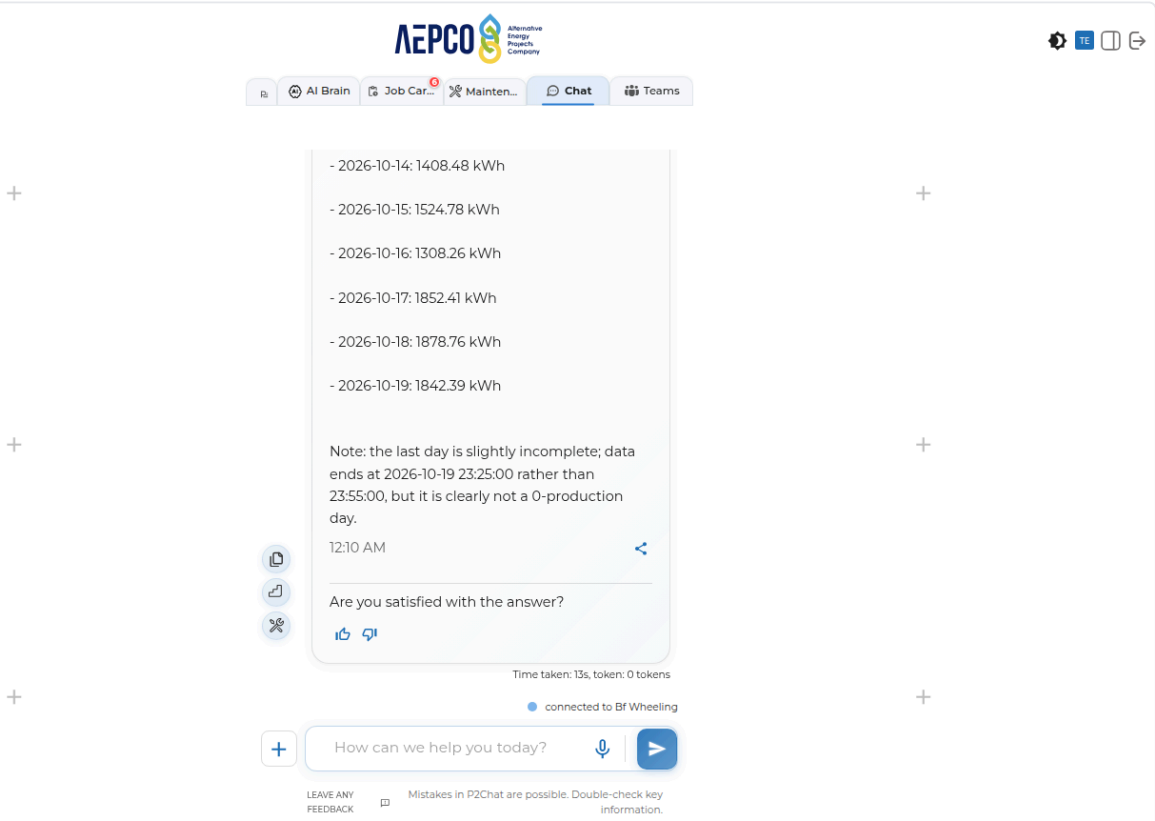
Production • 27.7s

Question: Was there any 0 production days in the last 7 days?

Response: No. There were no 0-production days in the last 7 days. Daily plant energy for 2026-10-13 to 2026-10-19: - 2026-10-13: 1576.57 kWh - 2026-10-14: 1408.48 kWh - 2026-10-15: 1524.78 kWh - 2026-10-16: 1308.26 kWh - 2026-10-17: 1852.41 kWh - 2026-10-18: 1878.76 kWh - 2026-10-19: 1842.39 kWh Note: the last day is slightly incomplete; data ends at 2026-10-19 23:25:00 rather than 23:55:00, but it is clearly not a 0-production day. 12:10 AM Are you satisfied with the answer? Time taken: 13s, token: 0 tokens connected to Bf Wheeling LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`functions.total_energy_func_ag`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



126. BF_Wheeling

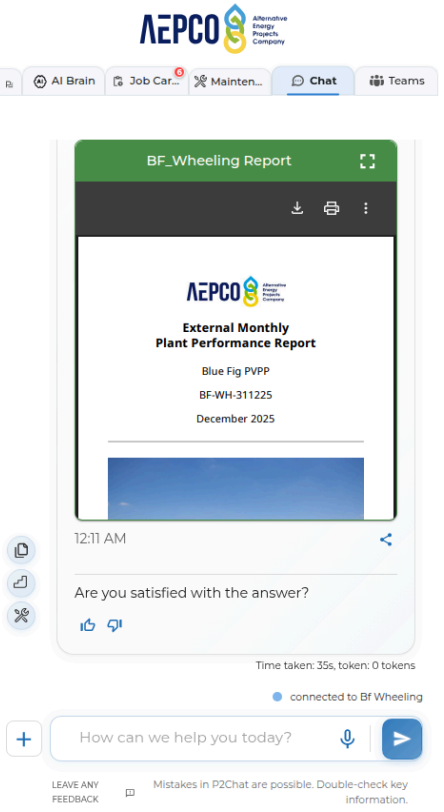
Production • 50.6s

Question: Please generate December 2025 performance report

Response: I've generated the external report for December 2025. You can view it here: 'https://report-aepco-sydney.s3.ap-southeast-2.amazonaws.com/BF_Wheeling_Solar_Plant_December_2025_external_report.pdf' BF_Wheeling Report 12:11 AM Are you satisfied with the answer? Time taken: 35s, token: 0 tokens connected to Bf Wheeling LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`report_agent`
Tools used	`agent_external_report_tool`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



127. BF_Wheeling

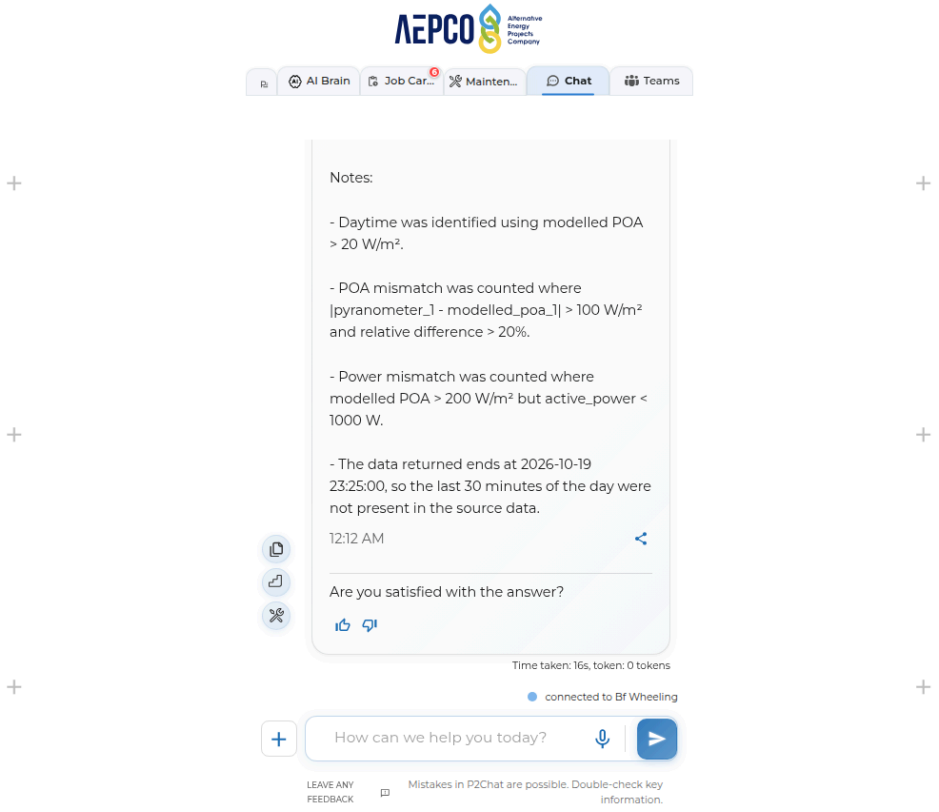
Production • 30.8s

Question: How is my data quality for POA and power? How many of them show missing data during daytime or mismatches with satellite prediction?

Response: For 2026-10-19: - Daytime points assessed: 132 - Missing POA data during daytime: 0 - Missing plant power data during daytime: 0 - POA mismatches vs satellite/modelled prediction: 100 - Power mismatches vs irradiance expectation: 0 Notes: - Daytime was identified using modelled POA > 20 W/m². - POA mismatch was counted where |pyranometer_1 - modelled_poa_1| > 100 W/m² and relative difference > 20%. - Power mismatch was counted where modelled POA > 200 W/m² but active_power < 1000 W. - The data returned ends at 2026-10-19 23:25:00, so the last 30 minutes of the day were not present in the source data. 12:12 AM Are you satisfied with the answer? Time taken: 16s, token: 0 tokens connected to Bf Wheeling LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`functions.extract_timeseries_data_ag, functions.execute_custom_code`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



128. BF_Wheeling

Production • 37.5s

Question: Please plot last 3 days hourly performance ratio for my MPPTs for inverter 2

Response: I've created the visualization for you. View Fullscreen 12:12 AM ★ Are you satisfied with the answer? Time taken: 21s, token: 0 tokens connected to Bf Wheeling LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.perf_ratio_ag, functions.generate_graphs_via_code`
API calls made	2
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	0
Network failures	0

129. BF_Wheeling

Production • 33.8s

Question: Please plot last 3 days hourly performance ratio for my Strings for inverter 2

Response: I've created the visualization for you. View Fullscreen 12:13 AM ★ Are you satisfied with the answer? Time taken: 18s, token: 0 tokens connected to Bf Wheeling LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.perf_ratio_ag, functions.extract_timeseries_data_ag, functions.execute_custom_code, functions.generate_graphs_via_code`
API calls made	`2`
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	`0`
Network failures	`0`

130. BF_Wheeling

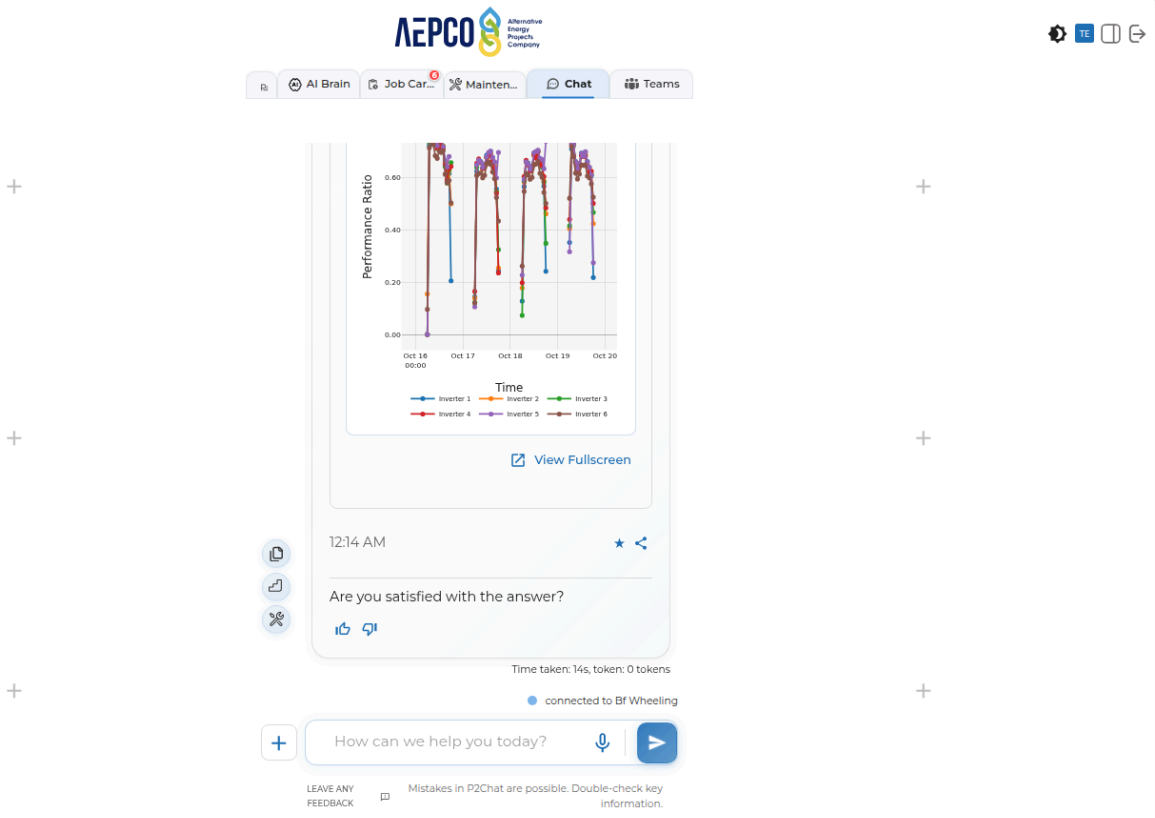
Production • 29.2s

Question: Please plot the last 4 days hourly performance ratio for all my inverters

Response: I've created the visualization for you. View Fullscreen 12:14 AM ★ Are you satisfied with the answer? Time taken: 14s, token: 0 tokens connected to Bf Wheeling LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.perf_ratio_ag, functions.generate_graphs_via_code`
API calls made	`1`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



131. BF_Wheeling

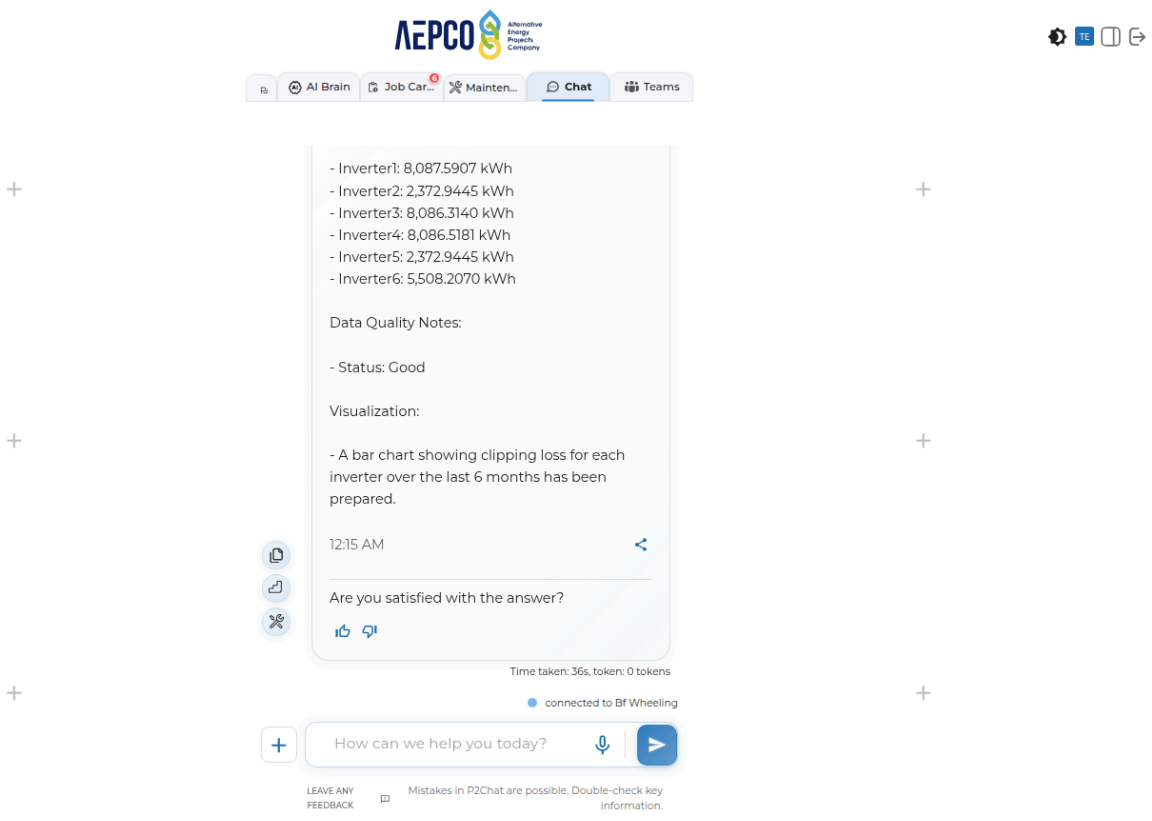
Production • 52.1s

Question: Please figure out how much clipping loss was found in the last 6 months broken by inverters, list them down and plot it as well

Response: Clipping Loss Summary (Last 6 Months): - Inverter1: 8,087.5907 kWh - Inverter2: 2,372.9445 kWh - Inverter3: 8,086.3140 kWh - Inverter4: 8,086.5181 kWh - Inverter5: 2,372.9445 kWh - Inverter6: 5,508.2070 kWh Data Quality Notes: - Status: Good Visualization: - A bar chart showing clipping loss for each inverter over the last 6 months has been prepared. 12:15 AM Are you satisfied with the answer? Time taken: 36s, token: 0 tokens connected to Bf Wheeling LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent, graph_agent`
Tools used	`functions.clipping_loss_via_requests, functions.calculate_clipping_or_curtailment_losses_ag, functions.inspect_xpool`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



132. BF_Wheeling

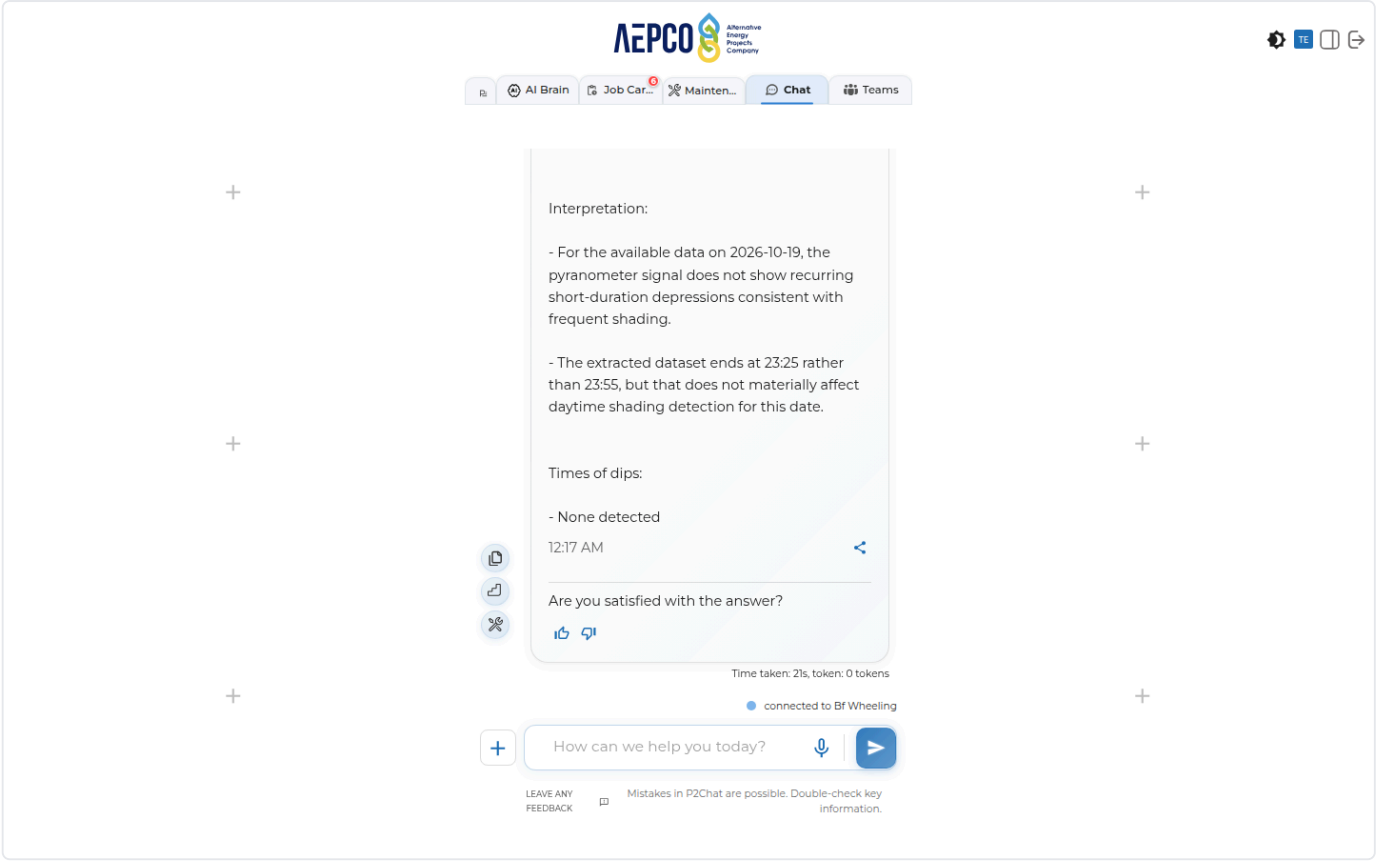
Production • 35.4s

Question: Is there shading happening with my plant? Please analyze the pyranometer and check for frequent dips at certain times

Response: I analyzed the pyranometer data for 2026-10-19 and did not find frequent dip events suggestive of shading. Results: - Daytime points analyzed: 95 - Dip points found: 0 - Dip events found: 0 - Frequency of dips: 0 events/day - Frequency by start hour: none Method used: - Compared measured POA irradiance (`pyranometer_1`) against `modelled_poa_1` - Focused on daytime periods with modelled POA > 200 W/m² - Flagged dips where measured irradiance fell at least 15 percentage points below the local rolling median ratio and the measured/modelled ratio was below 0.85 - Grouped consecutive 5-minute dip points into events Interpretation: - For the available data on 2026-10-19, the pyranometer signal does not show recurring short-duration depressions consistent with frequent shading. - The extracted dataset ends at 23:25 rather than 23:55, but that does not materially affect daytime shading detection for this date. Times of dips: - None detected 12:17 AM Are y

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`functions.extract_timeseries_data_ag, functions.execute_custom_code`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



133. BF_Wheeling

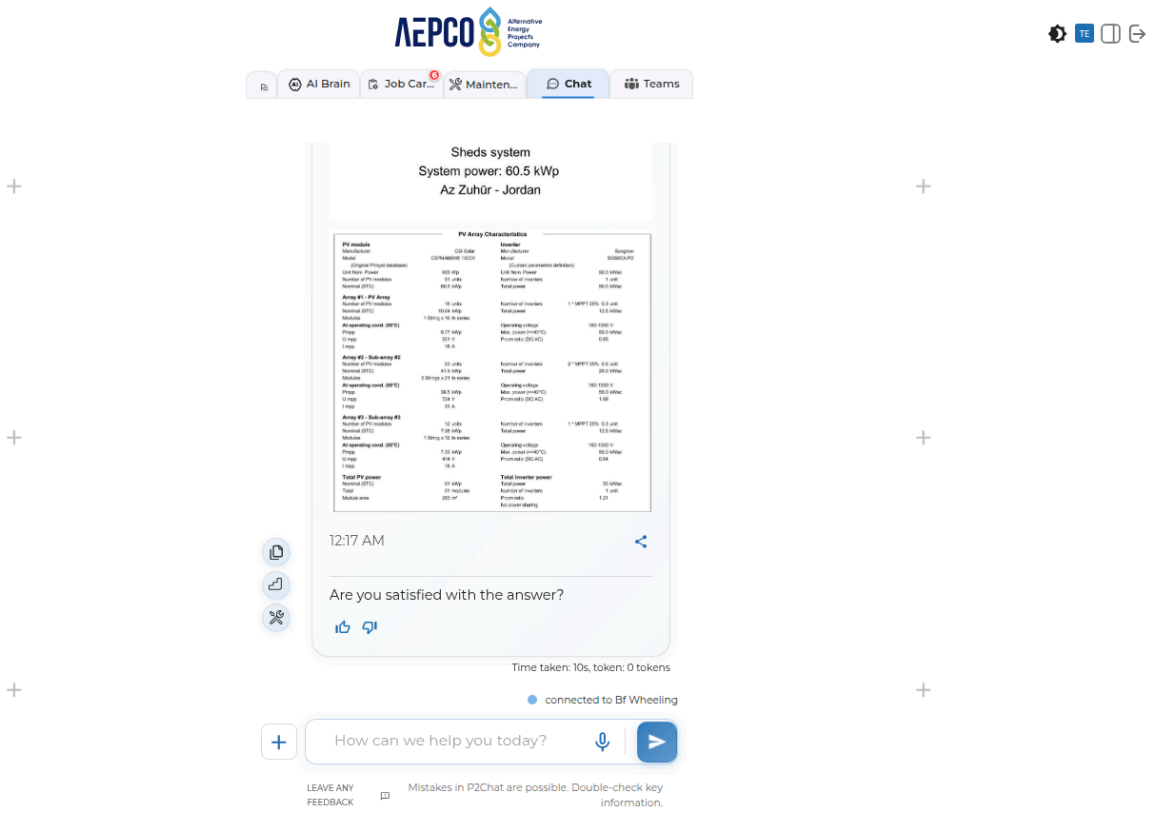
Production • 26.3s

Question: Based on pvsyst report, what is the rated power of this plant and array configuration

Response: Found 4 relevant PVsyst report sections: 1. [Blue_Fig] Front Page - System Title and Rated Power (Page 1, Table 1) 2. [Blue_Fig] PV Array Characteristics (Page 3, Table 2) 3. [Blue_Fig_Extension] Front Page - System Title and Rated Power (Page 1, Table 1) 4. [Blue_Fig_Extension] PV Array Characteristics (Page 3, Table 2) Please access the URLs provided to view the detailed rated power information. Here is the reference: https://image-pvsyst-aepco-sydney.s3.ap-southeast-2.amazonaws.com/testuser1/BF_Wheeling/Blue_Fig/page_1_image_1.png https://image-pvsyst-aepco-sydney.s3.ap-southeast-2.amazonaws.com/testuser1/BF_Wheeling/Blue_Fig/page_3_image_2.png https://image-pvsyst-aepco-sydney.s3.ap-southeast-2.amazonaws.com/testuser1/BF_Wheeling/Blue_Fig_Extension/page_1_image_1.png https://image-pvsyst-aepco-sydney.s3.ap-southeast-2.amazonaws.com/testuser1/BF_Wheeling/Blue_Fig_Extension/page_3_image_2.png 12:17 AM Are you satisfied with the answer? Time taken: 10s, token

TECHNICAL EVIDENCE

Agents used	`pvsyst_agent`
Tools used	`pvsyst_tool`
API calls made	`4`
Last API call	`GET /testuser2/chat/ → 404`
Console errors	`3`
Network failures	`1`



134. BF_Wheeling

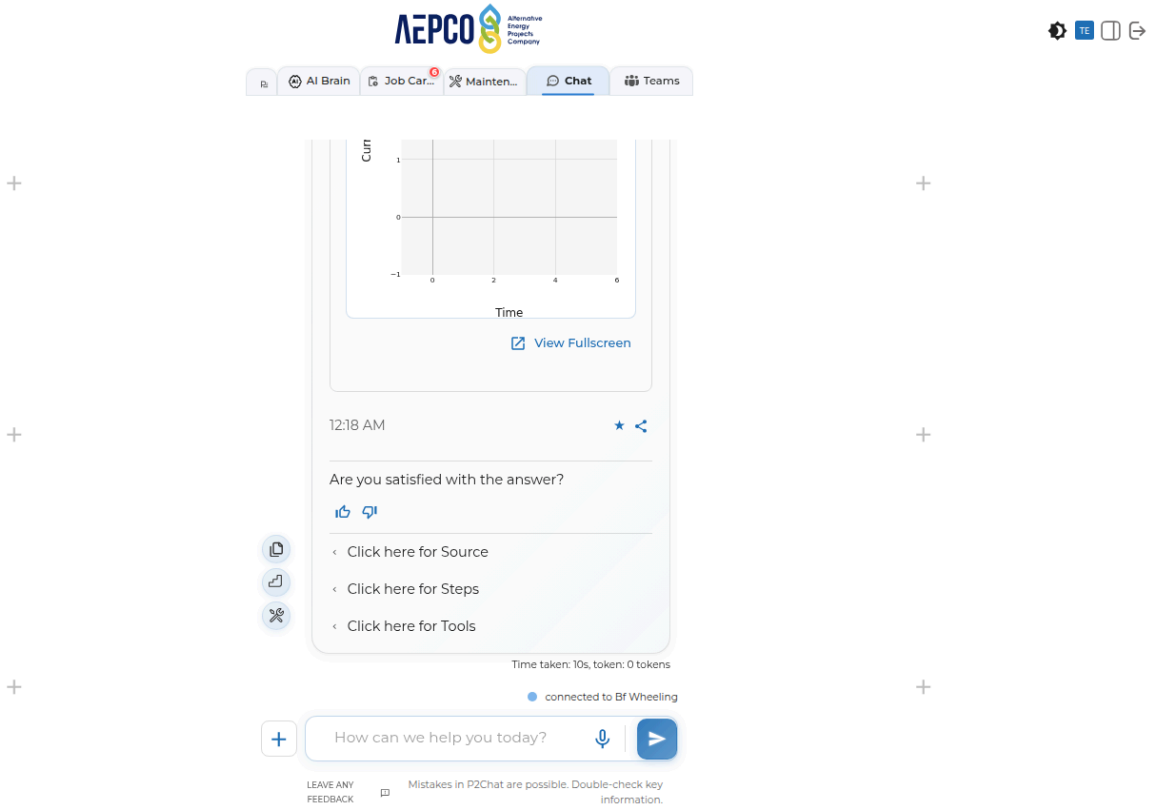
Production • 26.2s

Question: Please plot my current and voltage in a dual axes plot on the deepest possible granularity level for inverter3

Response: I've created the visualization for you. View Fullscreen 12:18 AM ★ Are you satisfied with the answer? Click here for Source Click here for Steps Click here for Tools Time taken: 10s, token: 0 tokens connected to Bf Wheeling LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.generate_graphs_via_code, functions.extract_timeseries_data_ag`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	1
Network failures	0



135. BF_Wheeling

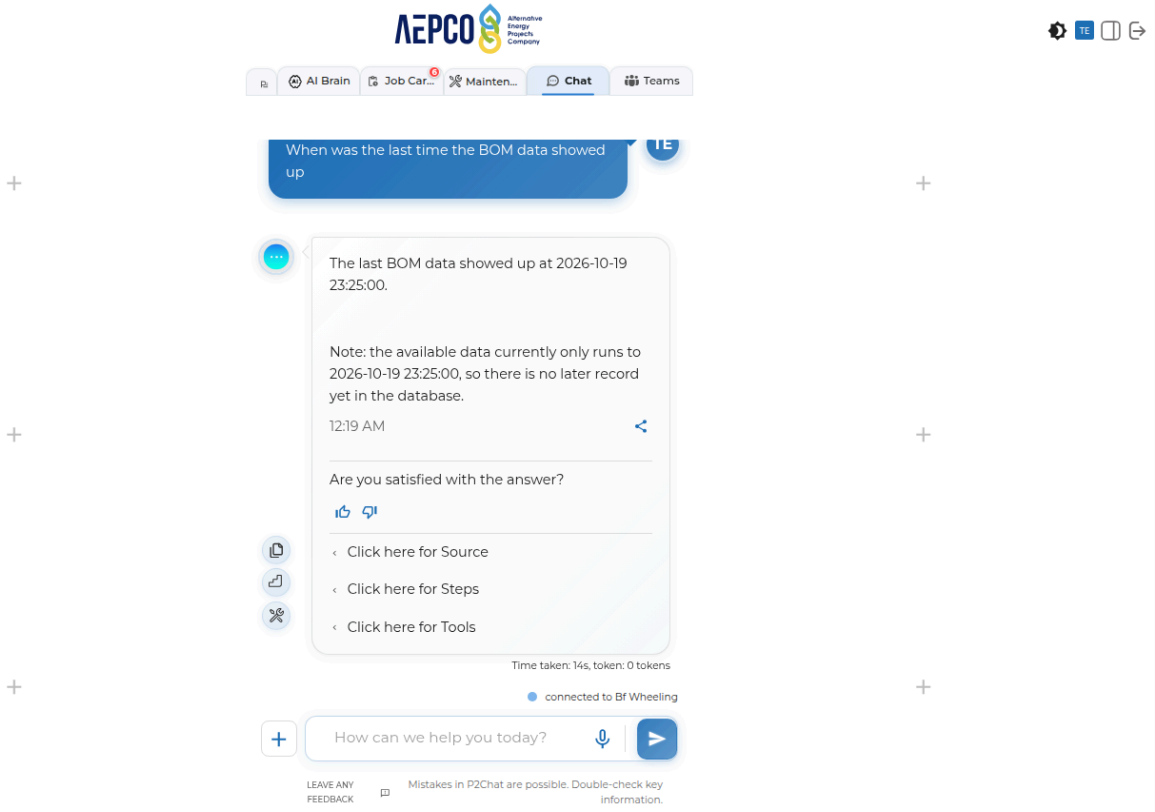
Production • 29.2s

Question: When was the last time the BOM data showed up

Response: The last BOM data showed up at 2026-10-19 23:25:00. Note: the available data currently only runs to 2026-10-19 23:25:00, so there is no later record yet in the database. 12:19 AM Are you satisfied with the answer? Click here for Source Click here for Steps Click here for Tools Time taken: 14s, token: 0 tokens connected to Bf Wheeling LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`execute_custom_code, extract_timeseries_data_ag`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



136. BF_Wheeling

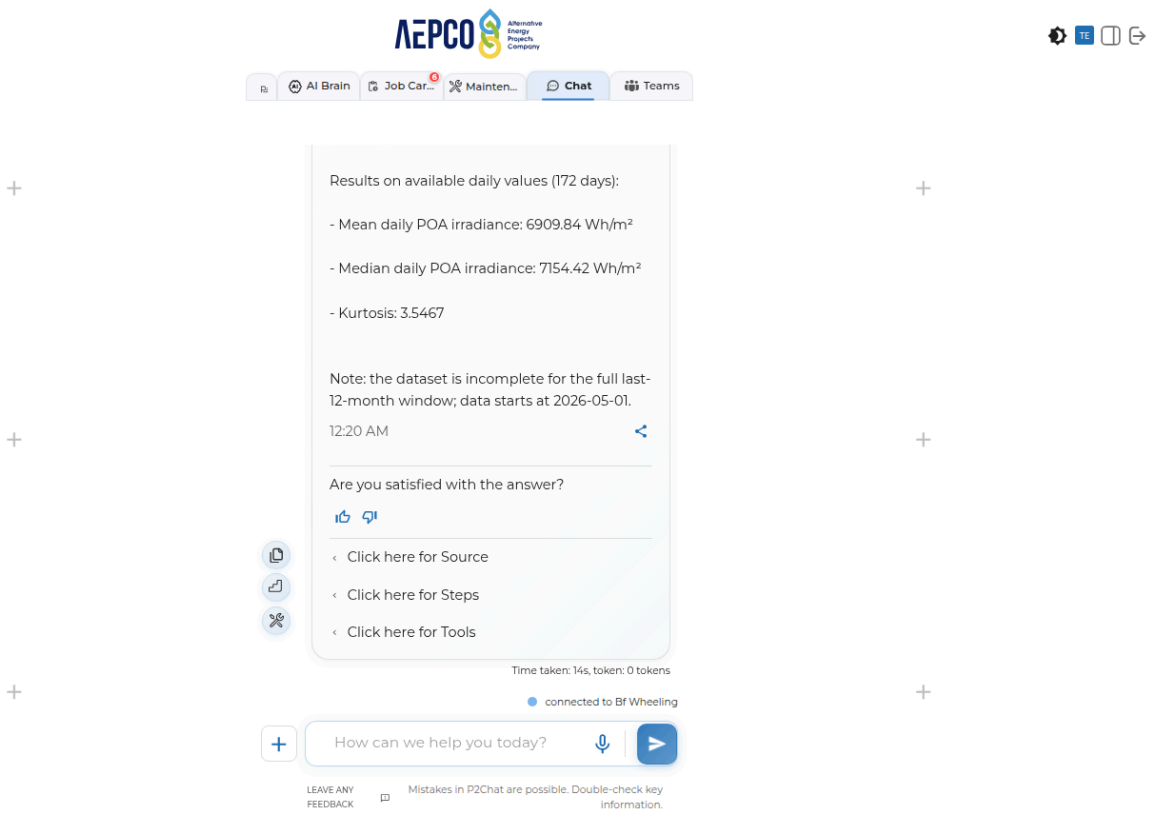
Production • 29.1s

Question: Please find out the daily POA poa summed up as a Wh/m^2, daily for the last 12 months and then calculate the median, mean and kurt

Response: Daily summed POA irradiance from `pyranometer_1` for the requested last 12 months could only be computed for the period with available data: 2026-05-01 to 2026-10-19. Results on available daily values (172 days): - Mean daily POA irradiance: 6909.84 Wh/m² - Median daily POA irradiance: 7154.42 Wh/m² - Kurtosis: 3.5467 Note: the dataset is incomplete for the full last-12-month window; data starts at 2026-05-01. 12:20 AM Are you satisfied with the answer? Click here for Source Click here for Steps Click here for Tools Time taken: 14s, token: 0 tokens connected to Bf Wheeling LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`functions.execute_custom_code, functions.extract_timeseries_data_ag`
API calls made	2
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	0
Network failures	0



137. BF_Wheeling

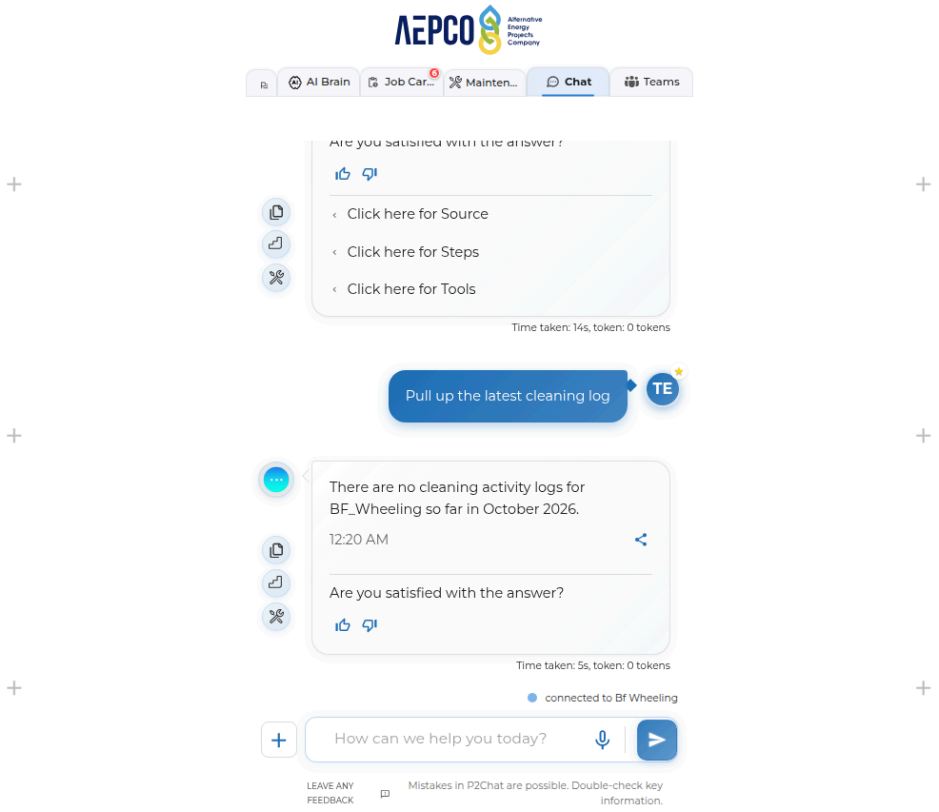
Production • 20.0s

Question: Pull up the latest cleaning log

Response: There are no cleaning activity logs for BF_Wheeling so far in October 2026. 12:20 AM Are you satisfied with the answer? Time taken: 5s, token: 0 tokens connected to Bf Wheeling LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_entry_agent`
API calls made	`1`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



138. BF_Wheeling

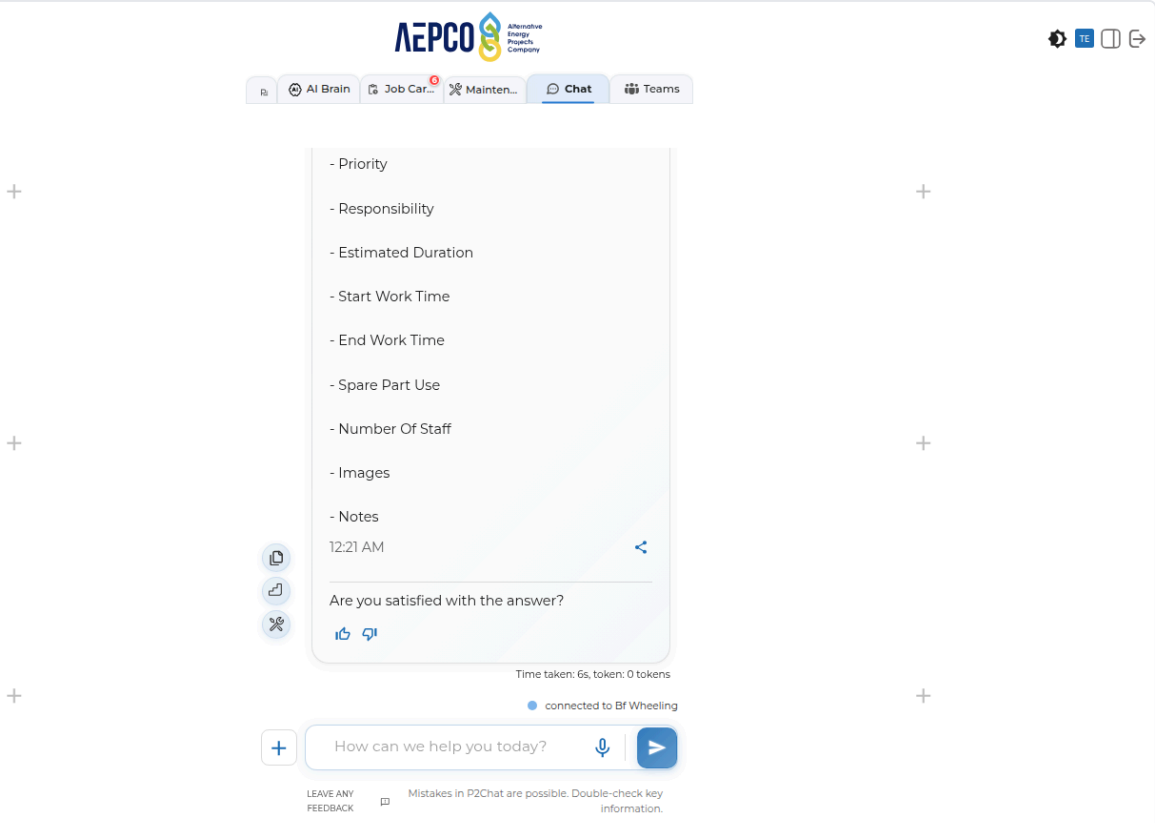
Production • 21.5s

Question: Jim cleaned the pyranometer today, can you add it to the cleaning log?

Response: To add a cleaning activity log, please provide the following required details: - Date - Fault Cause - Action Taken - Status Site will be set automatically from the connected plant. Optional fields you can also provide, but they are not required: - Activity Type (Inspection, Cleaning, HSE Incident, Spare Parts Usage, General Site Activity) - Priority - Responsibility - Estimated Duration - Start Work Time - End Work Time - Spare Part Use - Number Of Staff - Images - Notes 12:21 AM Are you satisfied with the answer? Time taken: 6s, token: 0 tokens connected to Bf Wheeling LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_entry_agent`
API calls made	2
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	0
Network failures	0



139. BF_Wheeling

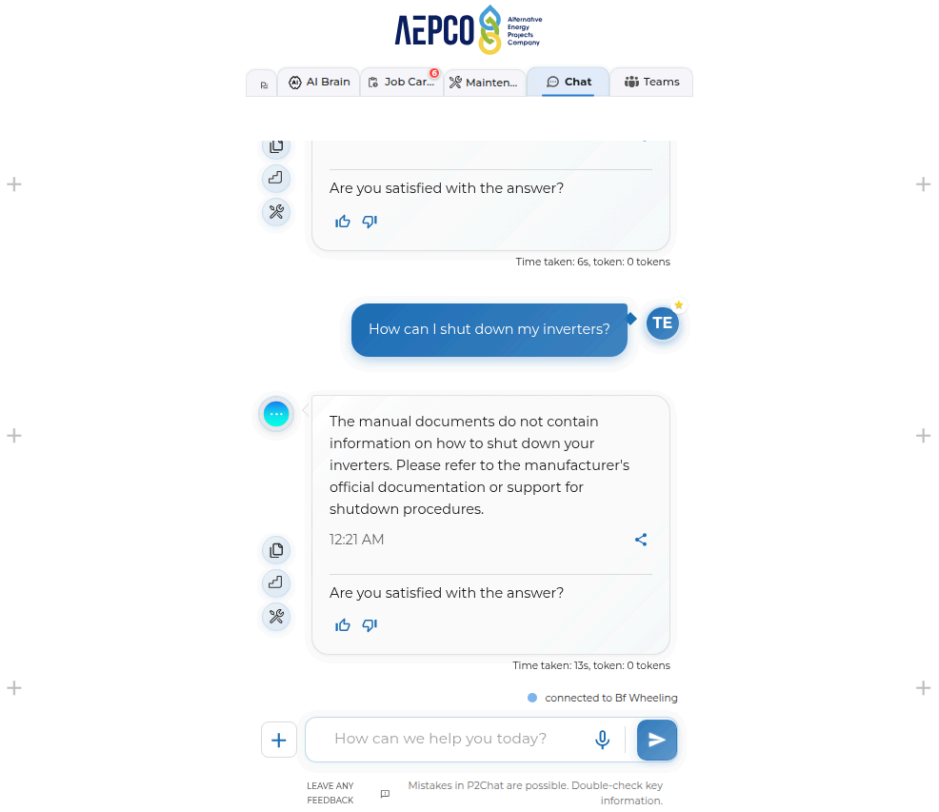
Production • 27.7s

Question: How can I shut down my inverters?

Response: The manual documents do not contain information on how to shut down your inverters. Please refer to the manufacturer's official documentation or support for shutdown procedures. 12:21 AM Are you satisfied with the answer? Time taken: 13s, token: 0 tokens connected to Bf Wheeling LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`manual_agent`
Tools used	`functions.manual_rag_tool`
API calls made	`1`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



140. BF_Wheeling

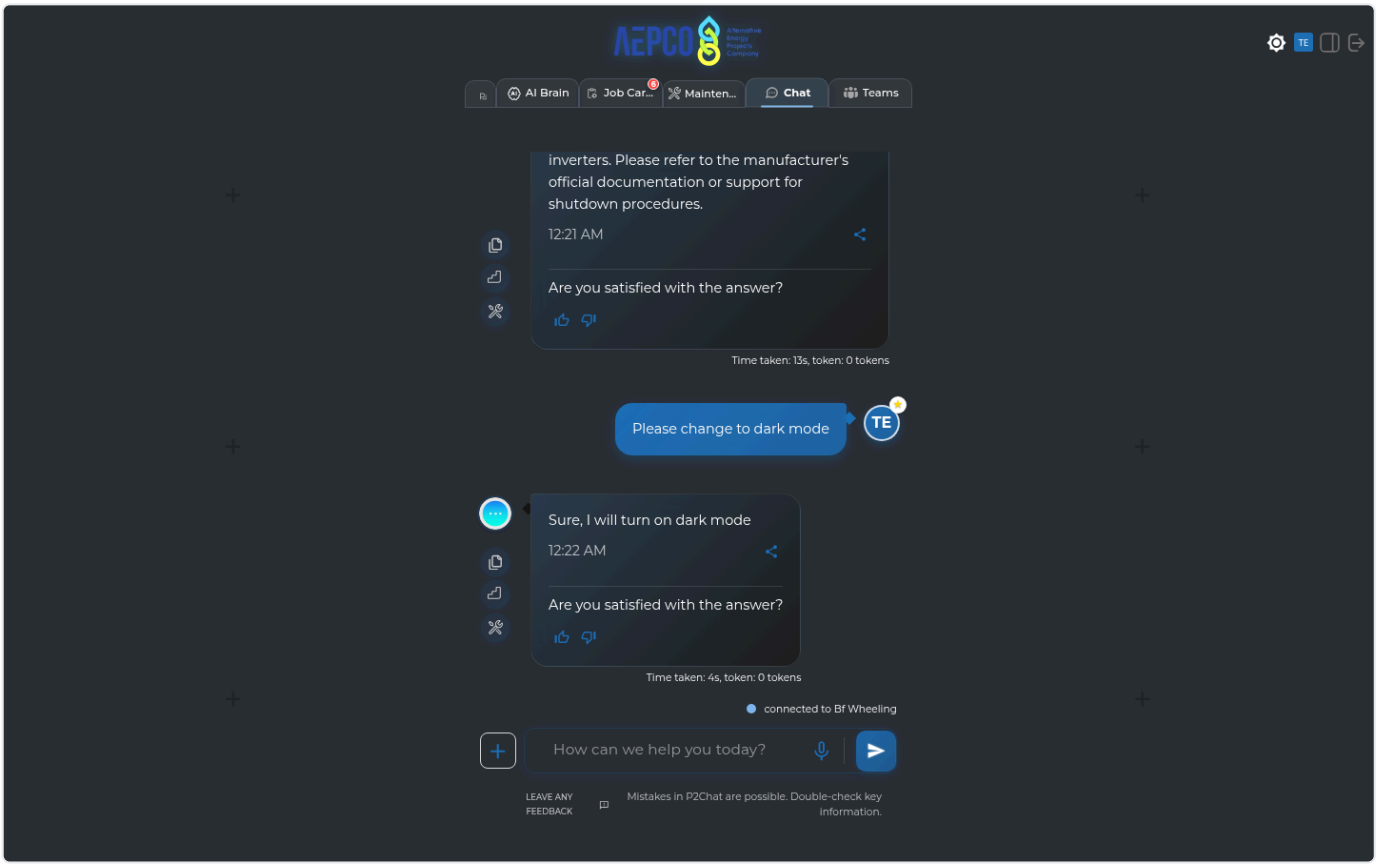
Production • 20.0s

Question: Please change to dark mode

Response: Sure, I will turn on dark mode 12:22 AM Are you satisfied with the answer? Time taken: 4s, token: 0 tokens connected to Bf Wheeling LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`ui_agent`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



141. BF_Wheeling

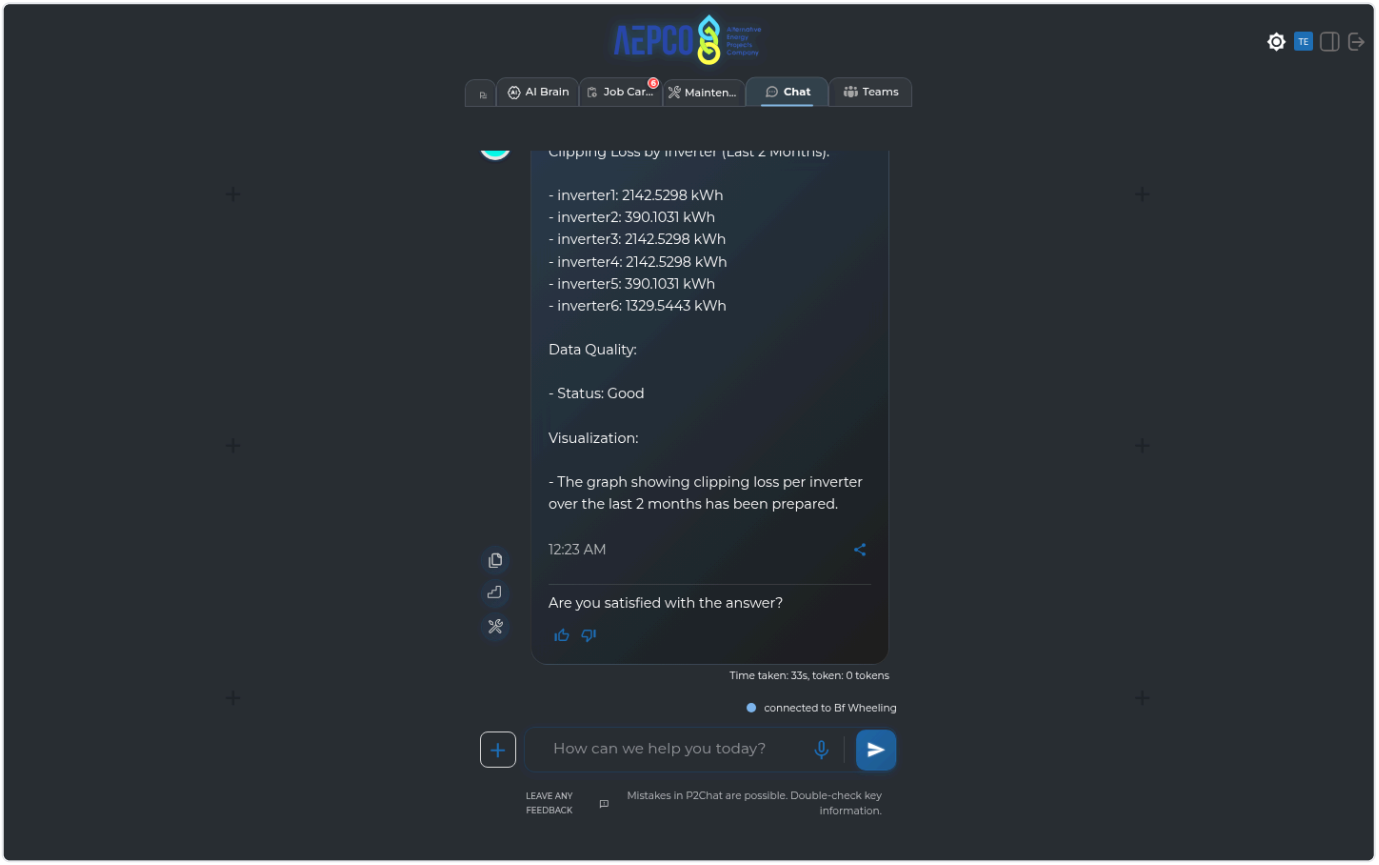
Production • 47.5s

Question: Please figure out how much clipping loss was found in the last 2 months broken by inverters, list them down and plot it as well

Response: Clipping Loss by Inverter (Last 2 Months): - inverter1: 2142.5298 kWh - inverter2: 390.1031 kWh - inverter3: 2142.5298 kWh - inverter4: 2142.5298 kWh - inverter5: 390.1031 kWh - inverter6: 1329.5443 kWh Data Quality: - Status: Good Visualization: - The graph showing clipping loss per inverter over the last 2 months has been prepared. 12:23 AM Are you satisfied with the answer? Time taken: 33s, token: 0 tokens connected to Bf Wheeling LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent, graph_agent`
Tools used	`functions.summarize_data_problems_ag, functions.inspect_xpool, functions.clipping_loss_via_requests, functions.calculate_clipping_or_curtailment_losses_ag`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



142. BF_Wheeling

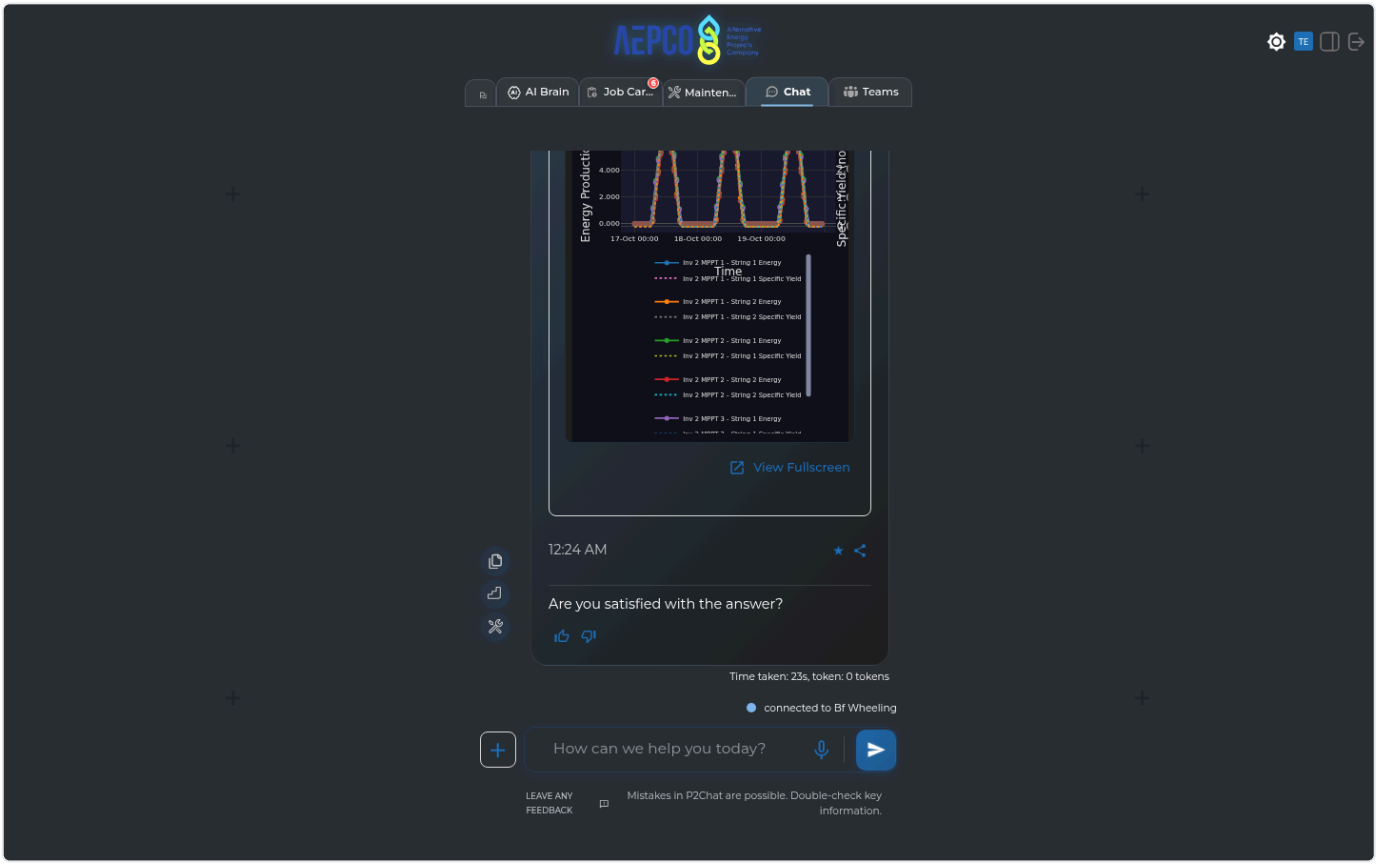
Production • 38.3s

Question: Please repeat the first plotting question, but interchange it with Specific yield on 1 axes, and energy production on the other axes on a single dual axes plot (Memory Test)

Response: I've created the visualization for you. View Fullscreen 12:24 AM ★ Are you satisfied with the answer? Time taken: 23s, token: 0 tokens connected to Bf Wheeling LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.extract_timeseries_data_ag, functions.execute_custom_code, multi_tool_use.parallel, functions.generate_graphs_via_code`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



143. Abdali_Mall_Lahoun

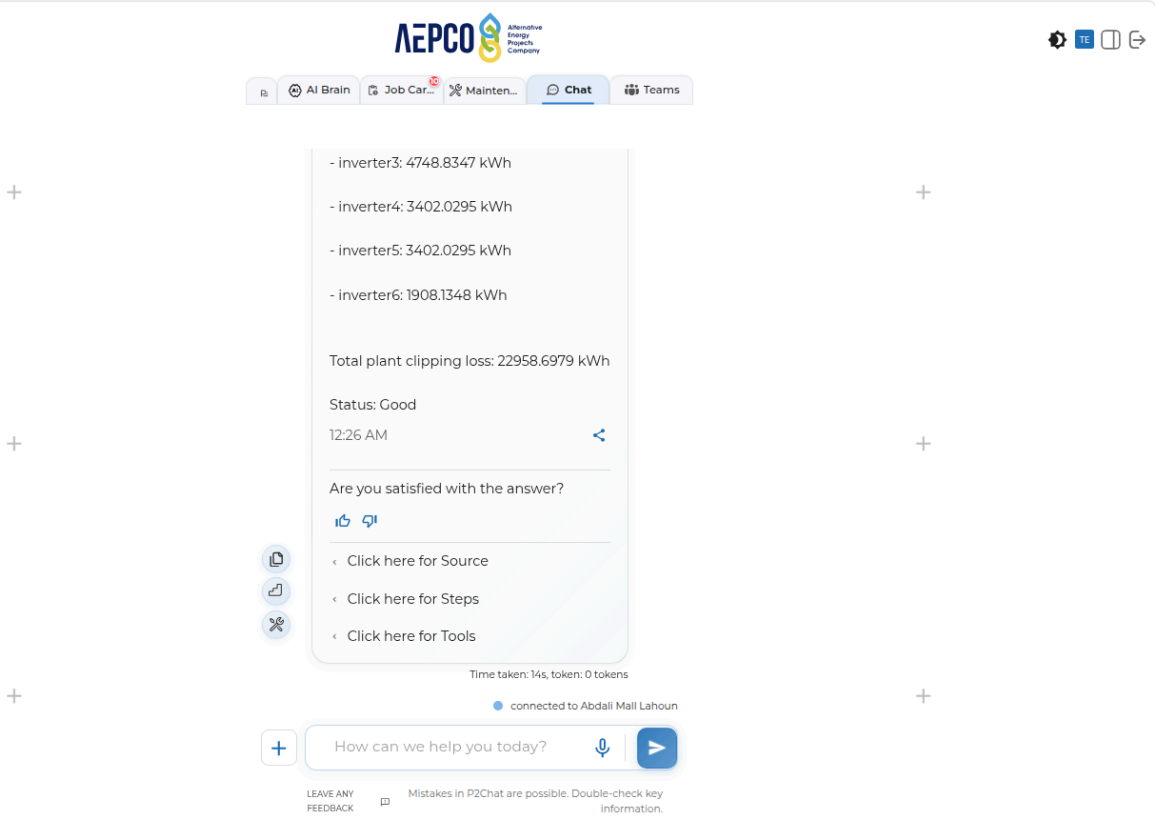
Production • 29.5s

Question: Use 'clipping_loss_via_requests' in order to calculate clipping loss if any for July 2026

Response: Clipping loss for July 2026: - inverter1: 4748.8347 kWh - inverter2: 4748.8347 kWh - inverter3: 4748.8347 kWh - inverter4: 3402.0295 kWh - inverter5: 3402.0295 kWh - inverter6: 1908.1348 kWh Total plant clipping loss: 22958.6979 kWh Status: Good 12:26 AM Are you satisfied with the answer? Click here for Source Click here for Steps Click here for Tools Time taken: 14s, token: 0 tokens connected to Abdali Mall Lahoun LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`functions.clipping_loss_via_requests`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



144. Abdali_Mall_Lahoun

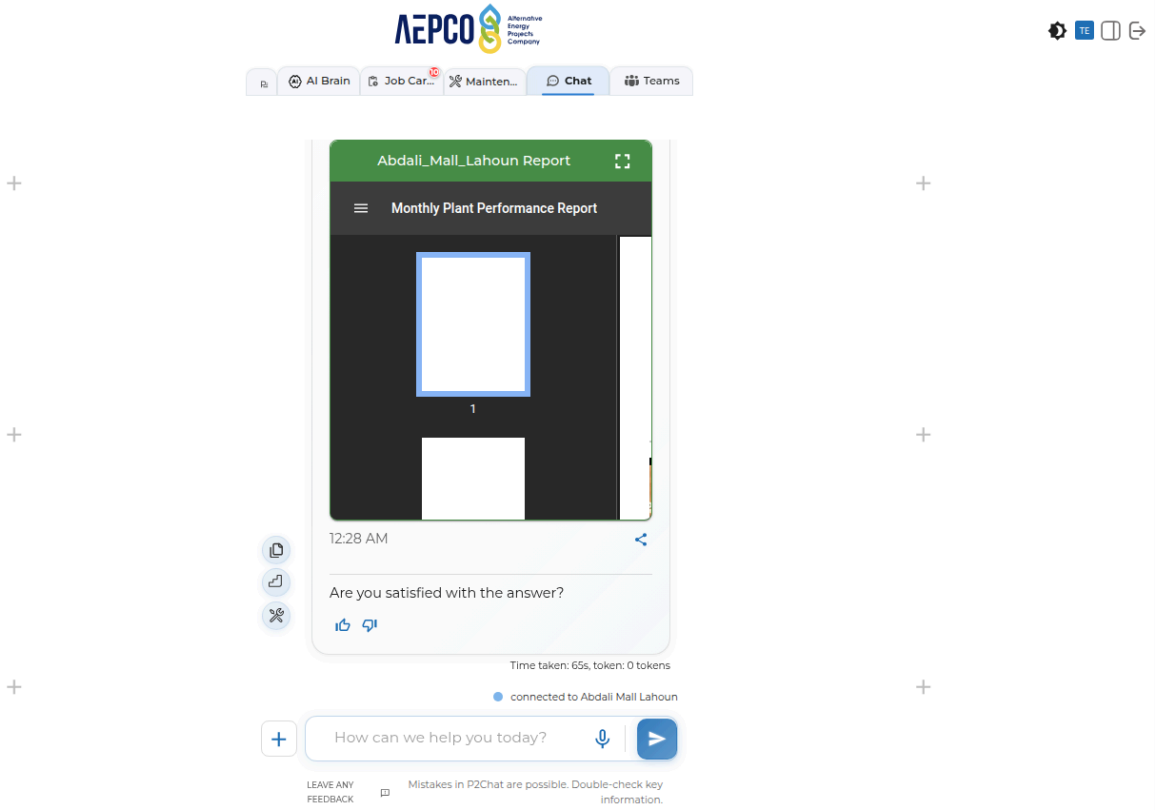
Production • 81.1s

Question: Please generate last months report

Response: I've generated the external report for September 2026. You can view it here: 'https://report-aepco-sydney.s3.ap-southeast-2.amazonaws.com/ABDM-L2_Project_September_2026_external_report.pdf' Abdali_Mall_Lahoun Report 12:28 AM Are you satisfied with the answer? Time taken: 65s, token: 0 tokens connected to Abdali Mall Lahoun LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`report_agent`
Tools used	`agent_external_report_tool`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



145. Abdali_Mall_Lahoun

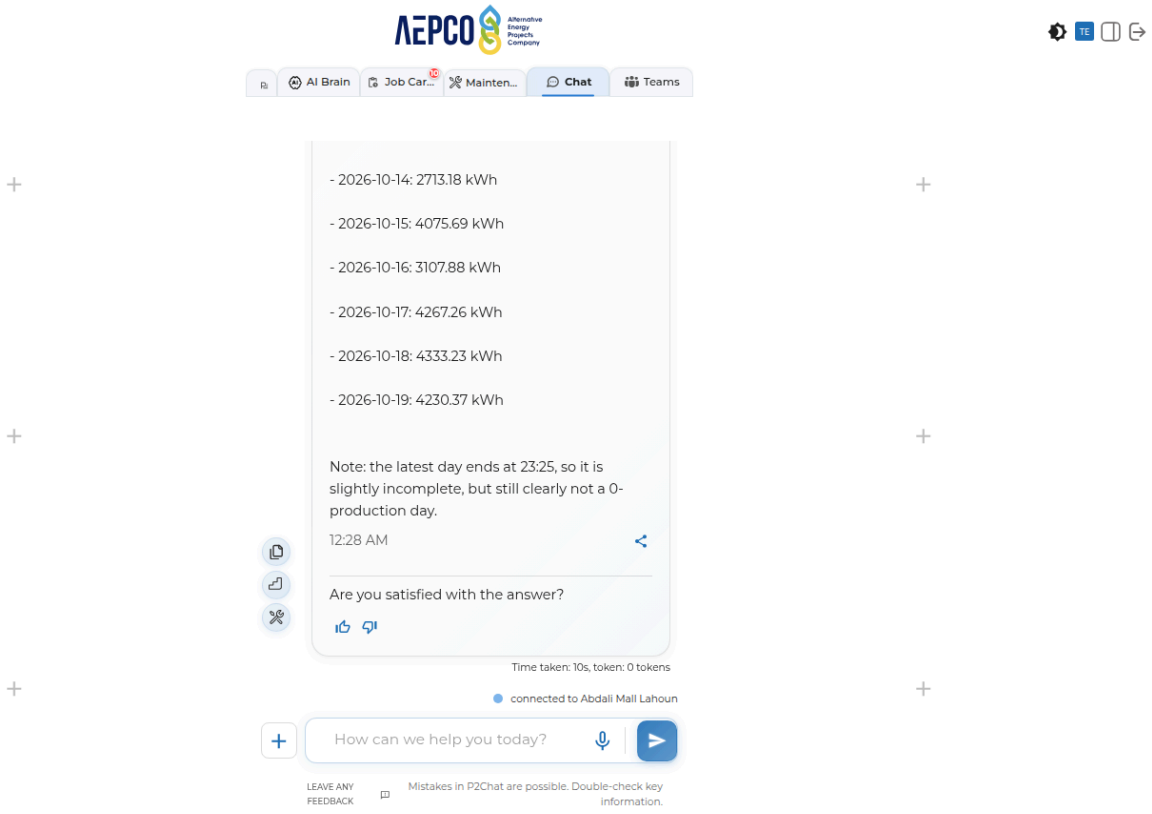
Production • 26.1s

Question: Was there any 0 production days in the last 7 days?

Response: No. There were no 0-production days in the last 7 days. Daily plant energy for 2026-10-13 to 2026-10-19: - 2026-10-13: 3479.91 kWh - 2026-10-14: 2713.18 kWh - 2026-10-15: 4075.69 kWh - 2026-10-16: 3107.88 kWh - 2026-10-17: 4267.26 kWh - 2026-10-18: 4333.23 kWh - 2026-10-19: 4230.37 kWh Note: the latest day ends at 23:25, so it is slightly incomplete, but still clearly not a 0-production day. 12:28 AM Are you satisfied with the answer? Time taken: 10s, token: 0 tokens connected to Abdali Mall Lahoun LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`functions.total_energy_func_ag`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



146. Abdali_Mall_Lahoun

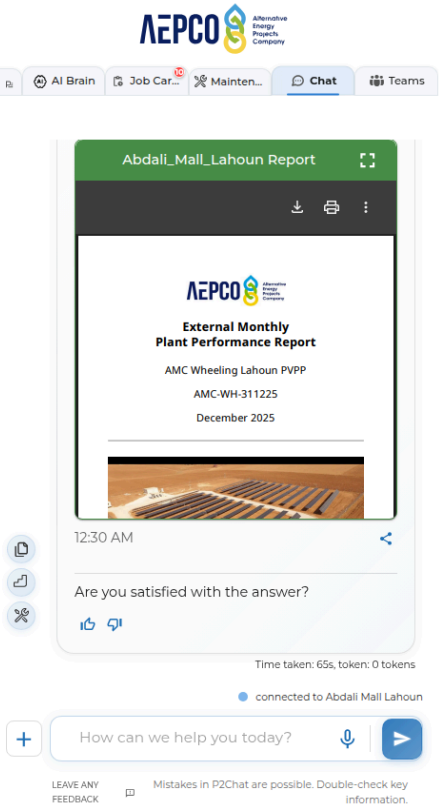
Production • 79.4s

Question: Please generate December 2025 performance report

Response: I've generated the external performance report for December 2025. You can view it here: 'https://report-aepco-sydney.s3.ap-southeast-2.amazonaws.com/ABDM-L2_Project_December_2025_external_report.pdf' Abdali_Mall_Lahoun Report 12:30 AM Are you satisfied with the answer? Time taken: 65s, token: 0 tokens connected to Abdali Mall Lahoun LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`report_agent`
Tools used	`agent_external_report_tool`
API calls made	`3`
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	`0`
Network failures	`0`



147. Abdali_Mall_Lahoun

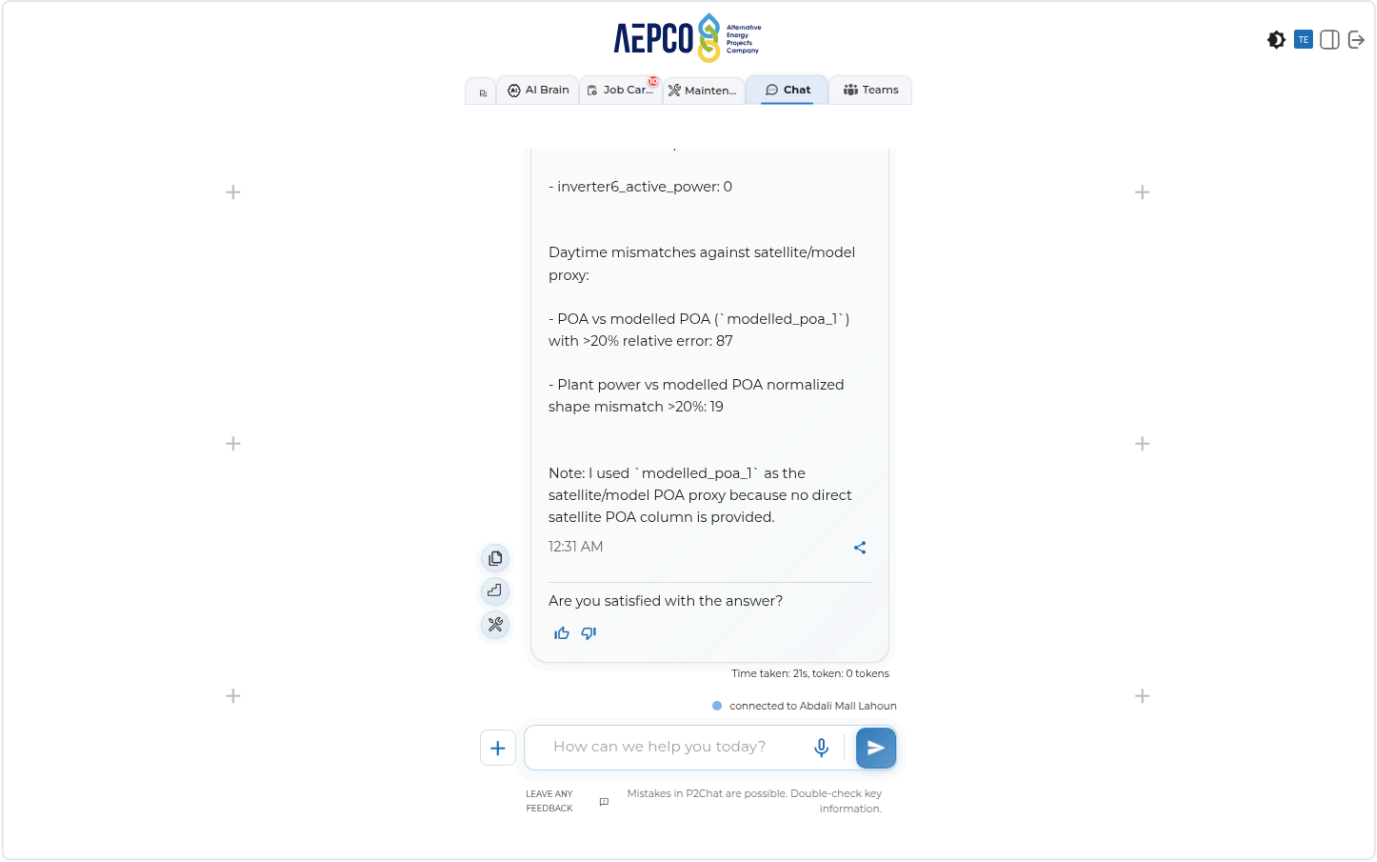
Production • 36.8s

Question: How is my data quality for POA and power? How many of them show missing data during daytime or mismatches with satellite prediction?

Response: For 2026-10-19: - Daytime data points evaluated: 134 - Missing daytime POA points (`pyranometer_1`): 0 - Missing daytime plant power points (`active_power`): 0 - Missing daytime combined plant power points (plant power missing or all inverter powers missing): 0 Daytime missing inverter power points: - inverter1_active_power: 0 - inverter2_active_power: 0 - inverter3_active_power: 0 - inverter4_active_power: 0 - inverter5_active_power: 0 - inverter6_active_power: 0 Daytime mismatches against satellite/model proxy: - POA vs modelled POA (`modelled_poa_1`) with >20% relative error: 87 - Plant power vs modelled POA normalized shape mismatch >20%: 19 Note: I used `modelled_poa_1` as the satellite/model POA proxy because no direct satellite POA column is provided. 12:31 AM Are you satisfied with the answer? Time taken: 21s, token: 0 tokens connected to Abdali Mall Lahoun LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`extract_timeseries_data_ag, execute_custom_code`
API calls made	`1`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



148. Abdali_Mall_Lahoun

Production • 34.7s

Question: Please plot last 3 days hourly performance ratio for my MPPTs for inverter 2

Response: I've created the visualization for you. View Fullscreen 12:31 AM ★ Are you satisfied with the answer? Time taken: 19s, token: 0 tokens connected to Abdali Mall Lahoun LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.generate_graphs_via_code, functions.perf_ratio_ag`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0

149. Abdali_Mall_Lahoun

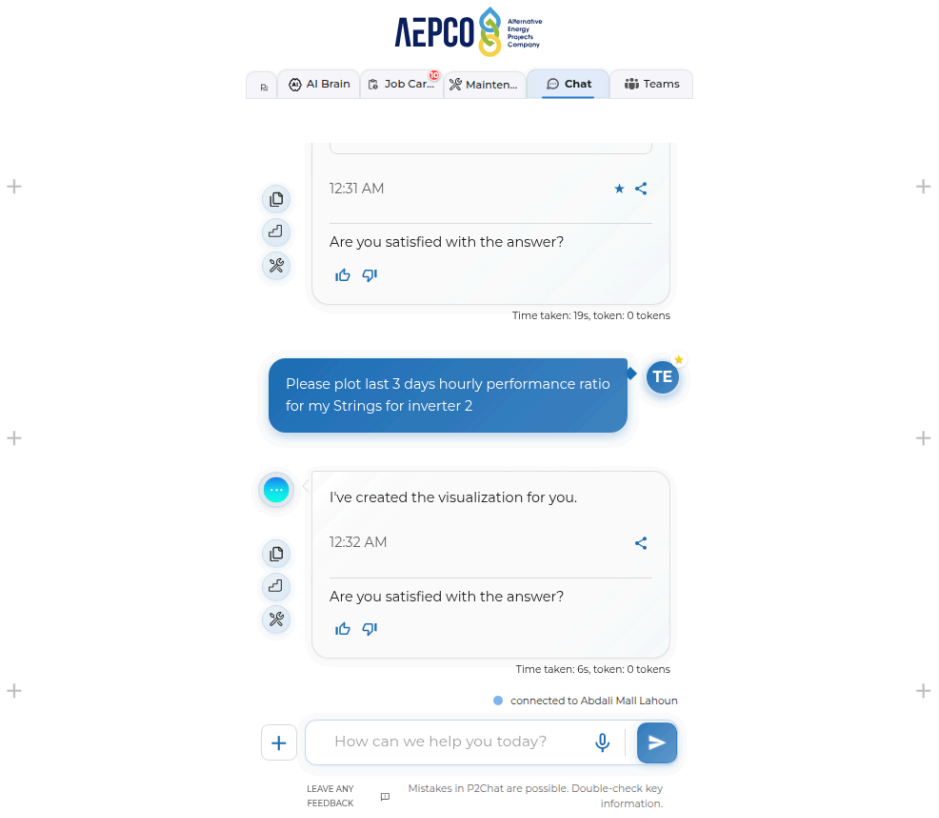
Production • 21.6s

Question: Please plot last 3 days hourly performance ratio for my Strings for inverter 2

Response: I've created the visualization for you. 12:32 AM Are you satisfied with the answer? Time taken: 6s, token: 0 tokens connected to Abdali Mall Lahoun LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.perf_ratio_ag`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



150. Abdali_Mall_Lahoun

Production • 30.9s

Question: Please plot the last 4 days hourly performance ratio for all my inverters

Response: I've created the visualization for you. View Fullscreen 12:33 AM ★ Are you satisfied with the answer? Time taken: 15s, token: 0 tokens connected to Abdali Mall Lahoun LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.generate_graphs_via_code, functions.perf_ratio_ag`
API calls made	2
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	0
Network failures	0

151. Abdali_Mall_Lahoun

Production • 65.8s

Question: Please figure out how much clipping loss was found in the last 6 months broken by inverters, list them down and plot it as well

Response: Clipping Loss Summary (2026-04-20 to 2026-10-20): - Inverter 1: 19,735.0715 kWh - Inverter 2: 19,700.5409 kWh - Inverter 3: 19,740.8899 kWh - Inverter 4: 14,005.5831 kWh - Inverter 5: 14,004.6920 kWh - Inverter 6: 7,908.7403 kWh Data Quality Status: Good Visualization: - A bar chart showing monthly clipping loss for each inverter over the last six months has been created. Clipping Loss Summary (2026-04-20 to 2026-10-20): - Inverter 1: 19,735.0715 kWh - Inverter 2: 19,700.5409 kWh - Inverter 3: 19,740.8899 kWh - Inverter 4: 14,005.5831 kWh - Inverter 5: 14,004.6920 kWh - Inverter 6: 7,908.7403 kWh Data Quality Status: Good Visualization: - A bar chart showing monthly clipping loss for each inverter over the last six months has been created. View Fullscreen 12:34 AM ★ Are you satisfied with the answer? Time taken: 51s, token: 0 tokens connected to Abdali Mall Lahoun LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key infor

TECHNICAL EVIDENCE

Agents used	`data_agent, graph_agent`
Tools used	`functions.calculate_clipping_or_curtailment_losses_ag, functions.clipping_loss_via_requests, functions.generate_graphs_via_code`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`

152. Abdali_Mall_Lahoun

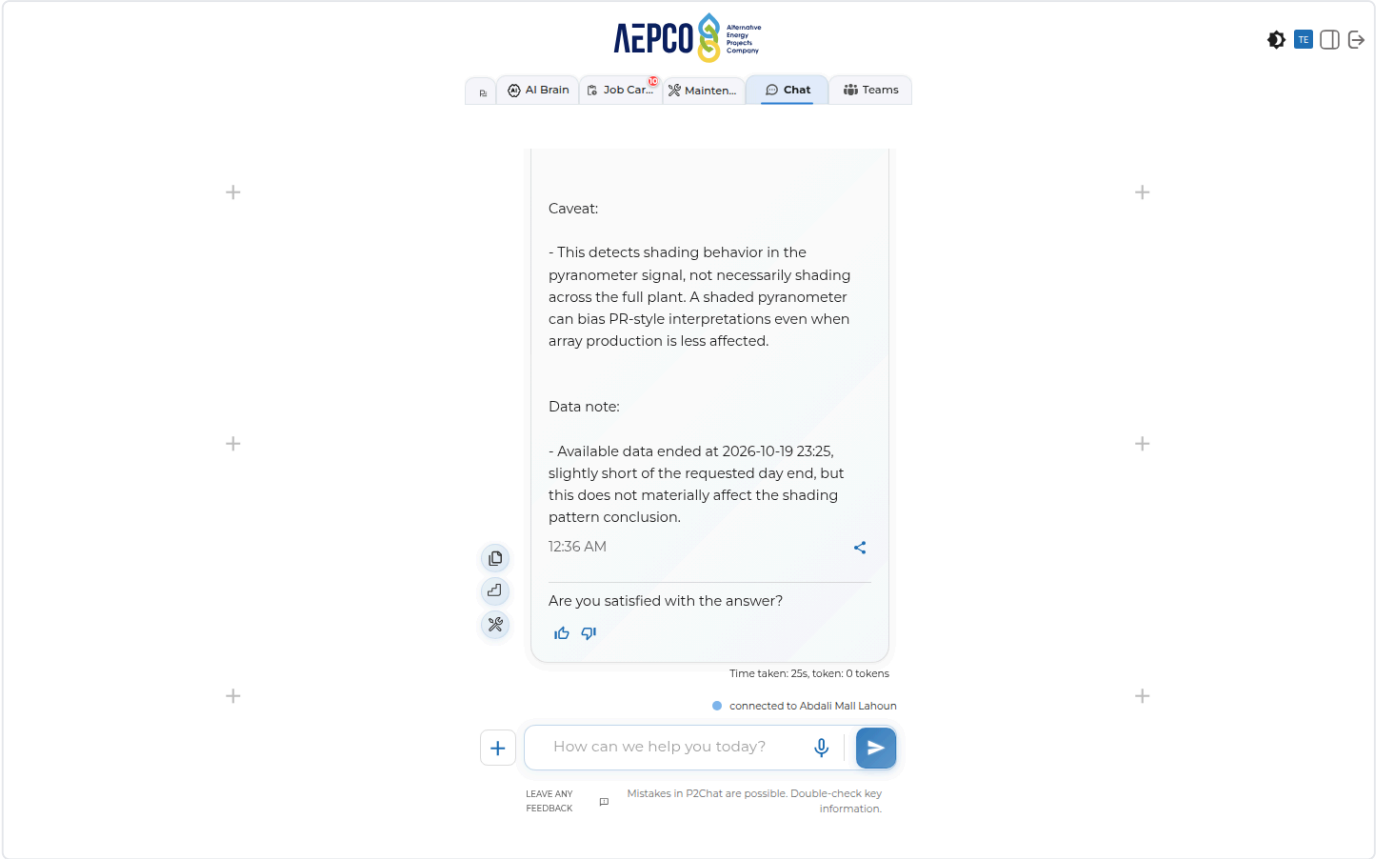
Production • 40.3s

Question: Is there shading happening with my plant? Please analyze the pyranometer and check for frequent dips at certain times

Response: I analyzed the last 30 days of pyranometer data (2026-09-20 to 2026-10-19, raw 5-min data) against modelled POA and normalized by solar azimuth to look for recurring time-of-day dips consistent with fixed-object shading. Findings: - There is a clear recurring morning dip window from 07:10 to 08:20. - In that window, the average dip frequency is 35.0% of sampled days/timestamps. - The worst 10th-percentile normalized irradiance in that window is 0.583, meaning the pyranometer sometimes drops to about 58% of its typical value for that sun position. - The median normalized irradiance across the full window is 0.952, so this is intermittent/moderate rather than constant severe shading. There is also a recurring late-afternoon dip window: - 15:55 to 17:05 - Average dip frequency: 33.7% - Worst 10th-percentile normalized irradiance: 0.426 - Median normalized irradiance across the window: 0.977 Top recurring dip times in the morning: - 07:10: dip frequency 39.1%, p1

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`functions.execute_custom_code, functions.extract_timeseries_data_ag`
API calls made	2
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	0
Network failures	0



153. Abdali_Mall_Lahoun

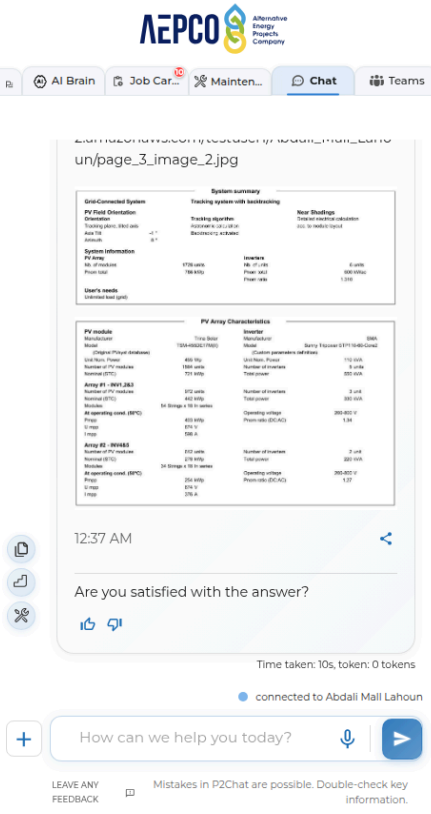
Production • 26.2s

Question: Based on pvsyst report, what is the rated power of this plant and array configuration

Response: Found 2 relevant PVsyst report sections: 1. [Abdali_Mall_Lahoun] System information (Page 2, Table 2) 2. [Abdali_Mall_Lahoun] PV Array Characteristics (Page 3, Table 2) Please access the URLs provided to view the detailed information. Here is the reference: https://image-pvsyst-aepco-sydney.s3.ap-southeast-2.amazonaws.com/testuser1/Abdali_Mall_Lahoun/page_2_image_2.jpg https://image-pvsyst-aepco-sydney.s3.ap-southeast-2.amazonaws.com/testuser1/Abdali_Mall_Lahoun/page_3_image_2.jpg 12:37 AM Are you satisfied with the answer? Time taken: 10s, token: 0 tokens connected to Abdali Mall Lahoun LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`pvsyst_agent`
Tools used	`pvsyst_tool`
API calls made	3
Last API call	`GET /testuser2/chat/ → 404`
Console errors	3
Network failures	1



154. Abdali_Mall_Lahoun

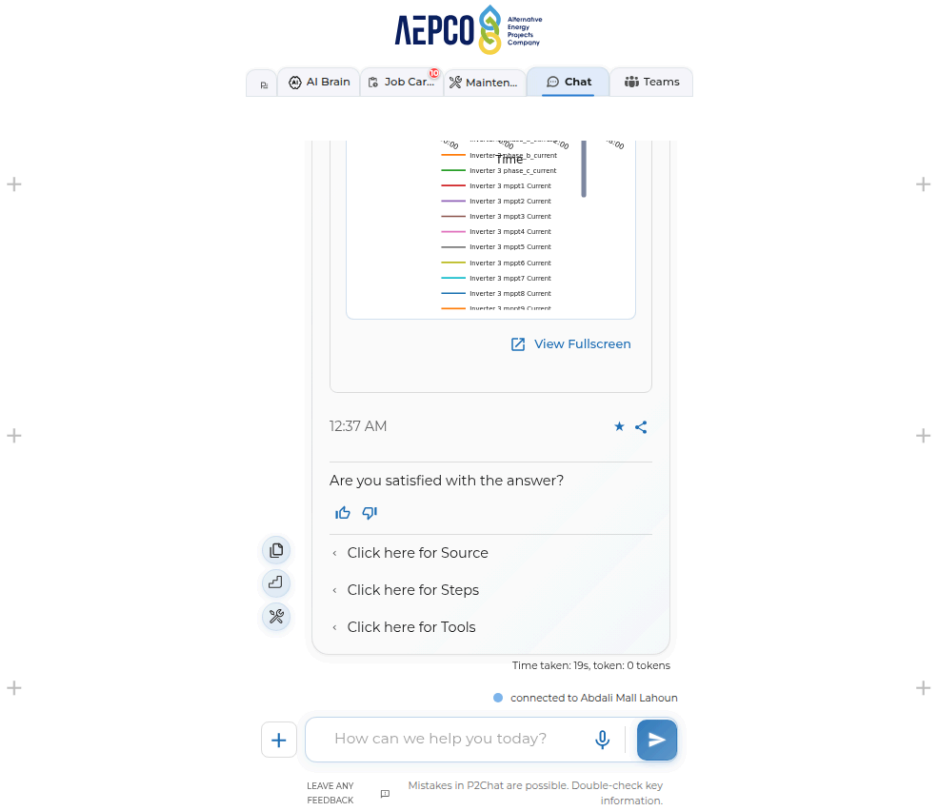
Production • 34.6s

Question: Please plot my current and voltage in a dual axes plot on the deepest possible granularity level for inverter3

Response: I've created the visualization for you. View Fullscreen 12:37 AM ★ Are you satisfied with the answer? Click here for Source Click here for Steps Click here for Tools Time taken: 19s, token: 0 tokens connected to Abdali Mall Lahoun LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.generate_graphs_via_code, functions.execute_custom_code, functions.extract_timeseries_data_ag`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



155. Abdali_Mall_Lahoun

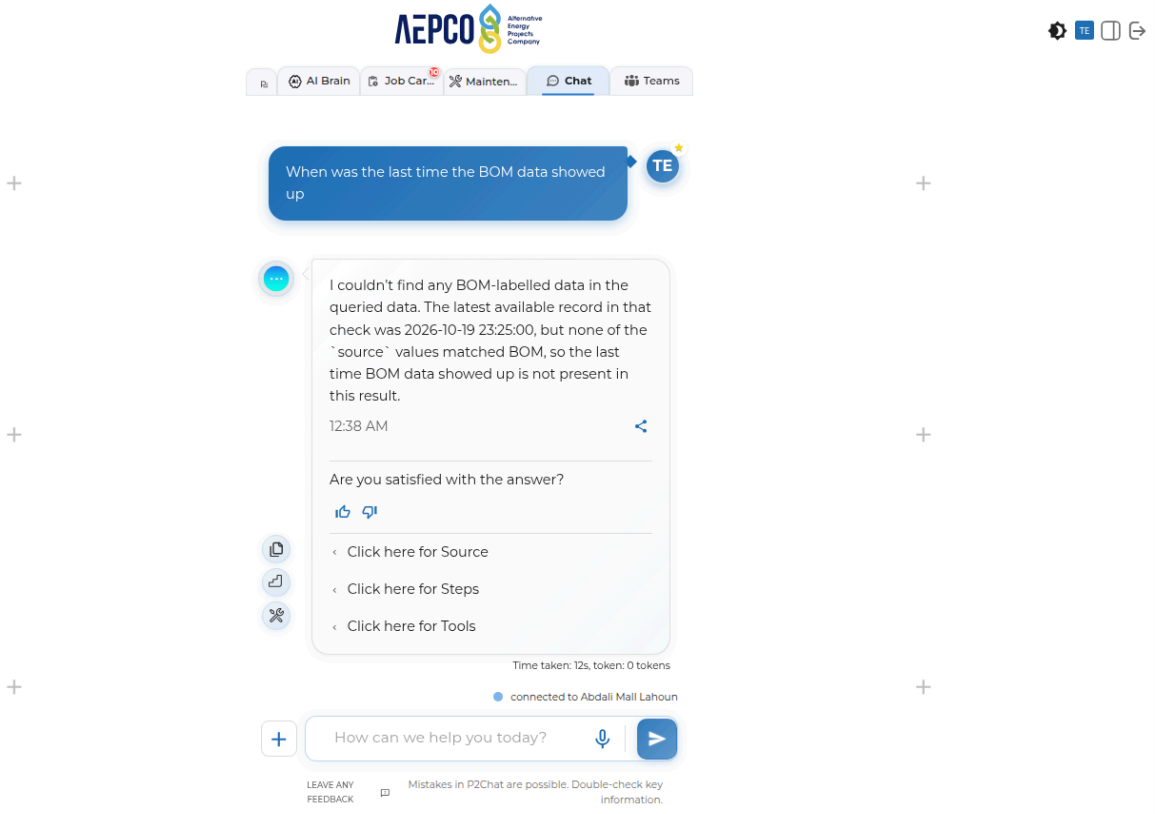
Production • 26.1s

Question: When was the last time the BOM data showed up

Response: I couldn't find any BOM-labelled data in the queried data. The latest available record in that check was 2026-10-19 23:25:00, but none of the `source` values matched BOM, so the last time BOM data showed up is not present in this result. 12:38 AM Are you satisfied with the answer? Click here for Source Click here for Steps Click here for Tools Time taken: 12s, token: 0 tokens connected to Abdali Mall Lahoun LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`execute_custom_code, extract_timeseries_data_ag`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



156. Abdali_Mall_Lahoun

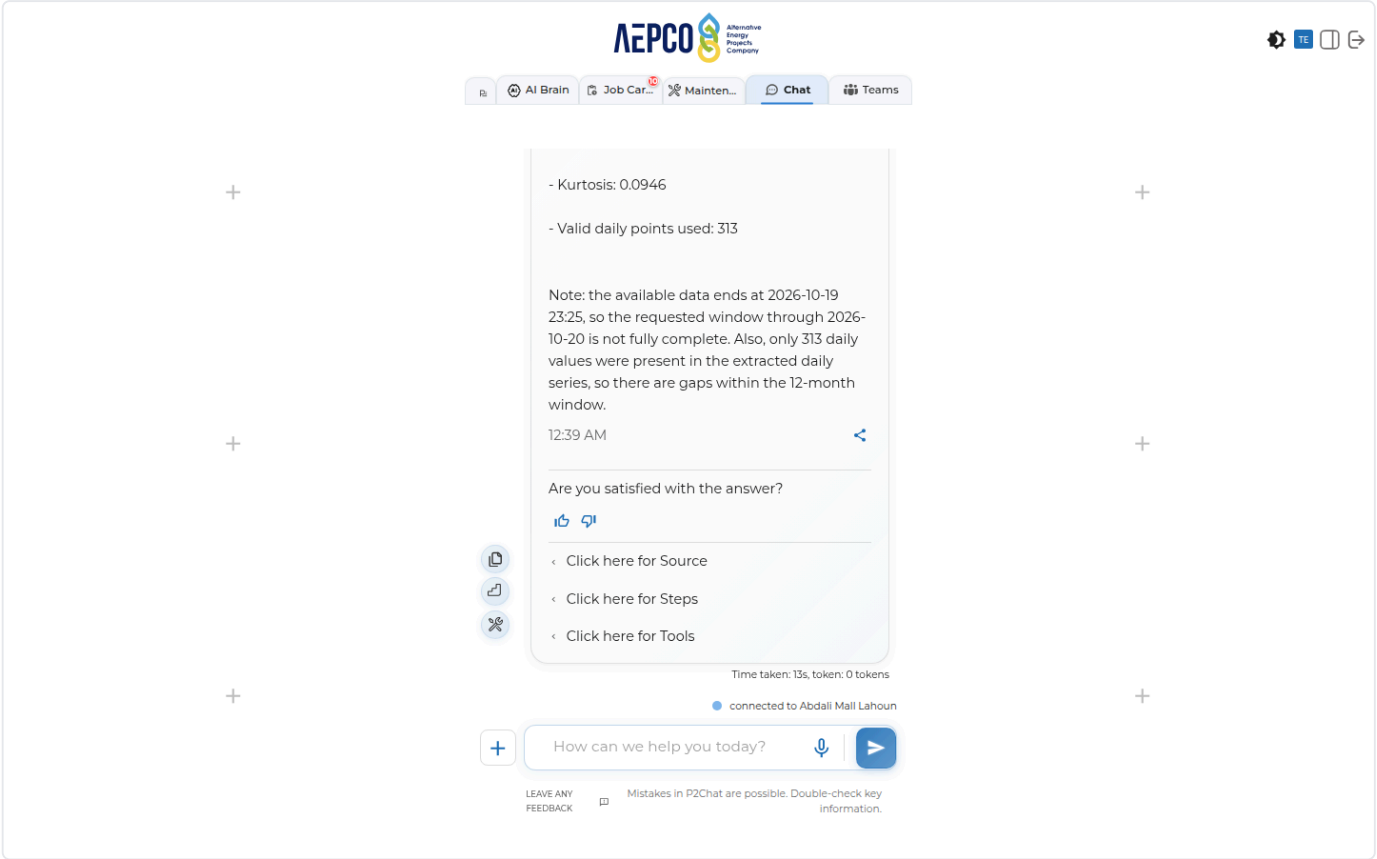
Production • 29.1s

Question: Please find out the daily POA poa summed up as a Wh/m^2, daily for the last 12 months and then calculate the median, mean and kurt

Response: Daily POA summed over the last 12 months (using `pyranometer_1`) gives: - Median: 5869.16 Wh/m²/day - Mean: 5860.43 Wh/m²/day - Kurtosis: 0.0946 - Valid daily points used: 313 Note: the available data ends at 2026-10-19 23:25, so the requested window through 2026-10-20 is not fully complete. Also, only 313 daily values were present in the extracted daily series, so there are gaps within the 12-month window. 12:39 AM Are you satisfied with the answer? Click here for Source Click here for Steps Click here for Tools Time taken: 13s, token: 0 tokens connected to Abdali Mall Lahoun LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`execute_custom_code, extract_timeseries_data_ag`
API calls made	2
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	0
Network failures	0



157. Abdali_Mall_Lahoun

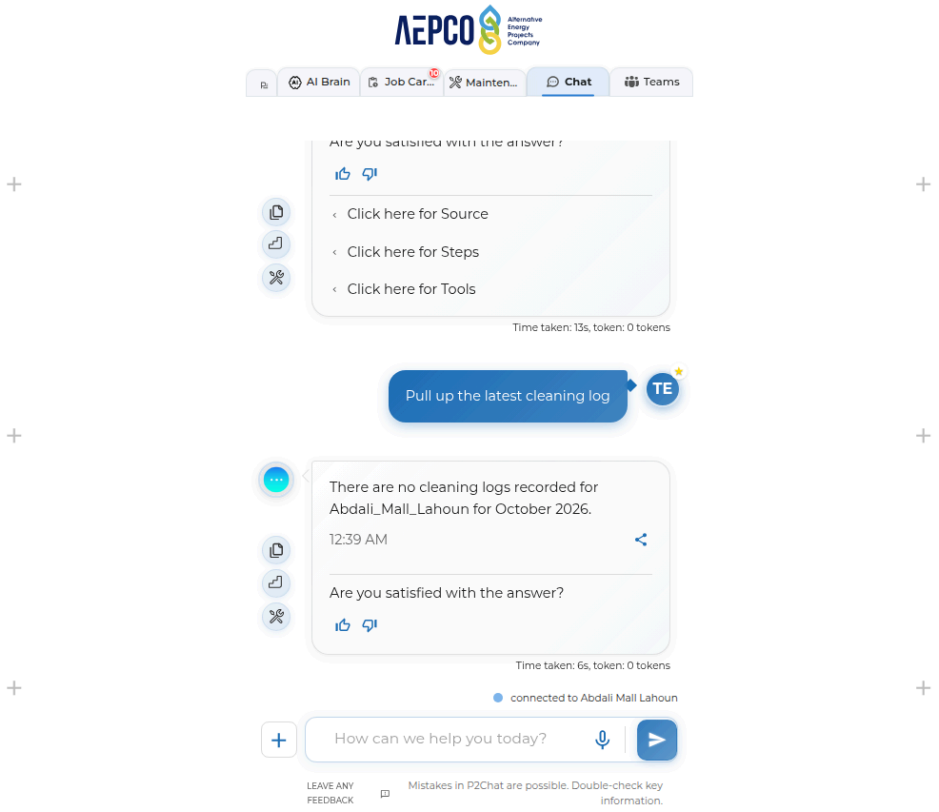
Production • 21.5s

Question: Pull up the latest cleaning log

Response: There are no cleaning logs recorded for Abdali_Mall_Lahoun for October 2026. 12:39 AM Are you satisfied with the answer? Time taken: 6s, token: 0 tokens connected to Abdali Mall Lahoun LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_entry_agent`
API calls made	`1`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



158. Abdali_Mall_Lahoun

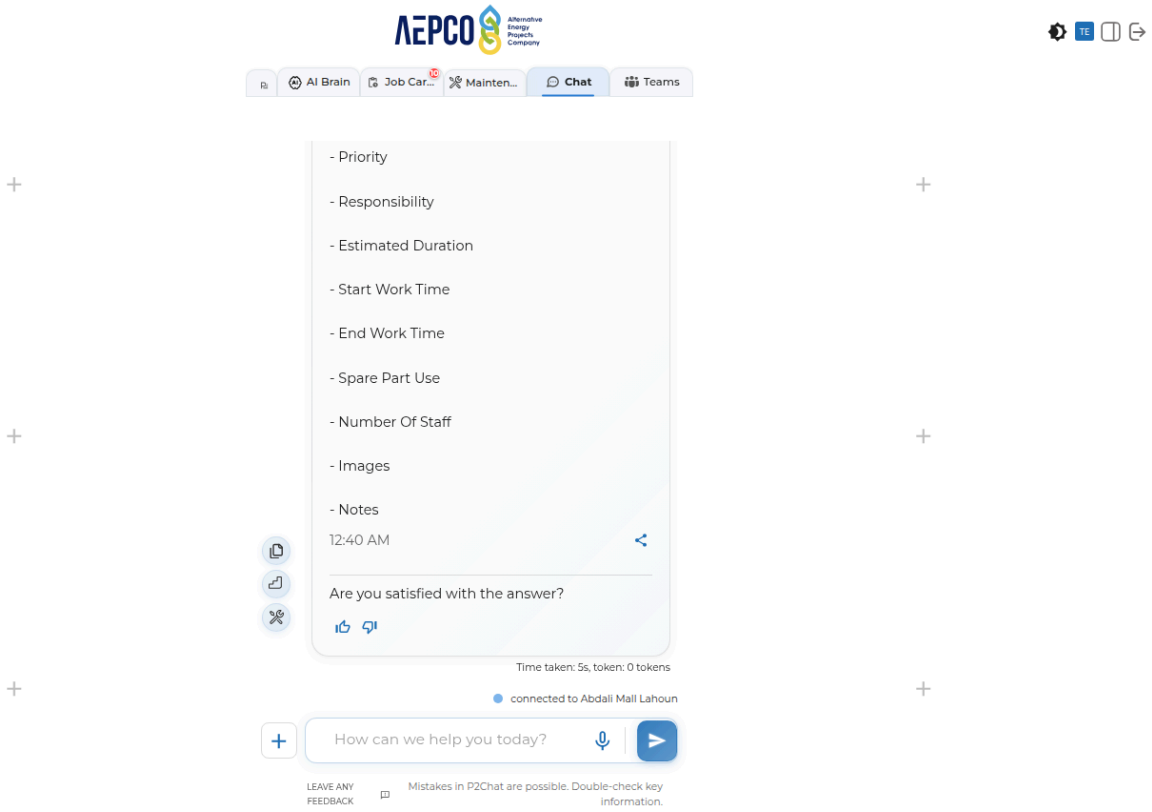
Production • 20.1s

Question: Jim cleaned the pyranometer today, can you add it to the cleaning log?

Response: To add a cleaning activity log, please provide the following required details: - Date - Fault Cause - Action Taken - Status Site will be set automatically from the connected plant. Optional fields you can also provide, but they are not required: - Activity Type (Inspection, Cleaning, HSE Incident, Spare Parts Usage, General Site Activity) - Priority - Responsibility - Estimated Duration - Start Work Time - End Work Time - Spare Part Use - Number Of Staff - Images - Notes 12:40 AM Are you satisfied with the answer? Time taken: 5s, token: 0 tokens connected to Abdali Mall Lahoun LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_entry_agent`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



159. Abdali_Mall_Lahoun

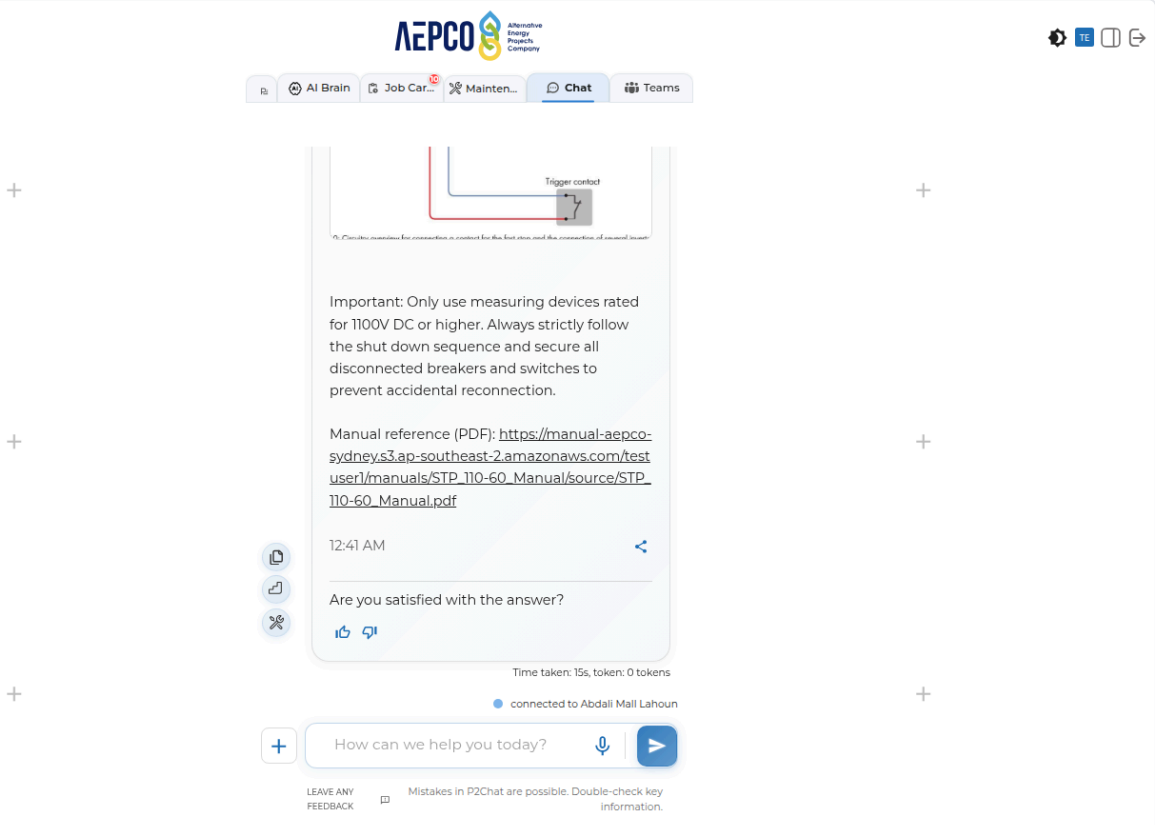
Production • 30.7s

Question: How can I shut down my inverters?

Response: To safely shut down your inverters, follow these steps as outlined in the manual: 1. Disconnect the AC miniature circuit breaker and secure it against reconnection. - See typical electrical set-up and breaker location: 2. Switch off all 4 DC load-break switches on the inverter and secure them against reconnection. - View labeled DC switches and connector positions: 3. Wait until all LEDs on the inverter have turned off, indicating it is no longer powered. 4. Use a current clamp to check that no current is present in the DC cables (reading should be 0A); refer to the current measuring position in the diagram above. 5. If required, disconnect the DC connector according to the alignment method shown here: If your system has a local or emergency stop function, verify the wiring of the LOCAL STOP input on the COM assembly using the digital input terminal diagram: - LOCAL STOP wiring and switch settings: Important: Only use measuring devices rated for 1100V DC or high

TECHNICAL EVIDENCE

Agents used	`manual_agent`
Tools used	`functions.manual_rag_tool`
API calls made	`2`
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	`0`
Network failures	`0`



160. Abdali_Mall_Lahoun

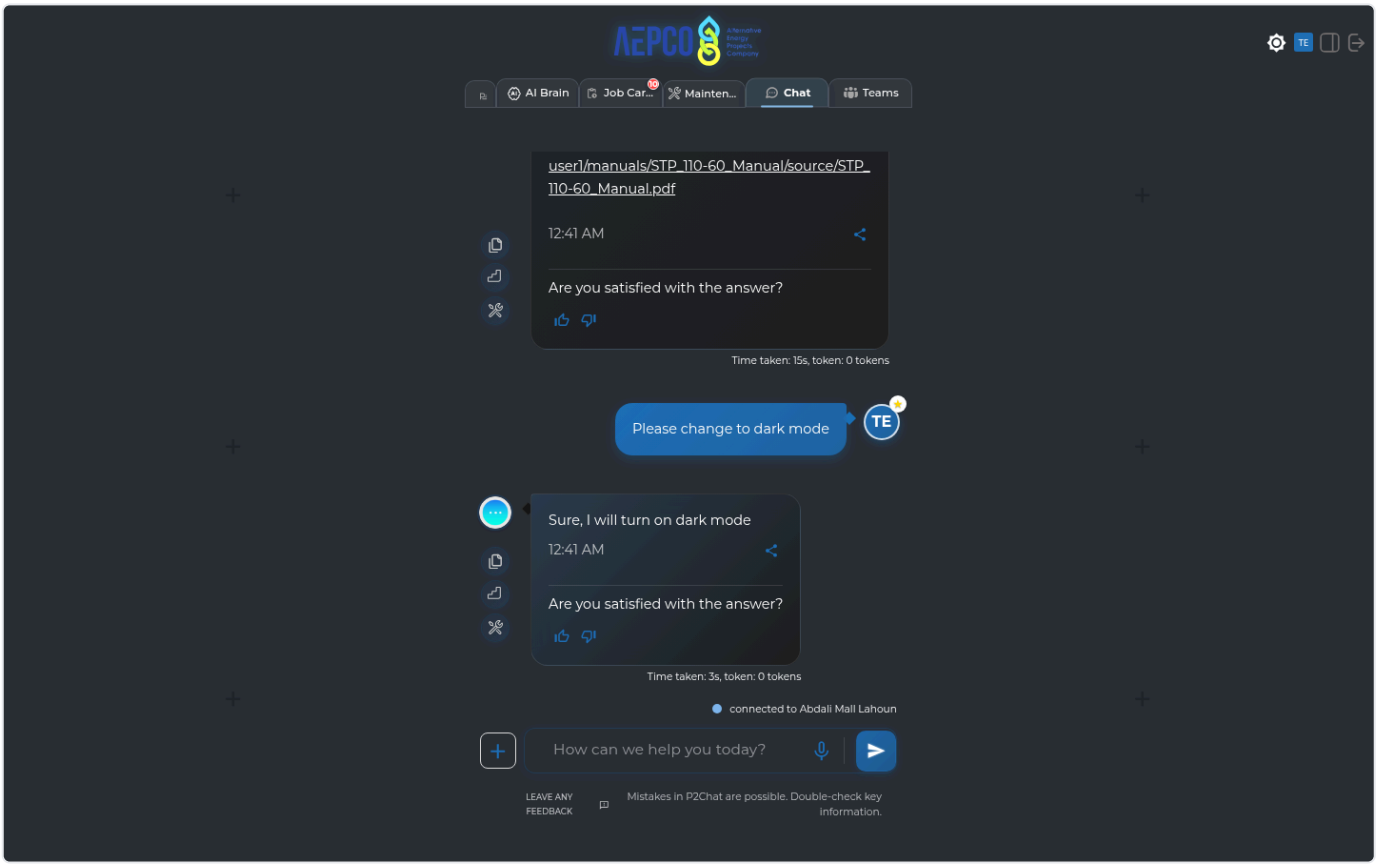
Production • 19.1s

Question: Please change to dark mode

Response: Sure, I will turn on dark mode 12:41 AM Are you satisfied with the answer? Time taken: 3s, token: 0 tokens connected to Abdali Mall Lahoun LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`ui_agent`
API calls made	`1`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



161. Abdali_Mall_Lahoun

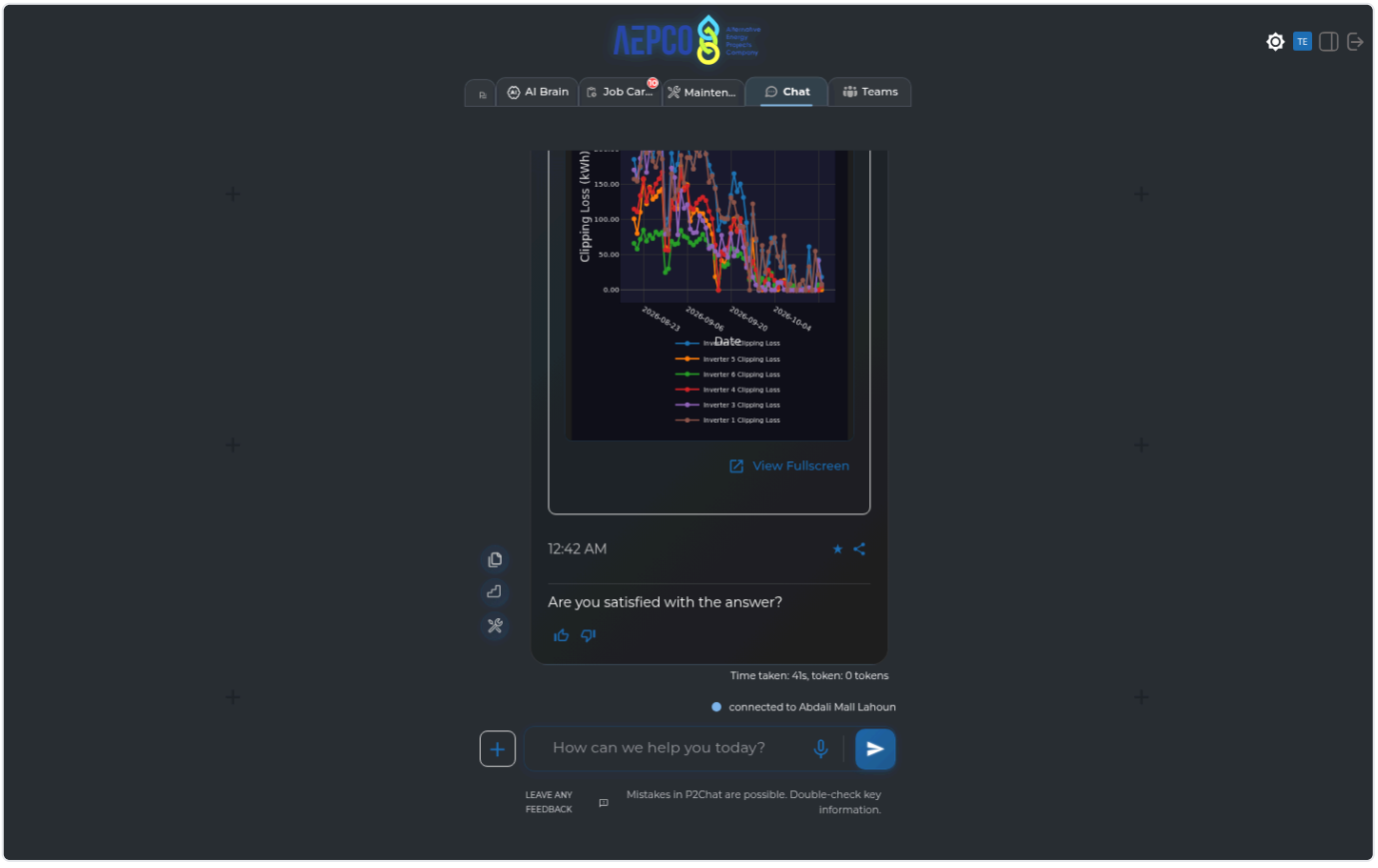
Production • 56.6s

Question: Please figure out how much clipping loss was found in the last 2 months broken by inverters, list them down and plot it as well

Response: Here are the total clipping losses over the last two months (2026-08-20 to 2026-10-20), broken down by inverter: - Inverter 1: 1544.19 kWh - Inverter 2: 1544.19 kWh - Inverter 3: 1544.19 kWh - Inverter 4: 866.78 kWh - Inverter 5: 865.89 kWh - Inverter 6: 565.52 kWh Data Quality Status: Good I've also created a line chart showing daily clipping loss per inverter for the last two months. Here are the total clipping losses over the last two months (2026-08-20 to 2026-10-20), broken down by inverter: - Inverter 1: 1544.19 kWh - Inverter 2: 1544.19 kWh - Inverter 3: 1544.19 kWh - Inverter 4: 866.78 kWh - Inverter 5: 865.89 kWh - Inverter 6: 565.52 kWh Data Quality Status: Good I've also created a line chart showing daily clipping loss per inverter for the last two months. View Fullscreen 12:42 AM ★ Are you satisfied with the answer? Time taken: 41s, token: 0 tokens connected to Abdali Mall Lahoun LEAVE ANY FEEDBACK Mistakes in P2Chat are po

TECHNICAL EVIDENCE

Agents used	`data_agent, graph_agent`
Tools used	`functions.generate_graphs_via_code, functions.calculate_clipping_or_curtailment_losses_ag, functions.clipping_loss_via_requests`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



162. Abdali_Mall_Lahoun

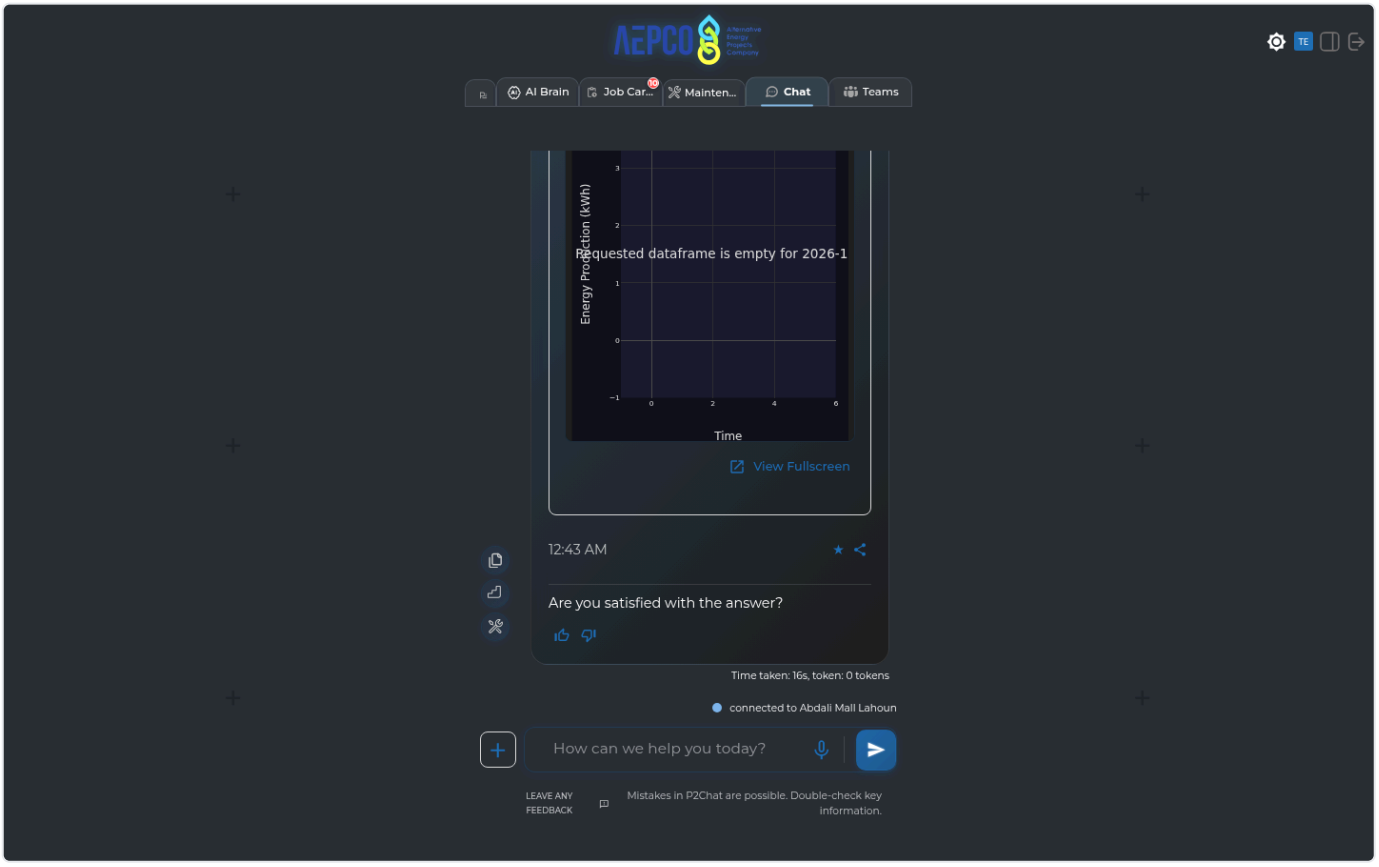
Production • 30.9s

Question: Please repeat the first plotting question, but interchange it with Specific yield on 1 axes, and energy production on the other axes on a single dual axes plot (Memory Test)

Response: I've created the visualization for you. View Fullscreen 12:43 AM ★ Are you satisfied with the answer? Time taken: 16s, token: 0 tokens connected to Abdali Mall Lahoun LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.generate_graphs_via_code, multi_tool_use.parallel`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



163. Abdali_Mall_Barza

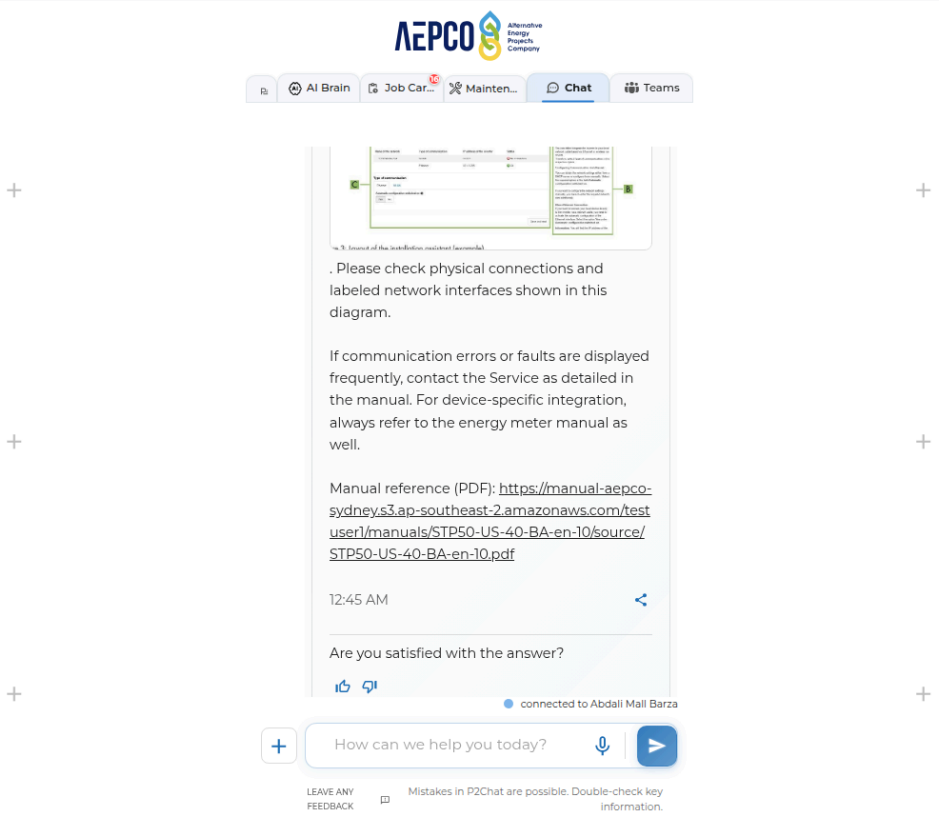
Production • 27.7s

Question: I have a COMMUNICATION_FAULT what do I do to my inverters?

Response: Troubleshooting steps for the COMMUNICATION_FAULT error code on inverters: 1. Ensure that the energy meter is correctly integrated into the same network as the inverter. Check all physical and network connections between the energy meter and the inverter, following the wiring or interface diagram. 2. For WLAN (wireless) connections: Improve the WLAN connection quality, for example, by positioning a WLAN repeater between the devices if needed. If possible, connect the inverter to the network using Ethernet for a more stable connection. 3. Confirm that the inverter is configured for the appropriate network settings. Refer to the user interface for network configuration shown in the diagram. For Ethernet, check the assigned IP address and that the status indicates "connected" and not "No connection." For WLAN, verify the connection status and assigned IP address as well. 4. If you are using automatic configuration, ensure the settings are correct. For direct Ethernet connections, veri

TECHNICAL EVIDENCE

Agents used	`manual_agent`
Tools used	`functions.manual_rag_tool`
API calls made	`11`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



164. Abdali_Mall_Barza

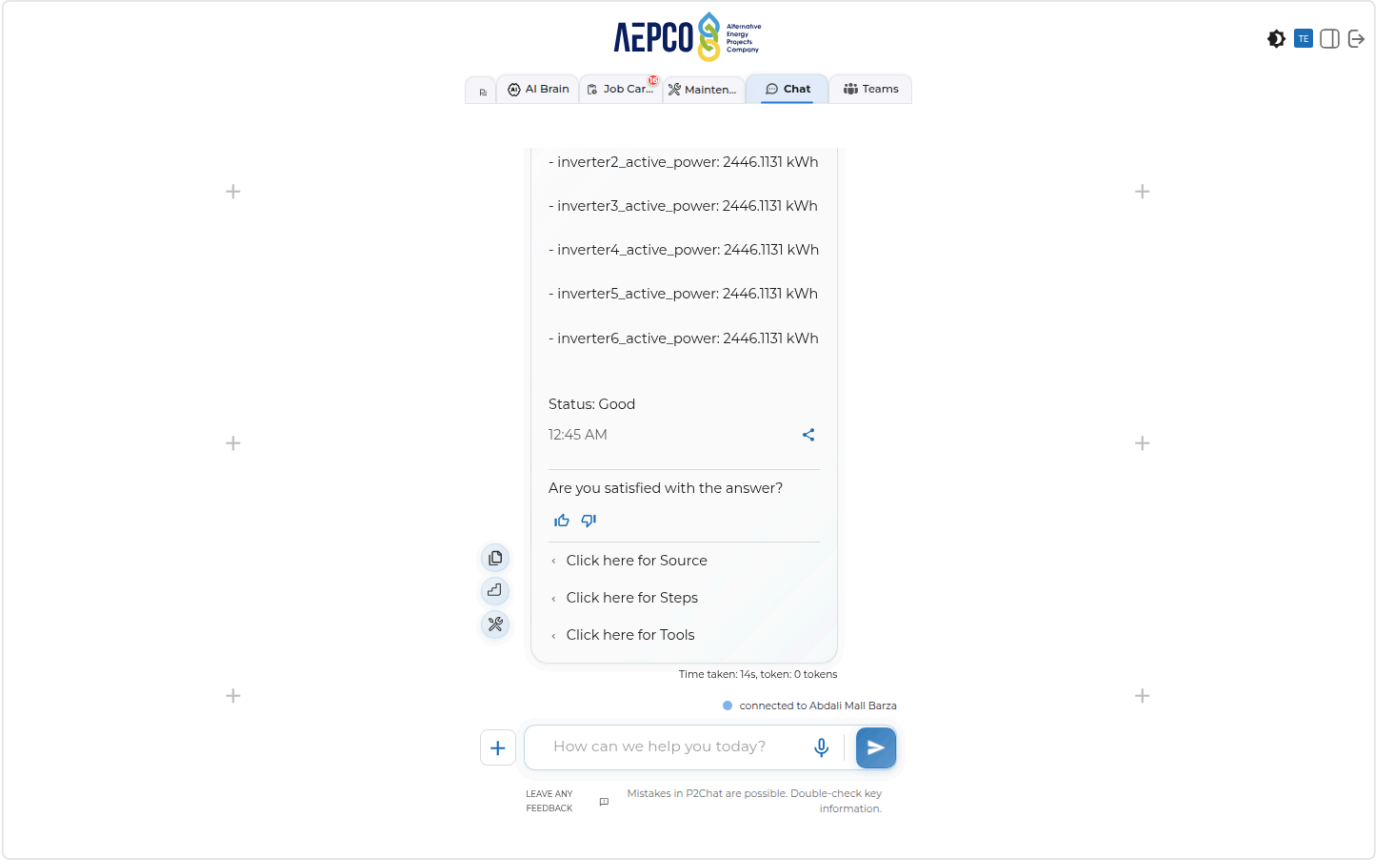
Production • 29.2s

Question: Use 'clipping_loss_via_requests' in order to calculate clipping loss if any for July 2026

Response: Clipping loss for July 2026: - inverter1_active_power: 2446.1131 kWh - inverter2_active_power: 2446.1131 kWh - inverter3_active_power: 2446.1131 kWh - inverter4_active_power: 2446.1131 kWh - inverter5_active_power: 2446.1131 kWh - inverter6_active_power: 2446.1131 kWh Status: Good 12:45 AM Are you satisfied with the answer? Click here for Source Click here for Steps Click here for Tools Time taken: 14s, token: 0 tokens connected to Abdali Mall Barza LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`functions.clipping_loss_via_requests`
API calls made	`2`
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	`0`
Network failures	`0`



165. Abdali_Mall_Barza

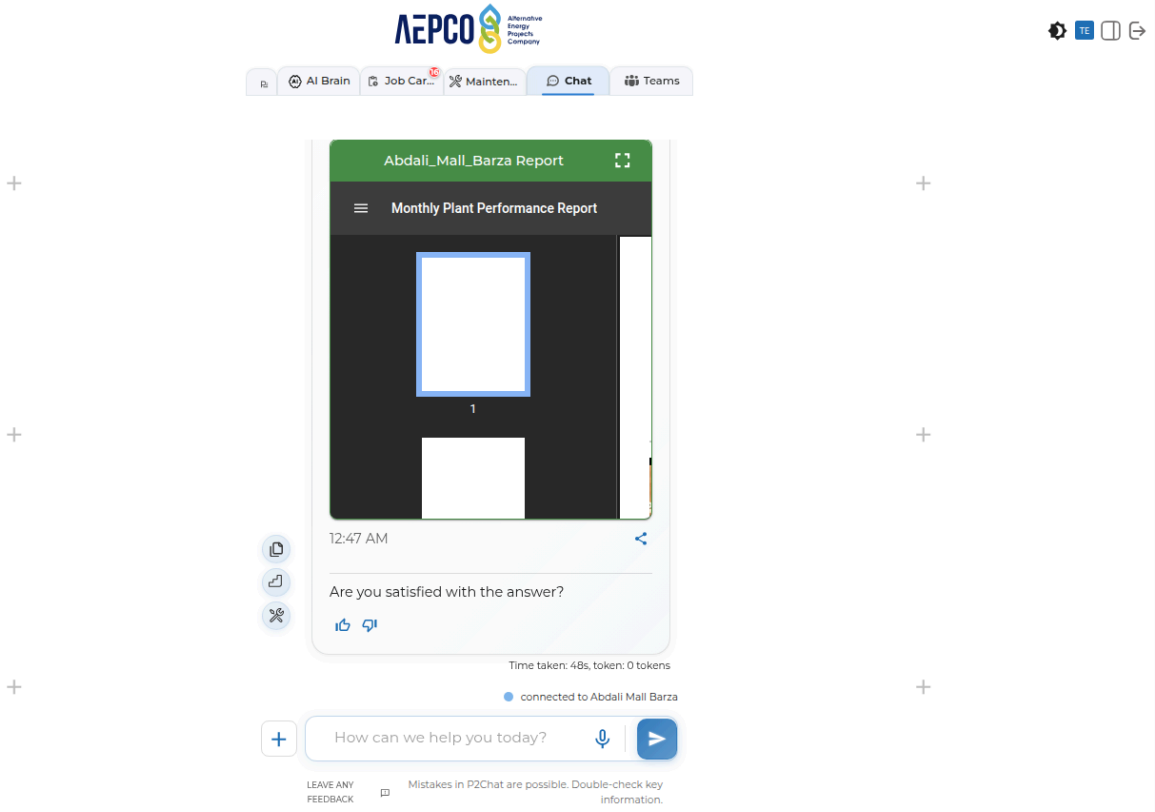
Production • 62.8s

Question: Please generate last months report

Response: I've generated the external report for September 2026. You can view it here: 'https://report-aepco-sydney.s3.ap-southeast-2.amazonaws.com/AMC-Abdali_Mall-Wheeling_September_2026_external_report.pdf' Abdali_Mall_Barza Report 12:47 AM Are you satisfied with the answer? Time taken: 48s, token: 0 tokens connected to Abdali Mall Barza LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`report_agent`
Tools used	`agent_external_report_tool`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



166. Abdali_Mall_Barza

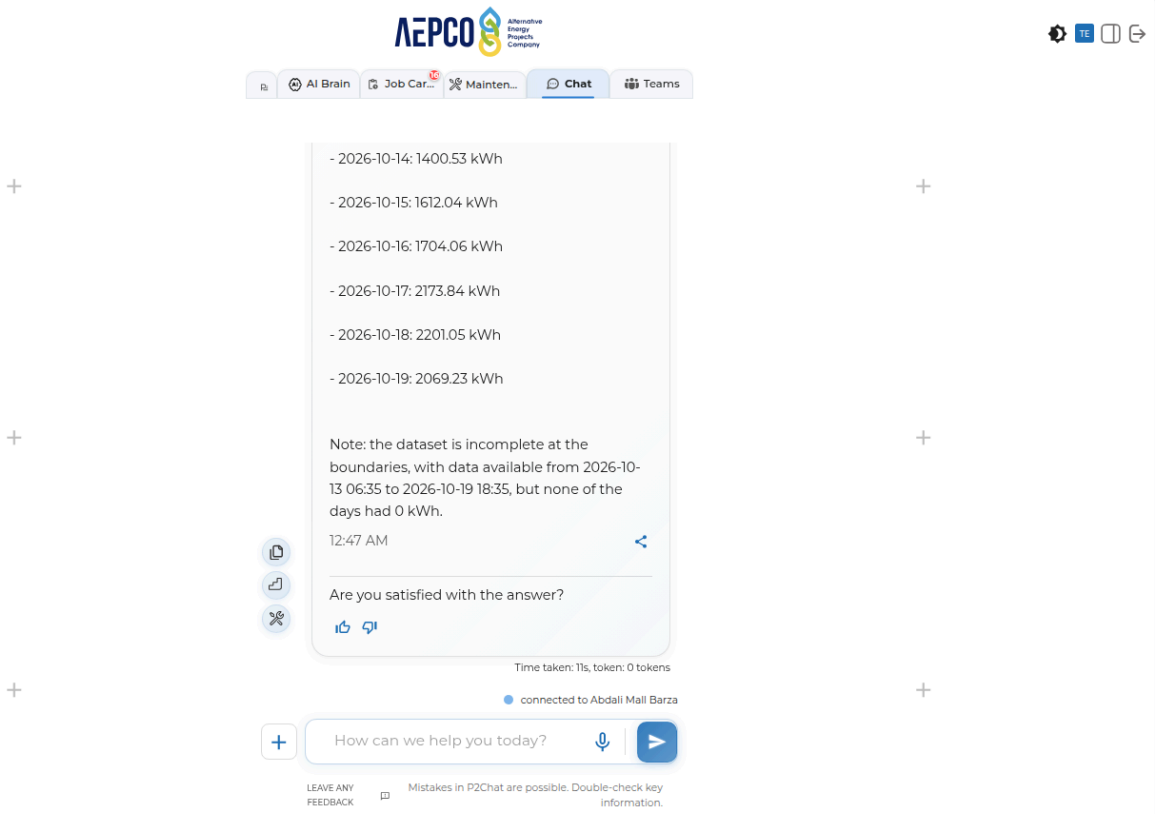
Production • 26.1s

Question: Was there any 0 production days in the last 7 days?

Response: No. There were no 0-production days in the last 7 days. Daily plant energy over 2026-10-13 to 2026-10-19: - 2026-10-13: 1904.29 kWh - 2026-10-14: 1400.53 kWh - 2026-10-15: 1612.04 kWh - 2026-10-16: 1704.06 kWh - 2026-10-17: 2173.84 kWh - 2026-10-18: 2201.05 kWh - 2026-10-19: 2069.23 kWh Note: the dataset is incomplete at the boundaries, with data available from 2026-10-13 06:35 to 2026-10-19 18:35, but none of the days had 0 kWh. 12:47 AM Are you satisfied with the answer? Time taken: 11s, token: 0 tokens connected to Abdali Mall Barza LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`functions.total_energy_func_ag`
API calls made	`2`
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	`0`
Network failures	`0`



167. Abdali_Mall_Barza

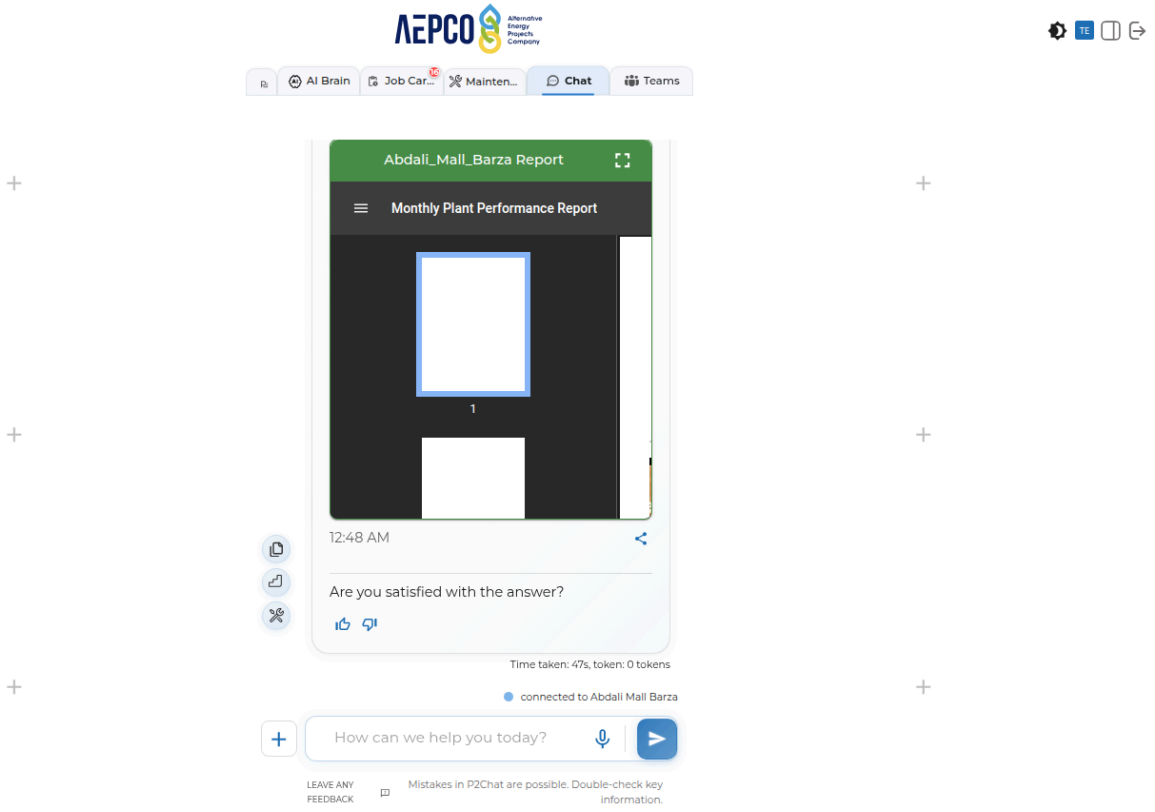
Production • 61.2s

Question: Please generate December 2025 performance report

Response: I've generated the external report for December 2025. You can view it here: 'https://report-aepco-sydney.s3.ap-southeast-2.amazonaws.com/AMC-Abdali_Mall-Wheeling_December_2025_external_report.pdf' Abdali_Mall_Barza Report 12:48 AM Are you satisfied with the answer? Time taken: 47s, token: 0 tokens connected to Abdali Mall Barza LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`report_agent`
Tools used	`agent_external_report_tool`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



168. Abdali_Mall_Barza

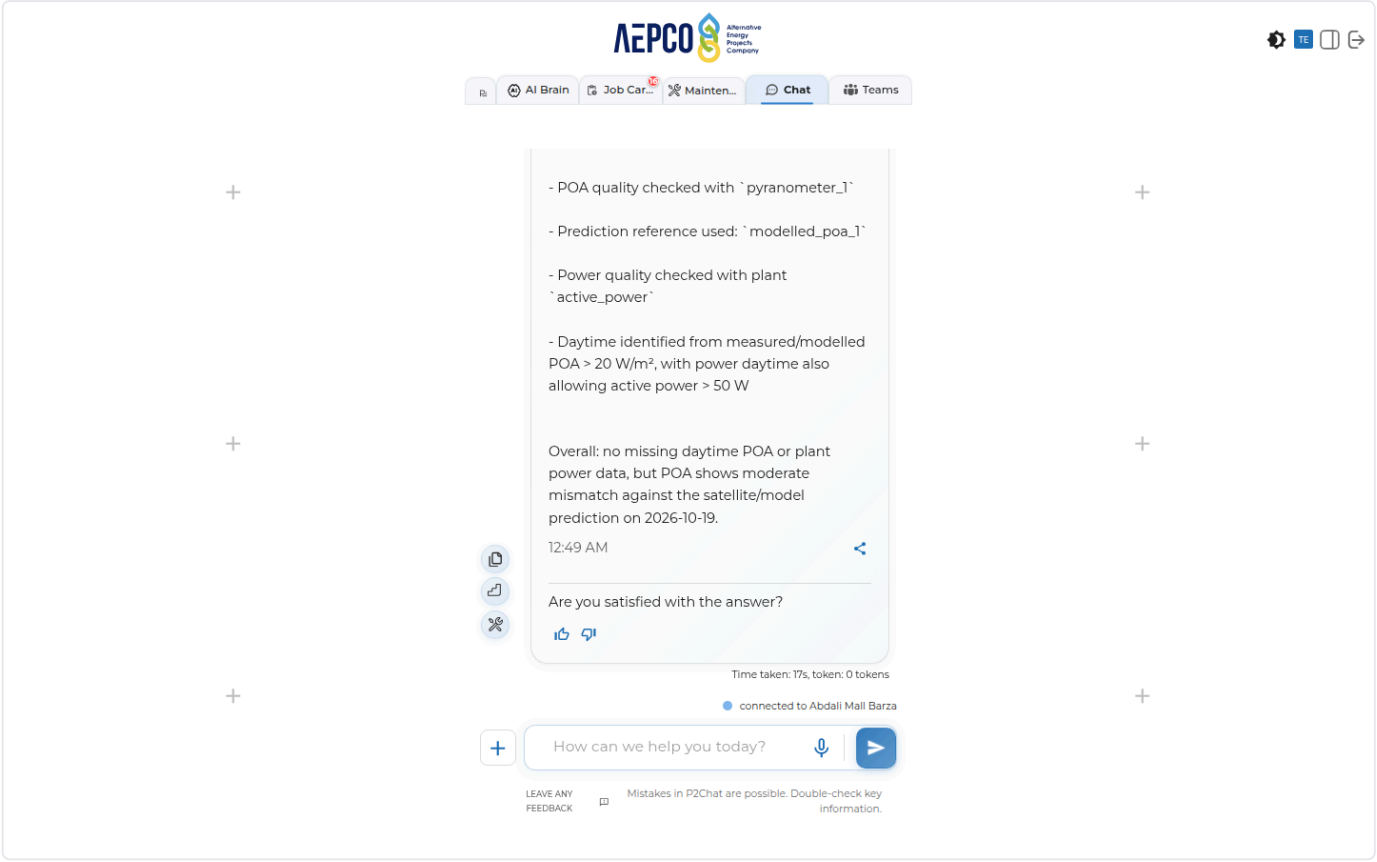
Production • 32.2s

Question: How is my data quality for POA and power? How many of them show missing data during daytime or mismatches with satellite prediction?

Response: For 2026-10-19: - Daytime missing POA points: 0 - Daytime missing plant power points: 0 POA vs satellite/model prediction mismatch: - Valid daytime comparison points: 132 - MAE: 146.3 W/m² - RMSE: 190.5 W/m² - Mean absolute percentage mismatch: 28.2% - Points with >20% mismatch: 71 - Points with >50% mismatch: 32 Basis used: - POA quality checked with `pyranometer_1` - Prediction reference used: `modelled_poa_1` - Power quality checked with plant `active_power` - Daytime identified from measured/modelled POA > 20 W/m², with power daytime also allowing active power > 50 W Overall: no missing daytime POA or plant power data, but POA shows moderate mismatch against the satellite/model prediction on 2026-10-19. 12:49 AM Are you satisfied with the answer? Time taken: 17s, token: 0 tokens connected to Abdali Mall Barza LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`functions.extract_timeseries_data_ag, functions.execute_custom_code`
API calls made	2
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	0
Network failures	0



169. Abdali_Mall_Barza

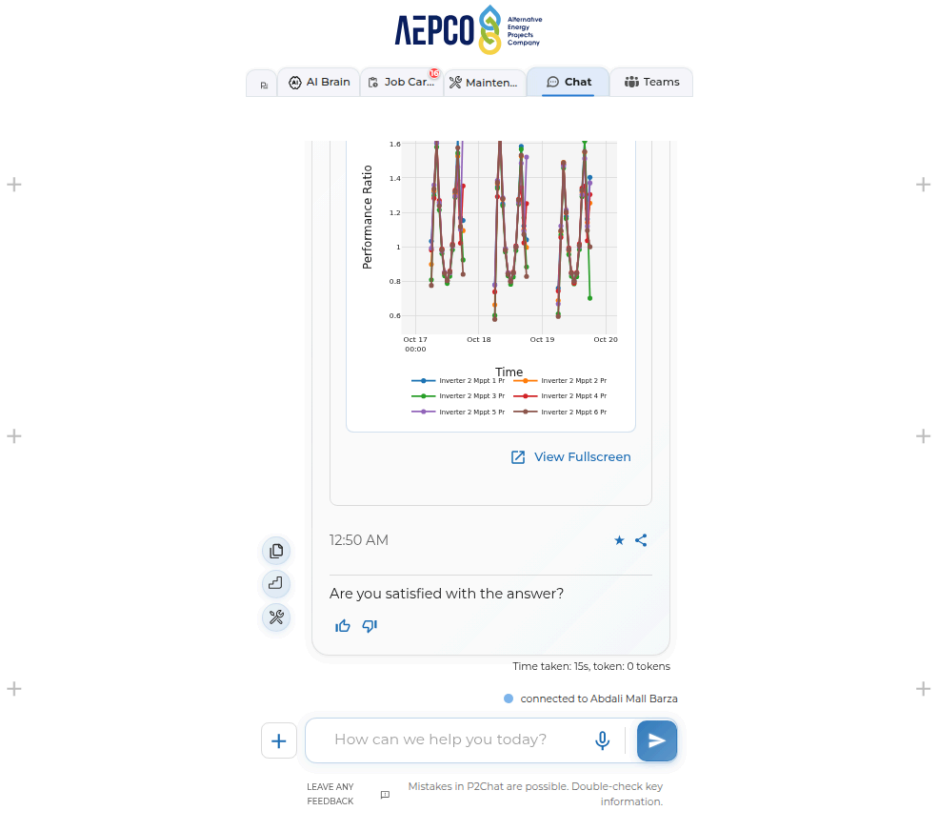
Production • 31.2s

Question: Please plot last 3 days hourly performance ratio for my MPPTs for inverter 2

Response: I've created the visualization for you. View Fullscreen 12:50 AM ★ Are you satisfied with the answer? Time taken: 15s, token: 0 tokens connected to Abdali Mall Barza LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.generate_graphs_via_code, functions.perf_ratio_ag`
API calls made	2
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	0
Network failures	0



170. Abdali_Mall_Barza

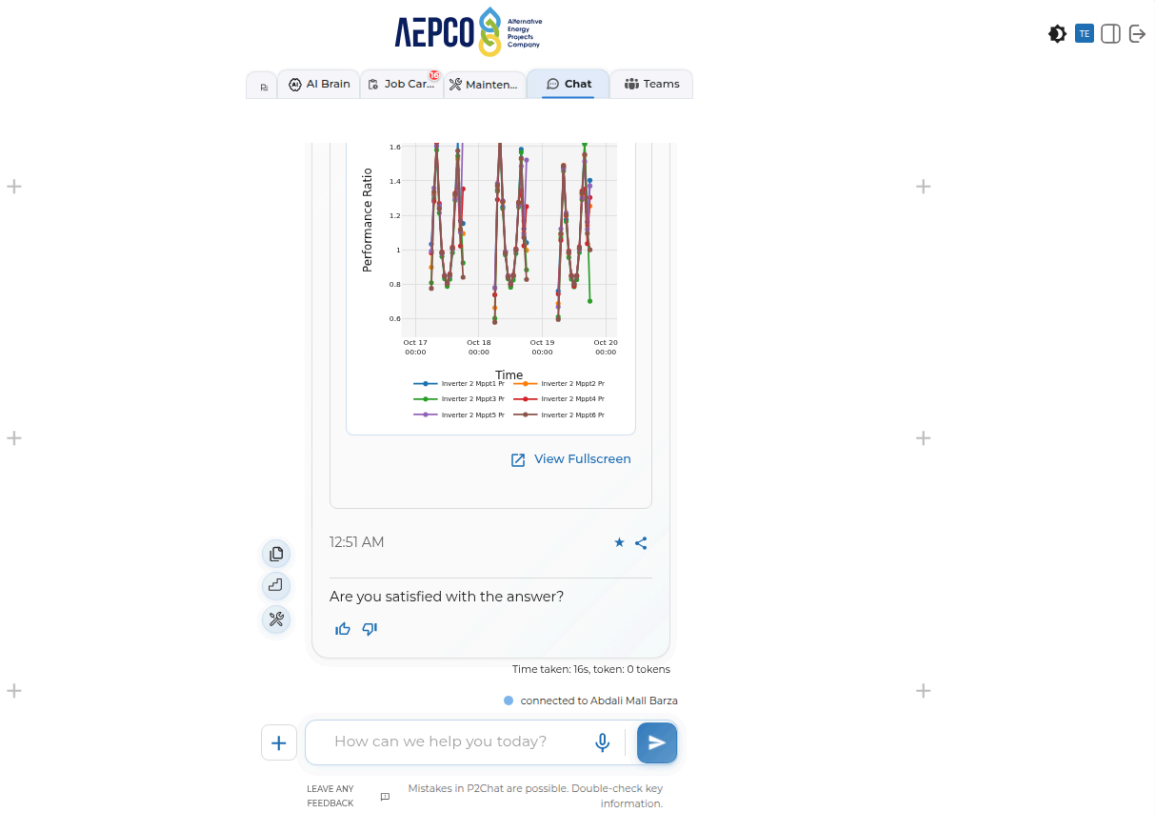
Production • 31.5s

Question: Please plot last 3 days hourly performance ratio for my Strings for inverter 2

Response: I've created the visualization for you. View Fullscreen 12:51 AM ★ Are you satisfied with the answer? Time taken: 16s, token: 0 tokens connected to Abdali Mall Barza LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.generate_graphs_via_code, functions.perf_ratio_ag`
API calls made	`1`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



171. Abdali_Mall_Barza

Production • 29.4s

Question: Please plot the last 4 days hourly performance ratio for all my inverters

Response: I've created the visualization for you. View Fullscreen 12:51 AM ★ Are you satisfied with the answer? Time taken: 15s, token: 0 tokens connected to Abdali Mall Barza LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.generate_graphs_via_code, functions.perf_ratio_ag`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0

172. Abdali_Mall_Barza

Production • 56.7s

Question: Please figure out how much clipping loss was found in the last 6 months broken by inverters, list them down and plot it as well

Response: Clipping Loss by Inverter (Last 6 Months: 2026-04-20 to 2026-10-20) - Inverter 1: 10,298.35 kWh - Inverter 2: 10,209.07 kWh - Inverter 3: 10,294.96 kWh - Inverter 4: 10,180.51 kWh - Inverter 5: 10,192.91 kWh - Inverter 6: 10,008.64 kWh Data Quality Status: Good The visualization has been prepared showing monthly clipping loss by inverter for the requested period. Clipping Loss by Inverter (Last 6 Months: 2026-04-20 to 2026-10-20) - Inverter 1: 10,298.35 kWh - Inverter 2: 10,209.07 kWh - Inverter 3: 10,294.96 kWh - Inverter 4: 10,180.51 kWh - Inverter 5: 10,192.91 kWh - Inverter 6: 10,008.64 kWh Data Quality Status: Good The visualization has been prepared showing monthly clipping loss by inverter for the requested period. View Fullscreen 12:53 AM ★ Are you satisfied with the answer? Time taken: 41s, token: 0 tokens connected to Abdali Mall Barza LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent, graph_agent`
Tools used	`functions.generate_graphs_via_code, functions.calculate_clipping_or_curtailment_losses_ag, functions.clipping_loss_via_requests`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`

173. Abdali_Mall_Barza

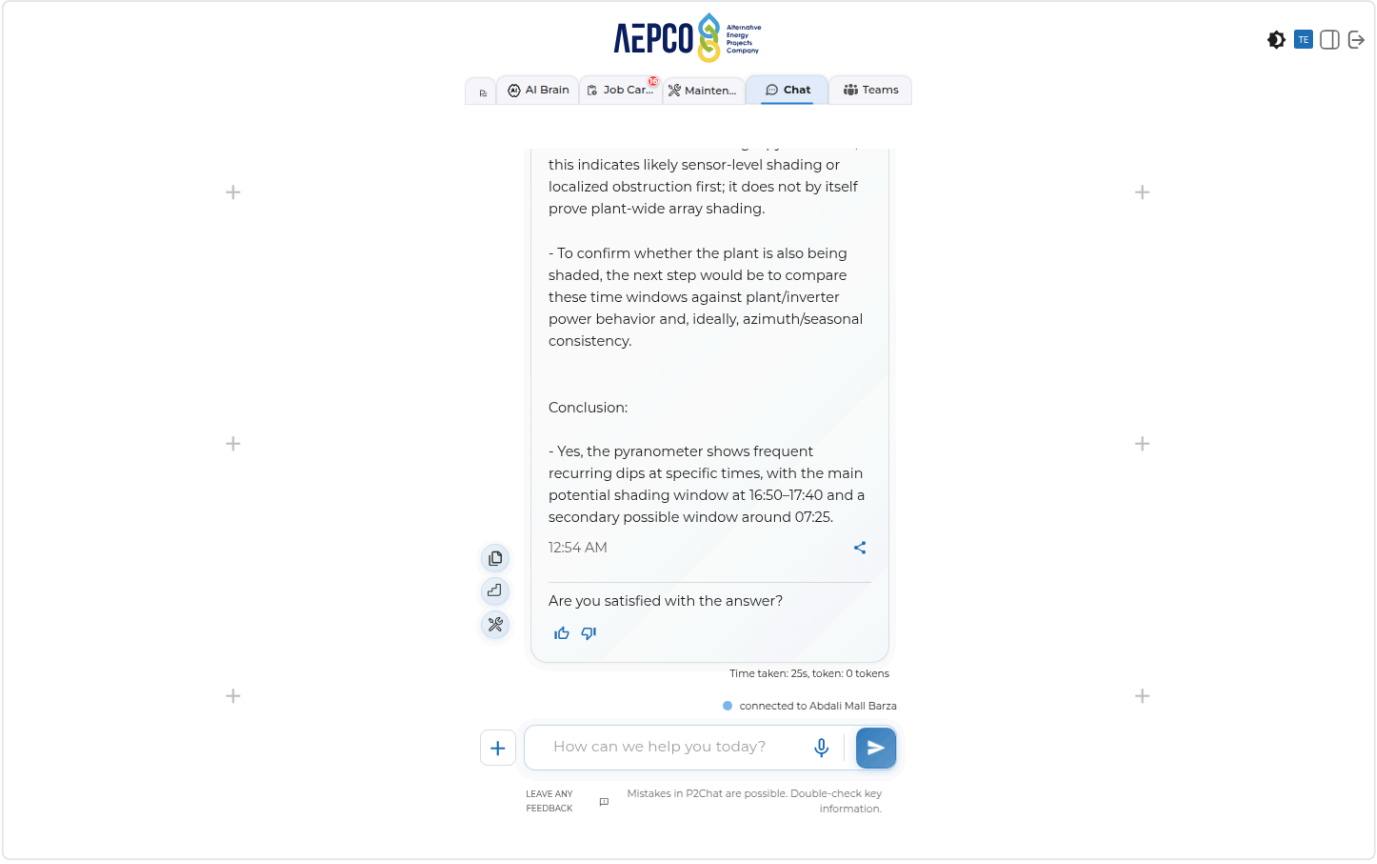
Production • 39.9s

Question: Is there shading happening with my plant? Please analyze the pyranometer and check for frequent dips at certain times

Response: I analyzed the last 30 days of 5-minute pyranometer data (2026-09-20 to 2026-10-19), comparing measured POA irradiance (`pyranometer_1`) against `modelled_poa_1` during daylight periods. Findings: - There is a clear recurring time-of-day dip pattern. - The strongest recurrent dip window is late afternoon: 16:50–17:40. - There is also a smaller recurrent morning dip around 07:25. Key evidence: - Overall median measured/modelled POA ratio: 0.755 - Morning median ratio: 0.717 - Afternoon median ratio: 0.802 - Worst individual recurring times by median ratio were: - 17:05: 0.367 - 17:10: 0.368 - 17:15: 0.369 - 17:20: 0.375 - 17:25: 0.376 - 17:00: 0.384 - 07:25: 0.421 Interpretation: - This is consistent with fixed-object shading affecting the pyranometer at low sun angles, especially near sunset. - The afternoon feature is much stronger and spans many consecutive 5-minute intervals, which makes it more likely to be true recurring shadi

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`functions.extract_timeseries_data_ag, functions.execute_custom_code`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



174. Abdali_Mall_Barza

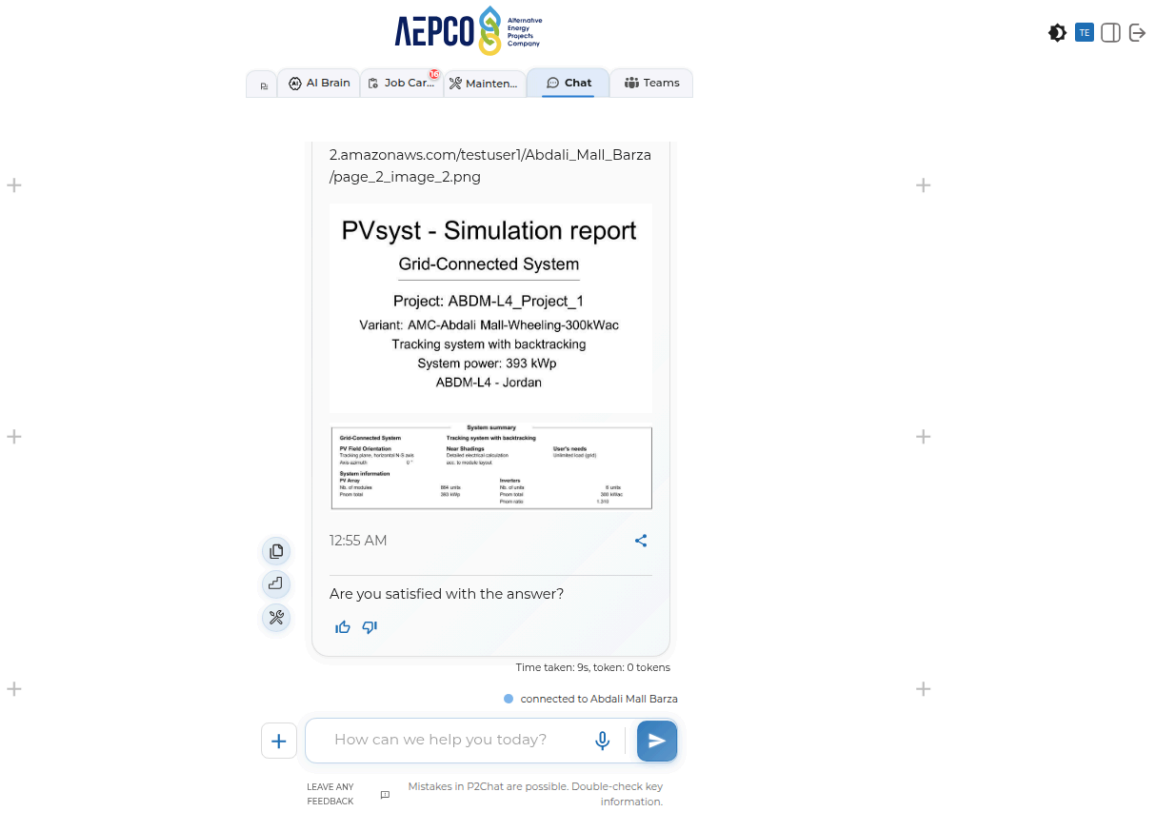
Production • 24.6s

Question: Based on pvsyst report, what is the rated power of this plant and array configuration

Response: Found 2 relevant PVsyst report sections: 1. [Abdali_Mall_Barza] Front page of PVsyst report with rated power and location (Page 1, Table 1) 2. [Abdali_Mall_Barza] System Summary with PV array details including module count and rated power (Page 2, Table 2) Please access the URLs provided to view the detailed information. Here is the reference: https://image-pvsyst-aepco-sydney.s3.ap-southeast-2.amazonaws.com/testuser1/Abdali_Mall_Barza/page_1_image_1.png https://image-pvsyst-aepco-sydney.s3.ap-southeast-2.amazonaws.com/testuser1/Abdali_Mall_Barza/page_2_image_2.png 12:55 AM Are you satisfied with the answer? Time taken: 9s, token: 0 tokens connected to Abdali Mall Barza LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`pvsyst_agent`
Tools used	`pvsyst_tool`
API calls made	`4`
Last API call	`GET /testuser2/chat/ → 404`
Console errors	`3`
Network failures	`0`



175. Abdali_Mall_Barza

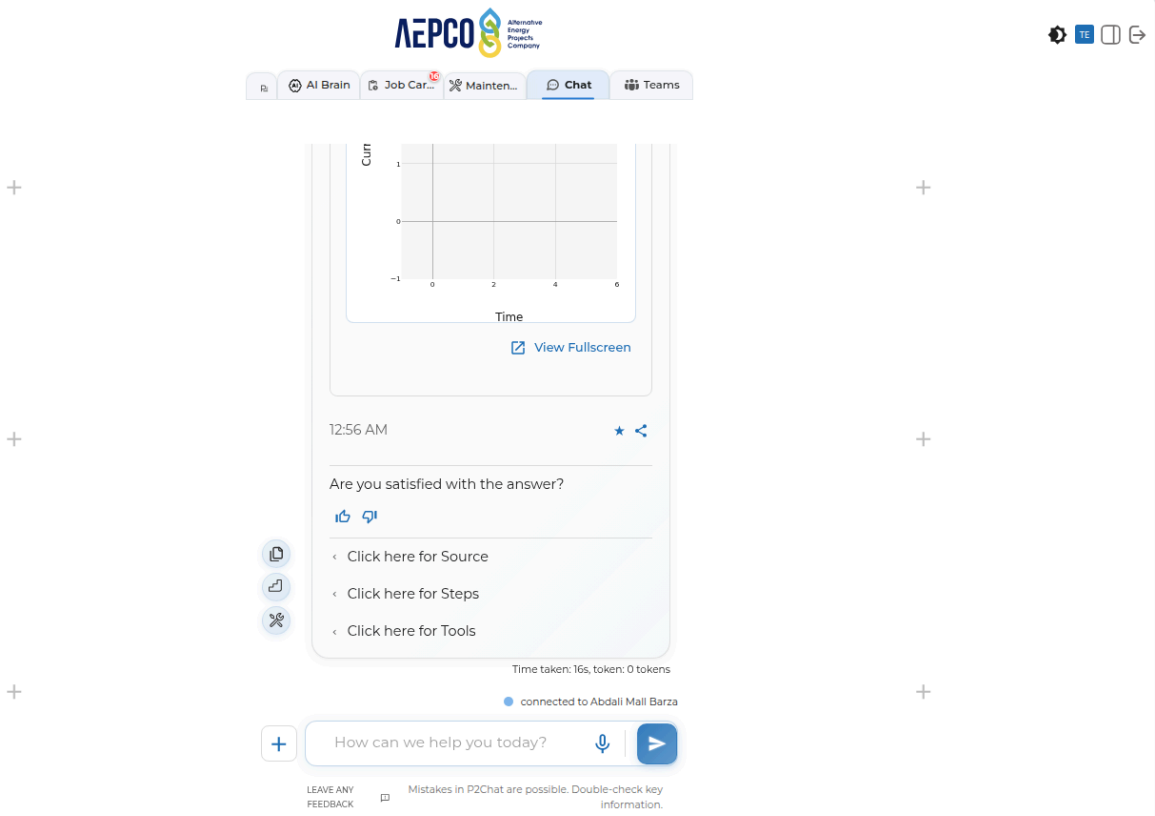
Production • 31.3s

Question: Please plot my current and voltage in a dual axes plot on the deepest possible granularity level for inverter3

Response: I've created the visualization for you. View Fullscreen 12:56 AM ★ Are you satisfied with the answer? Click here for Source Click here for Steps Click here for Tools Time taken: 16s, token: 0 tokens connected to Abdali Mall Barza LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.generate_graphs_via_code, functions.extract_timeseries_data_ag, functions.execute_custom_code`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`1`
Network failures	`0`



176. Abdali_Mall_Barza

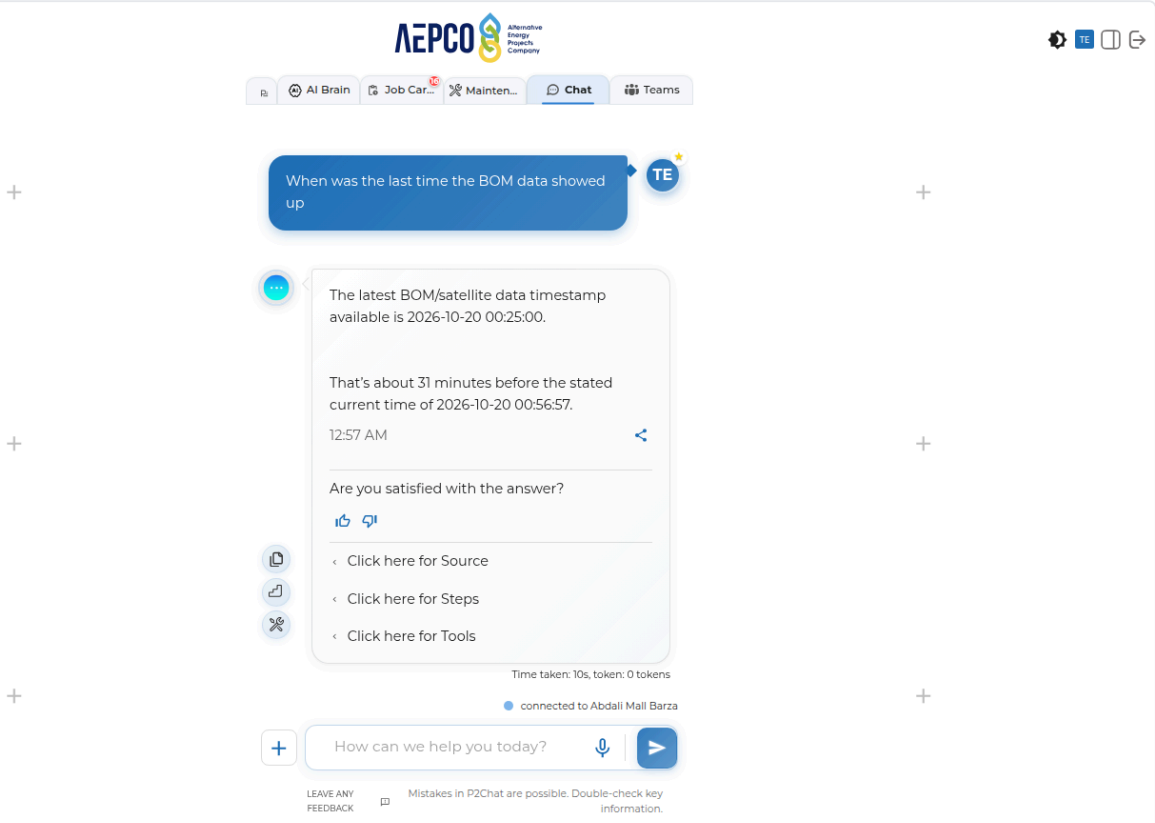
Production • 24.5s

Question: When was the last time the BOM data showed up

Response: The latest BOM/satellite data timestamp available is 2026-10-20 00:25:00. That's about 31 minutes before the stated current time of 2026-10-20 00:56:57. 12:57 AM Are you satisfied with the answer? Click here for Source Click here for Steps Click here for Tools Time taken: 10s, token: 0 tokens connected to Abdali Mall Barza LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`functions.extract_timeseries_data_ag`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



177. Abdali_Mall_Barza

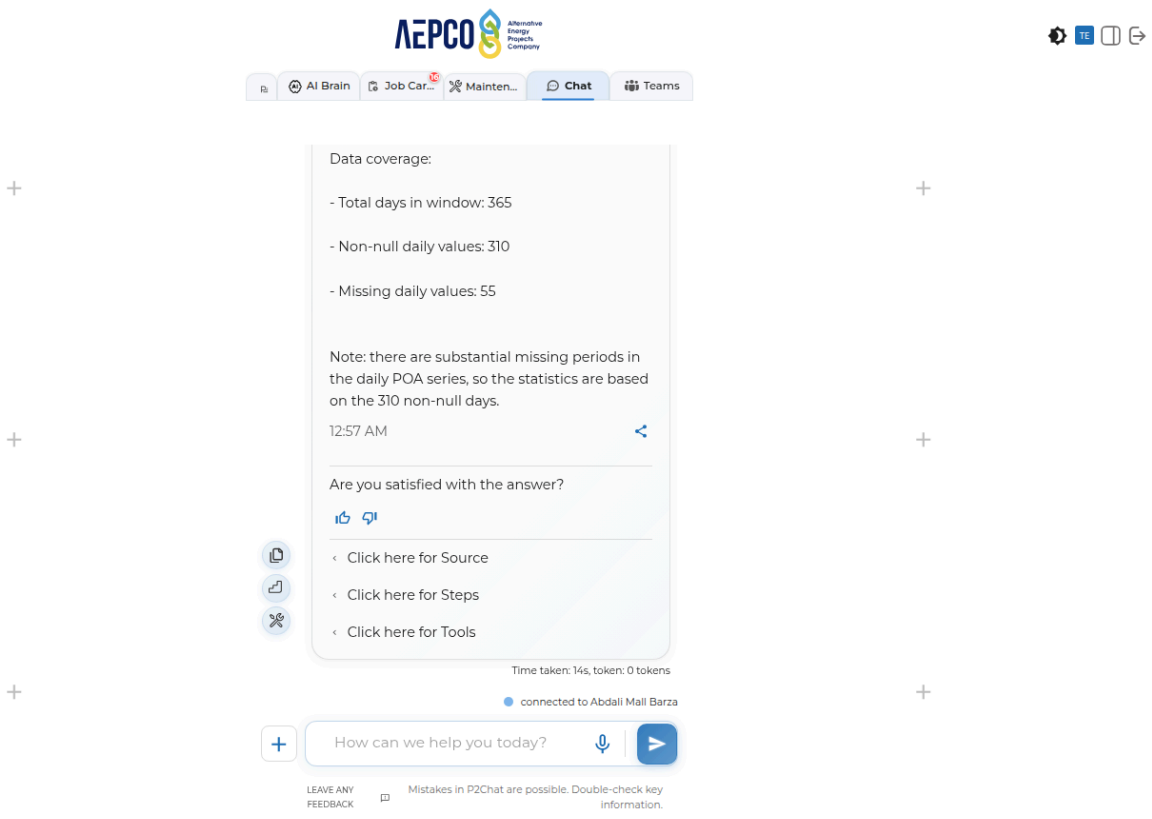
Production • 29.1s

Question: Please find out the daily POA poa summed up as a Wh/m^2, daily for the last 12 months and then calculate the median, mean and kurt

Response: Daily POA irradiation over the last 12 months (from `pyranometer_1`) was summed as Wh/m² using the 5-minute data. Summary statistics: - Median: 6439.50 Wh/m² - Mean: 6054.77 Wh/m² - Kurtosis: -0.333 Data coverage: - Total days in window: 365 - Non-null daily values: 310 - Missing daily values: 55 Note: there are substantial missing periods in the daily POA series, so the statistics are based on the 310 non-null days. 12:57 AM Are you satisfied with the answer? Click here for Source Click here for Steps Click here for Tools Time taken: 14s, token: 0 tokens connected to Abdali Mall Barza LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`functions.extract_timeseries_data_ag, functions.execute_custom_code`
API calls made	2
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	0
Network failures	0



178. Abdali_Mall_Barza

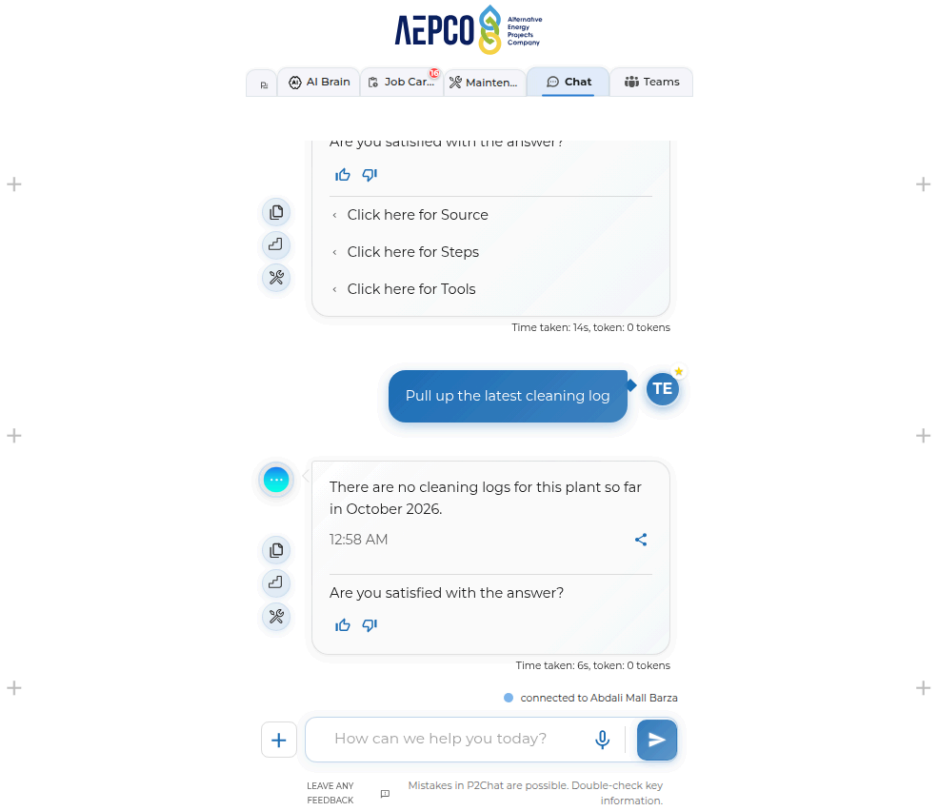
Production • 21.5s

Question: Pull up the latest cleaning log

Response: There are no cleaning logs for this plant so far in October 2026. 12:58 AM Are you satisfied with the answer? Time taken: 6s, token: 0 tokens connected to Abdali Mall Barza LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_entry_agent`
API calls made	`1`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



179. Abdali_Mall_Barza

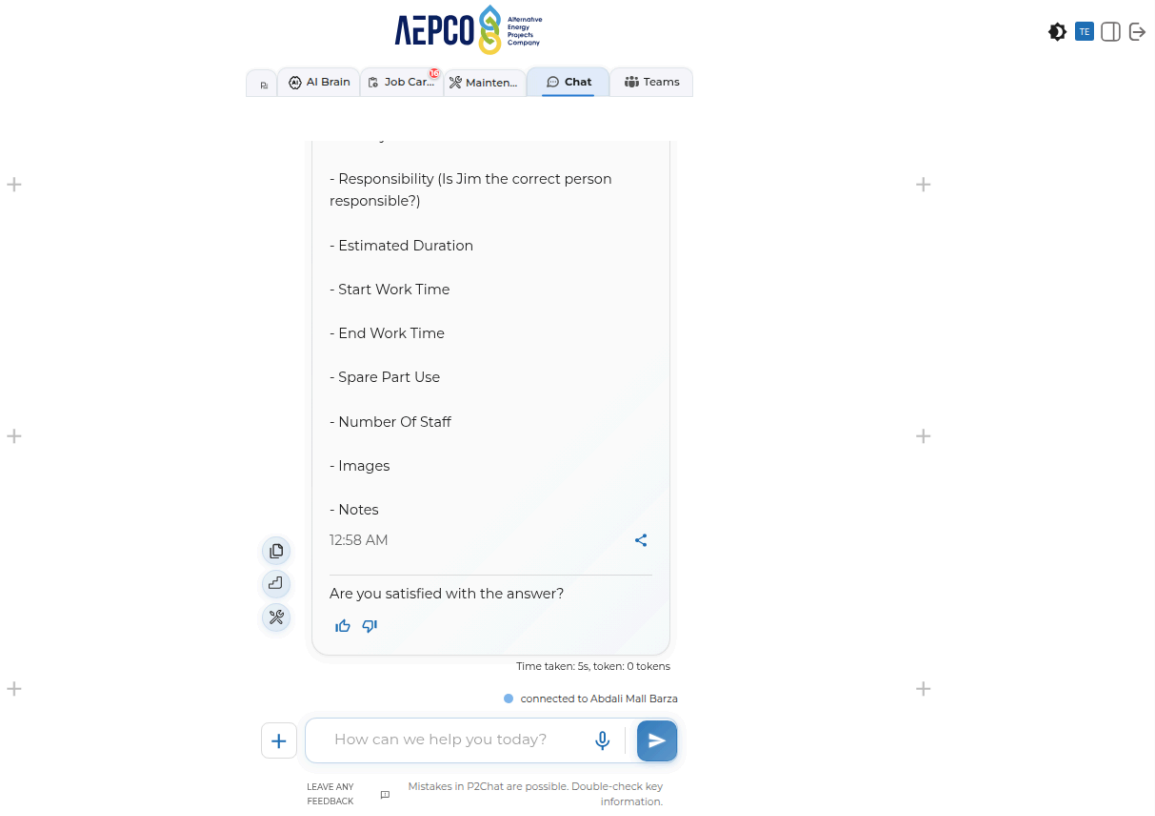
Production • 20.0s

Question: Jim cleaned the pyranometer today, can you add it to the cleaning log?

Response: To add a cleaning activity log, please provide the following required details: - Date (Is today, 2026-10-20, correct for this log?) - Fault Cause - Action Taken - Status Site will be set automatically as Abdali_Mall_Barza. Optional fields you can also provide, but they are not required: - Priority - Responsibility (Is Jim the correct person responsible?) - Estimated Duration - Start Work Time - End Work Time - Spare Part Use - Number Of Staff - Images - Notes 12:58 AM Are you satisfied with the answer? Time taken: 5s, token: 0 tokens connected to Abdali Mall Barza LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_entry_agent`
API calls made	`2`
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	`0`
Network failures	`0`



180. Abdali_Mall_Barza

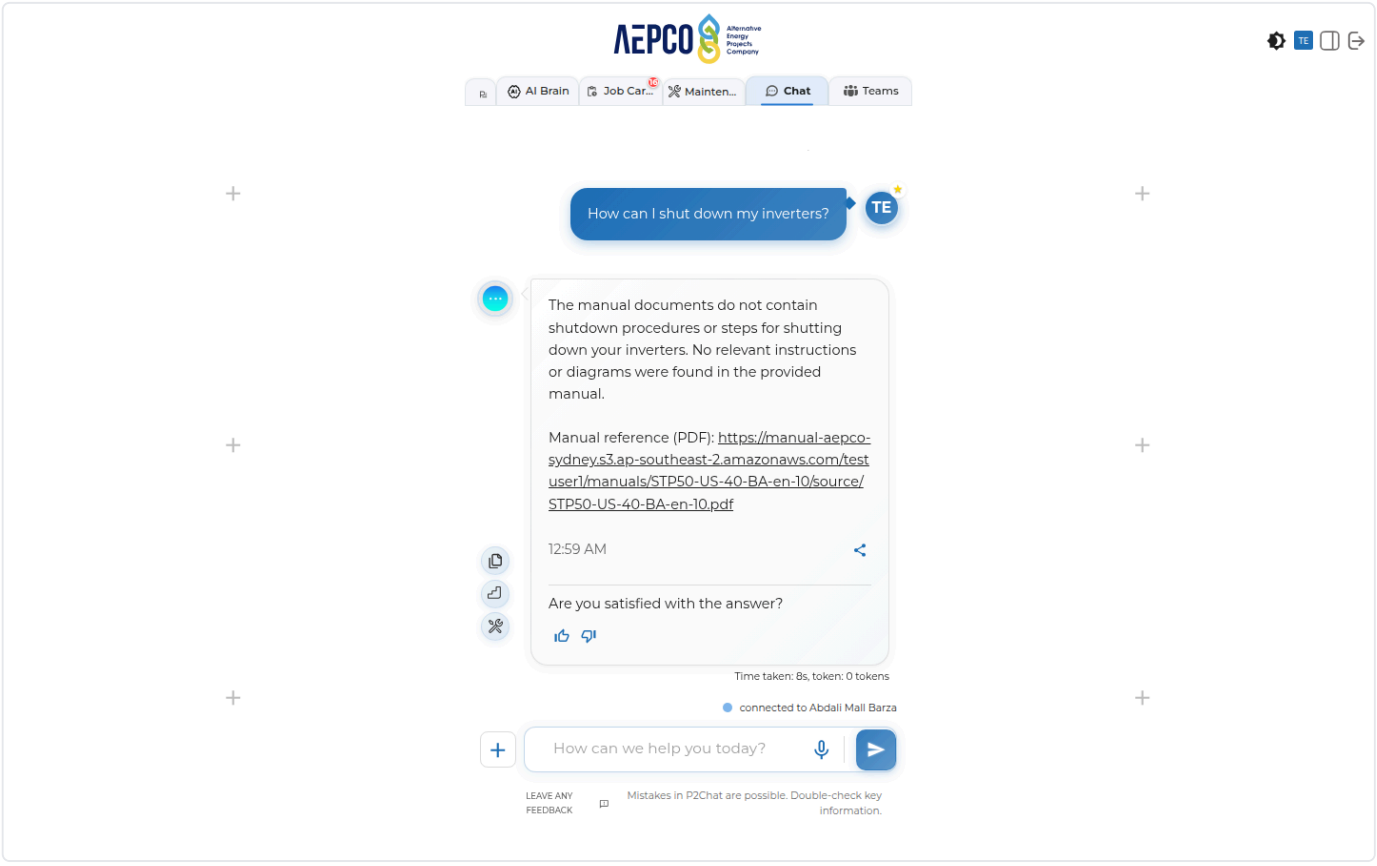
Production • 23.1s

Question: How can I shut down my inverters?

Response: The manual documents do not contain shutdown procedures or steps for shutting down your inverters. No relevant instructions or diagrams were found in the provided manual. Manual reference (PDF): https://manual-aepco-sydney.s3.ap-southeast-2.amazonaws.com/testuser1/manuals/STP50-US-40-BA-en-10/source/STP50-US-40-BA-en-10.pdf 12:59 AM Are you satisfied with the answer? Time taken: 8s, token: 0 tokens connected to Abdali Mall Barza LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`manual_agent`
Tools used	`functions.manual_rag_tool`
API calls made	`1`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



181. Abdali_Mall_Barza

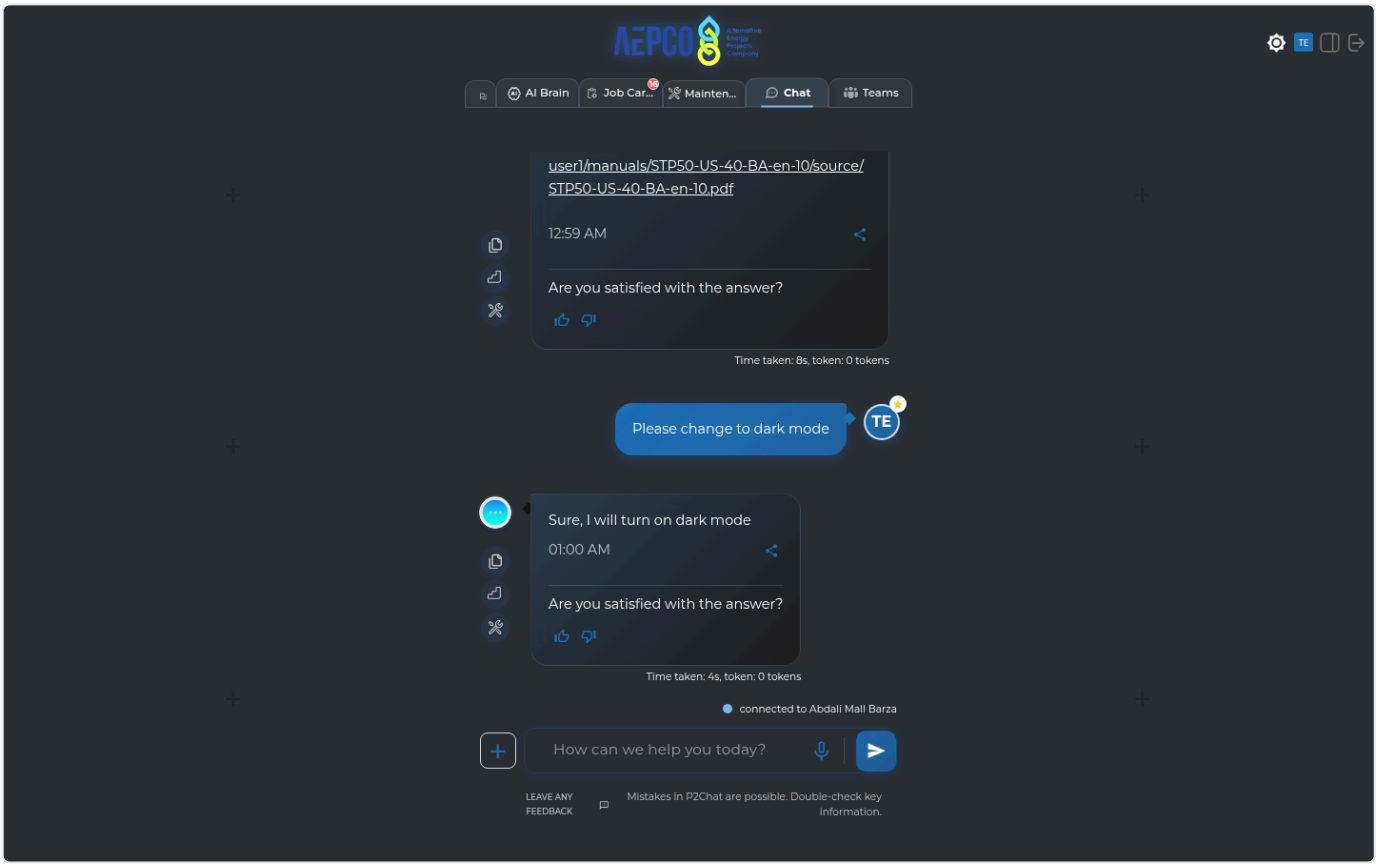
Production • 19.7s

Question: Please change to dark mode

Response: Sure, I will turn on dark mode 01:00 AM Are you satisfied with the answer? Time taken: 4s, token: 0 tokens connected to Abdali Mall Barza LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`ui_agent`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



182. Abdali_Mall_Barza

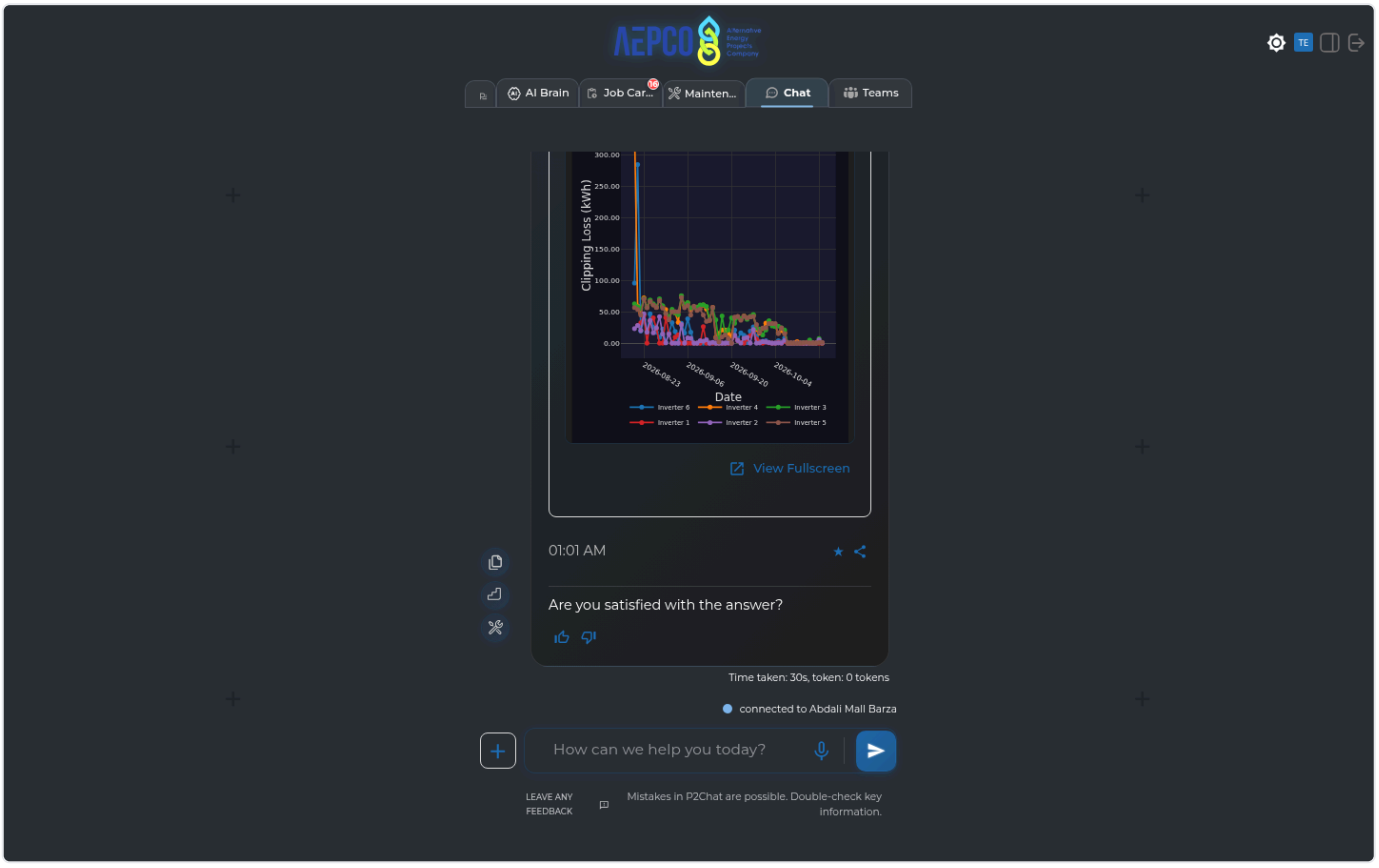
Production • 46.0s

Question: Please figure out how much clipping loss was found in the last 2 months broken by inverters, list them down and plot it as well

Response: Here is the summary of clipping losses by inverter for the period 2026-08-20 to 2026-10-20: - Inverter 1: 1378.02 kWh - Inverter 2: 1410.79 kWh - Inverter 3: 1380.72 kWh - Inverter 4: 1397.44 kWh - Inverter 5: 1409.84 kWh - Inverter 6: 1368.18 kWh Data Quality: Status Good I've created a line chart titled "Daily Clipping Loss by Inverter (Last 2 Months)" showing the day-to-day clipping loss for each inverter. Here is the summary of clipping losses by inverter for the period 2026-08-20 to 2026-10-20: - Inverter 1: 1378.02 kWh - Inverter 2: 1410.79 kWh - Inverter 3: 1380.72 kWh - Inverter 4: 1397.44 kWh - Inverter 5: 1409.84 kWh - Inverter 6: 1368.18 kWh Data Quality: Status Good I've created a line chart titled "Daily Clipping Loss by Inverter (Last 2 Months)" showing the day-to-day clipping loss for each inverter. View Fullscreen 01:01 AM ★ Are you satisfied with the answer? Time taken: 30s, token: 0 tokens connected to

TECHNICAL EVIDENCE

Agents used	`data_agent, graph_agent`
Tools used	`functions.generate_graphs_via_code, functions.calculate_clipping_or_curtailment_losses_ag, functions.clipping_loss_via_requests`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



183. Abdali_Mall_Barza

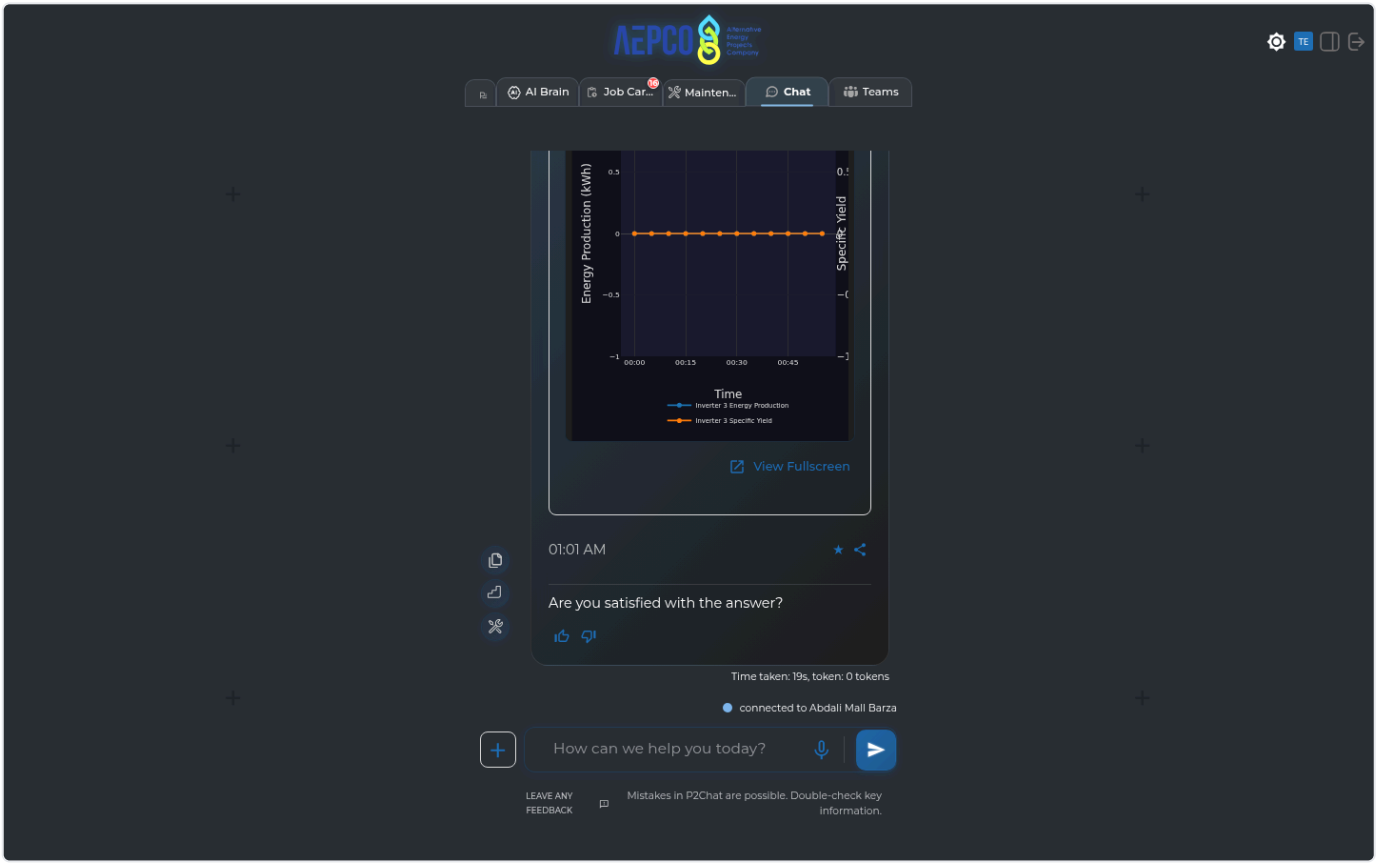
Production • 33.9s

Question: Please repeat the first plotting question, but interchange it with Specific yield on 1 axes, and energy production on the other axes on a single dual axes plot (Memory Test)

Response: I've created the visualization for you. View Fullscreen 01:01 AM ★ Are you satisfied with the answer? Time taken: 19s, token: 0 tokens connected to Abdali Mall Barza LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.specific_yield_ag, functions.total_energy_func_ag, functions.generate_graphs_via_code`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



184. SAK_1

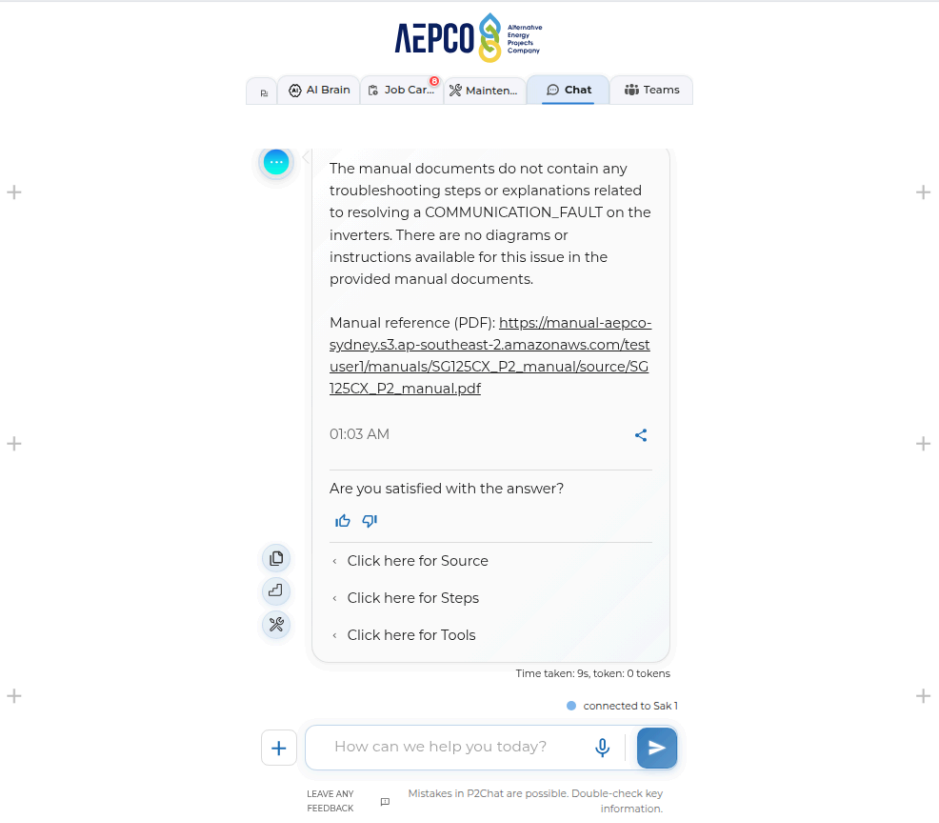
Production • 23.1s

Question: I have a COMMUNICATION_FAULT what do I do to my inverters?

Response: The manual documents do not contain any troubleshooting steps or explanations related to resolving a COMMUNICATION_FAULT on the inverters. There are no diagrams or instructions available for this issue in the provided manual documents. Manual reference (PDF): https://manual-aepco-sydney.s3.ap-southeast-2.amazonaws.com/testuser1/manuals/SG125CX_P2_manual/source/SG125CX_P2_manual.pdf 01:03 AM Are you satisfied with the answer? Click here for Source Click here for Steps Click here for Tools Time taken: 9s, token: 0 tokens connected to Sak 1 LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`manual_agent`
Tools used	`functions.manual_rag_tool`
API calls made	`12`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



185. SAK_1

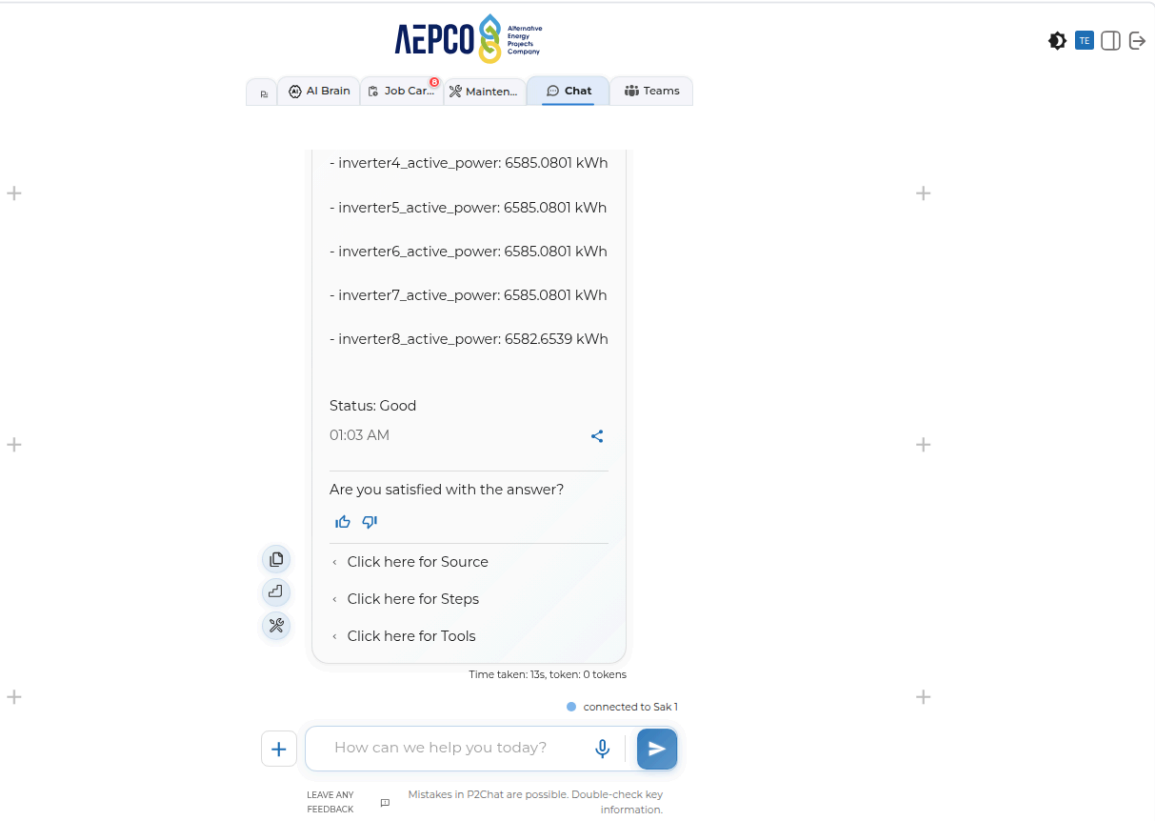
Production • 27.6s

Question: Use 'clipping_loss_via_requests' in order to calculate clipping loss if any for July 2026

Response: Clipping loss for July 2026: - inverter1_active_power: 6585.0801 kWh - inverter2_active_power: 6585.0801 kWh - inverter3_active_power: 6585.0801 kWh - inverter4_active_power: 6585.0801 kWh - inverter5_active_power: 6585.0801 kWh - inverter6_active_power: 6585.0801 kWh - inverter7_active_power: 6585.0801 kWh - inverter8_active_power: 6582.6539 kWh Status: Good 01:03 AM Are you satisfied with the answer? Click here for Source Click here for Steps Click here for Tools Time taken: 13s, token: 0 tokens connected to Sak 1 LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`functions.clipping_loss_via_requests`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



186. SAK_1

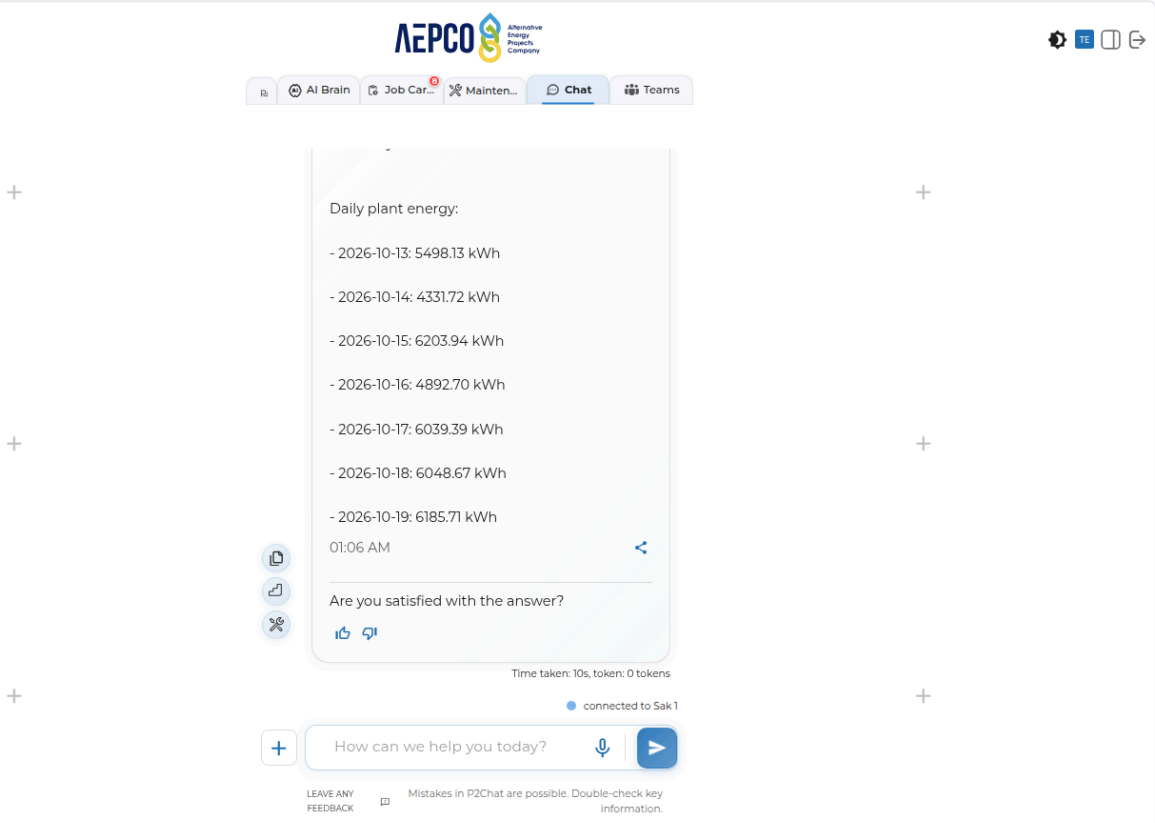
Production • 26.1s

Question: Was there any 0 production days in the last 7 days?

Response: No. There were no 0-production days in the last 7 days. Daily plant energy: - 2026-10-13: 5498.13 kWh - 2026-10-14: 4331.72 kWh - 2026-10-15: 6203.94 kWh - 2026-10-16: 4892.70 kWh - 2026-10-17: 6039.39 kWh - 2026-10-18: 6048.67 kWh - 2026-10-19: 6185.71 kWh 01:06 AM Are you satisfied with the answer? Time taken: 10s, token: 0 tokens connected to Sak 1 LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`functions.total_energy_func_ag`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



187. SAK_1

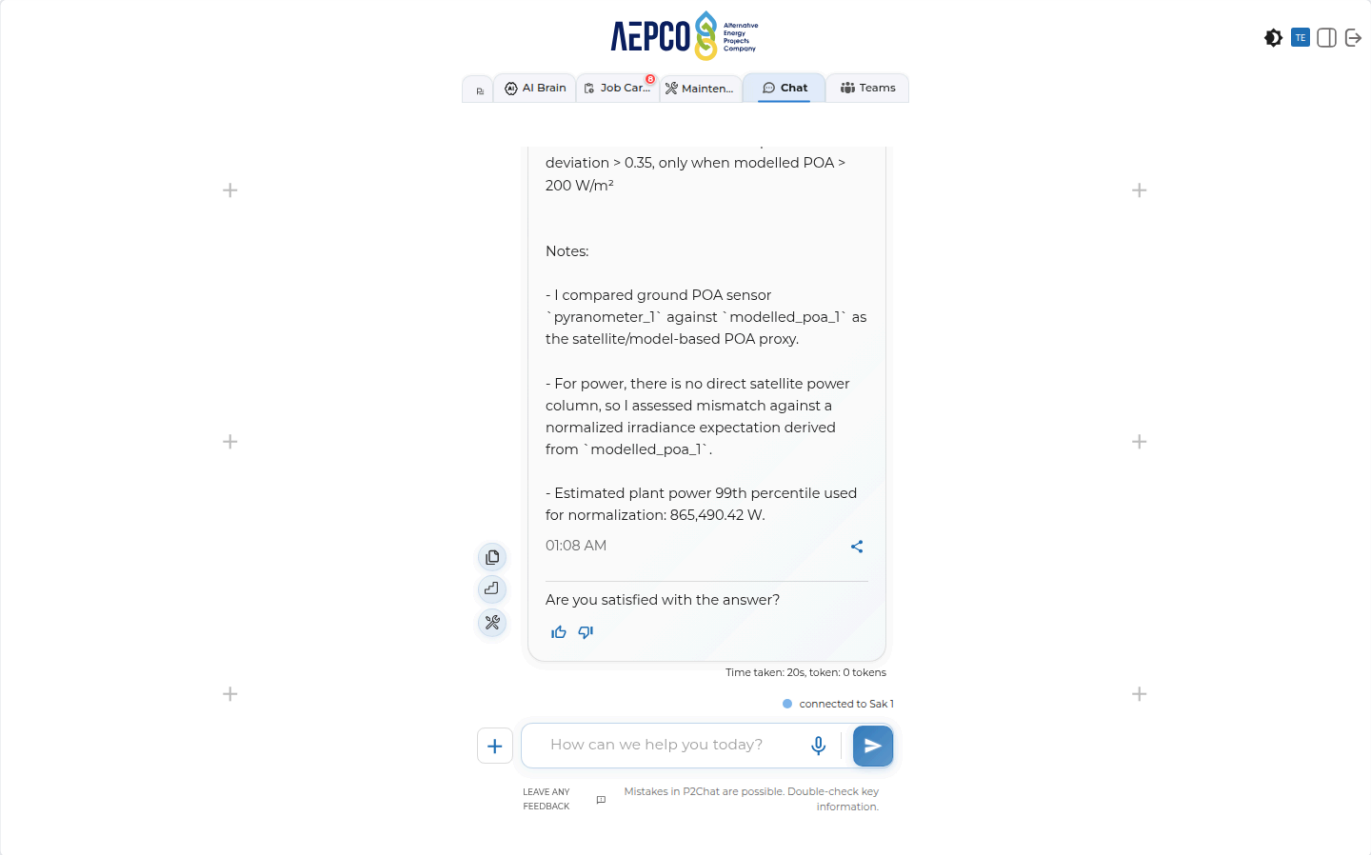
Production • 35.2s

Question: How is my data quality for POA and power? How many of them show missing data during daytime or mismatches with satellite prediction?

Response: For 2026-10-19, using daytime defined as modelled POA > 20 W/m²: - Daytime samples: 132 - Missing POA samples during daytime: 0 - Missing plant active power samples during daytime: 0 - POA mismatches vs satellite/modelled POA prediction: 54 - Power mismatches vs irradiance-based prediction proxy: 18 Thresholds used: - Daytime: modelled_poa_1 > 20 W/m² - POA mismatch: absolute difference > 100 W/m² and relative difference > 30% - Power mismatch: normalized power deviation > 0.35, only when modelled POA > 200 W/m² Notes: - I compared ground POA sensor `pyranometer_1` against `modelled_poa_1` as the satellite/model-based POA proxy. - For power, there is no direct satellite power column, so I assessed mismatch against a normalized irradiance expectation derived from `modelled_poa_1`. - Estimated plant power 99th percentile used for normalization: 865,490.42 W. 01:08 AM Are you satisfied with the answer? Time taken: 20s, token: 0 tokens connected to Sak 1 L

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`functions.execute_custom_code, functions.extract_timeseries_data_ag`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



188. SAK_1

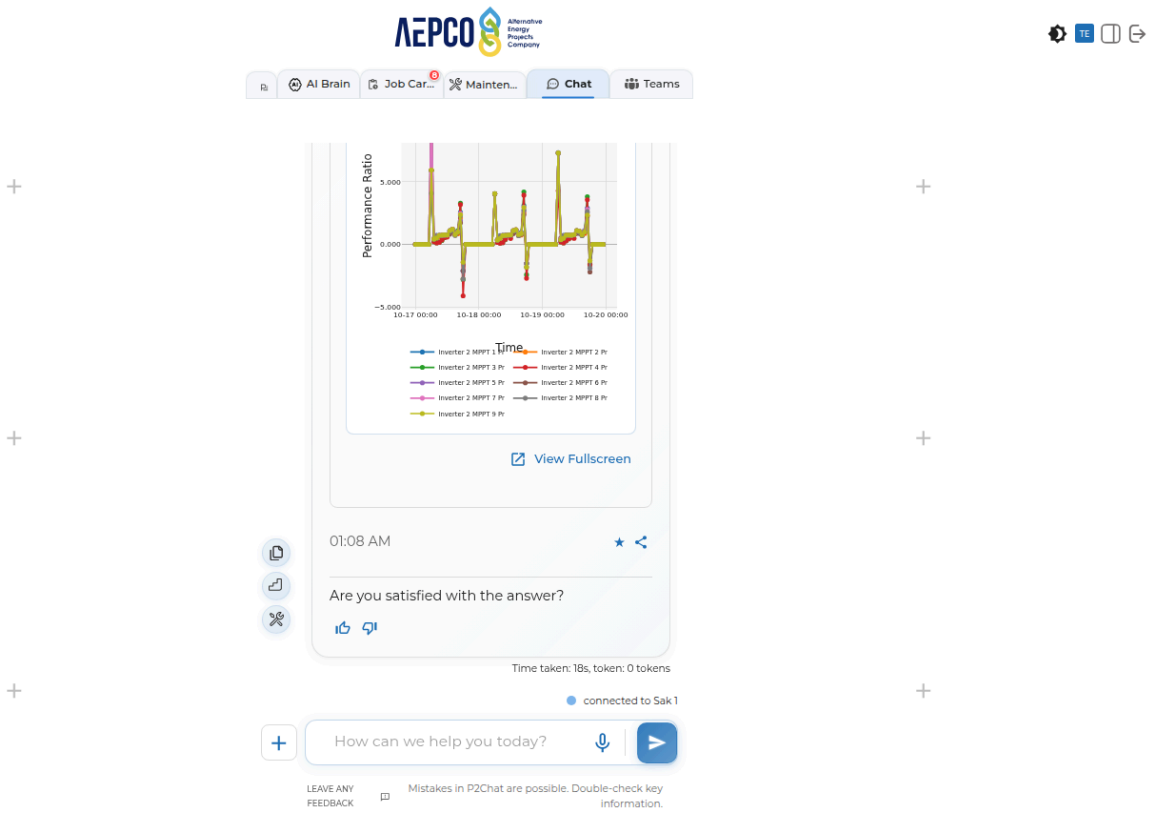
Production • 32.3s

Question: Please plot last 3 days hourly performance ratio for my MPPTs for inverter 2

Response: I've created the visualization for you. View Fullscreen 01:08 AM ★ Are you satisfied with the answer? Time taken: 18s, token: 0 tokens connected to Sak 1 LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.generate_graphs_via_code, functions.perf_ratio_ag`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



189. SAK_1

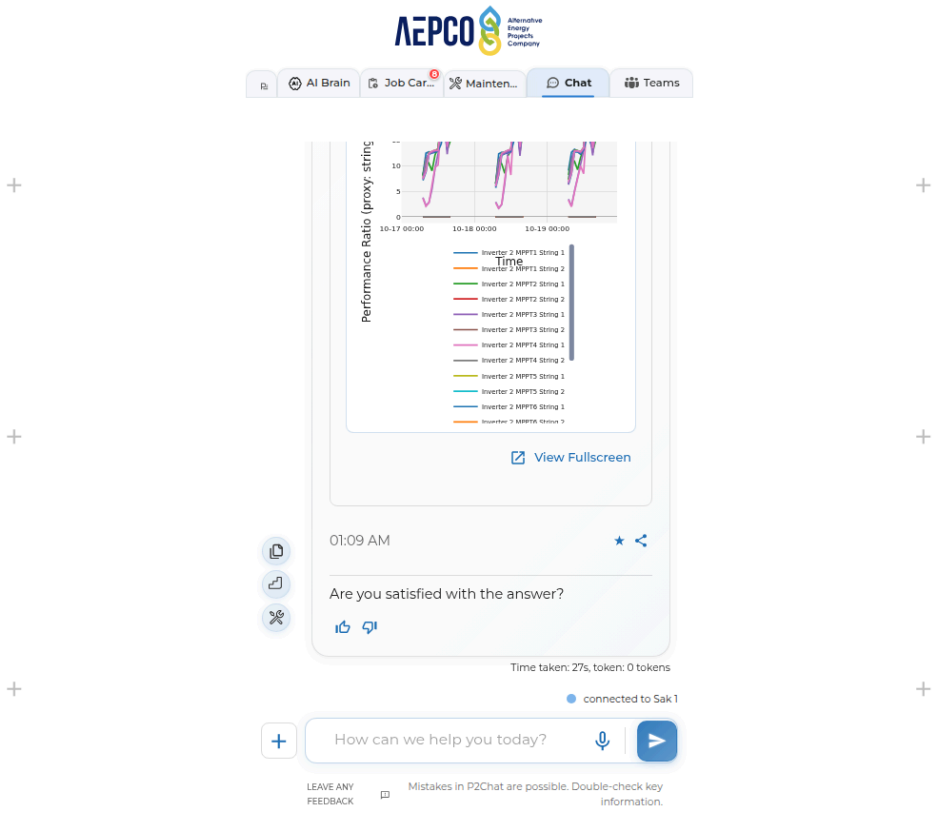
Production • 42.9s

Question: Please plot last 3 days hourly performance ratio for my Strings for inverter 2

Response: I've created the visualization for you. View Fullscreen 01:09 AM ★ Are you satisfied with the answer? Time taken: 27s, token: 0 tokens connected to Sak 1 LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.generate_graphs_via_code, functions.perf_ratio_ag, functions.execute_custom_code, functions.inspect_xpool, functions.extract_timeseries_data_ag`
API calls made	`2`
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	`0`
Network failures	`0`



190. SAK_1

Production • 33.8s

Question: Please plot the last 4 days hourly performance ratio for all my inverters

Response: I've created the visualization for you. View Fullscreen 01:10 AM ★ Are you satisfied with the answer? Time taken: 18s, token: 0 tokens connected to Sak 1 LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.generate_graphs_via_code, functions.perf_ratio_ag`
API calls made	2
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	0
Network failures	0

191. SAK_1

Production • 73.4s

Question: Please figure out how much clipping loss was found in the last 6 months broken by inverters, list them down and plot it as well

Response: Clipping Loss Summary (Last 6 Months: 2026-04-20 to 2026-10-20) - inverter1: 15,400.5209 kWh - inverter2: 15,489.1892 kWh - inverter3: 15,486.2119 kWh - inverter4: 15,489.1892 kWh - inverter5: 15,400.5209 kWh - inverter6: 15,489.1892 kWh - inverter7: 15,489.1892 kWh - inverter8: 15,395.3963 kWh Data Status: Good The visualization has been prepared: a grouped bar chart titled “Monthly Clipping Loss by Inverter (Last 6 Months)” with the y-axis representing Clipping Loss (kWh). Clipping Loss Summary (Last 6 Months: 2026-04-20 to 2026-10-20) - inverter1: 15,400.5209 kWh - inverter2: 15,489.1892 kWh - inverter3: 15,486.2119 kWh - inverter4: 15,489.1892 kWh - inverter5: 15,400.5209 kWh - inverter6: 15,489.1892 kWh - inverter7: 15,489.1892 kWh - inverter8: 15,395.3963 kWh Data Status: Good The visualization has been prepared: a grouped bar chart titled “Monthly Clipping Loss by Inverter (Last 6 Months)” with the y-axis representing Clipping Loss (

TECHNICAL EVIDENCE

Agents used	`data_agent, graph_agent`
Tools used	`functions.calculate_clipping_or_curtailment_losses_ag, functions.clipping_loss_via_requests, functions.generate_graphs_via_code`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`

192. SAK_1

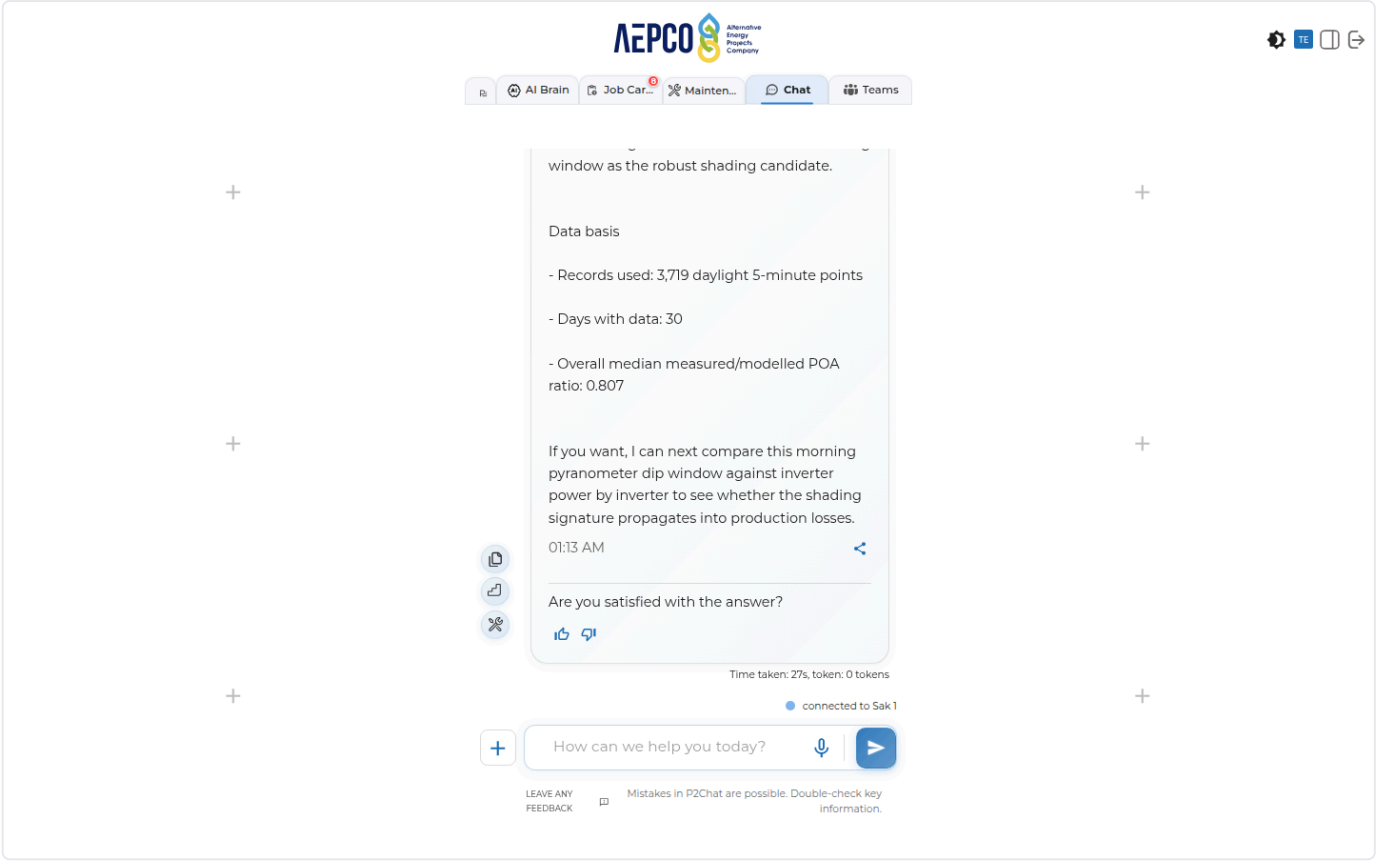
Production • 42.9s

Question: Is there shading happening with my plant? Please analyze the pyranometer and check for frequent dips at certain times

Response: Using the last 30 days of 5-minute pyranometer data (2026-09-20 to 2026-10-19), I found a clear repeated dip pattern consistent with time-of-day shading. Key finding - Strong recurring morning dip window: 07:15–08:45 - Mean dip frequency in that window: 45.0% of observed days - Worst sub-period: around 07:30–07:40 - Lowest normalized median level in that window: 0.590 at 07:30 - This means measured POA was typically only ~59% of that day's normal upper-envelope ratio at that time. Most suspicious timestamps - 07:30 — dip frequency 55.2%, median pyranometer/modelled POA ratio 0.622 - 07:35 — dip frequency 55.2%, median ratio 0.705 - 07:40 — dip frequency 51.7%, median ratio 0.704 - 08:15 — dip frequency 55.2%, median ratio 0.791 - 08:10 — dip frequency 51.7%, median ratio 0.798 Solar position during the main window - Approx azimuth range: ~99° to 113° across the strongest morning window - This points to an east to east-southeast obstruction signature.

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`functions.execute_custom_code, functions.extract_timeseries_data_ag`
API calls made	2
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	0
Network failures	0



193. SAK_1

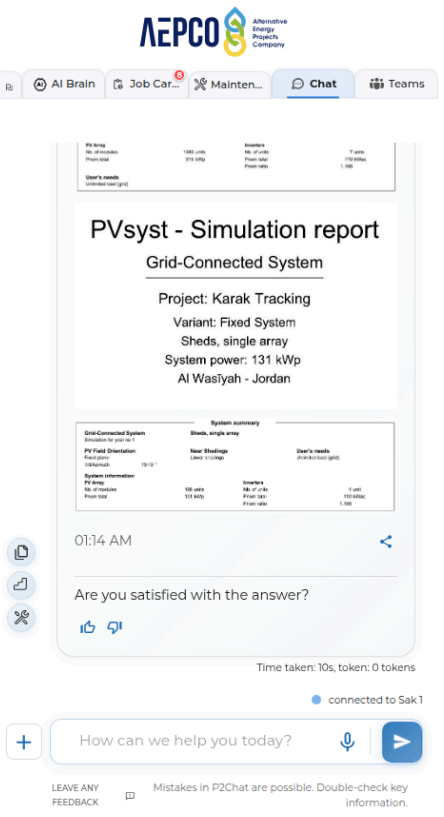
Production • 26.2s

Question: Based on pvsyst report, what is the rated power of this plant and array configuration

Response: Found 4 relevant PVsyst report sections: 1. [SR_Final_Part_1] Front page showing plant rated power and location (Page 1, Table 1) 2. [SR_Final_Part_1] System Summary including PV array details and rated power (Page 2, Table 2) 3. [SR_Final_Part_2] Front page showing plant rated power and location (Page 1, Table 1) 4. [SR_Final_Part_2] System Summary including PV array details and rated power (Page 2, Table 2) Please access the URLs provided to view the detailed information. Here is the reference: https://image-pvsyst-aepco-sydney.s3.ap-southeast-2.amazonaws.com/testuser1/SAK_1/SR_Final_Part_1/page_1_image_1.png https://image-pvsyst-aepco-sydney.s3.ap-southeast-2.amazonaws.com/testuser1/SAK_1/SR_Final_Part_1/page_2_image_2.png https://image-pvsyst-aepco-sydney.s3.ap-southeast-2.amazonaws.com/testuser1/SAK_1/SR_Final_Part_2/page_1_image_1.png https://image-pvsyst-aepco-sydney.s3.ap-southeast-2.amazonaws.com/testuser1/SAK_1/SR_Final_Part_2/page_2_image_2.png 01:14

TECHNICAL EVIDENCE

Agents used	`pvsyst_agent`
Tools used	`pvsyst_tool`
API calls made	3
Last API call	`GET /testuser2/chat/ → 404`
Console errors	3
Network failures	0



194. SAK_1

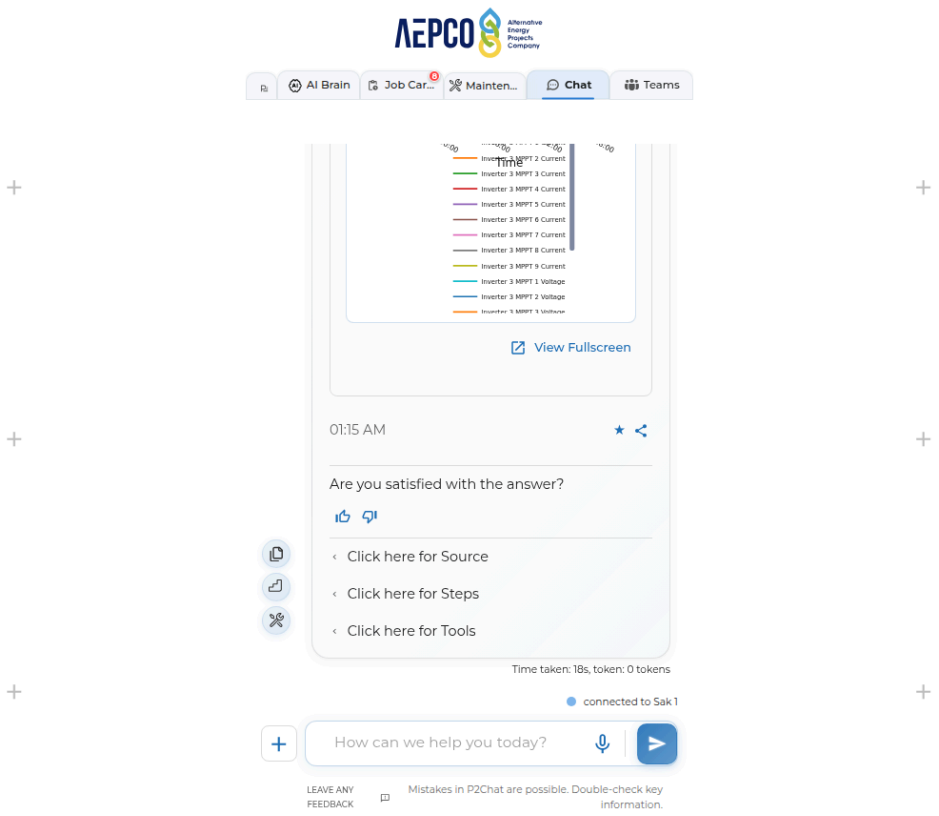
Production • 32.2s

Question: Please plot my current and voltage in a dual axes plot on the deepest possible granularity level for inverter3

Response: I've created the visualization for you. View Fullscreen 01:15 AM ★ Are you satisfied with the answer? Click here for Source Click here for Steps Click here for Tools Time taken: 18s, token: 0 tokens connected to Sak 1 LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.execute_custom_code, functions.generate_graphs_via_code, functions.extract_timeseries_data_ag`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



195. SAK_1

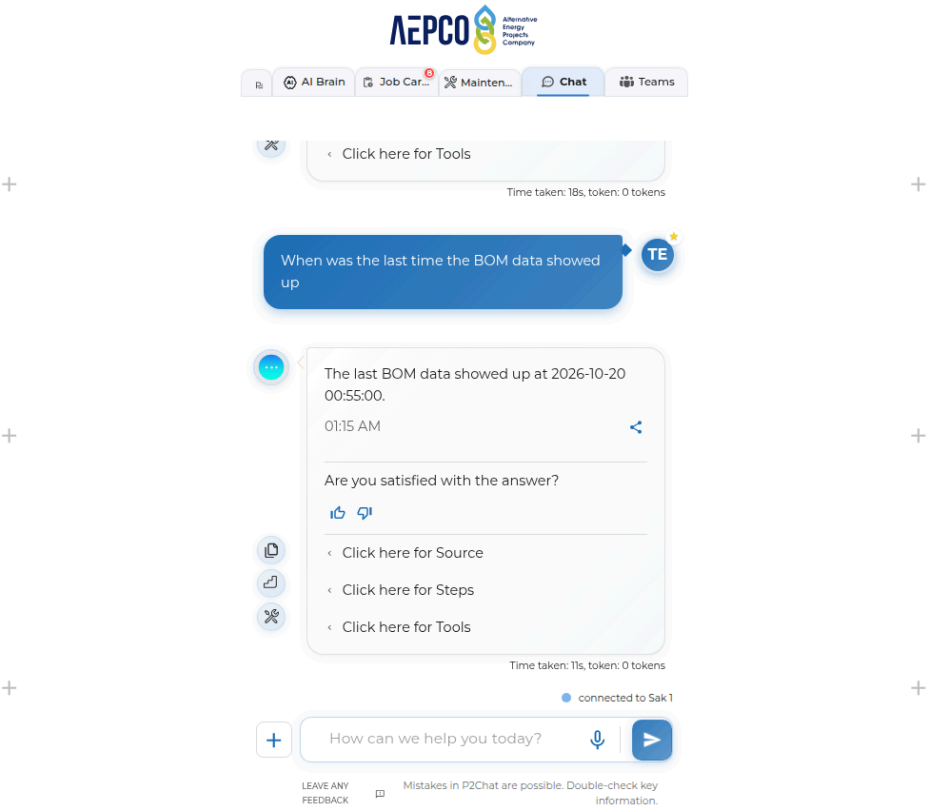
Production • 26.1s

Question: When was the last time the BOM data showed up

Response: The last BOM data showed up at 2026-10-20 00:55:00. 01:15 AM Are you satisfied with the answer? Click here for Source Click here for Steps Click here for Tools Time taken: 11s, token: 0 tokens connected to Sak 1 LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`execute_custom_code, extract_timeseries_data_ag`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



196. SAK_1

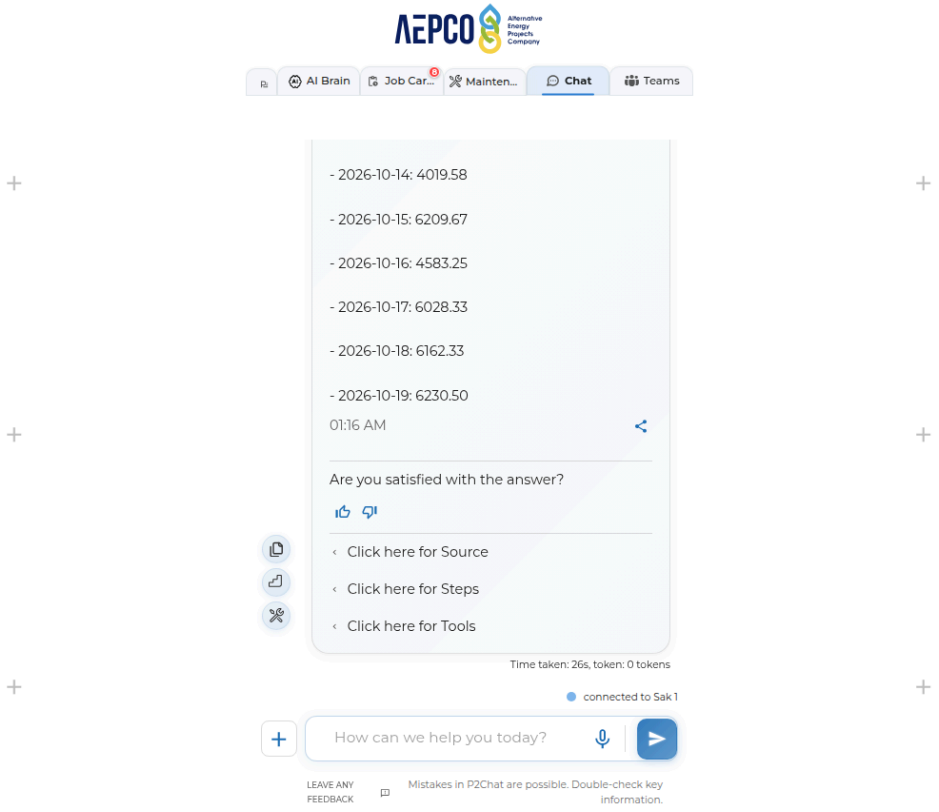
Production • 41.5s

Question: Please find out the daily POA poa summed up as a Wh/m^2, daily for the last 12 months and then calculate the median, mean and kurt

Response: Daily POA irradiance sums from `pyranometer_1` were computed in Wh/m² by summing 5-minute irradiance samples × 5/60 h for each day over the requested last 12 months window. Important note: the database only contains data from 2026-07-09 to 2026-10-19 within that window, so the result is based on 103 days, not a full 12 months. Summary statistics: - Median daily POA: 9129.75 Wh/m² - Mean daily POA: 8287.60 Wh/m² - Kurtosis (Fisher): -1.0852 - Days with data: 103 - Data coverage in window: 2026-07-09 to 2026-10-19 Daily POA sums (Wh/m²): - 2026-07-09: 10462.00 - 2026-07-10: 11383.92 - 2026-07-11: 11368.50 - 2026-07-12: 11239.83 - 2026-07-13: 11270.25 - 2026-07-14: 11337.83 - 2026-07-15: 11197.08 - 2026-07-16: 11263.00 - 2026-07-17: 11075.67 - 2026-07-18: 11304.83 - 2026-07-19: 10163.58 - 2026-07-20: 11082.50 - 2026-07-21: 11162.67 - 2026-07-22: 9375.42 - 2026-07-23: 10053.17 - 2026-07-24: 9706.75 - 2026-07-25: 9117.17 - 2026-07-26: 1

TECHNICAL EVIDENCE

Agents used	`data_agent`
Tools used	`execute_custom_code, extract_timeseries_data_ag`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



197. SAK_1

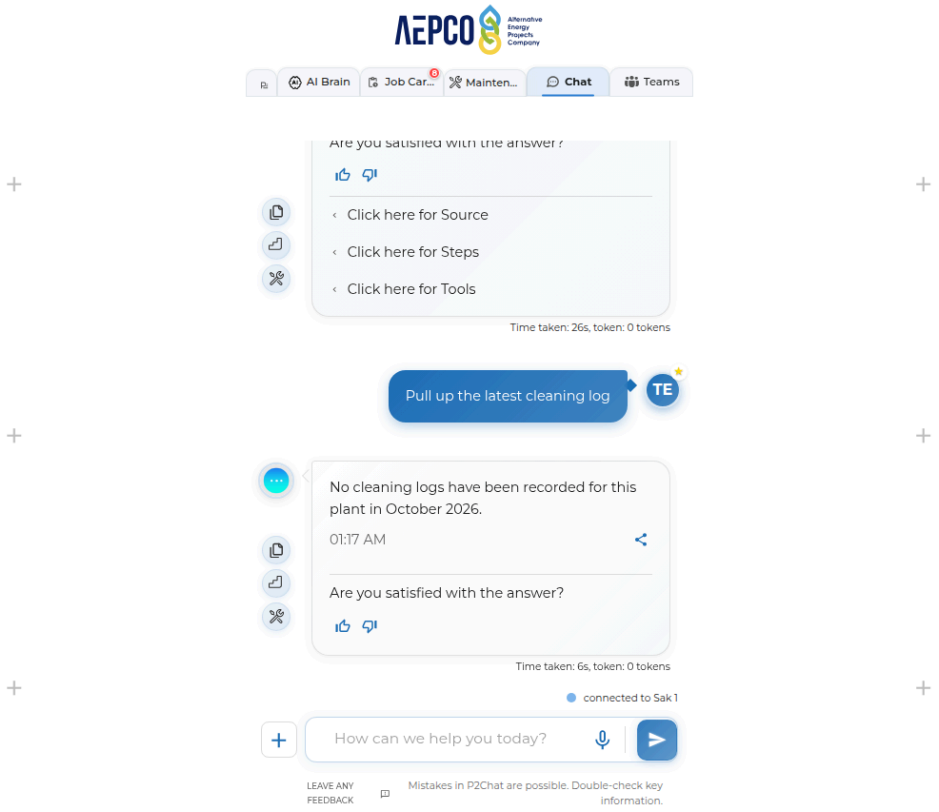
Production • 21.6s

Question: Pull up the latest cleaning log

Response: No cleaning logs have been recorded for this plant in October 2026. 01:17 AM Are you satisfied with the answer? Time taken: 6s, token: 0 tokens connected to Sak 1 LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_entry_agent`
API calls made	`1`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



198. SAK_1

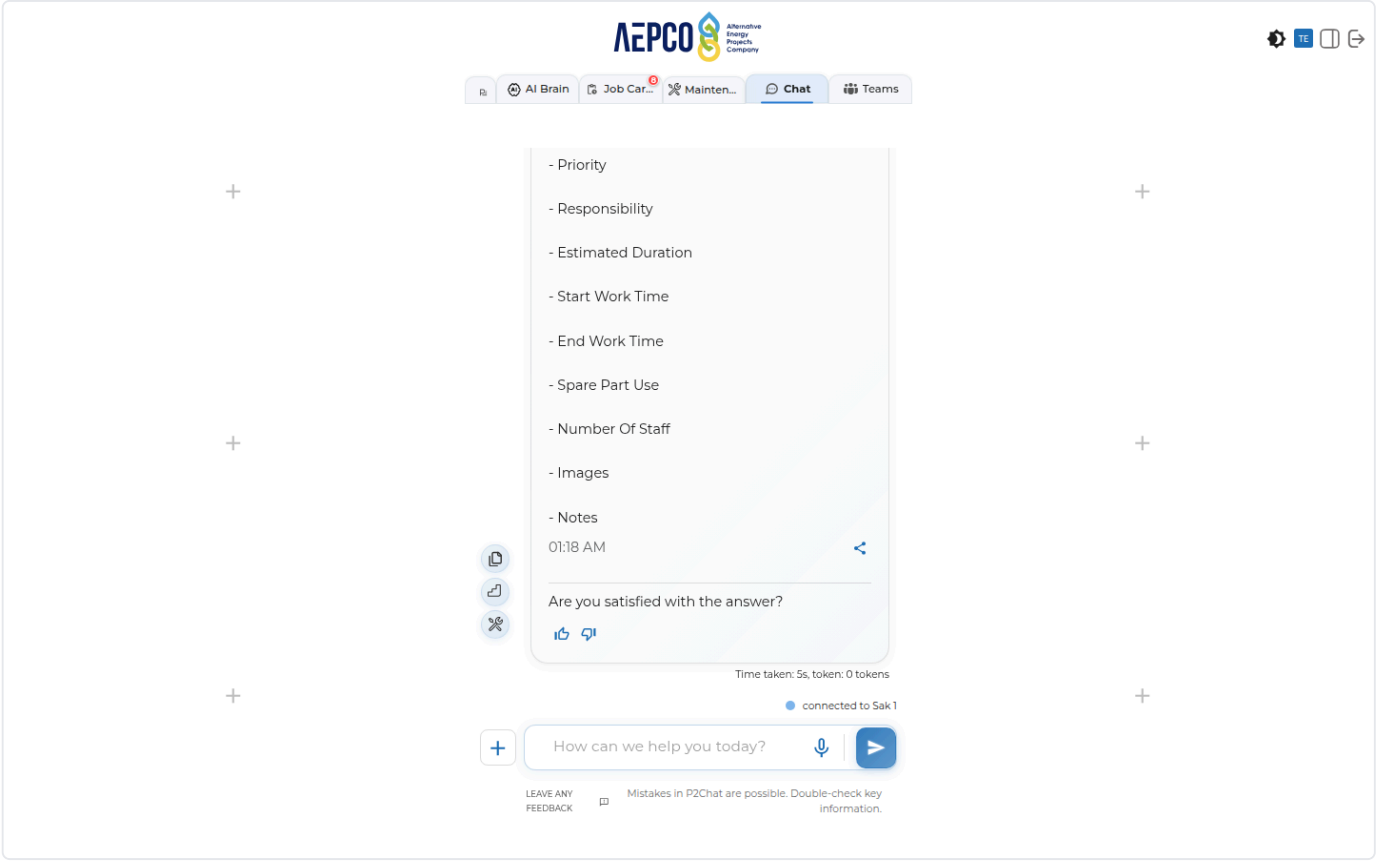
Production • 20.0s

Question: Jim cleaned the pyranometer today, can you add it to the cleaning log?

Response: To add this cleaning activity log, please provide the following required details: - Date - Fault Cause - Action Taken - Status Site will be set automatically from the connected plant. Optional fields you can also provide, but they are not required: - Activity Type (Inspection, Cleaning, HSE Incident, Spare Parts Usage, General Site Activity) - Priority - Responsibility - Estimated Duration - Start Work Time - End Work Time - Spare Part Use - Number Of Staff - Images - Notes 01:18 AM Are you satisfied with the answer? Time taken: 5s, token: 0 tokens connected to Sak 1 LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`data_entry_agent`
API calls made	2
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	0
Network failures	0



199. SAK_1

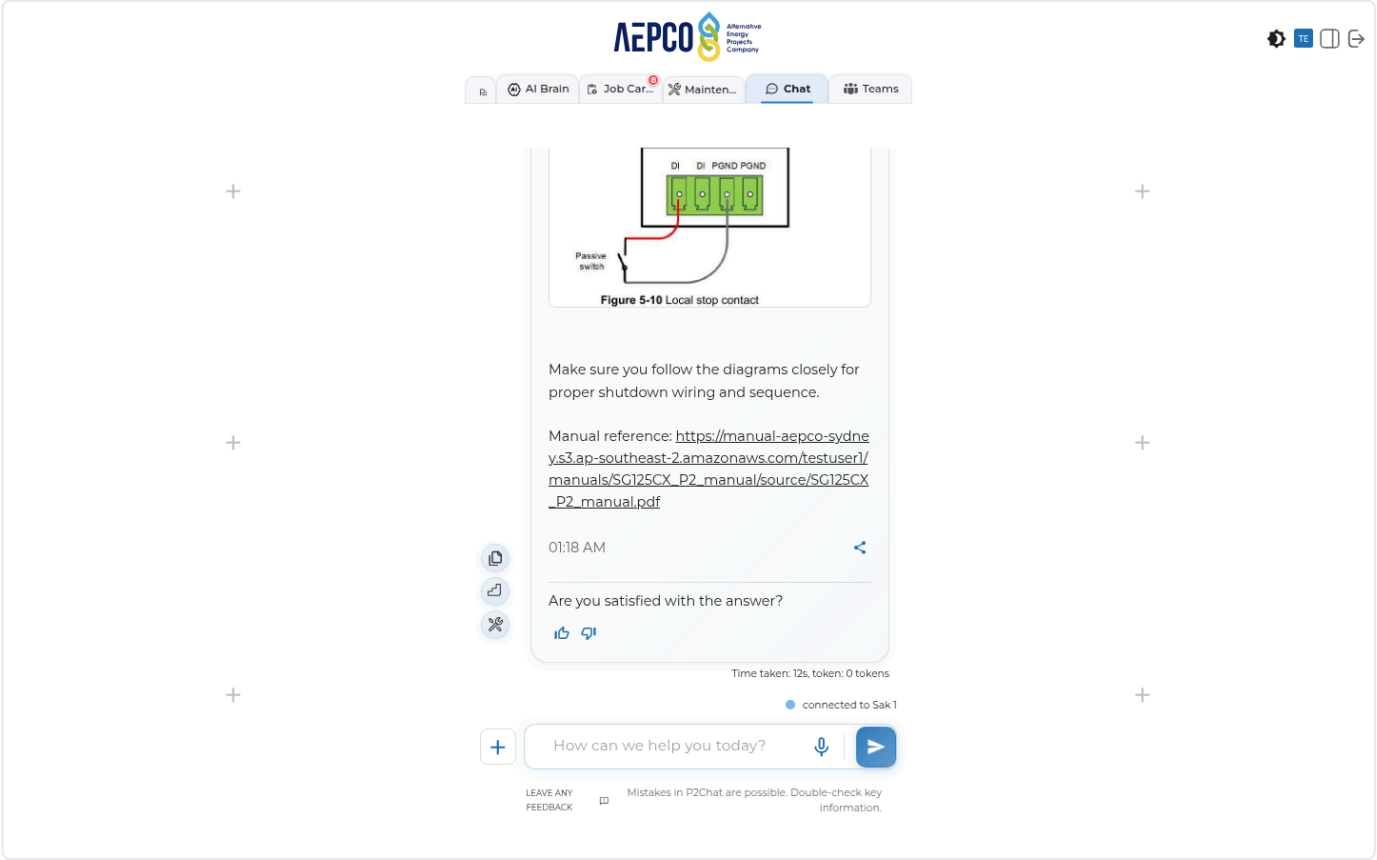
Production • 27.6s

Question: How can I shut down my inverters?

Response: To shut down your inverters, follow these steps from the manual: 1. Identify the local stop contact terminals on the inverter, labeled "DI" and "PGND." See this diagram for their location: 2. Connect a passive switch to the "DI" and "PGND" terminals to establish stop control. The wiring setup is illustrated here: 3. Once the stop switch is in place, activate it to halt inverter operation. Double-check the connection using this diagram: Make sure you follow the diagrams closely for proper shutdown wiring and sequence. Manual reference: https://manual-aepco-sydney.s3.ap-southeast-2.amazonaws.com/testuser1/manuals/SG125CX_P2_manual/source/SG125CX_P2_manual.pdf 01:18 AM Are you satisfied with the answer? Time taken: 12s, token: 0 tokens connected to Sak 1 LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`manual_agent`
Tools used	`functions.manual_rag_tool`
API calls made	2
Last API call	`GET /teamchat/testuser2/mentions/unread → 200`
Console errors	0
Network failures	0



200. SAK_1

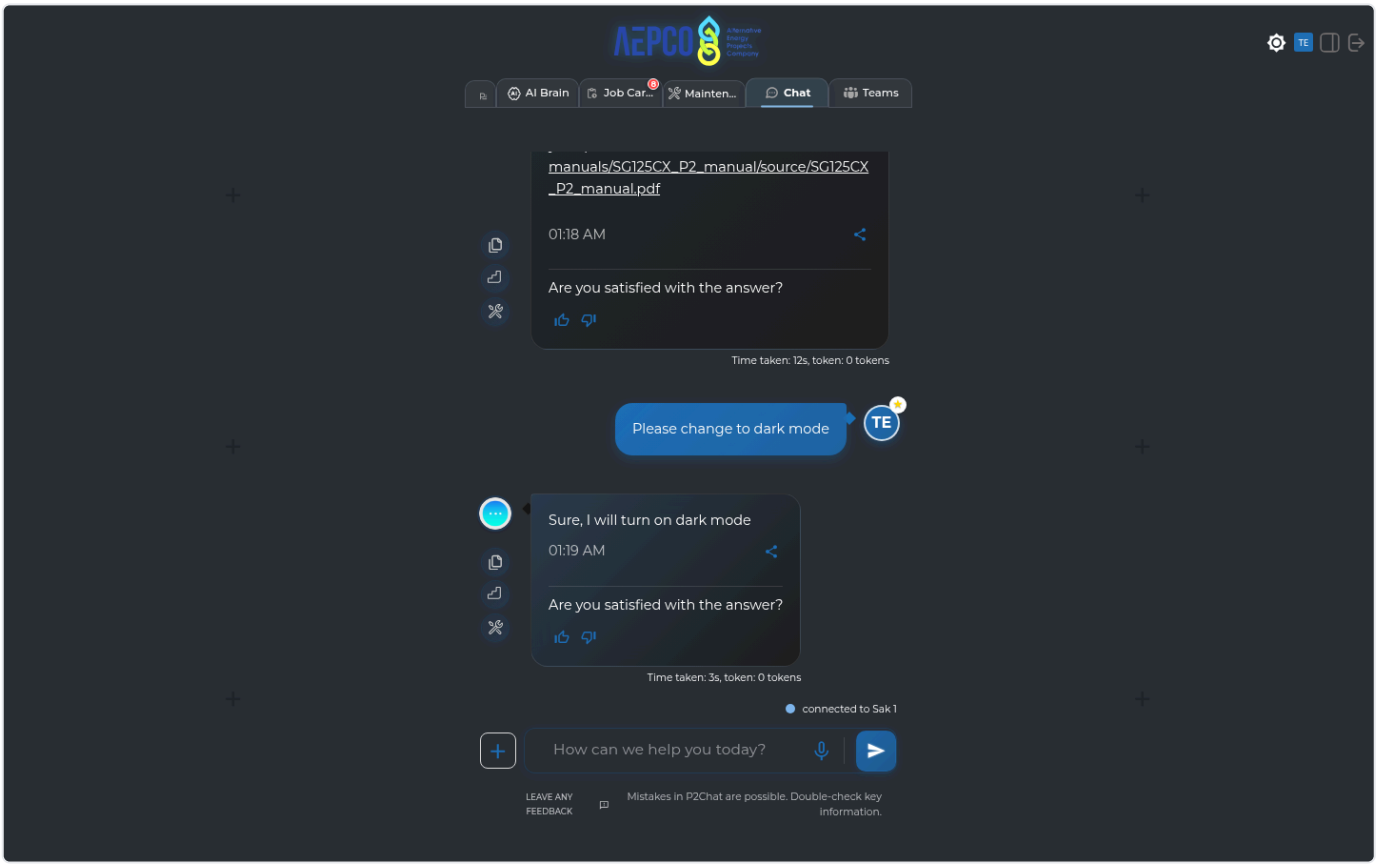
Production • 19.4s

Question: Please change to dark mode

Response: Sure, I will turn on dark mode 01:19 AM Are you satisfied with the answer? Time taken: 3s, token: 0 tokens connected to Sak 1 LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`ui_agent`
API calls made	`1`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



201. SAK_1

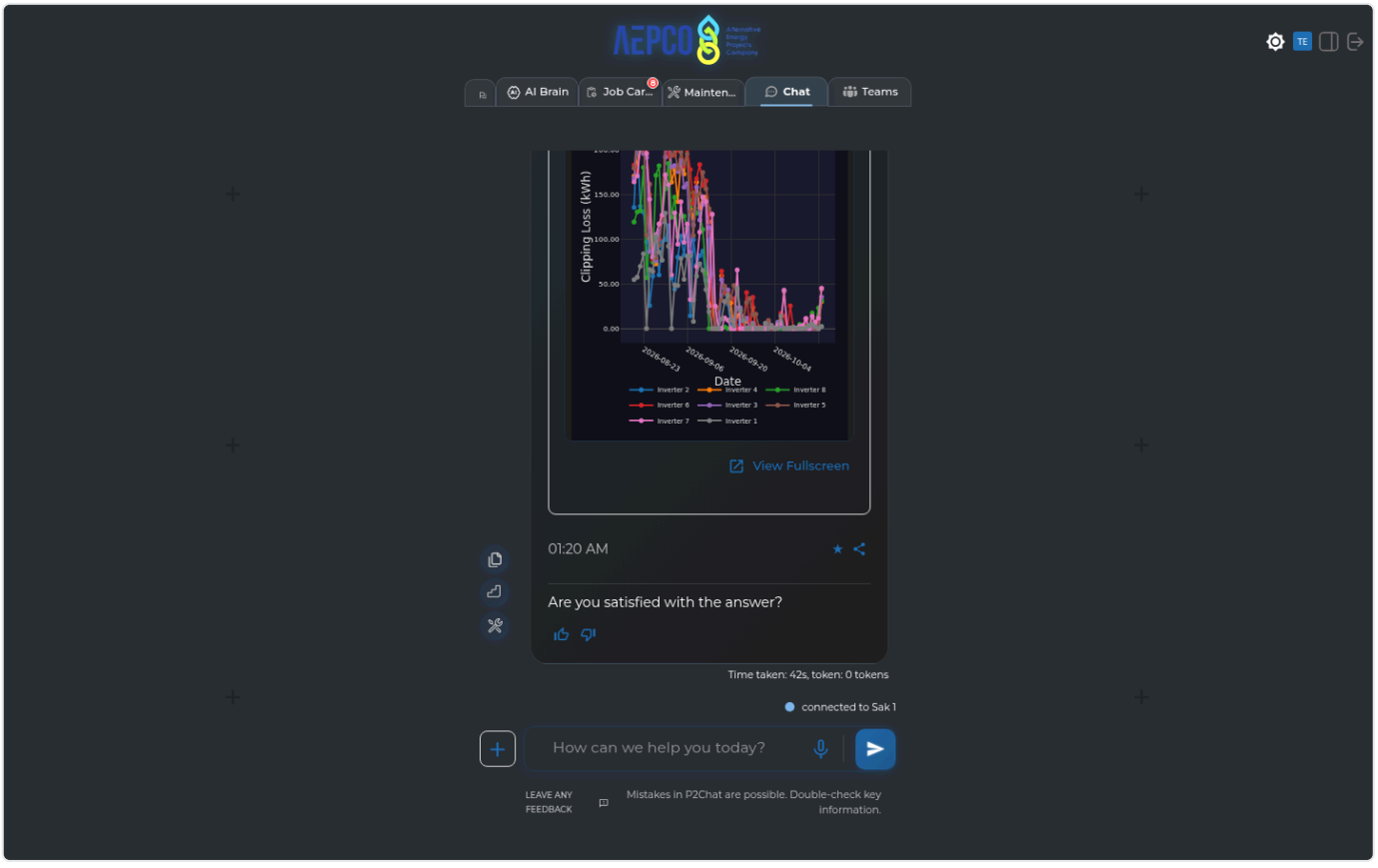
Production • 56.7s

Question: Please figure out how much clipping loss was found in the last 2 months broken by inverters, list them down and plot it as well

Response: Clipping Loss Summary (2026-08-20 to 2026-10-20): - Inverter 1: 5,084.623 kWh - Inverter 2: 5,172.793 kWh - Inverter 3: 5,170.314 kWh - Inverter 4: 5,172.793 kWh - Inverter 5: 5,084.623 kWh - Inverter 6: 5,172.793 kWh - Inverter 7: 5,172.793 kWh - Inverter 8: 5,084.623 kWh Data Quality Status: Good Visualization: - I've created a line chart showing daily clipping loss for each inverter over the last two months. Clipping Loss Summary (2026-08-20 to 2026-10-20): - Inverter 1: 5,084.623 kWh - Inverter 2: 5,172.793 kWh - Inverter 3: 5,170.314 kWh - Inverter 4: 5,172.793 kWh - Inverter 5: 5,084.623 kWh - Inverter 6: 5,172.793 kWh - Inverter 7: 5,172.793 kWh - Inverter 8: 5,084.623 kWh Data Quality Status: Good Visualization: - I've created a line chart showing daily clipping loss for each inverter over the last two months. View Fullscreen 01:20 AM ★ Are you satisfied with the answer? Time taken: 42s, token: 0 tokens connected to Sak 1

TECHNICAL EVIDENCE

Agents used	`data_agent, graph_agent`
Tools used	`functions.calculate_clipping_or_curtailment_losses_ag, functions.clipping_loss_via_requests, functions.generate_graphs_via_code`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`



202. SAK_1

Production • 32.4s

Question: Please repeat the first plotting question, but interchange it with Specific yield on 1 axes, and energy production on the other axes on a single dual axes plot (Memory Test)

Response: I've created the visualization for you. View Fullscreen 01:21 AM ★ Are you satisfied with the answer? Time taken: 18s, token: 0 tokens connected to Sak 1 LEAVE ANY FEEDBACK Mistakes in P2Chat are possible. Double-check key information.

TECHNICAL EVIDENCE

Agents used	`graph_agent`
Tools used	`functions.generate_graphs_via_code, multi_tool_use.parallel`
API calls made	`2`
Last API call	`GET /testuser2/ai_agent_chat → 200`
Console errors	`0`
Network failures	`0`